

Critical mixing in the Sherrington–Kirkpatrick model

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Abstract

At the critical inverse temperature $\beta = 1$, we determine the worst-start total-variation mixing exponent for single-site heat-bath dynamics in the zero-field Gaussian Sherrington–Kirkpatrick model. The mixing time is $n^{2/3+o(1)}$ when every site has rate one, or $n^{5/3+o(1)}$ attempted uniform-site updates, in probability over the disorder.

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1 Introduction

We determine the mixing exponent of single-site heat-bath dynamics at the critical temperature of the zero-field Gaussian Sherrington–Kirkpatrick model. Starting from the worst configuration, the dynamics mixes in time $n^{2/3+o(1)}$ when each site has rate one, or after $n^{5/3+o(1)}$ attempted uniform-site updates. The same per-site exponent governs the relaxation time and the inverse classical logarithmic Sobolev constant.

At criticality, equilibrium fluctuations are much larger than for independent spins. An upper mixing bound must hold from every initial configuration, while the functional bounds must hold for every test function. We obtain these uniform controls in two ways. The first relates a function’s heat-bath energy to the mass of its support. The second proves a classical logarithmic Sobolev inequality. Each argument has its own initialization and entropy estimates.

Let J° be a symmetric matrix with zero diagonal and independent $J_{ij}^\circ \sim N(0, 1/n)$ for $i < j$. On $\Omega_n = \{-1, 1\}^n$, put

$$\mu_J(x) = Z_J^{-1} \exp\left(\frac{1}{2} x^\top J^\circ x\right). \quad (1.1)$$

This is Gaussian SK at inverse temperature one and zero external field. Write Q_i for conditional expectation under μ_J given x_{-i} , and set

$$\mathcal{L}_J = \sum_{i=1}^n (Q_i - I), \quad P_J = \frac{1}{n} \sum_{i=1}^n Q_i, \quad \mathcal{D}_J(f) = \sum_{i=1}^n \mu_J[\text{Var}_{\mu_J}(f \mid x_{-i})]. \quad (1.2)$$

Thus $\mathcal{L}_J = n(P_J - I)$, and a discrete attempt may leave the spin unchanged. Let $t_{\text{mix},J}(\varepsilon)$ and $k_{\text{mix},J}(\varepsilon)$ denote worst-start total-variation mixing times for $e^{t\mathcal{L}_J}$ and P_J^k , respectively. The default threshold is $1/4$.

Define the relaxation time and the classical logarithmic Sobolev inverse constant by

$$R_J = \sup_{\text{Var}_{\mu_J}(f) > 0} \frac{\text{Var}_{\mu_J}(f)}{\mathcal{D}_J(f)}, \quad (1.3)$$

$$C_{\text{LS},J} = \sup_{\text{Ent}_{\mu_J}(f^2) > 0} \frac{\text{Ent}_{\mu_J}(f^2)}{\mathcal{D}_J(f)}, \quad (1.4)$$

where $\text{Ent}_\mu(g) = \mu(g \log g) - \mu(g) \log \mu(g)$ for $g \geq 0$. Both constants use the unscaled sum of heat-bath conditional variances. Thus R_J is the relaxation time for the rate-one-per-site generator; for one attempted update the relaxation time is nR_J . The conditional variance weights in $\mathcal{D}_J$ are retained in every comparison below.

Theorem 1.1 (Critical mixing and functional inequalities). *For every fixed $\varepsilon \in (0, 1/2)$, in probability over the Gaussian disorder,*

$$\begin{aligned} t_{\text{mix},J}(\varepsilon) &= n^{2/3+o(1)}, & k_{\text{mix},J}(\varepsilon) &= n^{5/3+o(1)}, \\ R_J &= n^{2/3+o(1)}, & C_{\text{LS},J} &= n^{2/3+o(1)}. \end{aligned}$$

In each expression, $X_n = n^{a+o(1)}$ means that for every fixed $\delta > 0$, $\mathbb{P}\{n^{a-\delta} \leq X_n \leq n^{a+\delta}\} \rightarrow 1$.

The two functional constants describe complementary forms of relaxation. For every centered observable f , the spectral theorem gives $\text{Var}_{\mu_J}(e^{t\mathcal{L}_J} f) \leq e^{-2t/R_J} \text{Var}_{\mu_J}(f)$, with equality for a slowest eigenfunction. The classical logarithmic Sobolev inequality supplies hypercontractive smoothing for arbitrary densities and implies the Poincaré inequality. Section 20 derives the worst-start bounds from these controls. Let KL denote relative entropy. Theorem 20.2 also gives the same per-site exponent for the inverse of the best rate c such that $\text{KL}(\nu e^{t\mathcal{L}_J} \|\mu_J) \leq e^{-ct} \text{KL}(\nu \|\mu_J)$ for every initial probability law ν on the cube and every $t \geq 0$.

The contribution of this paper is the upper bounds. To state precisely how they combine with the existing lower bounds, we record the critical companion's realized-start and linear-observable theorems [27, Theorems 1.2 and 1.3]. Their quantifiers are stronger than fixed exponent bounds and will be retained throughout.

Theorem 1.2 (Realized equilibrium starts and linear observables). *For every deterministic sequence $t_n \geq 0$ with $t_n = o(n^{2/3})$,*

$$\mu_J \left\{ x : \left\| e^{t_n \mathcal{L}_J}(x, \cdot) - \mu_J \right\|_{\text{TV}} > \frac{1}{4} \right\} \longrightarrow 1$$

in probability over disorder. For every deterministic nonnegative integer sequence $k_n = o(n^{5/3})$, the same assertion holds with $P_J^{k_n}$. Let $\Sigma_J = \text{Cov}_{\mu_J}(X)$ and

$$R_{\text{lin},J} = \sup_{a \in \mathbb{R}^n \setminus \{0\}} \frac{\text{Var}_{\mu_J}(a^\top X)}{\mathcal{D}_J(a^\top X)}.$$

For every positive deterministic $M_n \rightarrow \infty$, with probability tending to one both $\|\Sigma_J\|$ and $R_{\text{lin},J}$ belong to $[n^{2/3}/M_n, M_n n^{2/3}]$.

In the realized-start assertion, one samples an initial configuration from μ_J and holds it fixed before taking the transition law and its TV distance. Averaging that transition law over the initial configuration would instead give μ_J at every time. The last assertion concerns covariance and linear Rayleigh quotients. Its arbitrary-diverging-factor upper bound does not extend here to the full relaxation or logarithmic Sobolev constant: the upper estimates in Theorem 1.1 have every fixed positive exponent slack. For other fixed TV thresholds below $1/2$, the lower bound in Theorem 1.1 follows by testing the slowest gap eigenfunction, as shown in Section 20. The realized-start assertion itself retains its stated threshold $1/4$.

The result concerns exactly the critical, zero-field Gaussian model. It gives the mixing exponent and does not assert a cutoff, a cutoff window, or a leading constant. The two upper-bound arguments retain their separate intermediate conclusions and parameter choices.

The development of the dynamical question. Sherrington and Kirkpatrick introduced the infinite-range spin-glass model in 1975 [31]. Their later analysis with Gaussian couplings studied single-spin Glauber relaxation through a linearized mean-field calculation and Monte Carlo simulations [23, Section VII]. A parallel dynamical mean-field program described correlation and response functions in soft-spin Langevin models; the infinite-range specialization was developed by Sompolinsky and Zippelius [32]. These studies examined relaxation near the transition through correlation and response. The finite-size mixing question asks how long the full distribution, started from the most difficult configuration, takes to approach equilibrium.

Numerical work later examined how the critical time scale grows with the number of spins. Billoire’s study of slow samples used binary couplings $J_{ij} = \pm n^{-1/2}$ [5]. Billoire and Campbell’s subsequent critical study measured the equilibrium autocorrelation $q(t) = n^{-1} \sum_i \langle X_i(0) X_i(t) \rangle$ after equilibration. For heat-bath updates, with time measured in updates per spin, it tested the finite-size scaling form $q_c(n, t) \sim n^{-1/3} F(t/n^{2/3})$ [6, equation (7)]. That critical preprint does not restate the coupling law. We use its reported collapse as numerical evidence about equilibrium autocorrelation, keeping it separate from the Gaussian worst-start question here.

Rigorous high-temperature dynamics. For comparison with the critical law (1.1), inverse temperature β multiplies its exponent by β ; the resulting interaction entries have variance β^2/n . The SK statements below hold with high probability over disorder. The inverse temperature $\beta > 0$ is fixed inside each stated range, and mixing bounds use fixed total-variation accuracy unless ε is displayed. Spectral criteria gave functional inequalities well inside the high-temperature phase. Bauerschmidt and Bodineau proved a logarithmic Sobolev inequality under a spectral-width condition that applies to SK for fixed $\beta < 1/4$ [4, Theorem 1 and Corollary 2]. For Ising spins their energy is the unweighted full spin-flip gradient. The heat-bath form $\mathcal{D}_J$ instead retains conditional variance weights, so that result is not the heat-bath inequality used in this paper. Eldan, Koehler, and Zeitouni obtained a heat-bath spectral gap uniform over external fields under the corresponding shifted spectral condition. Their field-dependent worst-start bound gives polynomial mixing for zero-field SK in the range $\beta < 1/4$ [20, Theorems 1 and 11, Section 5]. Anari, Jain, Koehler, Pham, and Vuong then obtained $O_\beta(n \log n)$ random-site attempts in that range, uniformly over external fields, through a modified logarithmic Sobolev inequality [1, Theorem 12].

Subsequent work increased the temperature range. Anari, Koehler, and Vuong derived a criterion governed by a Volterra equation; it gives $O(n \log n)$ attempted updates for SK below a reported numerical threshold of approximately $\beta = 0.295$ [2, Theorem 3 and Section 1.3]. Wang proved $O_\beta(n \log(n/\varepsilon))$ attempted updates for every fixed $\beta < 1/2$, with high probability simultaneously over external fields and $\varepsilon \in (0, 1)$ [34, Theorem 1.1]. At zero field, Boban, Li, and Oveis Gharan obtained a constant per-site spectral gap and an $O(n^2)$ attempted-update bound for $\beta < 1/2 + \varepsilon_0$, with a universal $\varepsilon_0 > 0$ [7, Theorem 1.5 and Corollary 1.6]. The companion spectral-gap theorem covers each fixed zero-field $0 < \beta < 1$, with a constant that may depend on β [28, Theorem 1.1]. None of these fixed-subcritical bounds supplies a bound at $\beta = 1$ by taking a limit. At that endpoint the critical companion identifies the slow linear and realized-start scales in Theorem 1.2; the present upper bounds match their exponents for the full dynamics.

Static comparisons and the two critical mechanisms. The equilibrium geometry behind the critical scale is accessible through comparison with spins on a sphere. Comets conditioned on the eigenvalues of a GOE completion and averaged its eigenframe, obtaining an exact conditional first-moment identity between cube and spherical partition functions [14, equations (2.1)–(2.2)]. At criticality, Du and Huang developed sharper comparisons for free energy and overlap. Their local

overlap-cell estimates compare partition-function moments [16, Theorem 1.6 and Section 3.3], and their mixed cube–sphere tensor calculation cancels leading terms in a squared norm [17, Section 3.2, Lemma 3.5]. These are static comparisons. The upper bounds here need corresponding control after observations have changed the field and interaction, together with probability estimates strong enough for arbitrary test functions.

Gaussian stochastic localization supplies the framework for revealing those observations while tracking the remaining variance or entropy [12, 19]. The root calculation also develops conditional Kac–Rice methods used for TAP landscapes by Fan, Mei, and Montanari, and by Celentano, Fan, and Mei [11, 21]. Their cited applications have different positive-overlap or fixed-signal hypotheses; the shrinking-overlap critical estimates are proved here. The isotropic route extends the signed spin–Gaussian comparison to observation-dependent fields, then improves a typical-orientation estimate along the observation path. The cap route keeps the reduction of covariance in the field direction while fresh rotations reduce errors in orthogonal directions inside the active spectral subspace. The entropy arguments use these estimates only on their stated path or field domains and pay for their exceptions. Section 4 describes the inputs and outputs of the two mechanisms in detail.

2 Conventions and Gaussian observations

For a full-support law ν on Ω_n , we write

$$\mathcal{D}_\nu(f) = \sum_i \nu[\text{Var}_\nu(f \mid X_{-i})].$$

Conditional expectation given all spins except one is an orthogonal projection on $L^2(\nu)$. The average of these projections therefore has spectrum in $[0, 1]$. Its Dirichlet form is $\mathcal{D}_\nu/n$, a fact used later to pass directly from continuous time to deterministic numbers of attempted updates. For $f(x) = a^\top x$,

$$\mathcal{D}_\nu(f) = \sum_i a_i^2 \nu[\text{Var}_\nu(X_i \mid X_{-i})] \leq \|a\|^2. \quad (2.1)$$

All vector norms are Euclidean and all matrix norms are operator norms unless another norm is indicated. Empirical averages have a factor $1/n$.

We use the orthogonal invariance of a GOE completion. Adjoin independent $J_{ii} \sim N(0, 2/n)$ to J° and call the completed matrix J . The added energy is $\frac{1}{2} \sum_i J_{ii}$ because $x_i^2 = 1$; it is constant on the cube. Hence the Gibbs law, transition kernels, and all functional constants are unchanged. We may condition on the eigenvalues of J and use its Haar-distributed eigenframe in intermediate estimates. Once the conclusion is expressed in these intrinsic quantities, it is a statement about the original off-diagonal disorder.

For matrix A and field h , a tilted spin law is proportional to $\exp(x^\top Ax/2 + h^\top x)$. A Gaussian observation of positive semidefinite precision B has value

$$Y = BX + B^{1/2}G,$$

where G is standard Gaussian independent of X . Expanding the Gaussian likelihood shows that the posterior adds Y to the field and subtracts B from the interaction. Density calculations take place on the range of B ; null directions carry no observation. In the isotropic case $B = sI$, the subtraction is diagonal and the posterior is simply a field tilt of the original spin law. Spectral caps use the same identity with a matrix precision.

For increasing deterministic precisions, independent Gaussian increments give the same experiment. At time s let μ_s be the posterior law, $m_s = \mu_s X$ and $\Sigma_s = \text{Cov}_{\mu_s}(X)$. The isotropic experiment is $Y_s = sX + B_s$, with B_s standard Brownian motion. Its innovation $I_s := Y_s - \int_0^s m_u du$ is Brownian in the observation filtration. The finite-prior filtering identity is

$$d(\mu_s f) = \text{Cov}_{\mu_s}(f, X)^\top dI_s. \quad (2.2)$$

Conditioning also gives

$$\mathbb{E}\mathcal{D}_{\mu_s}(f) \leq \mathcal{D}_{\mu_0}(f). \quad (2.3)$$

This is the finite-prior form of Gaussian stochastic localization [19]. The posterior formula is given in [12, Section 2.4.2, Fact 14 and Equation (6), arXiv version 2]; the entropy drift and associated stability criteria appear in its Equation (27), Proposition 39 and Lemma 40. The contraction (2.3) is the heat-bath Dirichlet-form supermartingale of [20, Lemma 9, arXiv version 2]. The gap companion records these temperature-free identities with the same normalization [28, Section 2].

The critical task is to bound the posterior quantities in these identities on the observation paths actually used. Each route proves those estimates, including the cost of exceptional observations. In particular, a covariance estimate on one random path is not an all-tilts stability theorem, and the classical logarithmic Sobolev inequality will require its own entropy and layer argument.

For a nonnegative density g with $\mu g = 1$, $\text{KL}(g\mu\|\mu) = \text{Ent}_\mu(g)$. The classical inequality studied here is $\text{Ent}_\mu(f^2) \leq C_{\text{LS}}\mathcal{D}_\mu(f)$ for all real f . Here the energy acts on f , while the entropy acts on f^2 . An estimate using the logarithmic entropy-production form of g would be a different inequality and does not replace this one. For a fixed law on the finite cube every real function is admissible. Unless a varying parameter is explicitly specified, all constants and tolerances are chosen before n tends to infinity.

3 Companion inputs and the critical endpoint

The critical companion supplies exactly the two statements collected in [Theorem 1.2](#): its Theorems 1.2 and 1.3 [27]. The spectral, noncentral spherical, and spectral-cap estimates required by the upper bounds are proved below. The high-temperature gap companion supplies observation identities and deterministic field and residual tools, together with fixed-subcritical consequences under their stated hypotheses [28]. We describe these hypotheses before using the tools at a critical endpoint.

The field equations are of Thouless–Anderson–Palmer type [33]. We use the following normalization, including the rank-one derivative of the self-consistent overlap.

For an interaction of variance j/n , set

$$\begin{aligned} m &= \tanh y, & q &= n^{-1} \|m\|^2, \quad V_y = \text{diag}(1 - m_i^2), \\ F_{J,h}(y) &= y - h + [j(1 - q)I - J]m, \\ H_J(y, A) &= I + A^{1/2}[j(1 - q)I - J - 2jmm^\top/n]A^{1/2}. \end{aligned}$$

The stable-field hypothesis is uniform over a region: every y with $\|F_{J,h}(y)\| \leq \rho_0\sqrt{n}$ and every diagonal $0 \leq A \leq (1 + \eta)I$ sufficiently close to V_y in normalized Frobenius norm must satisfy $H_J(y, A) \succ cI$. The constants ρ_0, η, c are fixed positive margins. The whole-region assertion permits root continuation and spin-flip buffers; positivity at one selected root would not supply these uses.

The matrix diagnostics used below concern finite *words*: ordered products of bounded diagonal matrices and copies of the interaction matrix, with at most one controlled inverse factor. They bound the operator norm, the entrywise ℓ^4 norm off the diagonal, and the Euclidean norm of the

diagonal error from its Gaussian contraction prediction. The permitted diagonal classes and the actual inverse margins are stated in [Theorems 13.1](#) and [17.2](#); the distinction between adaptive diagonals and prescribed vector tests is recorded below.

Deterministic consequences. Under the specified finite-word diagnostics and stable-field hypotheses, the root r is unique. The gap paper’s Proposition 7.2 gives

$$\|\text{Cov}_{\mu_h}(G, X)\| \leq C(\text{Var}_{\mu_h}(G) + \mathcal{D}_{\mu_h}(G))^{1/2}. \quad (3.1)$$

Its proof also establishes $|\mu_h[G(X - \tanh r)^\top e]| \leq C(\mu_h G^2 + \mathcal{D}_{\mu_h}(G))^{1/2}$ for $\|e\| \leq 1$. In particular,

$$\|\mu_h X - \tanh r\| \leq C. \quad (3.2)$$

This is an unnormalized Euclidean mean estimate; it is not an approximation for the derivative of the mean map. For $N = \mu_h f^2 > 0$ and $\nu = f^2 \mu_h / N$, Proposition 7.3 gives, for each sufficiently small fixed $\delta > 0$,

$$\frac{\|\nu X - \tanh r\|}{\sqrt{n}} \leq C\delta + \frac{C_\delta}{\sqrt{n}} \left(1 + \frac{\mathcal{D}_{\mu_h}(f)}{N}\right). \quad (3.3)$$

The leading constant C is fixed by the stability margins. The constant C_δ and the finite diagnostic length may then depend on δ and on the resulting residual-recipe depth. This distinction will matter when we choose an integrated entropy-drift budget before choosing the residual tolerance that realizes it. The displayed statements and the uncentered proof consequence come from [\[28, Section 6, Lemma 7.1, Proposition 7.2 and its proof, Proposition 7.3\]](#).

Provider hypotheses. In the provider’s random word estimates, $j < 1$ and $j(1 + \eta)^2 < 1$ are fixed. When the interaction is β times a standard SK matrix, this variance parameter is $j = \beta^2$. Words have fixed length and at most one inverse factor; that inverse requires a separate positive margin for the actual matrix being inverted. Bilinear estimates apply to prescribed finite families of deterministic vectors. The random stable-field theorem is along ordinary Gaussian observations of a zero-field Gibbs sample. It is not a theorem for every field. The terminal observation-gap theorem also fixes $j < 1$, although its input spin law may be arbitrary [\[28, Proposition 5.1, Corollary 5.2, Proposition 8.3\]](#).

Some deterministic ingredients have broader domains. The root/inverse lemma permits every fixed $j > 0$ provided $H_J(y, V_y) \succeq cI$ throughout its entire small-residual region. That root lemma alone needs only $A = V_y$; the nearby-diagonal condition is an additional hypothesis for the implicit residual recipes. The local-volume lemma and the signed two-spin curvature inequality similarly have their explicit deterministic hypotheses [\[28, Lemmas 5.6, 5.7 and 8.1\]](#). We apply these deterministic tools at $j = 1$ only after verifying their hypotheses. The two routes prove the needed restricted norm–trace random diagnostics, actual adaptive-inverse margins, whole-region stability, and terminal extensions locally. Recipe depth is fixed before the number of allowed spin flips, buffer widths, and eventual large- n threshold are used. No critical random estimate is obtained by taking $\beta \uparrow 1$ in the fixed-temperature spectral-gap theorem.

4 Two upper-bound mechanisms

Both proofs use the geometry of the upper spectral edge, but transport different estimates. The limiting GOE spectral density decays as the square root of the distance to its upper edge 2, making $nu^{3/2}$ the characteristic dimension of an edge window of width u . The spectral estimates below

make this count precise at the scales where it is used. A spherical spin law can be represented as a Gaussian conditioned on $\|X\|^2 = n$, so its resolvent turns this edge geometry into explicit covariance estimates. The first route observes all directions at the same precision. The second observes only the soft edge directions and enlarges that set one spectral cap at a time.

All changes of law below use finite-dimensional likelihood identities with explicit partition normalizers. The planted law weights the disorder by its partition function and introduces an auxiliary spin. Estimates under that law, or under a further tilt, are returned to ordinary disorder with the stated probability cost. No critical contiguity assertion is used; the distinction at the spiked-Wigner threshold is discussed in [29, Appendix A].

4.1 Isotropic observations: from covariance to a support profile

The endpoint of the first route is [Theorem 14.2](#): for every fixed $\omega > 0$, on one disorder event of probability tending to one,

$$\mathcal{D}_J(f) \geq c_\omega n^{-2/3-\omega} \log(1/p) \mu_J(f^2), \quad 0 < p := \mu_J(f \neq 0) \leq \frac{1}{2},$$

simultaneously for every nonzero real f . The factor $\log(1/p)$ says that a function concentrated on a smaller set must pay more heat-bath energy relative to its squared norm. For an indicator, $\mathcal{D}_J(\mathbf{1}_A) = \sum_{x \in A} \mu_J(x) \sum_i Q_i(x, A^c)$ is the stationary probability current out of A per unit of per-site time. The profile controls the early evolution of a point-start density; the spectral gap controls the final decay.

A local covariance comparison. At isotropic precision s , write $r = \sqrt{s}$ and $k = nr^3 = ns^{3/2}$. The spectral and spherical estimates in [Section 5](#) locate the Gaussian saddle and control its length densities, projections, and replica overlaps. The unrestricted rotation comparison in [Section 6](#) then compares the spin and spherical partition functions and their replica moments. These estimates give the continuous reference and the probability budgets for the root analysis.

The root analysis studies solutions of

$$y + (1 - q)m - Jm = h_s, \quad m = \tanh y, \quad q = n^{-1} \|m\|^2.$$

The Kac–Rice formula in [Section 8](#) expresses planted expectations of sums over regular solutions as an integral against the observation density and the absolute Jacobian determinant. After gauging the planted spin to $\mathbf{1}$, the conditional matrix calculation in [Section 9](#) isolates three directions: m ; the component of $\mathbf{1}$ orthogonal to m ; and the component of the root field y orthogonal to both. Their span is $\mathcal{P}$. The last direction records the remaining cubic part of the root profile. Conditioning on these directions leaves Haar rotations on $\mathcal{P}^\perp$. [Section 10](#) compares their Haar weights with the determinant weights in the root count and locates the Gaussian trace. The exclusions are integrated over the root and conditional matrix data, with the imposed spectral restriction. The degree-one argument controls multiplicity when this root count is returned to the planted field law.

Here is the cancellation at the center of the comparison. Put $z = X - m$. For each replica, fix its projection on $\mathcal{P}$ and its translated length; for a pair, also record the overlap of their residual components in $\mathcal{P}^\perp$. The two replica types are a product spin prior with means m and a variable-variance Gaussian prior. Multiplication by the common quadratic weight $\exp\{z^\top (J - (1 - q)I)z/2\}$ turns the product spin prior into the posterior μ_s in the translated coordinates: the root equation cancels its linear term. The quantities $\mathcal{R}_{ij}$ in (11.4) are their conditional saddle-density ratios for this residual overlap, relative to a common continuous reference, for the four ordered type pairs. A projection multiplier first matches the one-replica saddle densities. The signed combination

$\mathcal{R}_{pp} - \mathcal{R}_{pg_0} - \mathcal{R}_{g_0p} + \mathcal{R}_{g_0g_0}$, with the common projection weight in (11.4), cancels common errors and errors depending on only one replica type. The actual spin statistics retain their lattice measures: Theorems 11.2 and 11.3 transfer only the smooth leading terms to common volume, after bounding each remainder on its own measure.

This signed estimate, including the centered matrix kernel of Theorem 11.4, yields Theorem 11.1. For $k \geq \log^{10} n$ and sufficiently small r , most Haar orientations on each retained root orbit satisfy $\|\Sigma_s\| \leq (1+\epsilon)/s$, together with the weighted covariance and mean bounds in (11.1). Those additional bounds control derivatives along rotations in the next stage.

A probability bound along the observation path. Theorem 12.3 combines short backward variance transfers with concentration around the local majority just obtained. Starting from fixed positive observation times, it bounds, at each deterministic time with $0 < s \leq s_0$ and $k \geq K_{\log} \log n$, the integrated planted mass of $\mathcal{G} \cap \{\|\Sigma_s\| > (1+\epsilon)/s\}$ by e^{-ak} . Here $\mathcal{G}$ is the rotation-invariant good-spectrum event in the proposition. The positive-time covariance and terminal gap inputs are proved at the critical parameter in Section 13; the subcritical spectral-gap theorem is not applied at its endpoint. In the intervening polylogarithmic window, the induction uses the unrestricted comparison of Section 7 in place of root-preserving rotations.

For the bottom range, put $\ell = \log \log(n + 1000)$. At each deterministic time with $k \leq K_{\log} \log n$, Theorem 12.4 bounds the integrated planted mass of $\mathcal{G} \cap \{\|\Sigma_s\| > Cn^{2/3}\ell^C\}$ by n^{-B} for any fixed $B > 0$. Its finite-replica expansion retains the central prefactors and lattice covolumes. After conversion to common volume, the alternating sum cancels every term that omits a replica label; increasing the fixed moment order then supplies the required probability exponent. These are estimates along the observation law at deterministic times, with their stated integrated exceptions.

Entropy and the uniform event. Section 14 first transfers the planted bounds to one typical ordinary-disorder event, before choosing f . After normalizing $\mu_J(f^2) = 1$, the restricted step assumes $0 < p := \mu_J(f \neq 0) \leq 1/2$ and $\|f\|_\infty^2 \leq 100p^{-2}$. It treats small, moderate, and order- n values of $\log(1/p)$ separately. Variance transport uses the bottom-scale estimate in the first range; replica moments and the sharp covariance bound retain entropy in the second; square-reweighted mean estimates control the last range. The product endpoint converts retained entropy to heat-bath energy, giving Theorem 14.1. Dyadic layers remove the bounded-test restriction and prove the support profile stated above.

4.2 Spectral caps: from a radial deficit to log-Sobolev

The second route proves Theorem 19.3: for every fixed $\rho > 0$, with probability tending to one,

$$\text{Ent}_{\mu_J}(f^2) \leq n^{2/3+\rho} \mathcal{D}_J(f) \quad \text{for every real } f.$$

This classical logarithmic Sobolev inequality controls the smoothing of arbitrary densities through hypercontractivity and also implies the Poincaré inequality.

The cap and the radial deficit. Let $d = 2I - J$. At scale p , observe with precision $B_p = (p^2 - d)_+$ and write $J_p = J - B_p$. On the spectral event used below, the active projection $P_p = \mathbf{1}_{\{d \leq p^2\}}$ has rank of order $k_p = np^3$; the cap makes J_p scalar on its range. For h in this range, the spherical comparison uses $R_p = (\max(d, p^2) + \alpha)^{-1}$, where α is chosen so that $\text{Tr } R_p + \|R_p h\|^2 = n$. Theorem 15.1 controls partition values and projected means uniformly on each prescribed ball $\|h\| \leq Rp^2 M_p$, where $M_p = \sqrt{np}$ and R is fixed.

Changing the field radially changes the length of the Gaussian mean, so the sphere constraint also changes α . This reduces the radial response relative to the frozen resolvent R_p . The induction records that reduction explicitly. The true potential is $\phi_p(h) = \log Z_{J_p}(h)$, whose Hessian is the capped Gibbs covariance. For this potential, $K_p = (P_p \nabla^2 \phi_p(h) P_p)|_{\text{ran } P_p}$. During initialization, K_p instead denotes the Hessian of an auxiliary potential on the same space. Restrict R_p there as well. With $e = h/\|h\|$, on the prescribed annulus the matrix $D = p^2(K_p - R_p)$ has a radial entry at most -0.30 , small transverse and mixed blocks, and a bounded lower error; see (16.2). This negative radial correction is the *radial deficit*. It provides the gain needed to propagate the covariance estimate to a smaller cap.

Transport and initialization. The curvature induction proceeds from larger caps to smaller caps. After the estimates have been proved, the entropy argument uses them along the forward observation process, in which p and B_p increase. For a child scale $c = \gamma p$, the exact Gaussian convolution (16.3) expresses the child potential through the parent. The exact increment formula conjugates the covariance of the parent field under the transition law by the inverse precision increment, then subtracts that inverse increment, to obtain the child's Hessian. The Brascamp–Lieb upper bound retains the radial correction through the inverse in (16.14); the variational Fisher lower bound controls the opposite side and the mixed block. Averaging the rank-one correction regenerates the radial deficit by (16.20). Meanwhile the still hidden eigenframe inside the flat parent space is Haar. Its compression makes the transverse error small enough at the child scale. This conditional rotation is used under the canonical Gaussian bridge before the change to the actual tilted transition. Theorem 16.1 is the resulting one-step statement.

The initial radial deficit comes from the small-field TAP expansion in Section 17, specifically Theorem 17.5. Theorem 18.2 uses an independent conditional completion of the first cap to construct an auxiliary potential with the curvature invariant and a bounded Euclidean gradient error from the true potential. The companion estimate used for that error is at one fixed $0 < \beta < 1$. Theorem 18.4 contracts the error over a finite number of cap steps, and Theorem 18.5 then transfers the invariant to the true potential. The induction continues down to $np^3 \asymp \sqrt{\log n}$. For fields $h \in \text{ran } P_p$, its final output, Theorem 18.6, gives a sharp $(1 + \epsilon)/p^2$ upper bound for the covariance projected onto $\text{ran } P_p$ on a thin saddle annulus, and a C/p^2 bound for the same projected covariance on any prescribed bounded field ball, at every scale in the stated interval.

Entropy retention and layers. Write $N = \mu_J(f^2) = e^{-L} > 0$ and $\nu = f^2 \mu_J / N$. The bounded-test reduction assumes $0 \leq f \leq 1$, $\mu_J(f > 0) \leq 1/2$, $\text{KL}(\nu \| \mu_J) \geq L/2$, and $L \leq Cn$. The disorder event is chosen before f . For $L \leq \lambda n$, where the small fixed $\lambda > 0$ is chosen after the curvature and ordinary-path tolerances, Section 19.3 chooses a first cap with dimension comparable to $\max\{\sqrt{\log n}, c_0 L\}$. The curvature bounds control the information revealed as the cap grows, retaining enough entropy for the product estimate in Section 19.5. For $\lambda n \leq L \leq Cn$, Section 19.6 argues by contradiction under $\mathcal{D}_J(f)/N \leq n^{-2/3}L$. Square-reweighted mean bounds and relative entropy of stopped observation paths retain order- n entropy until the product estimate contradicts that assumption. The noise diagnostics and stability tolerances are chosen before the test function. Finally, Theorem 19.1 passes from the bounded-test energy estimate to set capacities, then to arbitrary half-supported functions, and then by a median decomposition and the entropy shift (19.5) to the classical logarithmic Sobolev inequality.

The final dynamical transfer. Section 20 converts either functional endpoint to worst-start mixing: the support profile truncates large densities, while logarithmic Sobolev hypercontractivity

supplies the second smoothing argument. The identity $\mathcal{L}_J = n(P_J - I)$ and the nonnegative spectrum of P_J keep the factor n between per-site time and attempted updates explicit. The linear lower tests and [Theorem 1.2](#) supply the opposite exponents for mixing and the functional constants. The companion's realized-start and arbitrary-diverging-factor quantifiers remain those stated in [Theorem 1.2](#).

5 Spectral and spherical estimates for the support-profile argument

The support-profile argument begins with estimates that involve only the spectrum and unrestricted rotation orbits. These estimates have two uses. At positive observation times they provide fixed-rate overlap concentration and exponential-moment bounds with arbitrarily prescribed exceptional-set rates. At the smallest times a more accurate comparison with the spherical model gives covariance estimates and an independent lower-bound argument.

5.1 Conventions and the observation law

The matrix J has the GOE normalization: its independent off-diagonal entries have variance $1/n$, and its diagonal entries have variance $2/n$. The diagonal does not affect the spin law. Write π_c for uniform probability measure on $\{-1, 1\}^n$, and π_o for uniform probability measure on the sphere of radius $\sqrt{n}$. Define

$$Z(h) = \int \exp\{x^T J x / 2 + h^T x\} d\pi_c(x), \quad Z_o(h) = \int \exp\{x^T J x / 2 + h^T x\} d\pi_o(x).$$

Thus the partition functions use probability, rather than counting or surface-area, normalization. The spin law at field h is μ_h . Empirical averages are denoted by $\langle a \rangle = n^{-1} \sum_i a_i$; vector norms are Euclidean, matrix norms without a subscript are operator norms, and $\|\cdot\|_{\text{HS}}$ is the Hilbert–Schmidt norm.

Given J , sample X from μ_0 and observe

$$h_s = sX + B_s, \quad r = \sqrt{s}, \quad k = nr^3,$$

where B is an independent standard Brownian motion in $\mathbb{R}^n$. Completing the square in the Gaussian likelihood shows that the conditional spin law is $\mu_s := \mu_{h_s}$. Its mean and covariance will be denoted by m_s^{post} and Σ_s , respectively.

There are two probability laws for the disorder. The ordinary law is the centered GOE law. The planted law is its size bias by $Z(0)/\mathbb{E}Z(0)$, together with the Gibbs spin and observation. Under the present normalization $\mathbb{E}Z(0) = e^{n/4}$. Under the planted law, X is uniform on the cube and

$$J = W + XX^T/n,$$

where W is an independent centered GOE matrix. This follows directly by completing squares in its Gaussian density. A second exact identity will be used repeatedly. Fix the spectrum and write $J = U\Lambda U^T$, randomizing the finite eigenframe choices so that U belongs to the proper rotation group.

In eigen-coordinates $c = U^T h_s$, the planted joint law of (U, c) has density

$$\frac{Z(Uc)}{Z_o(Uc)}$$

relative to Haar measure in U and the spherical-model observation law in c . In this ratio both partition functions use the co-rotated matrix. Indeed the conditional density of U is $Z_U(0)/Z_o(0)$, whereas the observation density contributes $Z_U(Uc)/Z_U(0)$ times a rotation-invariant Gaussian factor. Averaging $Z_U(Uc)$ over U gives $Z_o(Uc)$. In particular, the marginal law of c is exactly the spherical observation law.

Constants may change from line to line. All rate budgets in this section are fixed before n tends to infinity. At a small scale, an $o(1)$ error may also be made small by decreasing a fixed upper bound on r ; the time itself need not tend to zero. Thus a statement with exceptional mass e^{-Bk} , for every fixed B , means that the allowed upper scale and the size threshold may depend on B and on the desired fixed accuracy. Polynomial factors are absorbed by rate slack when $k/\log n$ is sufficiently large.

5.2 Resolvents at the upper edge

Put

$$\ell = \log \log(n + 1000), \quad d_0 = n^{-1/3}\ell, \quad b_{\text{sc}}(z) = \frac{z - \sqrt{z^2 - 4}}{2}.$$

Let ρ_{sc} be the semicircle probability law on $[-2, 2]$. The next estimates are imposed on the *ordinary* disorder. Later planted probabilities will be intersected with this spectral event; the event need not have large planted probability.

Lemma 5.1 (Spectral estimates). *There are events of ordinary probability tending to one on which, uniformly for $d_0 \leq d \leq D_*$ with any fixed $D_* < \infty$,*

$$\lambda_{\max}(J) \leq 2 + o(d_0^2), \quad \frac{1}{n} \text{Tr}(2 + d^2 - J)^{-1} = b_{\text{sc}}(2 + d^2) + o\left(\frac{d}{\sqrt{nd^3}}\right). \quad (5.1)$$

For every fixed integer $p \geq 2$, the trace of the p th resolvent power is asymptotic to its semicircle value and is $O_p(nd^{3-2p})$. There are order nd^3 eigenvalues at depth comparable to d^2 below 2, and at most Cnd^3 within an upper-edge interval of that width. Moreover $\|J\| \leq 3$ and the empirical spectral measure converges to ρ_{sc} . Fixed-rank interlacings preserve the resolvent estimates when the evaluation point stays at least a fixed positive multiple of d^2 from every pole in both spectra.

Proof. The Gaussian–chi representation is the classical tridiagonal construction; see [18, Theorem 2.1 and Section 2.3] for the general beta-ensemble model. In the present normalization, successive orthogonal Gaussian elimination gives a tridiagonal matrix with the same spectrum as J . Its independent diagonal entries are $N(0, 2/n)$ and its adjacent entries are $\chi_{n-i}/\sqrt{n}$, $1 \leq i < n$. To see the independence, rotate the first column below the diagonal onto the second coordinate. Orthogonal invariance leaves a fresh independent GOE matrix on the remaining coordinates, and the procedure can then be iterated.

Let A be the deterministic Jacobi matrix with zero diagonal and adjacent entries $\sqrt{1 - i/n}$, and write the tridiagonal matrix as $A + \Delta$. Centered entries of Δ are sub-Gaussian at scale $n^{-1/2}$. Gaussian norm concentration and the second moment of a chi variable give adjacent-entry means $O((n(n - i))^{-1/2})$. For

$$R = (2 + d^2 - A)^{-1}, \quad a_i = (d^2 + i/n)^{1/2},$$

the positive walk expansion gives

$$(R^p)_{ii} \leq C_p a_i^{1-2p}, \quad p \geq 1. \quad (5.2)$$

Indeed the positive series

$$(R^p)_{ii} = \sum_{j \geq 0} \binom{p+j-1}{j} (2+d^2)^{-p-j} (A^j)_{ii}$$

is a sum over returning walks. Their count divided by 2^j is $O((j+1)^{-1/2})$. A walk staying above $i/2$ gains $e^{-cij/n}$ from the adjacent weights, so its contribution is bounded by

$$C_p \sum_{j \geq 0} (j+1)^{p-3/2} e^{-c(d^2+i/n)j} \leq C_p a_i^{1-2p}.$$

By reflection and the binomial tail bound, the relative count of walks that visit below $i/2$ is at most $Ce^{-ci^2/(j+1)}$. Their sum is at most $C_p d^{1-2p} e^{-cdi}$. Since $nd^3 \geq 1$, $di \geq i/(nd^2)$; exponential decay in this last quantity bounds the sum by $C_p d^{1-2p} (1+i/(nd^2))^{(1-2p)/2}$, which is again $C_p a_i^{1-2p}$.

The perturbation is small in the resolvent metric already at $d = n^{-1/3} \sqrt{\ell}$:

$$\|R^{1/2} \Delta R^{1/2}\| = o(1) \quad \text{with probability tending to one.} \quad (5.3)$$

The block-energy estimate below follows the partial-sum and summation-by-parts method used in [24, Proposition 7 and Lemmas 8–9]; see also [30, Lemma 5.6 and Section 6]. The resolvent precision needed here is verified explicitly. With $b_i = \sqrt{1-i/n}$ and $b_0 = b_n = 0$, the Jacobi energy is

$$u^\top (2+d^2-A)u = \sum_{i < n} b_i (u_{i+1} - u_i)^2 + \sum_i (2+d^2 - b_i - b_{i-1}) u_i^2.$$

It controls

$$\sum_i a_i^2 u_i^2 + \sum_{i < n/2} (u_{i+1} - u_i)^2.$$

The identity $2\Delta_{i,i+1} u_i u_{i+1} = \Delta_{i,i+1} (u_i^2 + u_{i+1}^2) - \Delta_{i,i+1} (u_{i+1} - u_i)^2$ separates each noisy adjacent term into a diagonal contribution and a squared difference.

The maximum entry of Δ is $o(1)$, so the difference terms and the terms on the bulk half are harmless. Divide the first half into consecutive blocks of length comparable to a_i^{-1} , merging a short last block. The values of a_i are comparable within each block. Sub-Gaussian maximal bounds, applied separately to each shifted entry sequence, show that centered diagonal partial sums from the beginning of a block are at most ϵa_i , except with probability $Ce^{-c\epsilon^2 n a_i^3}$ per block.

For a block B of length comparable to a_B^{-1} , summation by parts against a centered diagonal sequence ζ_i with these partial sums gives

$$\left| \sum_{i \in B} \zeta_i u_i^2 \right| \leq C\epsilon \left(a_B^2 \sum_{i \in B} u_i^2 + \sum_{i, i+1 \in B} (u_{i+1} - u_i)^2 \right).$$

To see the scale, average the endpoint value of u^2 over the block and bound the variation of u^2 by Cauchy–Schwarz. Multiplication by the partial-sum bound ϵa_B then gives the two terms displayed. The first-half means are $O(n^{-1})$, a relative $O((na_B^2)^{-1}) = o(1)$ contribution to the potential term.

A dyadic shell contains $O(na_B^3)$ blocks. At $d = n^{-1/3} \sqrt{\ell}$, the smallest value of na_B^3 is $\ell^{3/2}$. Choose $\epsilon \downarrow 0$ with $\epsilon^2 \ell^{3/2} \gg \log \ell$. The block counts and the exponents then grow geometrically on successive shells, so the exceptions are summable. Summing the block inequalities proves (5.3). Positivity of $R^{-1} - \Delta$ at this value of d also places the upper edge below $2 + n^{-2/3} \ell = 2 + o(d_0^2)$.

The relative perturbation estimate has placed the spectrum below the resolvent pole. The trace estimate requires more precision: its error must be small on the unnormalized trace scale. We first

estimate the linear and quadratic perturbation terms, then identify the deterministic Jacobi trace through its Hermite equation.

Expanding the resolvent and using (5.3) gives

$$\mathrm{Tr}(2 + d^2 - J)^{-1} = \mathrm{Tr} R + \mathrm{Tr} R^2 \Delta + O(\mathrm{Tr} R \Delta R^2 \Delta).$$

The quadratic remainder is nonnegative, so its expectation controls it by Markov's inequality. Its centered-noise expectation is $O(d^{-2})$, by (5.2), Cauchy–Schwarz for neighboring entries, and $n^{-1} \sum_i a_i^{-4} = O(d^{-2})$. For the mean contribution, first use $R \preceq C d^{-2} I$. On the bulk half the adjacent-entry means are $O(n^{-1/2})$ and the diagonals of R^2 are bounded; on the other half the means are $O(n^{-1})$ and the trace bound applies.

The linear term has standard deviation $O(d^{-2})$ and mean $O(d^{-1} + 1)$. These estimates extend to each fixed-ratio interval in d . In an eigenbasis of A , with $\rho_i = (2 + d^2 - \lambda_i(A))^{-1}$, the quadratic term is

$$Q(d) = \sum_{i,j} \rho_i \rho_j^2 |\tilde{\Delta}_{ij}|^2.$$

It is nonnegative and decreases with d , regardless of the signs in Δ . For the linear term $L(d) = \mathrm{Tr} R^2 \Delta$,

$$d \partial_d L(d) = -4d^2 \mathrm{Tr} R^3 \Delta.$$

The diagonal bounds give this logarithmic derivative the same $O(d^{-2})$ fluctuation scale as $L(d)$ on a fixed-ratio interval. Integrating it controls the supremum there. Markov's inequality for Q and the corresponding second-moment bound for L , at threshold $(nd^3)^{1/4} d^{-2}$, have summable failures over the dyadic intervals because nd^3 starts at ℓ^3 and grows geometrically. Finally

$$\frac{(nd^3)^{1/4} d^{-2}}{\sqrt{n/d}} = (nd^3)^{-1/4} \longrightarrow 0,$$

which gives the stated uniform trace error.

It remains to identify the deterministic trace. Reversing the Jacobi indices in its principal-determinant recurrence gives monic Hermite polynomials. The characteristic polynomial P_n satisfies

$$n^{-1} P_n'' - z P_n' + n P_n = 0.$$

Therefore $b_A = P_n'/(n P_n)$ obeys

$$b_A^2 - z b_A + 1 = -b_A'/n.$$

The right side is $O((nd)^{-1})$ by (5.2). Continuation from large z selects the smaller branch and gives $b_A(2 + d^2) = b_{\mathrm{sc}}(2 + d^2) + O((nd^2)^{-1})$. Together with the perturbation estimate this proves (5.1).

For a fixed integer $p \geq 2$, take the $(p-1)$ st finite difference at spacing hd^2 , with $\Delta_a f(u) = f(u+a) - f(u)$. The scalar identity is

$$\frac{(-1)^{p-1} \Delta_{hd^2}^{p-1} (t + d^2)^{-1}}{(p-1)!(hd^2)^{p-1}} = \prod_{j=0}^{p-1} (t + (1+jh)d^2)^{-1}$$

For an eigenvalue v of J , take $t = 2 - v$. The upper-edge bound already proved in (5.1) gives $t + d^2 \geq d^2 - o(d_0^2) \geq d^2/2$ for all sufficiently large n , uniformly for $d \geq d_0$. Hence

$$(1 + 2(p-1)h)^{-(p-1)} (t + d^2)^{-p} \leq \prod_{j=0}^{p-1} (t + (1+jh)d^2)^{-1} \leq (t + d^2)^{-p}.$$

The same comparison holds for semicircle depths $t \geq 0$. For fixed $h > 0$, (5.1) gives the semicircle limit for the finite difference with a relative error tending to zero. Letting h decrease after n tends to infinity now squeezes the p th power to its semicircle value.

For the edge counts at a shrinking scale, put $t_i = (2 - \lambda_i(J))/d^2$ and consider the finite measure

$$\nu_{n,d} = \frac{1}{nd^3} \sum_i (1 + t_i)^{-2} \delta_{t_i}.$$

To identify its limit, use the coordinate $x = (1 + t)^{-1}$. The upper-edge bound places the pushed-forward measures on $[0, 1 + o(1)]$, and for every fixed integer $j \geq 0$ their moments are

$$\int (1 + t)^{-j} d\nu_{n,d}(t) = \frac{d^{2j+1}}{n} \text{Tr}(2 + d^2 - J)^{-j-2}.$$

The trace-power estimates include the total mass, corresponding to $j = 0$. Polynomial approximation on a fixed compact interval therefore determines the weak limit of the pushed-forward finite measures. For $d \rightarrow 0$, the semicircle change of variables identifies its pullback as

$$\pi^{-1} \sqrt{t} (1 + t)^{-2} \mathbf{1}_{\{t > 0\}} dt.$$

The interval $[1, 2]$ has positive limiting mass and boundary of mass zero. Since its weight $(1 + t)^{-2}$ is bounded above and below, it contains order nd^3 eigenvalues. This conclusion is uniform over shrinking scales: any violating sequence would have the same moment limits and hence the same interval-mass limit. The second-power trace bound gives at most Cnd^3 eigenvalues in any fixed multiple of the d^2 upper-edge window. For d in a fixed interval bounded away from zero, ordinary semicircle convergence gives the same conclusions with constants depending on that interval. Applying the same argument to $-J$ gives the lower-edge bound.

For the interlacing assertion, the two eigenvalue counting functions differ by at most a fixed integer. Write z for the evaluation point and let $(z - cd^2, z + cd^2)$ be free of both spectra. The function $(z - v)^{-p}$, defined outside this gap, has an extension across the gap of total variation $O_p(d^{-2p})$. Integration by parts against the difference of the counting functions therefore bounds the raw trace difference by $O_p(d^{-2p})$. This argument also applies when z is interior to the full spectral range. For $p \geq 2$ the error is relative $O((nd^3)^{-1})$; for $p = 1$ it is $o(\sqrt{n/d})$, the unnormalized error scale in (5.1). $\square$

We will use these estimates under conditional matrix laws through a deterministic interlacing implication. If E has codimension one, $e \in E$, and

$$M = (P_E J P_E)|_E - ee^\top,$$

then the ordered eigenvalues satisfy

$$\lambda_j(J) \geq \lambda_j(M) \geq \lambda_{j+2}(J) \quad (1 \leq j \leq n - 2).$$

In particular $\lambda_{\max}(M) \leq \lambda_{\max}(J)$, cumulative upper-edge counts differ by at most two, and band counts differ by at most four. Thus the retained spectrum of the actual full matrix supplies the edge spread and trace estimates for its compressed block. The first-trace error $O(d^{-2})$ is also $o(\sqrt{n/d})$ because $nd^3 \rightarrow \infty$. When a conditional block has a Haar eigenframe, its concentration estimates can be proved uniformly on these inherited spectra before intersecting with the original full-matrix spectral event; this does not require conditioning on that event's probability.

5.3 Conditioning a Gaussian on the sphere

For a fixed spectrum and field, choose the unique $\lambda > \lambda_{\max}(J)$ such that

$$D = (\lambda - J)^{-1}, \quad m_o = Dh, \quad \text{Tr } D + \|m_o\|^2 = n. \quad (5.4)$$

The left side is strictly decreasing from infinity to zero, so the root is well defined. The spherical Gibbs law is precisely $N(m_o, D)$ conditioned on $\|x\|^2 = n$, in the sense of disintegration by its length density. Superscript 0 will denote the zero-field quantities.

Lemma 5.1 implies

$$|\lambda^0 - 2| \leq Cd_0^2, \quad \|D^0\| \leq Cnd_0. \quad (5.5)$$

Indeed at $2 + Cd_0^2$ the normalized trace deficit is $O(d_0)$. At distance $c/(nd_0)$ above the last pole, that pole contributes enough to close this deficit if c is small; monotonicity gives the second bound as well as the first.

The density estimates used in this conditioning are recorded with their proofs, since polynomial conditioning losses must be tracked at the lower edge.

Lemma 5.2 (Density at the mean). *Let g be a finite-dimensional standard Gaussian vector and let $Q = g^T Ag + b^T g$ with $A \succeq 0$ and $\sigma^2 = \text{Var } Q > 0$. The density of Q at $\mathbb{E}Q$ is at least c/σ , for an absolute $c > 0$. The same lower bound, with a smaller absolute constant, holds at a displacement $o(\sigma)$ along any sequence of such forms.*

Proof. After diagonalization and normalization it is enough to consider

$$\sum_i a_i(g_i^2 - 1) + 2 \sum_i b_i g_i, \quad a_i \geq 0, \quad 2 \sum_i a_i^2 + 4 \sum_i b_i^2 = 1.$$

Argue by subsequences. If an a_i stays bounded below, write the sum as $a_i(g_i^2 - 1) + 2b_i g_i + W$. Cantelli's inequality gives a positive lower bound for $\mathbb{P}(W \leq a_i/2)$, and Chebyshev's inequality removes $W < -C$ after a fixed large C is chosen. On the remaining event the one-coordinate target lies in $[-a_i/2, C]$, a compact interval strictly above the minimum $-a_i - b_i^2/a_i$ of the quadratic. Its density there has a positive lower bound. This proves the claim along the subsequence.

If $\max_i a_i \rightarrow 0$ but $\|b\|$ stays bounded below, write $g = z b/\|b\| + g_\perp$. The coefficient of z^2 tends to zero, and the random correction to its linear coefficient has variance $O((\max_i a_i)^2)$. Thus that linear coefficient stays bounded away from zero with high probability. The remaining term has bounded second moment. Restrict it to a fixed compact interval and the cross coefficient to a small interval; the change of variables in z then has a bounded Jacobian and a preimage in a fixed compact interval, where the Gaussian density is bounded below. In the remaining case Fourier inversion gives a local Gaussian limit. The quadratic-product formula supplies the integrable bound

$$(1 + 4(\max_i a_i)^2 t^2)^{-c/(\max_i a_i)^2}.$$

The same proof permits an $o(1)$ displacement of the target in standard-deviation units. $\square$

Suppose $\lambda - 2 \asymp d^2$, where $d_0 \leq d \lesssim 1$. The shell variable has standard deviation comparable to $\sqrt{n/d}$. The lower bound follows from $\text{Tr } D^2 \asymp n/d$, and its upper bound uses (5.4) to give $\|m_o\|^2 = O(nd)$ and $m_o^T D m_o = O(n/d)$. The mean-density lower bound controls the denominator of shell conditioning. For a projection test we also need an upper bound for the remaining length density. The spectral spread survives a compression of fixed codimension.

More explicitly, write $x = m_o + D^{1/2}g$, let $S = \|x\|^2$, and let $Z = V^\top g$ for a matrix V with a fixed number of orthonormal columns. Then

$$p_{Z|S=n}(z) = p_Z(z) \frac{f_{S|Z=z}(n)}{f_S(n)}.$$

Let $P = I - VV^\top$. After fixing Z , the quadratic matrix for the remaining white coordinates is the compression $PDP|_{\text{ran } P}$. The conditional covariance of x is $D^{1/2}PD^{1/2}$, whose nonzero eigenvalues agree with those of this compression. The surviving spectral block of the compression and the quadratic-product formula therefore give $f_{S|Z=z}(n) \leq C\sqrt{d/n}$ uniformly in the translated mean, hence uniformly in z . The mean-density lemma gives the matching lower scale for $f_S(n)$. This ratio supplies Gaussian moment and tail bounds for every fixed collection of white projections. In particular,

$$\left\| \mathbb{E}_{\text{shell}} \left[D^{-1/2}(x - m_o)(x - m_o)^\top D^{-1/2} \right] \right\| \leq 1 + o(1). \quad (5.6)$$

The density ratio gives uniform moment control; the sharp coefficient requires the limiting conditional law.

To obtain the coefficient 1, take any sequence of unit projection directions. The standardized shell variable and the projection have joint Gaussian subsequential limits, since

$$\frac{\|D\|}{\sqrt{\text{Tr } D^2}} = O((nd^3)^{-1/2}) = o(1).$$

Their possible limiting correlation comes only from the linear part of the shell variable. Fourier inversion conditional on the projection, with the integrable bound just obtained, gives local convergence at shell value zero against bounded projection tests.

The uniform projection tails make the squared projections uniformly integrable, so this local convergence also gives their second moments. Conditioning a joint Gaussian at its mean, or at an $o(1)$ displacement, can only decrease the second moment, which proves (5.6) uniformly over directions.

When $\lambda \leq 2 + Cd_0^2$, the same density-ratio argument costs at most ℓ^C . Indeed $D \preceq D^0$, $\|m_o\|^2 = O(nd_0)$, and the edge levels and remaining dyadic shells give

$$\text{Tr } D^2 \leq C[(nd_0^3)(nd_0)^2 + n/d_0], \quad m_o^\top D m_o = O((nd_0)^2).$$

The spread at depth comparable to d_0^2 supplies the required density upper bound after fixed-dimensional compression. Thus the ratio to the lower bound at the mean is bounded by a power of ℓ .

We will also condition coarse tail events on exact spheres. The rule used below is the following radial comparison. For a fixed number of Gaussian vectors, let $I_E(n, \dots, n)$ be the unnormalized joint density of their squared lengths at $(n, \dots, n)$, restricted to an angular event E . Suppose the Gaussian precision and its product with the mean have polynomial norms. If E_{buf} contains the radial images of E for $|s_j - n| \leq n^{-C_0}$, then

$$I_E(n, \dots, n) \leq n^C \mathbb{P}_G(E_{\text{buf}})$$

for a fixed C after C_0 is sufficiently large. Indeed, in polar coordinates the Gaussian log density and the radial log Jacobian vary by $O(1)$ on this interval, uniformly in the angular variables. Integrating along those rays proves the inequality. Dividing by a reference shell density bounded below by an inverse power of n gives the corresponding normalized bound. At each use below, E is specified by quadratic or fixed-degree Gram tests on a polynomial ball. Their polynomial derivative and inverse-tolerance bounds ensure that a slight relaxation of the tests contains this radial buffer.

Lemma 5.3 (Zero-field spherical normalization). *With ordinary probability tending to one,*

$$e^{n/4}n^{-C} \leq Z_o(0) \leq e^{n/4+o(n)}. \quad (5.7)$$

Proof. The Gaussian shell formula gives

$$\log Z_o(0) = \frac{n(\lambda^0 - 1)}{2} - \frac{1}{2} \log \det(\lambda^0 - J) + \log \frac{\text{length density for } N(0, D^0) \text{ at } n}{\text{length density for } N(0, I) \text{ at } n}.$$

The last term is $O(\log n)$ in the direction needed for the lower bound. Replacing λ^0 by $2 + Cd_0^2$ in the first two terms increases them by only $O(nd_0^3)$, by convexity and (5.1). At this new argument, concavity of log determinant gives the upper bound

$$\log \det(z - J) \leq \log \det(z - A) - \text{Tr}(z - A)^{-1} \Delta.$$

The linear correction has mean $O(1)$ and standard deviation $O(\sqrt{\log n})$, by (5.2).

Integrating the deterministic Hermite transform estimate shows that the deterministic log determinant differs from the semicircle value by $O(\log n)$. Beyond a fixed large z , the same Hermite equation and $\|A\| \leq 2$ give normalized-transform error $O(n^{-1}(z-2)^{-3})$, so the integral to infinity is harmless. Finally,

$$\frac{z-1}{2} - \frac{1}{2} \int \log(z-v) d\rho_{\text{sc}}(v)$$

has value $1/4$ at $z = 2$, as follows by integrating b_{sc} , and is increasing thereafter. This proves the lower bound. The upper bound follows from the same Gaussian representation at a fixed positive shift, using the length-density upper bound and then sending the shift to zero. $\square$

Lemma 5.4 (Spherical observations). *Assume the spectral event in Lemma 5.1. For every fixed tolerance $\epsilon > 0$, when $k \gg \log n$ and the allowed upper bound on r is sufficiently small, the spherical observation satisfies*

$$\lambda = 2 + (1 \pm \epsilon)s, \quad \|m_o\|^2 = (1 \pm 2\epsilon)nr,$$

except with probability $e^{-c_\epsilon k}$. For every fixed $B < \infty$ there are constants $c_B, C_B > 0$ such that

$$c_B s \leq \lambda - 2 \leq C_B s, \quad \|h_s\| \leq C_B r \sqrt{n},$$

except with probability e^{-Bk} . Polynomial conditioning factors can be accommodated already at $k > K \log n$, with K chosen after the tolerance and rate budget. For every deterministic actual time $0 \leq t \leq s = r^2$, the same upper test gives

$$\lambda(h_t) - 2 \leq C_B s, \quad \|h_t\| \leq C_B r \sqrt{n}$$

outside probability e^{-Bk} , where $k = nr^3$ is the nominal scale. The lower and sharp width bounds and the sharp mean estimate above concern the actual time s .

Proof. Test (5.4) at $\lambda = 2 + \gamma s$. Before conditioning, $X \sim N(0, D^0)$ and $h_s = rg + sX$ with an independent white Gaussian g . Put $\delta = \lambda - \lambda^0$. Since $k \gg \log n$, we have $r/d_0 \rightarrow \infty$ and $\delta = \gamma r^2(1 + o_n(1))$ for fixed $\gamma > 0$. The partial fraction identity

$$D^2 D^0 = \delta^{-2}(D^0 - D) - \delta^{-1} D^2$$

together with (5.1) and $\text{Tr } D^0 = n$ gives

$$\frac{\mathbb{E}\|D(rg + sX)\|^2}{nr} = \frac{1}{2\sqrt{\gamma}} + \frac{1}{2\gamma^{3/2}} + O_\gamma(r) + o_n(1), \quad \frac{n - \text{Tr } D}{nr} = \sqrt{\gamma} + O_\gamma(r) + o_n(1).$$

The leading expressions agree at $\gamma = 1$; their errors are small in the stated order of upper scale and dimension.

The signal square is $s^2 X^\top D^2 X$. Subtract $(s^2/\delta^2)(\|X\|^2 - n)$, which has mean zero and vanishes on the exact sphere. In a white eigen-coordinate with $t = \lambda^0 - \lambda_i(J) > 0$, the remaining quadratic coefficient is

$$\frac{s^2}{t} \left(\frac{1}{(\delta + t)^2} - \frac{1}{\delta^2} \right) = -\frac{s^2(2\delta + t)}{\delta^2(\delta + t)^2}.$$

The pole in D^0 has disappeared. The spectral trace bounds give operator norm $O_\gamma(r^{-2})$ and squared Hilbert–Schmidt norm $O_\gamma(n/r)$ for this quadratic matrix. The noise square has the same bounds. Gaussian exponentiation at a deviation of order nr therefore gives exponent $\min\{(nr)^2/(n/r), (nr)/r^{-2}\} \asymp k$. Conditional on X , the cross term is Gaussian with variance $O(n/r)$ when the signal square is $O(nr)$. The radial conditioning comparison preserves these rates after polynomial loss. Testing on either side of $\gamma = 1$ proves the sharp assertions.

For a prescribed rate, first choose a large fixed L . At a lower trial point $2 + \gamma r^2$, the $L^3 k$ modes at depth comparable to $L^2 r^2$ contribute noise-square mean comparable to nr/L , whereas the trace deficit is comparable to $n\sqrt{\gamma}r$. Choose $\sqrt{\gamma}$ smaller than a fixed multiple of L^{-1} . A fixed-fraction Gaussian small-ball bound on these modes then has failure $e^{-cL^3 k}$, uniformly in X because translation decreases the probability of a centered ball.

For the upper width, test at $2 + L^2 r^2$. The trace deficit is of order nLr , while the noise and signal means are $O(nr/L)$ and $O(nr/L^3)$. The noise quadratic matrix has norm $O(L^{-4}r^{-2})$ and squared Hilbert–Schmidt norm $O(n/(L^5 r))$; the canceled signal matrix has the corresponding bounds $O(L^{-6}r^{-2})$ and $O(n/(L^9 r))$. Their Gaussian tail exponents at the deficit scale are at least $cL^3 k$. For an actual time $t \leq s$, write $h_t = \sqrt{t}g + tX$ and use this same trial point. Both noise and signal prefactors decrease, so the upper-width estimate retains its nominal k -scale rate. Choose L from the requested rate, then the lower trial constant γ , then the upper scale so that these spectral windows are available. Finally choose the logarithmic threshold to absorb the polynomial shell cost. On the exact sphere $\|X\| = \sqrt{n}$; Gaussian norm concentration applied to $h_t = \sqrt{t}g + tX$ gives the stated field bound uniformly over the deterministic choices $0 \leq t \leq s$. $\square$

6 Comparison on unrestricted rotation orbits

In this section the matrix, field, and spherical mean rotate together. The spectral data and the field in eigen-coordinates are fixed. The comparison first controls unnormalized replica integrals. A controlled Brownian argument then supplies the lower partition-function tail needed to normalize those integrals.

Proposition 6.1 (Unrestricted orbit comparison). *Assume Lemma 5.1, let $d = \sqrt{\lambda - 2}$ be comparable to $r > 0$, and suppose $\|h\| \leq Cr\sqrt{n}$ and $k = nr^3 \gg \log n$. Fix the width-comparison factors. For every fixed B and every requested fixed relative accuracy, the following holds outside Haar mass e^{-Bk} once the allowed upper bound on r is sufficiently small:*

$$\left| \log \frac{Z(h)}{Z_o(h)} \right| \leq o(k). \quad (6.1)$$

For each fixed replica number p and fixed symmetric zero-diagonal $p \times p$ matrix P of sufficiently small norm, chosen before the upper scale and exceptional set, the same Haar bound holds for

$$\log \mathbb{E}_{\mu_h^{\otimes p}} \exp \left\{ \frac{s}{2} \sum_{a,b} P_{ab} (X^a - m_o)^T (X^b - m_o) \right\} \leq C \operatorname{Tr}(P^2)k + o(k). \quad (6.2)$$

The leading constant and the norm threshold depend only on the width-comparison factors; the allowed upper scale may also depend on the fixed replica data. The joint Haar-posterior probability that a pair overlap differs from $\|m_o\|^2/n$ by more than any prescribed fixed tolerance times r is at most e^{-ck} , with $c > 0$ depending on that tolerance.

The same conclusions are available at $k > K \log n$ with explicit polynomial losses. For any prescribed $\eta > 0$, the errors $o(k)$ in (6.1) and (6.2) may be replaced by $\eta k + C_* \log n$. The unabsorbed exceptional masses have the form $n^{C_*} e^{-B'k}$ for the partition and moment bounds and $n^{C_*} e^{-c'k}$ for the joint pair bound. Here C_* may depend on the fixed replica data, accuracy, and preliminary rate budget. Choose $B' > B$ and then K to absorb these losses; for the pair bound retain a smaller positive rate than c' . At fixed K the logarithmic loss is a chosen small fraction of k , rather than an error vanishing with n . The exceptional-mass conclusions also hold after weighting Haar measure by $Z(h)/Z_o(h)$, with an arbitrarily small decrease of the rate exponents.

6.1 Replica integrals and their tails

For independent uniform signs, write $R_{ab} = (X^a)^T X^b / n$. On polynomially small boxes near the identity Gram matrix, their off-diagonal overlaps have probability at most the corresponding spherical probability times

$$n^{C_p} \exp \left\{ C_p n \max_{a \neq b} |R_{ab}|^4 \right\}. \quad (6.3)$$

The spherical density is proportional to $(\det R)^{(n-p-1)/2}$, with polynomial normalization obtained by successive projections. For signs, the sum Chernoff bound has an analytic finite-dimensional Legendre expansion near zero, with identity covariance.

Its quadratic and cubic terms agree with $-\frac{1}{2} \log \det R$: the only nonzero third products are triangles, and each has expectation one. The difference thus begins at fourth order, which proves (6.3). Conditional on the Gram matrix, Haar rotation supplies exactly the same angular integral for both priors. The boxes can have any sufficiently small inverse-polynomial side length. Their Gram eigenvalues remain bounded below, and a smooth Gram factor changes the energy and the additional tilts in (6.2) by a polynomially bounded amount times the box width.

Under the spherical Gibbs energies, pair overlaps are $O(r)$. For $a \gg r$ up to a small fixed constant, the event $\max_{i \neq j} |R_{ij}| \geq a$ costs at most a polynomial times e^{-cna^3} , for every fixed number of replicas. To prove this, one of $(X^i + X^j)/\sqrt{2}$ and $(X^i - X^j)/\sqrt{2}$ has squared length at most $n(1-a)$. Translations decrease the relevant Gaussian small-ball probability. Add a negative square tilt of size ηa^2 , with $\eta > 0$ small. By (5.1), the normalized trace lost under this tilt is $O(r + \sqrt{\eta}a)$. The logarithmic Gaussian integral then gives the asserted na^3 exponent; exact length conditioning costs only a polynomial.

For the tilted calculation write $\xi^a = X^a - m_o$. Before shell conditioning these are independent Gaussians of precision D^{-1} . Choose the norm threshold on P so that $s\|P\| \leq \theta(\lambda - \lambda_{\max}(J))$ for a fixed $\theta < 1$. This is possible uniformly under the width comparison. Factoring the Gaussian integral leaves the determinant $\det(I - sP \otimes D)^{-1/2}$ and a normalized centered Gaussian law with covariance

$$(I_p \otimes D^{-1} - sP \otimes I)^{-1}.$$

The precision remains positive. The determinant series has no linear term because $\text{Tr } P = 0$, and the remaining terms cost at most $Cs^2 \text{Tr}(P^2) \text{Tr}(D^2) \leq Ck \text{Tr}(P^2)$. The new covariance is bounded between fixed shifted resolvents. Under this law $X^a = m_o + \xi^a$; the preceding small-ball argument for their sums and differences still applies because it is uniform in translations. The joint length numerator is estimated under this normalized correlated law, while the original single-shell densities

remain the denominators. The radial comparison gives the stated polynomial conditioning cost. Without a tilt, Gaussian quadratic and linear tails show concentration about $\|m_o\|^2/n$ within any fixed tolerance times r .

Consequently the annealed unnormalized products, divided by the corresponding powers of $Z_o(h)$ and restricted to all overlaps in a sufficiently small fixed interval, have logarithmic cost at most $Ck \operatorname{Tr} P^2 + o(k)$. On overlaps of order r , the cost $nr^4 = rk = o(k)$ in (6.3) is negligible. For a fixed replica number, choose the overlap cutoff η so that $C_p a^4 \leq (c/2)a^3$ for $0 < a \leq \eta$. On the rest of the small interval, the negative na^3 exponent under the tilted Gaussian law then absorbs the quartic prior-comparison error. The argument applies to repetitions in disjoint replica groups. For the proposition's joint Haar-posterior pair conclusion, take $P = 0$. For any prescribed fixed tolerance, restrict the pair to the complement of the interval centered at $\|m_o\|^2/n$ with that tolerance times r as its radius. The no-tilt concentration estimate above, combined with the same prior comparison, retains a strict negative rate of order k for this restricted pair integral.

Only the pair calculation requires a completely unrestricted unnormalized overlap tail. For every fixed $\eta > 0$, $|R_{12}| > \eta$ contributes at most $e^{-c_\eta n}$ relative to $Z_o(h)^2$. The field costs $O(nr)$; we will absorb it by choosing the upper scale after η . After rotating the orthogonal sum and difference, their angular energy is at most

$$\exp \left\{ \frac{n}{4} [(1 + R_{12})^2 + (1 - R_{12})^2] + o(n) \right\}.$$

At a parallel endpoint only the nonzero direction is needed. Here the elementary one-direction bound is that a uniform direction of squared radius n , tilted by strength $u \geq 0$, has logarithmic integral per coordinate at most $u^2/4 + o(1)$, uniformly on compact u -ranges, also in a fixed-codimension compression.

Indeed use a centered Gaussian with precision $z - uJ$, $z > 2u$, to obtain the upper bound

$$\frac{z-1}{2} - \frac{1}{2} \int \log(z - uv) d\rho_{\text{sc}}(v).$$

At a fixed shift the shell-density loss is subexponential. For $u \leq 1$, choose $z = 1 + u^2$; for $u \geq 1$, send $z \downarrow 2u$. The derivative of the latter continuation is $1 - 1/(2u) \leq u/2$, proving the claimed quadratic upper bound. Iterating over the two directions proves the display. Its exponent is $n/2 + nR_{12}^2/2 + o(n)$. For the denominator we need only $\log Z_o(0) \geq n/4 - o(n)$, which follows deterministically from the retained spectral event. Indeed (5.5) gives $\lambda^0 \rightarrow 2$ and a polynomial bound on $\|D^0\|$. The zero-field shell formula and the density-at-mean lower bound give

$$\log Z_o(0) \geq \frac{n(\lambda^0 - 1)}{2} - \frac{1}{2} \log \det(\lambda^0 - J) - O(\log n).$$

For every fixed $\delta > 0$, eventually $\lambda^0 \leq 2 + \delta$, so the log determinant is at most $\log \det((2 + \delta)I - J)$. Semicircle convergence identifies its normalized limit. Sending $\delta \downarrow 0$ gives the claimed lower bound. Thus no additional normalization event is needed here.

By symmetry $Z_o(h) \geq Z_o(0)$, so the denominator removes the $n/2$ term. The Cramér rate of the overlap of independent signs exceeds $R_{12}^2/2$ by a fixed positive amount on $|R_{12}| \geq \eta$. Choosing the upper bound on r so that the field cost is smaller than this gap proves the global pair estimate.

6.2 The lower partition-function tail

Set $F(U) = \log(Z(h)/Z_o(h))$ on the rotation orbit. The Haar averaging behind this first-moment identity follows Comets [14, Equations (2.1)–(2.2)]; the present field-dependent identity follows from

the rotation convention above. Haar averaging and the pair bounds give

$$\mathbb{E}_U e^F = 1, \quad \mathbb{E}_U e^{2F} \leq e^{o(k)}.$$

Paley–Zygmund therefore supplies a set of high anchors of mass $e^{-o(k)}$ on which $F \geq -o(k)$. These anchors may also be required to satisfy

$$\left\| \mathbb{E}_{\mu_h} X X^T \right\|_{\text{HS}} = o(n).$$

Indeed the square of this Hilbert–Schmidt norm is $\mathbb{E}_{\mu_h^{\otimes 2}} (X^1 \cdot X^2)^2$, and the global pair estimate removes the contrary event at exponentially small unnormalized mass.

For every fixed B' and $\epsilon > 0$,

$$\mathbb{P}_U(F > \epsilon k) \leq e^{-B'k}$$

once the upper bound on r is small enough. To see this, let $q_\eta(U) = \mu_h^{\otimes 2}(|R_{12}| > \eta)$. The global pair estimate is $\mathbb{E}_U[e^{2F} q_\eta] \leq e^{-c_\eta n}$. Hence

$$\mathbb{P}_U(F > \epsilon k, q_\eta > \delta) \leq \delta^{-1} e^{-2\epsilon k} \mathbb{E}_U[e^{2F} q_\eta].$$

Fix a replica number p and choose $\delta < (2(\frac{p}{2}))^{-1}$. On the remaining orientations at least half of the posterior p -replica mass has all overlaps below η . The annealed bound for that truncated integral then gives upper-event mass at most $2e^{-(p\epsilon - o_p(1))k}$. Choose p from B' , then the truncation tolerance and the upper scale. This obtains the strong upper tail from truncated moments.

We now turn the upper-tail bound into a lower-tail bound. The tangent inequality at a high anchor compares nearby endpoints. A reflected third rotation will cancel their first-order displacement, while its entropy cost allows us to control its endpoint by the upper tail just proved.

At a high anchor, Jensen’s inequality gives the deterministic lower tangent estimate

$$F(UR) \geq F(U) + \langle H_U, R - I \rangle_{\text{HS}} - o(n) \|R - I\|_{\text{HS}}^2, \quad \|H_U\|_{\text{HS}} = o(n). \quad (6.4)$$

To verify it, expand the rotated quadratic and linear energies under the Gibbs law at U . The linear coefficient is bounded by a constant times $\|J\| \|\mathbb{E} X X^T\|_{\text{HS}} + \sqrt{n} \|h\| = o(n)$. The quadratic remainder is bounded by $C \|J\| \|\mathbb{E} X X^T\| \|R - I\|_{\text{HS}}^2$. The denominator is rotation invariant. This proves (6.4), with arbitrarily small fixed leading errors at sufficiently small upper scale.

Suppose, toward a contradiction, that the low set $\{F < -\epsilon k\}$ has Haar mass at least e^{-Bk} . The following controlled reflection produces three nearby rotations: one at a high anchor, one in the low set, and one whose upper tail can be bounded by entropy.

Use rotational Brownian motion started at the identity, with right-translated antisymmetric noise dW whose upper entries have variance dt/n . Its Itô drift is $-(1 - 1/n)I/2$. Choose a deterministic terminal time sufficiently large that its endpoint law approximates Haar as accurately as needed. The heat kernel on the compact connected group permits this even for the exponentially small target sets at each fixed n .

The control cost is the entropy–drift representation associated with Föllmer’s process; the Euclidean form and its converse are given in [25, Proposition 1 and Theorem 2]. The group normalization needed here follows directly from the likelihood martingale. For a positive smooth terminal likelihood of free mean one, its backward heat extension is the density-process function. Itô’s formula under the Doob transform shows that the expected integrated squared drift in standard noise coordinates is twice the endpoint relative entropy.

An antisymmetric matrix velocity E corresponds to standard-coordinate drift with squared norm $n\|E\|_{\text{HS}}^2/2$. Thus its entropy cost is exactly

$$\frac{n}{4}\mathbb{E}\int\|E_t\|_{\text{HS}}^2 dt. \quad (6.5)$$

Bounded positive smooth approximations to normalized target indicators attain endpoint hit probabilities arbitrarily close to one and entropies within an additive constant of the indicator entropy. Conversely Girsanov's formula and data processing bound the endpoint entropy of any adapted control above by (6.5); stopping at finite energy justifies this statement without further integrability assumptions.

Choose feedback controls for two motions U, V so that their endpoints hit the high-anchor and low sets, respectively, with probability at least 0.95. Their expected joint matrix energy is $O_B(k/n)$. Stop all controls when the joint energy exceeds a sufficiently large fixed multiple of k/n . Markov's inequality makes this a small probability loss, so the two endpoint requirements still hold jointly with probability greater than 0.7.

Couple their noises as follows, and introduce a third motion V' . If $D = U^T V$, conjugate the noise for V to $D^{-1/2}dW D^{1/2}$; use the analogous conjugation for V' . These are Brownian marginal noises because conjugation is orthogonal on antisymmetric matrices. In base coordinates the three controls are

$$E_U, \quad E_V, \quad 2E_U - E_V.$$

Here E_V is obtained by undoing the conjugation on the prescribed feedback drift for V , which preserves its norm. The first two motions therefore retain their prescribed marginals up to the energy stopping. Initially use the local square roots near the identity; the estimates below show that their chart is never left.

The identity

$$dW S dW = (S^T - \text{Tr}(S)I) dt/n$$

gives the relative equation

$$\begin{aligned} dD = & -dW D + \sqrt{D} dW \sqrt{D} \\ & + \left[-(1 - 1/n)D + \frac{\text{Tr} \sqrt{D}}{n} \sqrt{D} - \frac{I}{n} \right] dt + (\sqrt{D} E_V \sqrt{D} - E_U D) dt. \end{aligned}$$

The same equation holds for $D' = U^T V'$ with the reflected control. The noise has trace zero, so q_D below has finite variation. For $q_D = \|D - I\|_{\text{HS}}^2 = 2(n - \text{Tr} D)$, expansion in rotation angles gives

$$-(1 - 1/n) \text{Tr} D + \frac{(\text{Tr} \sqrt{D})^2}{n} - 1 = \left(\frac{1}{4} - \frac{1}{2n} + O(\|D - I\|^2) \right) q_D.$$

The control contribution to q_D is bounded by $C\sqrt{q_D}(\|E_U\|_{\text{HS}} + \|E_V\|_{\text{HS}})$.

Hence, while in the chart,

$$q'_D \leq -cq_D + C\sqrt{q_D} e_t \leq -\frac{c}{2}q_D + Ce_t^2, \quad e_t = \|E_U\|_{\text{HS}} + \|E_V\|_{\text{HS}}.$$

The reflected motion obeys the same inequality with a constant multiple of e_t . The energy cutoff therefore gives

$$\sup_t (\|D_t - I\|_{\text{HS}} + \|D'_t - I\|_{\text{HS}}) \leq C_B \sqrt{k/n}. \quad (6.6)$$

Taking r sufficiently small in terms of B prevents the first exit and verifies the use of the local square roots.

The two relative rotations have opposite first-order displacements. Their sum therefore begins with the quadratic error:

$$\mathbb{E}\|D_t + D'_t - 2I\|_{\text{HS}}^2 \leq C_B(k/n)^2, \quad (6.7)$$

uniformly in the deterministic time t . This uniformity is needed when the terminal Brownian time is taken large to approximate Haar measure. We retain the damping term in the error equation to obtain it. Put $A = D - I$, $A' = D' - I$, and $H = A + A'$. The linear noise in A is $[A, dW]/2$ and its linear drift is $-(1/2 - 1/n)A$.

The term $(\text{Tr } A)I/(2n)$ is quadratic because $\text{Tr } A = -\|A\|_{\text{HS}}^2/2$. The constant control terms cancel upon adding the two equations. If $a^2 = C_B k/n$, the resulting equation is

$$dH = \frac{1}{2}[H, dW] - (1/2 - 1/n)H dt + dM_R + R_t dt,$$

where the Taylor remainders satisfy

$$\frac{d\langle M_R \rangle_{\text{HS}}}{dt} \leq Ca^4, \quad \|R_t\|_{\text{HS}} \leq Ca^2 + Ca e_t.$$

These follow from (6.6) and the convergent matrix power series for the square root.

For every real matrix H (the sum here need not be antisymmetric), the conditional noise variance is

$$\frac{\mathbb{E}\|[H, dW]/2\|_{\text{HS}}^2}{dt} = \frac{1 - 1/n}{2}\|H\|_{\text{HS}}^2 + \frac{\text{Tr}(H^2) - (\text{Tr } H)^2}{2n} \leq \frac{1}{2}\|H\|_{\text{HS}}^2.$$

Itô's formula and Young's inequality, with small fixed constants to absorb the quadratic-covariation cross term, yield

$$\frac{d}{dt}\mathbb{E}\|H\|_{\text{HS}}^2 \leq -c\mathbb{E}\|H\|_{\text{HS}}^2 + Ca^4 + Ca^2\mathbb{E}e_t^2.$$

Integrating against the exponential damping and using the pathwise bound $\int e_t^2 dt \leq Ca^2$ proves (6.7).

It remains to put the endpoint events and the matrix error on one positive-probability event. Let P_T be the free endpoint law and Q_T the reflected endpoint law. By (6.5), $\text{KL}(Q_T\|P_T) \leq C_B k$. For an event E with $0 < P_T(E) < 1$, binary relative entropy gives

$$Q_T(E) \leq \frac{\text{KL}(Q_T\|P_T) + \log 2}{\log(1/P_T(E))}.$$

If $P_T(E) = 0$, finite relative entropy gives $Q_T(E) = 0$. Choose the upper-tail rate B' large relative to C_B , and choose the terminal Brownian time so that its Haar approximation error is smaller than $e^{-B'k}$. The upper-tail estimate then gives $P_T(F > \epsilon k/2) \leq 2e^{-B'k}$, making the reflected failure probability as small as required. Markov's inequality in (6.7) likewise gives its stated order with any desired fixed high probability.

There is consequently positive probability of the two endpoint hits, this bound for V' , and both relative-matrix bounds, simultaneously. On that event, sum (6.4) from the high anchor U to V and V' . The linear error is at most $o(n)\|D + D' - 2I\|_{\text{HS}} = o(k)$, and the two quadratic errors are $o(k)$ by (6.6). Thus $F(V) + F(V') \geq -o(k)$. This contradicts $F(V) < -\epsilon k$ and $F(V') \leq \epsilon k/2$ after fixing the little-oh accuracies. The lower tail, and hence (6.1), follows.

Completion of Proposition 6.1. Normalize the replica integral bounds from Section 6.1 using (6.1). To handle large overlaps between replica groups after normalization, put $q_\eta(U) = \mu_h^{\otimes 2}(|R_{12}| > \eta)$. The global pair bound gives

$$\mathbb{E}_U[e^{2F} q_\eta] \leq e^{-c_\eta n}.$$

On the retained lower-partition event $F \geq -\epsilon k$, Markov's inequality shows that $q_\eta \leq e^{-c_\eta n/2}$ outside Haar mass $e^{-c_\eta n/2+2\epsilon k}$. A union bound applies to any fixed number of replica pairs. Their fixed tilts cost at most $C_p n r^2$, since $\|X^a\| = \sqrt{n}$ and $\|m_o\| \leq \sqrt{n}$. Choosing the upper scale after η therefore makes the tilted exterior exponentially small in n as well.

On the remaining replica configurations the truncated integral estimates apply. Markov's inequality with repeated tilted replica groups gives (6.2), choosing the denominator accuracy arbitrarily small first. The pair concentration estimate transfers in the same way, taking the denominator error below its strict rate. The polynomial Gram-box and shell-density factors cost $C_* \log n$ in these logarithmic estimates and in the exponents of the raw failure bounds. Choosing K after the rate and accuracy absorbs them when $k > K \log n$, as stated.

Finally, for an event with an arbitrarily prescribed raw rate,

$$\mathbb{E}_U[e^F \mathbf{1}_E] \leq (\mathbb{E}_U e^{2F})^{1/2} \mathbb{P}_U(E)^{1/2}$$

gives any desired weighted rate by starting with more than twice that rate. For a bounded posterior failure probability $0 \leq q(U) \leq 1$ having only a fixed strict rate, split instead at $F = \delta k$:

$$\mathbb{E}_U[e^F q] \leq e^{\delta k} \mathbb{E}_U q + (\mathbb{E}_U e^{2F})^{1/2} \mathbb{P}_U(F > \delta k)^{1/2}.$$

The second term has any fixed rate by the strong upper tail, and the first loses only δ from its rate. Taking δ small proves the stated weighted conclusions. $\square$

7 The polylogarithmic edge window

The unrestricted comparison above has exponential accuracy on the k -scale. In the lowest spectral window a different second-moment calculation gives convergence of the partition-function ratio to one and an absolute error smaller than the critical covariance scale. This calculation also supplies an independent, supplementary lower-bound argument. At zero field the partition-function comparison follows the quantitative critical comparison of [16, Theorem 1.6 and Section 3.3]. The mixed cube-sphere cancellation for matrix moments is the tensor-barycenter method of [17, Section 3.2, Lemma 3.5]. The field range and the weighted moment errors required here are proved below.

Proposition 7.1 (Edge comparison). *Assume the spectral event of Lemma 5.1. Suppose*

$$r \leq n^{-1/3} \log^{C_1} n, \quad \lambda - \lambda^0 \leq n^{-2/3} \log^{C_2} n, \quad \|h\| \leq \sqrt{n} r \log n,$$

where C_1, C_2 are fixed. Set $L = n^{2/3}$ and $A_0 = 2I - J$. Outside Haar mass $o(1)$,

$$\frac{Z(h)}{Z_o(h)} = 1 + o(1).$$

The cube and spherical raw second moments about m_o differ in operator norm by at most $n^{1/2+o(1)}$. The cube raw second moment, sandwiched by A_0 , has operator norm at most $n^{1/6+o(1)}$. Each exponent bound can be read with any fixed positive exponent slack. Unspecified powers of logarithms may depend on C_1, C_2 .

Proof. For two independent prior vectors, the sphere–sphere and cube–sphere overlap distributions agree exactly. The cube–cube overlap has lattice spacing $2/n$. Stirling’s formula and the spherical density $c_n(1 - R_{12}^2)^{(n-3)/2}$ show that its mass, divided by the lattice spacing and the spherical density, equals

$$1 + O(n^{-1/3} \log^C n) \quad \text{for } |R_{12}| \leq L \log^{C'} n/n. \quad (7.1)$$

Choose C' sufficiently large. Outside this window the second integrals, including any fixed bounded-degree polynomial insertions, are negligible.

Indeed the negative-square-tilt argument of Section 6.1 applies at depths much larger than $(\lambda - 2)_+^{1/2} + d_0$; combine it with (6.3), then with the global pair estimate. A cube-only first moment averages exactly to the spherical first moment.

The following cancellation is needed to turn (7.1) into an operator estimate. Before length conditioning, write independent Gaussian replicas as

$$x_i = m_o + D^{1/2} \eta_i, \quad i = 1, 2,$$

with white η_i , and use Fourier inversion for

$$\frac{\|x_1\|^2 - n}{L}, \quad \frac{\|x_2\|^2 - n}{L}, \quad \frac{x_1^T x_2}{L}. \quad (7.2)$$

The reciprocals of the two length densities at zero, in these units, are at most $\log^C n$.

The Fourier inversions, including any fixed number of derivatives, cost only powers of $\log n$ apart from the insertion size. Choose an edge width $d_* = n^{-1/3} \log^M n$ larger than the polylogarithmic saddle width. The spectral count supplies $N \asymp \log^{3M} n$ modes on which $D_i/L \geq \log^{-C} n$. A frequency vector for the three statistics is a symmetric 2×2 matrix T . At least one eigenvalue of T has size comparable to $\|T\|$, so the quadratic-product formula on those modes gives

$$|\varphi(T)| \leq (1 + c\|T\|^2 \log^{-C} n)^{-cN}.$$

The noncentral factor does not increase the modulus, because the real part of the inverse complex precision is positive. This bound is integrable with any prescribed fixed frequency polynomial, with a polylogarithmic integral. It remains so after deleting finitely many modes, as required below.

We now assemble the signed comparison before estimating it. Let π_c and π_o be the uniform cube and sphere priors, put $E_U(x) = x^T J x / 2 + h^T x$, and write $H_U(x) = (x - m_o)(x - m_o)^T - D$. Set $Z_c = Z$. For $i \in \{c, o\}$ define

$$z_i = \frac{Z_i}{Z_o}, \quad \widehat{M}_i(U) = \frac{1}{Z_o} \int H_U(x) e^{E_U(x)} d\pi_i(x), \quad z_o = 1.$$

All matrix data rotate with U . For a symmetric A commuting with D , expansion of the square gives

$$\mathbb{E}_U \|A(\widehat{M}_c - \widehat{M}_o)A\|_{\text{HS}}^2 = \mathcal{I}_{cc} - \mathcal{I}_{co} - \mathcal{I}_{oc} + \mathcal{I}_{oo},$$

where

$$\mathcal{I}_{ij} = \frac{1}{Z_o^2} \iint \mathbb{E}_U \left[e^{E_U(x) + E_U(y)} \text{Tr}(H_U(x) A^2 H_U(y) A^2) \right] d\pi_i(x) d\pi_j(y).$$

Conditional on the prior overlap, Haar integration gives the same angular integral in all four terms. The three pair laws containing a sphere use the same continuous overlap density; only the cube pair uses the lattice and its multiplier in (7.1).

After whitening, the common signed angular insertion is

$$\text{Tr}\left[(\eta_1\eta_1^\top - I)K(\eta_2\eta_2^\top - I)K\right], \quad K = D^{1/2}A^2D^{1/2}. \quad (7.3)$$

Under the complex Fourier tilt its covariance differs from identity by

$$\Delta_T = (I - 2iT \otimes D/L)^{-1} - I.$$

Its Hilbert–Schmidt norm, and the tilted mean using $\|D^{1/2}m_o\|/L \leq \log^C n$, are bounded by frequency polynomials times powers of $\log n$. Centering in (7.3) cancels the baseline and the terms belonging to only one replica. With zero tilted mean the remaining Wick terms have the forms

$$\text{Tr}(K\Delta_{22}K\Delta_{11}), \quad \text{Tr}(K\Delta_{21}K\Delta_{21}), \quad (\text{Tr } K\Delta_{12})^2.$$

The first two are bounded using $\|K\|^2$ and the Hilbert–Schmidt bounds on Δ_T . For the last, $|\text{Tr } K\Delta_{12}| \leq C\|T\| \text{Tr}(|K|D)/L$. Terms containing the tilted mean obey the same envelope. Fourier inversion therefore bounds the signed density of the insertion and its overlap derivative by a power of $\log n$ times

$$\|K\|^2 + \left(\frac{\text{Tr}(|K|D)}{L}\right)^2. \quad (7.4)$$

The frequency decay above absorbs all fixed derivative polynomials.

The spectral estimates give

$$\|D\| \leq L \log^C n, \quad \text{Tr } D^2 \leq L^2 \log^C n, \quad \|A_0^2 D\| \leq \log^C n, \quad \text{Tr}(A_0^2 D^2) \leq n \log^C n.$$

The last two follow from $A_0 D = I - (\lambda - 2)D$ and $\|A_0\| \leq 5$. Thus (7.4) is at most $L^2 \log^C n$ for $A = I$ and $n^{2/3} \log^C n$ for $A = A_0$. Insert a smooth cutoff in the central overlap window. The lattice spacing in the last statistic of (7.2) is $2/L$, so its Riemann-sum error is the same envelope times $L^{-1} \log^C n$. The leading continuous integrals cancel in the four-term sum, and (7.1) leaves a squared error bounded by $n^{-1/3}$ times the envelope, up to logarithms. The mass calculation has envelope one.

Consequently the root-mean-square errors are

$$\begin{aligned} (\mathbb{E}_U |z_c - 1|)^{1/2} &\leq n^{-1/6} \log^C n, \\ (\mathbb{E}_U \|\widehat{M}_c - \widehat{M}_o\|_{\text{HS}}^2)^{1/2} &\leq n^{1/2} \log^C n, \\ (\mathbb{E}_U \|A_0(\widehat{M}_c - \widehat{M}_o)A_0\|_{\text{HS}}^2)^{1/2} &\leq n^{1/6} \log^C n. \end{aligned}$$

Here $n^{-1/3}L^2 = n$ and $n^{-1/3}n^{2/3} = n^{1/3}$ are the two squared matrix scales. Markov’s inequality gives the stated bounds outside Haar mass $o(1)$, with any fixed exponent slack.

Finally let Q_i be the normalized raw second moment about m_o . Then

$$Q_c = D + \widehat{M}_c/z_c, \quad Q_o = D + \widehat{M}_o,$$

so the common D cancels and

$$Q_c - Q_o = z_c^{-1}(\widehat{M}_c - \widehat{M}_o) + (z_c^{-1} - 1)\widehat{M}_o.$$

The shell density-ratio bound controls $\widehat{M}_o$ in operator norm by $L \log^C n$, and its A_0 sandwich by $\log^C n$. The scalar mass error therefore costs at most $n^{1/2+o(1)}$ in the unweighted comparison and less in the weighted one. The spherical weighted raw moment itself is polylogarithmically bounded. This proves all claims. $\square$

7.1 An independent covariance and slab lower bound

The realized-equilibrium-start and linear-test results in [27, Theorems 1.2–1.3] have stronger quantifiers: they apply to every deterministic time sequence below the critical scale and every deterministic diverging multiplicative slack, respectively. This supplementary subsection gives an independent covariance obstruction and a worst-start slab proof with fixed exponent slack.

At zero field, Proposition 7.1 and (5.7) give, with ordinary probability tending to one,

$$Z(0) \geq n^{-C} e^{n/4}. \quad (7.5)$$

This is also the lower partition-function event used later to undo the planted size bias at polynomial cost.

On the same type of event the zero-field Gibbs covariance has an eigenvalue at least $n^{2/3-o(1)}$. Choose a spherical spectral mode at depth of order $n^{-2/3}\ell^{10}$ below 2, available by Lemma 5.1. Its unconditioned variance is $n^{2/3}\ell^{-O(1)}$. More precisely, the chosen mode has variance $O(n^{2/3}\ell^{-10})$, whereas the other modes at depth comparable to d_0^2 give remaining length standard deviation at least $cn^{2/3}\ell^{-1/2}$. Conditioning the chosen standardized projection to lie between 1 and 2 therefore shifts the remaining length target by $O(\ell^{-19/2})$ of that standard deviation. Lemma 5.2 and the length-density upper bound show that this mode retains its variance within an $\ell^{O(1)}$ factor on the sphere.

Symmetry at zero field and the $n^{1/2+o(1)}$ cube–sphere error then give

$$\|\text{Cov}_{\mu_0} X\| \geq n^{2/3-o(1)}. \quad (7.6)$$

For a unit linear test $f_v(x) = v^T x$, the unscaled heat-bath form satisfies

$$\mathcal{D}(f_v) = \sum_i \mathbb{E}_{\mu_0} \text{Var}(f_v \mid X_{[n]\setminus\{i\}}) \leq \sum_i v_i^2 = 1.$$

Thus the inverse unscaled spectral gap is at least $n^{2/3-o(1)}$. A top nonconstant eigenfunction gives the same exponent lower bound for fixed mixing accuracies less than 1/2.

The comparison also proves that worst-start distance tends to one below this exponent. Let v be the same unit spectral mode, and put

$$Y = \frac{v^T X}{\sigma_Y}, \quad \sigma_Y = n^{1/3}\ell^{-O(1)},$$

where σ_Y is its unconditioned spherical Gaussian standard deviation. For every fixed $K > 0$ and $w = \ell^{-K}$, the spherical probabilities of both $|Y| \leq w$ and $|Y| \leq 2w$ lie between $cw\ell^{-C}$ and $Cw\ell^C$. The density estimates just used are uniform when this standardized projection lies in the indicated range, which proves these bounds.

For these events, the unnormalized cube probabilities, measured in units of $Z_o(0)$, differ from their spherical counterparts by at most n^{-c} , outside ordinary Haar mass $o(1)$, for an absolute small $c > 0$. This follows from the same mass second-moment comparison, with the projection indicator on each copy. The angular integrals at fixed Gram matrix continue to agree because v co-rotates with the matrix. In the central overlap window, the bounded Stirling error is sufficient.

For the lattice quadrature, condition first on the two tested white Gaussian coordinates, each of absolute value at most one. Deleting this one spectral mode leaves the eigenvalue spread for the remaining two lengths and overlap. Their joint density and its overlap derivative are therefore bounded by powers of $\log n$ in the units of (7.2), uniformly in the conditioned coordinates. The

quadrature error is again at most a power of $\log n$ divided by L . The exterior of the window has the same absolute tail bound as before. Markov's inequality proves the asserted n^{-c} comparison.

Since the total mass ratio tends to one, choose $K > C + 1$. With ordinary probability tending to one, the narrow slab $A = \{|Y| \leq w\}$ then has Gibbs mass at least $n^{-o(1)}$, while the wider slab $A^+ = \{|Y| \leq 2w\}$ has mass tending to zero. For the stationary rate-one-per-site chain, the spectral theorem and $\mathcal{D}(f_v) \leq 1$ give

$$\mathbb{E}_{\mu_0}(f_v(X_t) - f_v(X_0))^2 \leq 2t.$$

For m single-site attempts the same bound is $2m/n$, since the attempt kernel is an average of conditional-expectation projections and hence has spectrum in $[0, 1]$.

Conditioning the stationary chain at its initial point to lie in A increases the bound by at most $1/\mu_0(A)$. Consequently, for every fixed $\delta > 0$ and $t \leq n^{2/3-\delta}$ in rate-one-per-site time,

$$\mathbb{P}_{\mu_0(\cdot|A)}(X_t \notin A^+) \leq \frac{2t}{\mu_0(A) w^2 \sigma_Y^2} = o(1).$$

The stationary mass of A^+ is $o(1)$, so this conditional-start law has total variation distance $1 - o(1)$ from equilibrium. Averaging over its starting points, after the disorder and time have been fixed, yields a starting configuration with the same conclusion. In attempts, the corresponding time range is $m \leq n^{5/3-\delta}$.

8 Counting exact field roots

We express expected counts of field roots as integrals. Conditioning at a root leaves a Gaussian matrix on its orthogonal complement, while the scalar part of the integral determines the root's empirical moments. An excluded root count will later bound the probability of an excluded field. All integrated estimates below use the planted law, with the original spectral restriction imposed when indicated.

The organization by a Gaussian root density and a conditional determinant follows the Kac–Rice analysis of TAP equations in Fan et al. [21, Sections 3–4] and its local-convexity refinement in Celentano et al. [11, Lemma 4.8 and Appendix A.5]. The critical shrinking scale, polynomial determinant bound, and nq^3 exclusion rates are proved here.

At observation precision $s = r^2$, the planted experiment may be gauged so that its spin is $\mathbf{1}$. If $S_X = \text{diag } X$, this gauge sends J to $S_X J S_X$ and sends h, y, m to $S_X h, S_X y, S_X m$. In the gauged coordinates $J = W + \mathbf{1}\mathbf{1}^\top/n$ and $h_s = s\mathbf{1} + \sqrt{s}g$, with W a GOE matrix and g an independent standard Gaussian. We study the roots of

$$\Phi_J(y) := y + (1 - q)m - Jm = h_s, \quad m = \tanh y, \quad q = \langle m^2 \rangle.$$

Write

$$b = 1 - q, \quad v = 1 - m^2, \quad V = \text{diag } v, \quad M = \langle m \rangle,$$

and, for the scalar Gaussian regression, put

$$\nu = q + s, \quad d = M + s, \quad a = \langle (y - d\mathbf{1})m \rangle, \quad C_a = a/\nu, \quad j_0 = q/\nu, \quad \alpha = (C_a - b)/(1 + j_0).$$

For fixed y , let $\mathbf{Q}_y$ be the conditional law of $J = W + \mathbf{1}\mathbf{1}^\top/n$ given

$$Wm + \sqrt{s}g = y - d\mathbf{1} + bm.$$

The field represented by (y, J) is $\Phi_J(y)$. All root-counting arguments in this section use $k = nr^3 \geq \log^{10} n$ and small r .

Lemma 8.1 (The exact counting density). *For every nonnegative Borel test $T(y, J)$,*

$$\mathbb{E}_{\text{pl}} \sum_{\substack{y: \Phi_J(y)=h_s \\ \det D\Phi_J(y) \neq 0}} T(y, J) = \int (1 + j_0)^{-1/2} G(y) \mathbb{E}_{\mathbf{Q}_y} [T(y, J) \mathcal{W}(y, J)] dy,$$

where

$$G(y) = \prod_i \varphi_{d,\nu}(y_i) \exp\{n(1 + j_0)\alpha^2/2\}$$

and

$$\mathcal{W}(y, J) = e^{-nb^2/2} |\det(I + (bI - J - 2mm^T/n)V)|. \quad (8.1)$$

Here $\varphi_{d,\nu}$ is the density of a real Gaussian with mean d and variance ν . Matrix restrictions are included in T .

Proof. In the conditioning above, y is fixed and $m = \tanh y$ is its deterministic profile. The root map is proper, since $\Phi_J(y) - y$ is bounded for fixed J, n . Its critical target values have Lebesgue measure zero, and the Gaussian field has an everywhere positive smooth density. The area formula therefore expresses the expected root sum as the integral of the observation density at $\Phi_J(y)$ times $|\det D\Phi_J(y)|$; exhaustion by compact balls and monotone convergence handle arbitrary nonnegative tests.

The derivative is

$$D\Phi_J(y) = I + (bI - J - 2mm^T/n)V.$$

The conditioned observation has covariance $\nu I + mm^T/n$, determinant $\nu^n(1 + j_0)$, and inverse $\nu^{-1}I - mm^T/[n\nu^2(1 + j_0)]$. After dividing its density by $\prod_i \varphi_{d,\nu}(y_i)$, the log ratio per site, apart from the determinant prefactor, is

$$-bC_a - \frac{b^2 j_0}{2} + \frac{(C_a + bj_0)^2}{2(1 + j_0)} = \frac{(1 + j_0)\alpha^2}{2} - \frac{b^2}{2}.$$

This proves the formula, including the conditional matrix law and all nonnegative restrictions. $\square$

Lemma 8.2 (A determinant square bound). *For each fixed D , the conditional second moment $\mathbb{E}_{\mathbf{Q}_y} \mathcal{W}(y, J)^2$ is at most n^{C_D} on $\|y\| \leq n^D$, uniformly over the parameters under consideration. Outside these balls the bound has a fixed polynomial envelope in $\|y\|$.*

Proof. The identity $\det(I + AB) = \det(I + BA)$ writes the determinant in $\mathcal{W}$ as the symmetric determinant $\det(I + \sqrt{V}(bI - J - 2mm^T/n)\sqrt{V})$. First compare the conditional matrix with an unconditioned GOE matrix W_0 . For $q > 0$, put $u = m/\sqrt{nq}$ and $E = u^\perp$. The conditional law leaves $P_E W P_E$ as a fresh GOE block. Its remaining row and scalar are Gaussian; with $e = P_E \mathbf{1}/\sqrt{n}$ they have the representation

$$\begin{aligned} P_E J u &= \frac{M}{\sqrt{q}} e + \frac{\sqrt{q}}{\sqrt{n}\nu} P_E (y - d\mathbf{1}) + \sqrt{\frac{s}{n\nu}} g_E, \\ u^\top J u &= \frac{M^2}{q} + \frac{2(a + bq)}{2q + s} + \sqrt{\frac{2s}{n(2q + s)}} \xi, \end{aligned}$$

where g_E, ξ are independent standard Gaussians, also independent of the E block. Share that block with W_0 . The row change, the plant $ee^\top$, and the TAP spike are a perturbation of fixed rank. Its norm is polynomial in $n, \|y\|$ and the fresh Gaussian norms: $|M|/\sqrt{q} \leq 1$, $a^2 \leq q\langle (y - d)^2 \rangle$, and

$\nu, 2q + s \geq s$ with s^{-1} polynomial on the present range. When $q = 0, m = 0$ and the conditioning is independent of W , so the same conclusion is immediate.

Here is a useful bound for this reduction. For symmetric H and rank- j symmetric N ,

$$|\det(H + N)| \leq (1 + \|N\|/\varepsilon)^j |\det(H + i\varepsilon I)|.$$

It follows from singular-value interlacing after factoring $H + i\varepsilon I$: the factor $H(H + i\varepsilon I)^{-1}$ is a contraction and the remaining change has rank j . Take $\varepsilon = n^{-5}$. Markov's polynomial derivative inequality on $[-1/n, 1/n]$, followed by integration on a small interval around a maximizer, gives for every real polynomial P of degree at most n

$$|P(i\varepsilon)|^2 \leq n^C \int_{-1/n}^{1/n} |P(z)|^2 dz.$$

Indeed $\|P'\|_\infty \leq Cn^3 \|P\|_\infty$ on this interval; iterating this bound controls the imaginary displacement, and an interval of length cn^{-3} controls the maximum by the integral. Truncate enormously large fresh entries at negligible cost by crude determinant bounds.

It remains to bound the unconditioned square. For real z , put $x_i = 1 + z + bv_i$. The exact identity is

$$\mathbb{E}_{W_0} \det((1+z)I + bV - \sqrt{V}W_0\sqrt{V})^2 = \mathbb{E}_H \prod_i \left[(x_i + iv_i H_1)(x_i + iv_i H_2) + v_i^2(|h_1|^2 + |h_2|^2) \right],$$

where H_1, H_2 are independent real normals of variance $1/n$ and h_1, h_2 are independent circular normals with $\mathbb{E}|h_j|^2 = 1/n$. For completeness, represent each determinant as the top coefficient of the exponential in one set of exterior variables $\bar{\psi}, \psi$. Averaging W_0 adds $-[X_{11}^2 + X_{22}^2 + 2X_{12}X_{21} + 2PQ]/(2n)$ to the exponent, where $X_{ab} = \sum_i v_i \bar{\psi}_i^a \psi_i^b$, $P = \sum_i v_i \bar{\psi}_i^1 \bar{\psi}_i^2$, and $Q = \sum_i v_i \psi_i^1 \psi_i^2$. Decoupling these four quadratic terms with the displayed Gaussian variables and extracting the site coefficients gives the identity.

Move each H_j contour to $H_j + ib$. The factors become $1 + z + iv_i H_j$, while the two Gaussian densities cost e^{nb^2} in modulus. This exactly cancels the factor e^{-nb^2} in $\mathcal{W}^2$. The contour move is valid because the integrand is an entire polynomial times a Gaussian. If e_1, e_2 are the eigenvalues of the two-by-two Hermitian matrix with diagonals H_1, H_2 and squared off-diagonal modulus $|h_1|^2 + |h_2|^2$, the remaining modulus is at most $\prod_{j=1}^2 [(1+z)^2 + e_j^2]^{n/2}$, since $0 \leq v_i \leq 1$.

More explicitly, write $R_H^2 = H_1^2 + H_2^2 + 2(|h_1|^2 + |h_2|^2) = e_1^2 + e_2^2$. The Gaussian exponent is $-nR_H^2/2$. For $R_H \leq 1$, the inequality $\log(1+x) \leq x - cx^2$ on a fixed bounded interval bounds the remaining integrand by $C \exp(-cnR_H^4)$; for $R_H > 1$, split off a fixed compact annulus and then use $\log(1+x) \leq x - cx$ for x sufficiently large, obtaining an integrable $Ce^{-cn \min(R_H^4, R_H^2)}$ bound. There are six real Gaussian variables, so their normalized integral is at most $Cn^3 \int_{\mathbb{R}^6} e^{-cn \min(|u|^4, |u|^2)} du \leq Cn^{3/2}$. The bounded-rank and real-shift reductions multiply this by another fixed polynomial. This proves the second-moment bound. $\square$

Proposition 8.3 (Scalar root diagnostics). *Fix an exclusion budget $A > 0$ and an accuracy for the little-oh terms. After choosing a sufficiently small upper bound on r and then a sufficiently large n , the following statements hold for $k = nr^3 \geq \log^{10} n$.*

The total integral of $G(y) dy$ is at most polynomial in n . Outside mass $n^C e^{-Ak}$ under this scalar integral,

$$q \asymp_A r, \quad |M| = O_A(q), \quad |\alpha| = O_A(q).$$

There are absolute $L, c > 0$ such that the part with $q \geq q_{\min} \geq Lr$ has mass at most $n^C e^{-cnq_{\min}^3}$. For a sufficiently small fixed exclusion budget, the retained parameters instead satisfy $q, M = (1 \pm \epsilon)r$ and $|\alpha| \leq \epsilon q$, for any requested fixed $\epsilon > 0$.

Under either budget the retained empirical moments satisfy

$$\langle m^4 \rangle = 3q^2 - 8q^3 + o(q^3), \quad \langle m^6 \rangle = 15q^3 + o(q^3), \quad \langle m^3 \rangle = 3qM + O_A(q^3). \quad (8.2)$$

For $l = \langle ym \rangle / q$, $|\langle y - lm \rangle| \leq C_A q^3$, with an arbitrarily small leading constant under a sufficiently small budget. For any fixed sufficiently small $\eta > 0$, one may also retain

$$\langle y^2 \mathbf{1}_{\{|y| > q^{1/2-\eta}\}} \rangle \leq C_A q^4, \quad \frac{1}{n} \|P_{\text{span}(m, \mathbf{1})^\perp} y\|^2 \asymp q^3.$$

Finally, fix $\tau > 0$ and $D < \infty$. There are constants $c, C > 0$ such that, for every (t_1, t_3) with $\tau \leq \|(t_1, t_3)\| \leq n^D$ and every $\theta \in \mathbb{R}$, at least cn indices satisfy

$$|m_i| \leq C\sqrt{q}, \quad |y_i - lm_i| \leq Cq^{3/2}, \quad \text{dist} \left(t_1 \frac{m_i}{\sqrt{q}} + t_3 \frac{y_i - lm_i}{q^{3/2}} + \theta, \pi\mathbb{Z} \right) \geq c.$$

The determinant-weighted exclusions follow from Lemma 8.2 after increasing the preliminary budget. Through Lemma 8.1, they bound the expected number of excluded regular roots. After removing the null set of critical target fields, the probability that any root is excluded is at most that expected count. This is the simultaneous-root meaning of the diagnostics.

Proof. Scalar comparison and exclusion of remote parameters. First restrict to a fixed polynomial ball in y . The bound $a^2 \leq q\langle (y - d)^2 \rangle$ and $j_0/(1 + j_0) \leq 1/2$ give a Gaussian tail envelope outside it. On the ball, the logarithmic likelihood and its parameter derivatives are polynomially bounded, since $\nu \geq s$ and s^{-1} is polynomial. Divide q, M, a into a polynomial number of bins with sufficiently fine inverse-polynomial mesh. Replacing their empirical values by a bin representative changes the log likelihood by $O(1)$. In the transport and variance comparisons through the retained-range bound below, the symbols $q, M, a, b, \nu, d, C_a, j_0, \alpha$ denote these frozen representatives; y and $m = \tanh y$ remain the variables being integrated. In the variance comparison, j_0^{-1} is used only where $C_a > 1$; Cauchy–Schwarz and the polynomial ball then also bound q^{-1} polynomially.

Suppose $\alpha < 1$ stays a fixed distance from one. The map $z'_i = y_i - \alpha \tanh y_i$ has Jacobian $\prod_i (1 - \alpha v_i)$; here α is the frozen bin value. Direct expansion of the two Gaussian exponents gives

$$\frac{G(y) dy}{\prod_i \varphi_{d, \nu}(z'_i) dz'} = e^{O(1) - n(b\alpha + \alpha^2/2)} \prod_i (1 - \alpha v_i)^{-1} \leq e^{O(1) - nH(\alpha)},$$

where

$$H(u) = \frac{u^2}{2} - b[-\log(1 - u) - u].$$

The inequality uses $\log(1 - uv) \geq v \log(1 - u)$ for $0 \leq v \leq 1$. In particular $H(u) \geq qu^2/6$ for $u \leq q$, with a cubic or stronger penalty on the negative side when q is small.

The remaining branch is $C_a > C_0 := 1 + j_0 q$, equivalently $\alpha > q$. Compare there with the product Gaussian on y having variance $\nu C_a^2 / j_0$. Cauchy–Schwarz on a gives the remaining rate $f(C_a)/2$, where

$$f(C_a) = C_a^2 / j_0 - 1 - \log(C_a^2 / j_0) - (C_a - b)^2 / (1 + j_0).$$

The bounds $f(C_0) \geq -2q^3/3$, $f'(C_0) \geq 2q/(1 + q)$, and $f'' \geq 1$ exclude $\alpha > q + C_A R_*^{3/2}$ at budget AnR_*^3 , where $R_* := \max(r, q)$. The H comparison excludes $\alpha < -C_A R_*$. On the retained range

$$-C_A R_* \leq \alpha \leq q + C_A R_*^{3/2}, \quad H(\alpha) \geq -\delta R_*^3,$$

where δ can be made arbitrarily small by decreasing the upper scale.

The frozen-symbol convention for those two comparisons ends here. We also need a fixed negative rate away from $q = 0$ as $s \downarrow 0$. Fix $q_0 > 0$. For a probability law P with finite second moment and $\mathbb{E}_P \tanh^2 Y \geq q_0$, define

$$q = \mathbb{E}_P \tanh^2 Y, \quad M = \mathbb{E}_P \tanh Y, \quad d = M + s, \quad \nu = q + s, \\ a = \mathbb{E}_P[(Y - d) \tanh Y], \quad b = 1 - q, \quad j_0 = q/\nu, \quad \alpha = (a/\nu - b)/(1 + j_0).$$

Put $\psi_s(P) = (1 + j_0)\alpha^2/2$ and

$$\mathcal{R}_s(P) = \psi_s(P) - \text{KL}(P \| N(d, \nu)).$$

The endpoint scalar inequality of Lemma 13.3, proved independently below, gives $\mathcal{R}_s \leq 0$, with equality only for the indicated Gaussian. At $s = 0$, a consistent equality law with $q > 0$ would be $N(M, q)$. Its mean equation $M = \mathbb{E}_{N(M, q)} \tanh Y$ forces $M = 0$, since the difference of the two sides is strictly increasing and vanishes at zero. But then $\mathbb{E}_{N(0, q)} \tanh^2 Y < \mathbb{E} Y^2 = q$, a contradiction.

For completeness, this pointwise exclusion has the uniformity needed by the integral. Fix a bound on $\mathbb{E}_P Y^2$. The resulting set of laws with $q \geq q_0$ is weakly compact. Its first moments are uniformly integrable, so the average of $y \tanh y$ and hence ψ_s are continuous there; $\nu \geq q_0$ keeps the denominators separated from zero. Relative entropy is lower semicontinuous, making $\mathcal{R}_s$ upper semicontinuous. Its maximum at $s = 0, q \geq q_0$ is therefore strictly negative, and remains so for sufficiently small s . The second-moment bound also makes replacement of nearby Gaussian parameters uniformly $e^{o(n)}$ in the product density. A finite cover by neighborhoods on which one bounded continuous entropy test has this slack bounds the scalar integral by e^{-cn} . This is the compact-set integral bound in the proof of Lemma 13.3, with its smoothing variance set to zero and $\nu = q + s \geq q_0$; that proof records the finite-cover argument of [28, Lemma 5.4]. The Gaussian envelope $G(y) \leq e^{Cn-c \sum_i y_i^2}$ on $q \geq q_0$ removes the second-moment restriction. This proves the fixed- q_0 exclusion; the cubic rate for shrinking q is derived next.

Empirical moments at the cubic scale. The transport comparison handles the negative and moderate scalar parameters, and variance inflation excludes the remaining positive range. On the retained range we can now impose product-Gaussian moment diagnostics and use root consistency to locate the overlap scale.

Return to the root integral. For the shifted-coordinate reference of the bin containing y , write $\alpha_{\text{bin}}, d_{\text{bin}}, \nu_{\text{bin}}$ for its frozen values and put

$$z'_{\text{bin}, i} = y_i - \alpha_{\text{bin}} \tanh y_i, \quad Z_i = (z'_{\text{bin}, i} - d_{\text{bin}})/\sqrt{\nu_{\text{bin}}}.$$

The rate parameters in this product-reference estimate also use the frozen bin values. For any fixed $A' > 0$, exponential Markov bounds give the following outside product probability $e^{-A' n R_*^3}$: the second moment on $|Z_i| > R_*^{-\eta}$ is at most $C A' R_*^3$, and clipped moments through any prescribed fixed degree differ from their Gaussian values by $O(\sqrt{A'} R_*^{3/2})$. Choose η sufficiently small for that degree. These estimates use a fixed exponential tilt for the tail square and a tilt of order $\sqrt{A'} R_*^{3/2}$ for each clipped polynomial; their constants stay bounded as $A' \downarrow 0$.

The scalar change of measure costs at most $e^{\delta n R_*^3}$ on a retained bin. Hence a reference exception of size $e^{-A' n R_*^3}$ contributes at most $e^{-(A' - \delta) n R_*^3}$ to the scalar integral, up to the polynomial bin factor. For a small target budget choose A' proportional to that budget and then make $\delta < A'/2$ by shrinking the upper scale. This preserves the small $O(\sqrt{A'})$ moment errors needed below.

For the consistency and residual conclusions below, the symbols $q, M, a, b, \nu, d, C_a, j_0, \alpha$ again denote the exact empirical quantities defined at the start of the section. Expand the exact identity

$m = \tanh(z'_{\text{bin}} + \alpha_{\text{bin}} m)$ on the clipped region and use the tail second moment on its complement. Replacing the bin parameters in the resulting expressions by the empirical quantities is included in the displayed errors, using the initially chosen sufficiently fine mesh. The first consistency estimate is

$$q = \nu + d^2 + O_A(R_*)(\nu + d^2).$$

It implies $s + d^2 \leq C_A R_* q$. If $q \leq r$, the inequality $r^2 = s \leq C_A r q$ forces $q \geq c_A r$; otherwise this is automatic. Thus $R_* = O_A(q)$ and $|d| = O_A(q)$. On this narrower range the retained H is nonnegative. Keeping the next terms gives

$$0 = s + d^2 + 2\alpha q - 2q^2 + o_A(q^2), \quad M = d + \alpha d - qd + O_A(q^2).$$

Apart from an $o(q^2)$ error, the last constant tends to zero with the small budget. The H and f penalties then force $|\alpha| \ll q$. Since $d = M + s$, division of the two consistency equations by q^2 gives

$$\frac{s}{q^2} = \frac{d}{q} + o_{\text{small}}(1), \quad \frac{s}{q^2} + \left(\frac{d}{q}\right)^2 = 2 + o_{\text{small}}(1).$$

The first ratio is nonnegative. These two relations therefore force s/q^2 and d/q close to one, proving the small-budget location. In particular, bins with $q \gg r$ have a fixed negative rate cnq^3 . This tail also pays a prescribed budget Ak in the large-budget calculation: choose q_{min}/r so that $c(q_{\text{min}}/r)^3 > A$, with slack for the polynomial bins.

Finally the total scalar integral is polynomial. The exceptional bins just considered have strict negative rates. On the retained bins the reference product integral is one and $H \geq 0$; indeed $H(q) \geq q^3/6$ and this sign persists up to $q + O_A(q^{3/2})$. Summing over the polynomial grid proves the claim.

Fourth and sixth moments and the cubic residual. The scale comparison is not yet enough for the matrix estimates: they also use the first correction to the fourth moment. We therefore keep the cubic-order terms in the same expansion. The cubic residual will require a separate cancellation at small exclusion budget.

The same expansion has linear coefficient $1 + \alpha + O(q^2)$ and cubic coefficient $-1/3 + O(q)$. The consistency equation for q gives

$$3q^2 = 3\nu^2 + 6\nu d^2 + 12\alpha\nu^2 - 12\nu^3 + o(q^3),$$

whereas direct expansion of the fourth moment gives

$$\langle m^4 \rangle = 3\nu^2 + 6\nu d^2 + 12\alpha\nu^2 - 20\nu^3 + o(q^3).$$

Since $\nu = q + O(q^2)$, their difference is $-8q^3 + o(q^3)$. Likewise $\langle m^6 \rangle = 15\nu^3 + o(q^3)$ and $\langle m^3 \rangle = 3\nu d + O_A(q^3) = 3qM + O_A(q^3)$. The clipped empirical errors are $o(q^3)$ at this scale even for a fixed large budget. On the complement, boundedness of $\tanh$ and the original-scale tail second moment $O_A(q^4)$ control the remainders.

For the residual, use the identity

$$\langle y - lm \rangle = \langle y - m \rangle - \frac{\langle (y - m)m \rangle}{q} M.$$

Expand it in z'_{bin} through degree six, including the occurrences of M and q in this expression. Replacing empirical moments by their Gaussian values costs $O(\sqrt{A'} q^3) + o(q^3)$. The order- q^2 terms cancel: the cubic Gaussian moment starts with $3\nu d$ and the fourth with $3\nu^2$. After division by

q^3 , the remaining leading coefficient is continuous in the bounded ratios $\nu/q, d/q, \alpha/q$, with ν/q bounded below.

At the small-budget limit these ratios approach 1, 1, 0. The leading coefficient vanishes there by an exact reflection identity. If $Y \sim N(\nu, \nu)$, its density obeys $\varphi(-y) = e^{-2y}\varphi(y)$, so pairing y and $-y$ gives

$$\mathbb{E}F(Y) = \mathbb{E}[F(Y) \tanh Y] \quad \text{for every integrable odd } F.$$

With $F(y) = y$ and $F(y) = \tanh y$, this gives $\mathbb{E}Y = \mathbb{E}[Y \tanh Y]$ and $\mathbb{E} \tanh Y = \mathbb{E} \tanh^2 Y$. The integrated residual is therefore exactly zero at $\alpha = 0, d = \nu$. Continuity and the empirical error prove the stated small leading constant.

Nonlattice and spanning diagnostics. On bounded standardized inputs, the normalized profiles $m/\sqrt{q}$ and $(y - lm)/q^{3/2}$ converge to the nonsingular pair z and $z^3/3 - z$. Their Gram matrix on a fixed bounded input set is consequently bounded above and below. Put $w = y - lm$. Then $\langle wm \rangle = 0$, $\langle w^2 \rangle \asymp q^3$, and $1 - M^2/q = 1 - O_A(q)$ is bounded below. The normalized squared norm of the projection of w onto $\text{span}(m, \mathbf{1})$ is $\langle w \rangle^2 / (1 - M^2/q) = O_A(q^6)$. Removing both directions therefore leaves squared norm comparable to nq^3 , as asserted.

For phase dispersion we need derivative control of the actual profiles, because the frequencies may grow with n . In a frozen reference bin, write $Y(z)$ for the inverse of

$$Y - \alpha \tanh Y = d + \sqrt{\nu} z, \quad Y'(z) = \frac{\sqrt{\nu}}{1 - \alpha \text{sech}^2 Y(z)}.$$

The retained parameters satisfy $d = O_A(q)$, $\nu = q + O_A(q^2)$, $\alpha = O_A(q)$, and $l = (a + dM)/q = 1 + q + O_A(q^2)$. The last expansion follows by expanding $ym = m^2 + m^4/3 + m^6/5 + \dots$ on the clipped region and using (8.2) and the tail-square bound. On every fixed bounded z interval, the functions

$$f_q(z) = \frac{\tanh Y(z)}{\sqrt{q}}, \quad g_q(z) = \frac{Y(z) - l \tanh Y(z)}{q^{3/2}}$$

therefore satisfy $f_q(z) = z + O_A(\sqrt{q})$ and $g_q(z) = z^3/3 - z + O_A(\sqrt{q})$ in C^1 . For example,

$$g'_q(z) = \frac{(1 - l \text{sech}^2 Y(z))\sqrt{\nu}}{q^{3/2}(1 - \alpha \text{sech}^2 Y(z))},$$

which converges uniformly to $z^2 - 1$ on that interval.

Put $R = \|(t_1, t_3)\|$ and $v = (t_1, t_3)/R$. Each polynomial $v_1 z + v_3(z^3/3 - z)$, with $\|v\| = 1$, has an interval of fixed positive length on which its derivative is bounded above and away from zero in absolute value. Compactness in v permits a finite choice of such intervals and uniform constants. The C^1 bounds give the same derivative bounds for the actual function $v_1 f_q + v_3 g_q$ when the upper scale is small. On its chosen interval this function is monotone. For all sufficiently large R , counting periods of $R(v_1 f_q + v_3 g_q) + \theta$ shows that a fixed positive fraction of the interval stays a fixed distance from $\pi\mathbb{Z}$, uniformly in θ . For $\tau \leq R$ in a fixed bounded range, compactness in $R, v, \theta \bmod \pi$ gives the same conclusion with possibly smaller constants. Gaussian density is bounded below on the bounded intervals used. No approximation error is multiplied by the growing frequency in this argument.

Use a smooth periodic test supported a little inside this separated phase set and a smooth cutoff inside the bounded input interval. For each fixed choice, product concentration gives failure e^{-cn} . All parameter derivatives on the retained bins are polynomially bounded, since q^{-1} is polynomially bounded. The parameter l is determined by the existing bins through $(a + dM)/q$. Refine their mesh, and mesh frequencies and phase, at inverse-polynomial accuracy chosen after D , so that every

test changes by less than the fixed separation slack even for $R \leq n^D$. There are only polynomially many tests. Their union retains failure $e^{-c'n}$, and the scalar change of measure preserves that rate after the upper scale is decreased. The same bounded-input concentration gives the Gram estimates used above. $\square$

9 Conditioned matrix estimates

The scalar diagnostics now determine the small eigenvalues of the conditional matrix. The point is to retain the coefficient 1 in the lower edge q^2 , which later yields the coefficient 1 in s^{-1} .

Put

$$u = \frac{m}{\sqrt{nq}}, \quad E = u^\perp, \quad e = \frac{P_E \mathbf{1}}{\sqrt{n}}, \quad c_3 = \frac{P_{\text{span}(m, \mathbf{1})^\perp} y}{\|P_{\text{span}(m, \mathbf{1})^\perp} y\|}.$$

The scalar diagnostics make c_3 well defined. The plant component e is not normalized: $\|e\|^2 = 1 - M^2/q$. Let $\mathcal{P} = \text{span}(u, e, c_3)$ and let $V_{\mathcal{P}} = (P_{\mathcal{P}} V P_{\mathcal{P}})|_{\mathcal{P}}$. Operators defined on $\mathcal{P}$ are extended by zero on its orthogonal complement. Consider

$$\begin{aligned} K &= V^{-1} + bI - J - 2quu^\top / \langle v^2 \rangle, \\ K_o &= (b + b^{-1})I - J + [V_{\mathcal{P}}^{-1} - b^{-1}I_{\mathcal{P}}] - 2quu^\top / b^2, \\ K_T &= V^{-1} + bI - J - 2quu^\top. \end{aligned}$$

Proposition 9.1 (Conditioned precision bounds). *Fix a sufficiently small $\eta > 0$, an accuracy $\epsilon > 0$, and a budget $B > 0$. On retained scalar data with the required accuracies, the following estimates hold with the integrated exclusion specified below. For $K \in \{K, K_o, K_T\}$,*

$$K \succeq (1 - \epsilon)q^2 I, \quad \text{Tr } K^{-2} \leq Cn/q, \quad \|A_0 K^{-1/2}\| \leq Cq^{-\eta}, \quad A_0 = 2I - J. \quad (9.1)$$

For K_o the last bound holds with $\eta = 0$. Each precision has at least $c_A n q^3$ eigenvalues below $C_A q^2$, and $\text{Tr } K_o^{-1} = nb + o(\sqrt{n/q})$. For $K \in \{K, K_o\}$, the three projection bounds are

$$u^\top K^{-1} u \leq C_A q^{-2}, \quad e^\top K^{-1} e \leq C_A q^{-1}, \quad c_3^\top K^{-1} c_3 \leq C_A.$$

Let $\mathcal{S}$ be the retained scalar set, let $\mathcal{G}_{\text{spec}}$ be the original spectral event, and let $\mathcal{F}_y$ be failure of any estimate above. In the conditional law $\mathbb{Q}_y$ of Lemma 8.1, the bound is

$$\int_{\mathcal{S}} G(y) \mathbb{Q}_y(\mathcal{G}_{\text{spec}} \cap \mathcal{F}_y) dy \leq n^C e^{-Bk}.$$

The same bound with the determinant weight follows after increasing the preliminary budget. The spectral restriction stays inside the event being estimated; the argument does not condition on its probability.

Proof. Conditional law and Haar concentration. We use Haar concentration in the Frobenius metric on proper rotations: an L -Lipschitz function on $SO(N)$ has tail $2e^{-cNz^2/L^2}$ about its mean. See [26, Sections 5.2–5.3, Proposition 5.13 and Theorem 5.16]. A constant-mesh net of size $e^{O(j)}$ then gives an operator bound on a j -dimensional space, using polarization for bilinear forms. All such tests below condition first on deterministic eigenvalues, so they also apply to resolvents defined separately on spectral bins.

In $\mathbf{Q}_y$, the scalar data and the vectors determined by y are fixed. The regression in Lemma 8.1 gives

$$J|_E = W_E + ee^\top, \quad p := P_E J u = p_0 + \sqrt{s/\nu} w, \quad p_0 = \frac{q}{\nu} \frac{P_E(y - d\mathbf{1})}{\sqrt{nq}} + \frac{M}{\sqrt{q}} e,$$

where W_E is a centered GOE block and $w \sim N(0, I_E/n)$ is independent of it. The mean p_0 is the projected regression mean of $Wm + \sqrt{s}g$, plus the plant column. We may next condition on the eigenvalues of W_E and retain its Haar eigenframe. On the intersection with $\mathcal{G}_{\text{spec}}$, interlacing of W_E , $W_E + ee^\top = J|_E$, and J supplies the required counts and $\lambda_{\max}(W_E) \leq \lambda_{\max}(J)$. The concentration probabilities are estimated before that intersection is imposed.

Diagonal truncation and the compensating moments. Compare with $D = (b + b^{-1})I - J$. In K and K_T , replace the positive diagonal entry $v_i^{-1} - 1$ by $\min\{v_i^{-1} - 1, q^{1-\eta}\}$. This gives a lower precision. Let B be the difference of either truncated precision, or of K_o , from D , and set $B_E = P_E B|_E$ and $f = P_E B u$.

The scalar moments give the quantities needed by the Schur complement. With $\mu_B = n^{-1} \text{Tr } B_E$ and $\nu_B = n^{-1} \text{Tr } B_E^2$,

$$\begin{aligned} \|B\| &= O(q^{1-\eta}), & \mu_B, \nu_B &= O(q^2), & \mu_B - \nu_B &= o(q^2), \\ u^\top B u &\geq 2q^2 + o(q^2), & \|f\|^2 &= 6q^2 + o(q^2), & p_0 &= f/3 + O_A(q^{3/2}), \\ e^\top f &= O_A(q^2), & \|B_E e\| &= O(q), & e^\top B_E e &= O_A(q^2), & e^\top p_0 &= O_A(q^{3/2}). \end{aligned}$$

The error in the vector relation is in Euclidean norm.

For the diagonal precisions, the truncated entry expands as $m^2 + m^4 + \dots$ on $|y| \leq q^{1/2-\eta_0}$, with $0 < \eta_0 < \eta/2$. The missed bounded powers on the complement are $O(m^2)$ and have average $O_A(q^4)$ by the tail-square diagnostic. The bulk mean is therefore $3q^2 - q^2 = 2q^2$ to leading order, equal to the centered variance $\langle (m^2 - q)^2 \rangle = 2q^2 + o(q^2)$. In the root direction,

$$\frac{\langle m^4 + m^6 \rangle}{q} = 3q + 7q^2 + o(q^2), \quad \frac{2q}{\langle v^2 \rangle} = 2q + 4q^2 + o(q^2).$$

Subtracting the scalar $b^{-1} - 1 = q + q^2 + O(q^3)$ leaves at least $2q^2 + o(q^2)$. Also

$$\|P_E \text{diag}(m^2)u\|^2 = \frac{\langle m^6 \rangle}{q} - \left(\frac{\langle m^4 \rangle}{q} \right)^2 = 6q^2 + o(q^2).$$

This is the leading value of $\|f\|^2$.

The root identity $y = m + m^3/3 + O(m^5)$ on the clipped region gives

$$\frac{P_E y}{\sqrt{nq}} = \frac{1}{3} P_E \text{diag}(m^2)u + O_A(q^{3/2}).$$

The tail contributes $O_A(q^4/q)$ to its squared error. In the formula for p_0 , the extra e coefficient is $(M - (q/\nu)d)/\sqrt{q} = O_A(q^{3/2})$, giving $p_0 = f/3 + O_A(q^{3/2})$. The bounds involving e follow similarly from $\langle m^3 \rangle = 3qM + O_A(q^3)$.

For K_o , put $L_B = (P_{\mathcal{P}}(\text{diag}(m^2) - qI)P_{\mathcal{P}})|_{\mathcal{P}}$. Its norm is $O_A(q)$: in the cubic direction the numerator with an extra factor m^2 is $O(q^4)$ and the normalization is of order q^3 . Since $V_{\mathcal{P}} = bI_{\mathcal{P}} - L_B$,

$$V_{\mathcal{P}}^{-1} = b^{-1}I_{\mathcal{P}} + b^{-2}L_B + b^{-3}L_B^2 + O_A(q^3).$$

Here $(L_B)_{uu} = 2q - 8q^2 + o(q^2)$ and $(L_B^2)_{uu} = 10q^2 + o(q^2)$. Projection onto $\mathcal{P}$ in the second identity loses only $O_A(q^3)$, by the preceding approximation of the cubic column. Substitution gives the same root coefficient and column estimates; the e entries again use the third moment.

Finally, the diagonal restored in K and K_T is positive and is supported where $v_i^{-1} - 1 > q^{1-\eta}$. The tail-square bound limits this support to $O(nq^{3+\eta'})$ sites for some fixed $\eta' > 0$. Thus restoring it is a positive perturbation of rank $o(nq^3)$.

Resolvents away from the edge. Augment by one scalar β for the negative spike $-ee^\top$, so the augmented form contains $\beta^2 - 2\beta e^\top x_E$. Put $\Delta_{\text{sp}} = q^{8/5}$ and define

$$N = \text{ran } \mathbf{1}_{\{2-W_E \leq \Delta_{\text{sp}}\}}, \quad H = E \ominus N.$$

Then $\dim N = O(n\Delta_{\text{sp}}^{3/2})$. The high-block resolvent for the unperturbed comparison, also after subtracting $(1-\epsilon)q^2$, is $R = (2 + O(q^2) - W_E)_H^{-1}$.

Its traces divided by n for the first and second powers are $1 + o(1)$, $\asymp \Delta_{\text{sp}}^{-1/2}$. On the conditional Haar orientation with the spectra satisfying the interlacings, except at arbitrarily high rate,

$$\|R^{1/2}B_ER^{1/2}\| = o(1).$$

Indeed dyadic blocks at depths $t_1, t_2 \geq \Delta_{\text{sp}}$ give centered bilinear deviations at most $Cq^{1-\eta} \max(t_1, t_2)^{3/4}$ by Haar concentration and nets. After division by $\sqrt{t_1 t_2}$, the dyadic sum is bounded by a logarithmic factor times $q^{1-\eta} \Delta_{\text{sp}}^{-1/2} = q^{1/5-\eta}$. The mean costs $O(q^2/\Delta_{\text{sp}}) = O(q^{2/5})$. Both vanish for small η ; the sum includes a constant-width bulk bin.

Uniform tests on the remaining coordinates. For remaining variables γ, x_N, β , normalized by $\gamma^2 + \|x_N\|^2 + \beta^2/\sqrt{\Delta_{\text{sp}}} \leq 1$, write $x = x_N + R(p\gamma + e\beta)$. The needed simultaneous Haar estimates are

$$\begin{aligned} \|x\| &= O_A(1), \quad x^\top B_E x = \mu_B \|x\|^2 + o(q^2), \quad f^\top x = (2q^2 + o(q^2))\gamma + o(q^2), \\ \|R^{1/2}(B_E x + f\gamma)\|^2 &= \nu_B \|x\|^2 + 6q^2 \gamma^2 + o(q^2). \end{aligned} \quad (9.2)$$

Here little-oh can mean any specified fixed leading error.

For a fixed symmetric weight T , normalized Gaussian quadratic tails and polarization give error

$$C_B \frac{\sqrt{k}}{n} \sqrt{\text{Tr } T^2} + C_B \frac{k}{n} \|T\|$$

for a Haar quadratic test on fixed orthonormal axes, at failure e^{-Bk} . The same bound holds for an independent Gaussian column with covariance I_E/n . A subspace test instead pays its dimension through a net. Since $q \asymp_A r$, fixed changes of the constants allow nq^3 in place of k .

Fix, before sampling W_E and w , a fixed-dimensional space S containing e, c_3, p_0, f . On S , compression of R differs from its trace mean by $O_A(q^{3/2} \Delta_{\text{sp}}^{-1/4} + q^3/\Delta_{\text{sp}})$; compression of R^2 has relatively vanishing error. Test w separately, with zero mixed means, so S stays deterministic. Also $\|P_N|_S\| = O(\Delta_{\text{sp}}^{3/4})$. These estimates give $\|x\| = O_A(1)$ and

$$f^\top R p = \frac{\text{Tr } R}{n} f^\top p_0 + o(q^2) = 2q^2 + o(q^2).$$

The terms $f^\top x_N$ and $\beta f^\top R e$ are $o(q^2)$ on the normalized variable ball. This proves the $f^\top x$ line of (9.2). The same tests give

$$P_S x = \frac{\text{Tr } R}{n} \beta e + O_A(q).$$

The fixed-axis estimates have identified the projection of the optimizing vector. We next expose the low frame and the two resolved columns. Rotations preserving these data control the remaining components and hence the final weighted quadratic test.

Now expose N, Rp, Re and put $Y = S + N + \text{span}(Rp, Re)$. Before this exposure, the part of Y beyond S is invariant under rotations fixing S , hence is a uniform subspace of $S^\perp$. Its compressions of B_E^j , $j = 1, 2$, have deviations from their scalar means of size $O_A(q^{j(1-\eta)}\Delta_{\text{sp}}^{3/4})$; mixed compressions with S have the same bound and mean zero. Together with the estimates on $B_E e$, $e^\top B_E e$, and $P_S x$, this gives

$$x^\top B_E x = \mu_B \|x\|^2 + o(q^2), \quad \|B_E x\|^2 = \nu_B \|x\|^2 + o(q^2),$$

as well as $\|P_S B_E x\| = o(q)$ and $\|P_{Y \cap S^\perp} B_E x\| = O_A(q^2)$.

After the exposure, only rotations fixing Y pointwise remain. Split $R = R_{\text{near}} + R_{\text{far}}$ at a small fixed depth ζ . The conditional mean of the far restriction on $Y^\perp$ is its trace on that complement divided by its dimension. Since $\dim Y = O(n\Delta_{\text{sp}}^{3/2})$, $\|R_{\text{far}}\| \leq C/\zeta$, and $n^{-1} \text{Tr } R_{\text{near}} = O(\sqrt{\zeta})$, this scalar is

$$1 + O(\sqrt{\zeta}) + O(\Delta_{\text{sp}}^{3/2}/\zeta) + o(1).$$

Residual Haar concentration and the low-dimensional nets make this the quadratic coefficient on the needed vectors in $Y^\perp$. The component of $B_E x$ in Y contributes $o(q^2)$ to the far quadratic and mixed terms, by its two projection bounds and the fixed far norm. The mixed term with Rf has zero leading conditional mean; $f^\top Rf$ is already covered by the fixed-axis test.

For the near part, $\|R_{\text{near}}\| \leq C/\Delta_{\text{sp}}$ and its compression to S is small by the fixed-axis tests. Its action on the $O_A(q^2)$ component in $Y \cap S^\perp$ is negligible at scale q^2 . On the remaining vectors, the conditional square-root norm is bounded by a constant times

$$\sqrt{\text{Tr } R_{\text{near}}/n} + \sqrt{\dim Y/n} \|R_{\text{near}}^{1/2}\| = O(\zeta^{1/4} + \Delta_{\text{sp}}^{1/4}).$$

This is the Haar Lipschitz bound plus its net cost. First choose ζ small, then the upper q scale. These estimates prove the last line of (9.2).

All these failures were estimated under the Gaussian/Haar conditional law before imposing $\mathcal{G}_{\text{spec}}$. On its intersection, interlacing supplies the spectra of W_E used in the trace and count bounds. Integration in y gives the stated exclusion.

Schur complements, counts, and projections. Substitution of the simultaneous tests into the high-block Schur complements gives the perturbed-minus-unperturbed difference

$$\gamma^2 u^\top B u + 2\gamma f^\top x + x^\top B_E x - \|R^{1/2}(B_E x + f\gamma)\|^2.$$

By (9.2) and $u^\top B u \geq 2q^2 + o(q^2)$, the displayed difference is bounded below on the normalized variable ball by

$$(2 + 4 - 6)q^2\gamma^2 + (\mu_B - \nu_B)\|x\|^2 - o(q^2) = -o(q^2).$$

This is the one-sided root-direction cancellation and the bulk cancellation.

To see the remaining positive margin, let $x_{\text{phys}} = \gamma u + x_E$. The shifted augmented base form is exactly

$$x_{\text{phys}}^\top (D - (1 - \epsilon)q^2 I) x_{\text{phys}} + (\beta - e^\top x_E)^2.$$

The original spectral bound makes its first term at least a fixed multiple of $\epsilon q^2 \|x_{\text{phys}}\|^2$. At the high optimum, $x_H = R(p\gamma + e\beta)$, and

$$\|x_H\|^2 \geq c\beta^2 \|Re\|^2 - C\gamma^2 \|Rp\|^2, \quad \|Re\|^2 \asymp \Delta_{\text{sp}}^{-1/2}, \quad \|Rp\| = o(1).$$

After absorption, the base complement therefore controls $ceq^2(\gamma^2 + \|x_N\|^2 + \beta^2/\sqrt{\Delta_{\text{sp}}})$. The $o(q^2)$ difference preserves this margin and proves the lower precision bound.

The one-sided estimate proves positivity. The trace and shell-density arguments also need enough eigenvalues near the critical edge. Removing the fixed-rank directions makes the bulk cancellation two-sided, so the same calculation can now be repeated at dyadic thresholds.

After removing the fixed-rank u, e interactions, the unaugmented N -Schur complement for a truncated precision differs from the base one by $o(q^2)$ on both sides, because $\mu_B - \nu_B = o(q^2)$. At a shifted threshold $a \leq \Delta_{\text{sp}}/2$ the high block remains positive. Schur inertia therefore identifies the number of full eigenvalues below a with the negative-eigenvalue count of the low complement, up to the removed fixed ranks. The two-sided comparison bounds this count by the W_E edge counts at thresholds displaced by $o(q^2)$.

The same estimates hold on a finite dyadic grid $Cq^2 \leq a \leq \Delta_{\text{sp}}/2$: the shift is at most half the high gap, so the high resolvents remain comparable and their normalized traces change by $O(\sqrt{\Delta_{\text{sp}}})$. This gives the upper count $Cna^{3/2}$ and at least cnq^3 eigenvalues below Cq^2 for a sufficiently large fixed C . Above a fixed multiple of Δ_{sp} , the weighted dyadic form bound gives the same upper count. Restoring the positive discarded diagonal preserves upper counts and removes at most its rank $O(nq^{3+\eta'}) = o(nq^3)$ from the lower spread. Dyadic integration of these counts gives all trace-square bounds.

For K_o , the correction to D has fixed rank. At $b + b^{-1}$, the semicircle transform is exactly b ; (5.1) and the fixed-rank trace error $O(q^{-2}) = o(\sqrt{n/q})$ give its stated inverse trace. For a truncated precision, write A_0 as that precision minus an $O(q^{1-\eta})$ term and use the lower bound at q^2 to obtain $\|A_0 K^{-1/2}\| \leq Cq^{-\eta}$. Restoring the positive diagonal preserves this bound by inverse order. The fixed-rank structure gives the stated $\eta = 0$ bound for K_o .

It remains to control variances in the distinguished directions. For projection bounds at e, c_3 first use D . At $R_1 = (b + b^{-1} - W_E)^{-1}$, deterministic fixed-axis compressions have means $1 + O(q)$ and errors $O_A(q)$, $\|R_1 e\|^2 \asymp_A 1/q$ (lower via modes at sufficiently enlarged width to get any desired rate). Thus $1 - e^T R_1 e \gtrsim_A q$ by the principal-block top eigenvalue bound at $2 + q^2/2$ and shifting. Also $e^T R_1 p = O_A(q^{3/2})$, $c_3^T R_1 p = O_A(q)$. Sherman-Morrison for the spike followed by inclusion of u (whose inverse variance bound after inclusion is the global operator bound) gives the orders stated.

For a truncated precision or K_o , the reduced Schur form controls the base reduced form up to a factor two. View a projection variance as the square of the dual norm of its physical source in the augmented positive form. The high precision stays comparable to R^{-1} , and its minimizing vector changes by $-(R^{-1} + B|_H)^{-1} P_H(B_E x + f\gamma)$. For a source at e or c_3 , (9.2) and its bounded $R^{1/2}$ norm bound the change in testing this optimum by

$$CAq(|\gamma| + \|x_N\| + \Delta_{\text{sp}}^{-1/4}|\beta|).$$

The reduced energy controls q^2 times the square of this parenthesis, so the additional squared dual norm is $O_A(1)$. The remaining high square also costs $O_A(1)$. Adding these costs to the base variances preserves the orders q^{-1} for e and 1 for c_3 ; the global lower precision bound supplies the q^{-2} order for u . Inverse order again permits restoration of the diagonal. $\square$

Corollary 9.2 (Uniqueness of the root on the retained event). *On the retained field event every root is regular and the root is unique. The excluded root-count mass, and hence the exceptional field probability under the spectral restriction, have the chosen exponential rate.*

Proof. First remove the null set of critical target fields from Lemma 8.1. For the remaining fields every root is regular, and the probability of an excluded root is bounded by its expected count. On

the complementary field event, $V^{1/2}K_TV^{1/2} = I + V^{1/2}(bI - J - 2mm^\top/n)V^{1/2}$ is positive. The Jacobian $D\Phi_J$ is similar to this matrix and thus has positive determinant at every root.

Properness and the identity homotopy outside a sufficiently large ball give degree one. Hence the signed root count is one. With all signs positive there is exactly one root. More generally, for a regular target the numbers of positive and negative roots obey $N_+ - N_- = 1$, so the excess multiplicity is $N_+ + N_- - 1 = 2N_-$. This identity also bounds multiplicity errors by the excluded negative-root count.

On the retained event the unique root is a measurable function of (J, h) ; extend it by zero elsewhere when a globally defined profile is needed. Conditional on (J, h) in the planted experiment, the planted spin has law μ_h . Thus an equivariant test involving this profile may replace the plant by a posterior replica, with the excluded event charged separately. This is a conditional posterior identity under planted disorder; the later likelihood comparison performs the transfer to ordinary disorder. $\square$

10 Changing orbit weights and locating the trace

This section changes from the counting measure to Haar measure on an orbit that fixes the three distinguished root directions. It then places the Gaussian trace within one length standard deviation, using three length densities rather than a direct trace expansion.

10.1 The determinant weight

Fix the root y and the conditional matrix data modulo proper rotations that fix $\mathcal{P}$ pointwise. For a proper rotation O of $\mathcal{P}^\perp$, put $R = I_{\mathcal{P}} \oplus O$. The orbit action is

$$J \mapsto RJR^\top, \quad h = \Phi_J(y) \mapsto Rh.$$

It fixes $y, m, \mathbf{1}$ and preserves the root equation. The original spectral restriction is constant on the orbit. Before the determinant weight, disintegrate the root integral as $d\pi(\omega) d\text{Haar}(O)$, where ω denotes the base data and $d\pi$ includes the scalar factors and, when imposed, the invariant spectral restriction. The conditional matrix law is Haar in O because all its fixed means and axes lie in $\mathcal{P}$.

Let $\mathcal{D}_\omega$ be the set of orientations satisfying all precision, trace, count, weighted, and projection diagnostics of Proposition 9.1 with the chosen tolerances. This is stronger than mere Jacobian regularity. Call an orbit suitable when $\text{Haar}(\mathcal{D}_\omega^c) \leq e^{-B_0 k}$ for a large fixed B_0 . The integrated matrix exclusion and Markov's inequality remove unsuitable base data at any desired smaller rate, by starting with a larger raw rate. Lemma 8.2 and Cauchy–Schwarz give the same conclusion after determinant weighting.

On $\mathcal{D}_\omega$, $K_T \succeq cq^2 I$. Extend scalar $\log t$ linearly below cq^2 , and let $F(O)$ be the trace of this extended function of K_T . It equals $\log \det K_T$ on $\mathcal{D}_\omega$. Concavity of the extension and the trace-norm bound on the second rotation derivative of J give the upper Taylor bound

$$F(O') \leq F(O) + \langle \nabla F(O), \dot{O} \rangle + Cq^{-2} \text{dist}(O, O')^2$$

along a minimizing geodesic, with the linear term evaluated on its initial tangent. On diagnostic orientations, $\|\nabla F\| \leq \|[J, K_T^{-1}]\|_{\text{HS}} = o(\sqrt{n})$. For the diagonally truncated precision, the commutator is bounded by $Cq^{1-\eta}\sqrt{n/q}$. Restoring its positive diagonal changes the inverse by a matrix of the discarded rank; the weighted inverse and operator bounds give commutator cost at most $Cq^{-1-\eta}\sqrt{\text{rank}}$. Since that rank is $O(nq^3)$ and η is small, both costs are $o(\sqrt{n})$.

Choose a median m_F of F on $\mathcal{D}_\omega$. Its high and low anchor sets each have a fixed positive Haar mass. Distance to either set has a sub-Gaussian tail at scale $n^{-1/2}$. The upper Taylor bound from a low anchor bounds $F - m_F$ above; applying it from a diagnostic orientation to a high anchor bounds $F - m_F$ below. At distance $C_B \sqrt{k/n}$, the two errors are

$$O_B \left(\sqrt{k} + \frac{k}{nq^2} \right) = o(k),$$

and the complementary Haar mass is e^{-Bk} . At distance $O(n^{-1/2})$ the errors are bounded. Integrating the same distance tail shows that $\mathbb{E}_{\text{Haar}} e^{t(F-m_F)}$ is bounded for every fixed $t > 0$. This also controls the weighted contribution of exceptional diagnostic orientations.

Write $w = \mathcal{W}(y, J)$ and

$$w_0 = e^{-nb^2/2} \det(V) e^{m_F}.$$

On $\mathcal{D}_\omega$, $w = e^{-nb^2/2} \det(V) e^F$, so the preceding bounds compare w with the constant orbit weight w_0 . A fixed positive Haar fraction has $w \geq cw_0$; the total root-count bound therefore gives $\int w_0 d\pi \leq n^C$ on suitable base data. For every $0 \leq p(O) \leq 1$, every small fixed $\xi > 0$, and every needed fixed B , the comparison yields

$$\int w_0 \mathbb{E}_{\text{Haar}} p d\pi \leq e^{\xi k} \int \mathbb{E}_{\text{Haar}}(wp) d\pi + e^{-Bk} \int w_0 d\pi.$$

Conversely,

$$\int \mathbb{E}_{\text{Haar}}(wp) d\pi \leq e^{\xi k} \int w_0 \mathbb{E}_{\text{Haar}} p d\pi + e^{-Bk} \int w_0 d\pi + \mathcal{E}_B,$$

where $\mathcal{E}_B \leq n^C e^{-Bk}$ is the globally excluded weighted mass of non-diagnostic orientations, after the raw matrix rate is chosen. The comparison is integrated over ω ; it does not require a lower weight on each spectral fiber.

These inequalities permit further Markov exclusions over base data. When the input is a field probability with a planted spin, the degree identity in Corollary 9.2 charges excess multiplicity to twice the negative-root count. Its integrated exclusion rate is already available. Thus the same comparisons apply to those posterior tests after this multiplicity error is included.

10.2 Median trace control

We need one further estimate on most orientations of each suitable orbit:

$$|\text{Tr } K^{-1} - nb| \leq C \sqrt{n/q}. \quad (10.1)$$

The precision estimates give the fluctuation scale of Gaussian length, but not the location of its mean. We compare its density at the target and one fluctuation on either side. The lemma below shows that these three densities locate the trace.

Put $L = \sqrt{n/q}$ and $y_w = nb + wL$ for $w = 0, \pm 1$; here y_w is a scalar squared length. On $\mathcal{P}^\perp$, let γ_V be the normalized law $N(0, V)$ conditioned on zero projection onto $\mathcal{P}$, and let $\gamma_b = N(0, bI_{\mathcal{P}^\perp})$. Denote their squared-length densities by ℓ_V and ℓ_b . Define on every orientation

$$A_w(O) = \ell_V(y_w) \mathbb{E}_{\gamma_V} \left[e^{z^T(J-bI)z/2} \mid \|z\|^2 = y_w \right].$$

This is an unnormalized weighted length density. On a diagnostic orientation the full weighted integral is finite; writing its mass as $Z_V^\perp(O)$ gives

$$A_w(O) = Z_V^\perp(O) f_{\hat{C}_O}(y_w), \quad \hat{C}_O = (K|_{\mathcal{P}^\perp})^{-1},$$

where f_C denotes the squared-length density under $N(0, C)$. The fixed-length definition of A_w remains meaningful even on the other orientations.

For the scalar prior, the normalized weighted covariance is

$$\hat{C}_o = ((b + b^{-1})I - J)|_{\mathcal{P}^\perp})^{-1} = (K_o|_{\mathcal{P}^\perp})^{-1}.$$

Let $Z_o^\perp$ be its unnormalized weighted mass and $f_o = f_{\hat{C}_o}$. Both are invariant under the remaining rotation. Haar averaging at each fixed length gives the exact identity

$$\bar{A}_w := \mathbb{E}_O A_w(O) = Z_o^\perp f_o(y_w) \frac{\ell_V(y_w)}{\ell_b(y_w)}.$$

The ratios $\bar{A}_{\pm 1}/\bar{A}_0$ are bounded above and below. For f_o , this follows from $\text{Tr } \hat{C}_o = nb + o(L)$, its trace-square scale L^2 , and the Fourier local limit supplied by the spread. For the prior ratio, both traces are $nb + O(1)$, the top eigenvalues are bounded, and the trace-square discrepancy is $O(nq^2 + 1)$. The scalar length-tilt expansion therefore gives

$$\log \frac{\ell_V(y_w)/\ell_b(y_w)}{\ell_V(y_0)/\ell_b(y_0)} = O\left(\frac{L^2 q^2}{n} + \frac{L^3}{n^2} + o(1)\right) = O(1).$$

The normalizer $Z_o^\perp$ is the same at all three lengths.

We next show $\mathbb{E}_O (A_w/\bar{A}_w)^2 \leq 1 + o(1)$. Let $\rho_V(T | y_w, y_w)$ and $\rho_b(T | y_w, y_w)$ be the overlap densities of two independent draws from the corresponding unweighted priors, conditioned on those lengths. Let $\mu_{o,w}$ be the scalar prior at length y_w , weighted by the common angular energy and normalized. Haar averaging of a pair at fixed overlap gives

$$\mathbb{E}_O \left(\frac{A_w}{\bar{A}_w} \right)^2 = \mathbb{E}_{\mu_{o,w}^{\otimes 2}} \frac{\rho_V(T | y_w, y_w)}{\rho_b(T | y_w, y_w)}, \quad T = z_1^\top z_2.$$

Thus the positive multiplier is a ratio of *prior* densities, while the expectation and its tails use the *energy-tilted* reference law.

To estimate the multiplier, tilt each prior length to y_w ; its scalar tilt is $O(L/n)$. At zero cross tilt the overlap is symmetric, and its covariance with either length is zero. Its conditional curvature is therefore $\text{Tr } \tilde{V}^2$, where $\tilde{V}$ is the length-tilted covariance. The two tilted traces are the same target, and their trace squares differ relatively by $O(q^2 + 1/n)$: the variable prior differs from scalar covariance by normalized Hilbert–Schmidt size $O(q + 1/\sqrt{n})$, and the scalar tilt changes eigenvalues by a uniformly Lipschitz function. Analytic expansion of the three quadratic log moments, with their nondegenerate Hessian, and Fourier inversion of the central prefactors give, for $|T|/n$ small,

$$\log \frac{\rho_V(T | y_w, y_w)}{\rho_b(T | y_w, y_w)} \leq o(1) + C(q^2 + 1/n) \frac{T^2}{n} + C \frac{T^4}{n^3}.$$

Odd cross derivatives vanish by symmetry; the prefactor error is $O(q + |T|/n)$ in the same small-scale convention. Conditioning the initial projections at zero changes only a fixed rank.

Here is the tail comparison that makes this local bound integrable. Put $U = T/L$ and $N = nq^3$. The positive quadratic and quartic errors become

$$C \left((q + (nq)^{-1})U^2 + \frac{U^4}{nq^2} \right).$$

Under $\mu_{o,w}^{\otimes 2}$, quadratic exponentiation and the two length-density bounds give a Gaussian tail in U up to $|U| \leq c\sqrt{N}$. There the quartic error is at most $Cc^2 q U^2$, so the Gaussian tail dominates after

the upper scale is decreased; for fixed U the multiplier tends to one. For $a = |T|/n$ from order q to a small fixed cutoff, the reference length-deficit estimate of Section 6.1 gives e^{-cna^3} . Dividing the positive exponent by na^3 leaves $O(q^2/a + a + 1/(na))$, which is small in this range after the cutoff and upper scale are chosen. These two ranges give uniform integrability through small overlaps.

For a fixed larger overlap, simultaneous quadratic Chernoff bounds for the unweighted priors, conditional on both lengths, give at least the spherical fixed-overlap rate $-\frac{1}{2}\log(1-a^2)$ up to an $o(1)$ loss per coordinate as the upper scale decreases. This rate is strictly larger than $a^2/2$ away from zero. Near the parallel endpoints, fixed tilts chosen before n supply the finite strict rate gap needed below. Exact length densities under these tilts cost at most a polynomial by Fourier inversion, using damping from typical sites or a bounded-eigenvalue block; the preceding $O(L)$ prior length deviations cost subexponentially. The angular pair upper bound of Section 6.1 costs only $a^2/2$ after normalization: the reference angular integral per draw is at least $\exp\{n(1/4 - o(1))\}$. To verify that lower bound, use the Gaussian formula on $\mathcal{P}^\perp$ at precision $((b + b^{-1})I - J)|_{\mathcal{P}^\perp}$. Its centered length density costs only a polynomial, its normalized log determinant is at most $1/2 + o(1)$ by the semicircle bound and fixed-rank interlacing, and $y_w/n = 1 + o(1)$. The strict rate gap controls the remaining overlaps. Thus the displayed second moment is $1 + o(1)$.

The first and second moments imply that all three ratios $A_w/\bar{A}_w$ are close to one outside as small a fixed Haar mass as desired. On diagnostic orientations the common factor $Z_V^\perp(O)$ cancels, so $f_{\hat{C}_O}(y_{\pm 1})/f_{\hat{C}_O}(y_0)$ is bounded above. The next lemma now applies. Its operator and trace-square hypotheses follow from Proposition 9.1 after removing three dimensions. Since $v_i \geq 1/2$ outside at most $2nq$ sites, the coordinate subspace supported on $\{i : v_i \geq 1/2\}$ has intersection with $\mathcal{P}^\perp$ of dimension at least $n - 2nq - 3$. The compressed precision has bounded norm on this intersection, so the min-max principle gives a fixed positive fraction of covariance eigenvalues bounded below. Finally, fixed-rank interlacing gives $|\text{Tr } K^{-1} - \text{Tr } \hat{C}_O| \leq Cq^{-2} = o(L)$. The lemma proves (10.1).

Lemma 10.1 (Three length densities locate a Gaussian trace). *Fix positive constants c_0, C_0, M . Let $0 < q \leq q_0$ and $nq^3 \geq \log^{10} n$, and put $L = \sqrt{n/q}$. Let C be a positive definite matrix of dimension $n - O(1)$ such that*

$$\|C\| \leq C_0 q^{-2}, \quad c_0 L^2 \leq \text{Tr } C^2 \leq C_0 L^2.$$

Assume also that at least $c_0 n$ eigenvalues of C exceed c_0 . Let f be the density of $\|C^{1/2}g\|^2$, with g standard Gaussian, and let $x_0 \in [c_0 n, C_0 n]$. If

$$f(x_0 + L) \leq M f(x_0), \quad f(x_0 - L) \leq M f(x_0),$$

then $|\text{Tr } C - x_0| \leq C_1 L$, where C_1 depends only on the fixed constants. The assertion holds for all sufficiently large n .

Proof. Write $m(\lambda) = \text{Tr}(C^{-1} + \lambda I)^{-1}$ on $\lambda > -\|C\|^{-1}$ and $C_\lambda = (C^{-1} + \lambda I)^{-1}$. The function m decreases continuously from infinity to zero, and

$$m'(\lambda) = -\text{Tr } C_\lambda^2.$$

If $|m(0) - x_0| \leq L$, there is nothing to prove. Otherwise set $\varepsilon = \text{sign}(m(0) - x_0)$ and $x_1 = x_0 + \varepsilon L$. There is a unique λ with $m(\lambda) = x_1$, and $\varepsilon \lambda > 0$. Exponential tilting of the squared length gives its density f_λ under covariance C_λ and the identity

$$\frac{f(x_1)}{f(x_0)} = e^{|\lambda|L/2} \frac{f_\lambda(x_1)}{f_\lambda(x_0)}.$$

This identity treats the two signs with exactly the same inequality.

First we bound the tilt without assuming a local central limit theorem near the final pole. This coarse bound will place the covariance in a region comparable to its original value; sharper density estimates can then improve the logarithmic tilt bound to a constant. Since $\text{Tr } C_\lambda = x_1 \asymp n$, its standard deviation is at most $\sqrt{2}x_1 \leq Cn$. The mean-density bound of Lemma 5.2 gives $f_\lambda(x_1) \geq c/n$. If $\lambda > 0$, the inequality $m(\lambda) \leq n/\lambda$ implies $\lambda \leq C$; if $\lambda < 0$, every eigenvalue increases.

In both cases at least cn eigenvalues of C_λ are bounded below by a fixed positive constant. The quadratic Gaussian product formula therefore gives

$$|\mathbb{E} e^{it\|C_\lambda^{1/2}g\|^2}| \leq (1 + ct^2)^{-cn}, \quad \sup_x f_\lambda(x) \leq Cn^{-1/2}.$$

Consequently $M \geq cn^{-1/2}e^{|\lambda|L/2}$, so $|\lambda|L \leq C \log n$. This is the promised control near a pole: no local central limit theorem is being assumed there, and the density loss is only polynomial.

Choose $a > 0$ so small that $aC_0 < 1/2$. Because $q^2L = \sqrt{nq^3} \geq \log^5 n$, the preceding bound forces $|\lambda| < aq^2$. Thus C_θ is comparable to C for every θ between zero and λ , and $\text{Tr } C_\theta^2 \asymp L^2$. We can now use the spectral spread at the tilted covariance. For these tilts, the largest eigenvalue divided by the square root of the trace square is $O((nq^3)^{-1/2}) = o(1)$. To check the Fourier bound explicitly, if a_i are the eigenvalues of C_λ and $\sigma^2 = 2 \sum_i a_i^2$, concavity of $x \mapsto \log(1 + ux)$ on $[0, \max_i a_i^2]$ gives

$$\prod_i (1 + 4t^2 a_i^2 / \sigma^2)^{-1/4} \leq \left(1 + \frac{4t^2 a_{\max}^2}{\sigma^2}\right)^{-\sigma^2/(8a_{\max}^2)}.$$

Since $a_{\max}^2/\sigma^2 \leq 1/16$ for large n , the right-hand side is bounded by $(1 + t^2/4)^{-2}$, an integrable function independent of the parameters.

Hence its density has supremum at most C/L . Its density at the mean is at least c/L by the same mean-density lemma. Applying these bounds to f_λ in the ratio identity improves the preceding estimate to $|\lambda|L \leq C$. Finally,

$$|m(0) - x_0| \leq L + |m(0) - m(\lambda)| \leq L + \int_0^{|\lambda|} CL^2 du \leq C_1 L.$$

□

11 Comparison on root-preserving rotation orbits

The next comparison transfers continuous-prior estimates to spins on most orientations of each retained root orbit. Its essential quantitative point is the signed cancellation between two spin and two Gaussian replica integrals. An absolute bound on the matrix insertion would be too large for the weighted covariance estimate.

The mixed cube–sphere tensor cancellation of Du and Huang [17, Lemma 3.5] is the close predecessor of this comparison. We retain the observation-dependent projection constraints and a weighted signed matrix kernel throughout the calculation below.

Proposition 11.1 (Covariance estimates on most root orientations). *Let $\mathcal{S}$ be suitable base orbit data retained by the preceding exclusions in Section 10. On $\mathcal{S}$, use the disintegration $d\pi d\text{Haar}$ before the determinant weight w . Here $d\pi$ includes the scalar root factors and the imposed rotation-invariant spectral restriction, Haar is normalized measure on rotations fixing $\mathcal{P}$, and w_0 is the median reference weight defined there.*

Fix $1/2 < \gamma < 1$ and positive accuracies ϵ, η , together with the requested implied small constants below. Choose a sufficiently small fixed rate $a_{\text{orb}} > 0$, then the upper bound on r sufficiently small. At each scale $k = nr^3 \geq \log^{10} n$, for sufficiently large n , one may discard a further set $\mathcal{E} \subseteq \mathcal{S}$ satisfying

$$\int_{\mathcal{E}} w_0 d\pi \leq Cn^C e^{-a_{\text{orb}} k}.$$

On each remaining orbit, a set of Haar orientations of mass $> 3/4$ satisfies simultaneously

$$\begin{aligned} \|\Sigma_s\| &\leq (1 + \epsilon)/s, & \|A_0 \Sigma_s A_0\| &\ll r^{-\gamma}, \\ \|m_s^{\text{post}} - m\| &\ll r^{-1-\eta}, & \|A_0(m_s^{\text{post}} - m)\| &\ll r^{-\gamma/2}. \end{aligned} \quad (11.1)$$

Here m_s^{post} and Σ_s are the posterior mean and covariance. Every implied small constant can be prescribed in advance by shrinking the upper bound on r and then taking n sufficiently large. The previously excluded unsuitable data retain their preceding integrated bounds; w_0 is used only on $\mathcal{S}$. No bound uniform conditional on each spectrum is asserted.

Proof. Projection cuts and the three priors. The plant is gauged to $\mathbf{1}$ as above. Let $\mathbf{P}$ be the $n \times 3$ matrix whose columns are the three orthogonalized profiles spanning $\mathcal{P}$, normalized by $n^{-1}\mathbf{P}^T\mathbf{P} = I_3$. Its first column is $m/\sqrt{q}$, and its second is the normalized vector $\mathbf{1} - (M/q)m$; its third is the normalized residual of y after projection onto those first two columns. We write $\mathbf{P}_j$ for column j , and empirical contractions of matrices always include the factor n^{-1} .

Put $z = X - m$, $N = nq^3$, $L = \sqrt{n/q}$, and use the statistics

$$a = \sqrt{n} \delta, \quad \delta = \langle \mathbf{P}^T z \rangle = Q\vartheta, \quad Q = \text{diag}(\sqrt{q}, q, q^{3/2}), \quad W = \sqrt{N} \vartheta.$$

Here ϑ denotes the scaled projection coordinate. For a pair write $t = \langle (z - \mathbf{P}\delta)(z' - \mathbf{P}\delta') \rangle$, $U = nt/L$, so $t = qU/\sqrt{N}$. The three-column notation always uses empirical contractions over sites. The exact length constraint is $\langle z^2 \rangle = b - 2\sqrt{q}\delta_1$. We tilt by the common energy $z^T(J - b)z/2$; with p the product of independent spins of means m (translated by $-m$), this gives precisely the posterior by the root equation.

We cut p smoothly to $|\vartheta| \leq \epsilon_0$ (one below $\epsilon_0/2$), $\epsilon_0 > 0$ an arbitrarily small fixed constant. Except at a small rate in the true counting law with good spectrum, the cut omits negligible normalized posterior mass (n^{-D} , D any fixed constant). Indeed a posterior spin can replace the plant in tests with the equivariant axes $m, y - lm$ by uniqueness and (8.2), so $|\langle mz \rangle| \ll q$, $|\langle (y - lm)z \rangle| \ll q^3$. The overlaps of plant and replica concentrate about q to any fixed tolerance times q by (5.1), the spherical observation estimates, (6.1)–(6.3) and $q/r \simeq 1$.

Thus $|\langle z \rangle|$ (overlap with the gauged plant minus M) is bounded by arbitrarily small fixed times q , and the subtraction $(M/q)\langle mz \rangle$ in the e axis is likewise small. Orthogonalizing the cubic axis uses $|\langle y - lm \rangle| = O(q^3)$ and residual norm asymptotic, in empirical units up to constant factors, to $q^{3/2}$, by the clipped Gaussian moment diagnostics and the tail-square bound. Here multi-root fields do not affect counting claims: the count of positive-sign regular roots is at most one plus the number of negative-sign roots by degree, whose expectation on good spectrum is exponentially small. Null singular fields may be ignored. The orbit weight comparisons thus allow us to discard further orbits at a small integrated rate measured by $w_0 d\pi$, so that the cut loss is negligible on most Haar orientations.

Use two continuous auxiliary priors, not necessarily of mass one. We fix their length-density units here. Put

$$E = \mathbf{P}/\sqrt{n}, \quad \mathcal{L}(z) = \|z\|^2 + 2m^T z, \quad S_{\mathbf{P}} = E^T V E, \quad \kappa_2 = \langle v^2 \rangle.$$

For a Gaussian prior, the notation $\delta_D(\mathcal{L}(z) - nb) dz$ means the coarea density at the raw squared length nb . Since this shell is $\|z + m\|^2 = n$, it is surface measure divided by $2\sqrt{n}$. In particular, if $T = (\mathcal{L} - nb)/L$, then $\delta_D(\mathcal{L} - nb) = L^{-1}\delta_D(T)$. The spin prior already lies on this shell.

With φ_C the density of $N(0, C)$, define the uncut pre-energy measures

$$\begin{aligned} g_0(dz) &= \mathbf{c}_{g_0} \mathbf{r}_{g_0}(a) \varphi_V(z) \delta_D(\mathcal{L}(z) - nb) dz, \\ o(dz) &= \mathbf{c}_o \mathbf{r}_o(a) \varphi_{bI}(z) \delta_D(\mathcal{L}(z) - nb) dz, \end{aligned}$$

where

$$\mathbf{r}_{g_0}(a) = e^{qa_1^2/\kappa_2}, \quad \mathbf{r}_o(a) = e^{qa_1^2/b^2 - \frac{1}{2}a^T(S_P^{-1} - b^{-1}I)a}.$$

The constants $\mathbf{c}_{g_0}, \mathbf{c}_o$ are fixed by the zero-projection saddle normalization below. Multiplication by the common energy $e^{z^T(J-bI)z/2}$ changes the underlying Gaussian precisions to K and K_o , respectively.

For $Z \sim N(0, C_o)$, $C_o = K_o^{-1}$, put $T_o(Z) = (\mathcal{L}(Z) - nb)/L$. The reference mass used throughout is projection-uncut but still shell-constrained:

$$Z_* := \int e^{z^T(J-bI)z/2} o(dz) = \mathbf{c}_o \det(bK_o)^{-1/2} \frac{f_{T_o}(0)}{L}.$$

Thus its normalized physical law is exactly the density disintegration of $N(0, C_o)$ conditional on $T_o = 0$. Both factors in the last formula are invariant under rotations fixing $\mathcal{P}$, so Z_* is an orbit invariant. We cut g_0 by the same projection cutoff as p , and multiply it there by a projection-only factor to form g . All cuts below are separable across replicas.

The three priors serve different parts of the comparison. The reference prior has an orbit-invariant tilted mass; the variable-variance Gaussian has the precision needed for covariance control. The projection multiplier aligns its single-replica marginal with the spin marginal. What remains is a pair-density calculation in which the signed sum cancels common errors and errors additive in the replica types.

Single-prior projection saddles. We begin by defining the single-prior saddle functions. Write $p_\ell = \mathbf{P}_{\ell\bullet}^T \in \mathbb{R}^3$ for the profile row at site ℓ . The independent site laws are

$$\begin{aligned} \gamma_{p,\ell} &= \frac{1+m_\ell}{2} \delta_{1-m_\ell} + \frac{1-m_\ell}{2} \delta_{-1-m_\ell}, \\ \gamma_{g_0,\ell} &= N(0, v_\ell), \quad \gamma_{o,\ell} = N(0, b). \end{aligned}$$

Their constraint vectors and per-site targets are

i	$D_{i,\ell}(z)$	$c_i(\delta)$	$d_i = \dim D_{i,\ell}$
p	$p_\ell z$	δ	3
g_0, o	$(p_\ell z, z^2)$	$(\delta, b - 2\sqrt{q}\delta_1)$	4.

For a spin, $z_\ell^2 + 2m_\ell z_\ell = v_\ell$ identically, so its shell constraint is automatic; adding a square coordinate would make the constraint covariance singular. For Gaussians the square coordinate and its target are equivalent to the translated length coordinate by a determinant-one triangular change.

Set

$$\Lambda_i(\theta) = \frac{1}{n} \sum_{\ell=1}^n \log \int e^{\theta^T D_{i,\ell}(z)} \gamma_{i,\ell}(dz).$$

On the local interior branch let $\nabla \Lambda_i(\theta_i) = c_i(\delta)$ and $\mathbf{H}_i = \nabla^2 \Lambda_i(\theta_i)$. The branch and its positive Hessian are established on the small projection cut below. With $\mathbf{c}_p = \mathbf{r}_p = 1$, define

$$f_i^a(a) = \mathbf{c}_i \mathbf{r}_i(a) n^{3/2} (2\pi n)^{-d_i/2} (\det \mathbf{H}_i)^{-1/2} e^{n[\Lambda_i(\theta_i) - \theta_i^T c_i(\delta)]}.$$

This is a smooth saddle function per ordinary d^3a , not an exact density for an atomic spin projection. The factor $n^{3/2}$ converts the three raw projection sums $\sqrt{n}a$ to a ; the Gaussian raw length-density coordinate keeps its original units. Primitive lattice covolumes are excluded here and will belong to the mixed counting measures.

At zero projection the constraint Hessians are S_P , $\text{diag}(S_P, 2\kappa_2)$, and $\text{diag}(bI_3, 2b^2)$. Hence the choices

$$\mathbf{c}_{g_0} = \sqrt{4\pi n \kappa_2}, \quad \mathbf{c}_o = \sqrt{4\pi n b^5 / \det S_P}$$

make all three saddle functions agree at zero. These are saddle normalizations, not divisions by the exact shell masses. The additional multiplier defining g on its cut is $f_p^a / f_{g_0}^a$.

These constant normalizations cost only polynomial for the unconditioned product Gaussian density integrals. On the cut we have

$$\log(f_i^a / f_o^a) = O(|\vartheta| + N|\vartheta|^3), \quad (11.2)$$

also with the corresponding Taylor bounds after up to two scaled derivatives (bounded derivatives for the prefactor log, and $O(N|\vartheta|^{3-j})$ for the rate-part log).

We include details of the expansions and bounds; all constants may use the scalar diagnostics above. In particular $S_P = I - O(q)$; $Q^{-1}S_PQ$ and its inverse are bounded. At typical sites, meaning the clipped region of those diagnostics with cutoff exponent sufficiently tiny, the profiles P have bounded empirical absolute moments through any prescribed fixed degree, and y such moments at scale $\sqrt{q}$. Outside that region $\langle y^2 \mathbf{1}_{\text{tail}} \rangle = O(q^4)$; the scaled profiles PQ are bounded there by $C|y|$.

These bounds follow directly from (8.2), $l = 1 + O(q)$, and the residual cubic normalization just noted. Centered moments involving P and exponentially damped site fluctuations of order through fixed $j \geq 2$ thus have tail bounds for their profile factors up to $C_j(1 + q^{4-3j/2})$.

For p solve the linear-projection saddle by tilting by $P\theta$, $\theta = O_Q(|\vartheta|)$, where O_Q denotes a bound in Q^{-1} units. Indeed on that parameter ball the response beyond linear $vP\theta$ has empirical L^2 size $O(q^{3/2}|Q^{-1}\theta|^2)$. Its scaled derivatives of order j through fixed orders have the bound $C_j q^{3/2}|Q^{-1}\theta|^{\max(2-j, 0)}$. This follows by the tanh derivatives (second bounded by a constant times the absolute small argument on typical sites, using the moment bounds), and on the tail the derivatives are exponentially damped since the new field differs from y by at most a small fraction of $|y|$.

Thus scaled smooth inversion applies at uniform small radii. Evaluating the Legendre exponent at $\theta = S_P^{-1}\delta$ instead of the minimizer costs $O(q^3|\vartheta|^4)$ per site, by the ordinary Hessian bounds. The density exponent per site, apart from the Gaussian projection term $-\delta^T S_P^{-1}\delta/2$, is

$$-\sqrt{q}\delta_1^3 - \delta_1^4/4 + O(q^3|\vartheta|^3). \quad (11.3)$$

Indeed the third log moment term is $-\langle mv(P\theta)^3 \rangle/3$, with 111 profile average $\langle mvP_1^3 \rangle = 3\sqrt{q} + O(q^{3/2})$, 112 average $O(q)$ by (8.2).

The fourth term is $-\langle v(1 - 3m^2)(P\theta)^4 \rangle/12$, with 1111 average $3 + O(q)$, 1112 average $O(\sqrt{q})$. The fifth coefficient has a factor $O(m)$, and the sixth remainder is bounded with tail damping. All other scaled monomials thus cost $O(q^3)$ by typical moments and the tail bounds. In the two leading powers θ_1 can be replaced by δ_1 . The estimates retain their Taylor derivative orders: the nonlinear Legendre correction does also by scaled smooth inversion and the stated derivative bounds (the unscaled shift in the minimizing coefficient has norm $O(q^{3/2}|\vartheta|^2)$ with the corresponding derivatives).

For g_0 , first optimize the projection coefficients in the formal saddle and then optimize the length coefficient. The Gaussian projection exponent is exact. At this canonical projection saddle, the length target differs from the mean by $-2\sqrt{q}\delta_1 - \langle (vPS_P^{-1}\delta)^2 \rangle$ per site. This is a saddle mean,

rather than the trace of an exactly projection-conditioned finite-dimensional Gaussian. The length curvature in the partial log moment is $2\langle v^2 \rangle + O(q|\vartheta|^2)$. Thus the additional exponent is the negative square of that deficit divided by $4\langle v^2 \rangle$, to $O(q^3|\vartheta|^3)$; its leading quadratic cancels the explicit rank factor.

Since $\langle (vPS_{\mathbf{p}}^{-1}\delta)^2 \rangle = |\delta|^2 + O(q^3|\vartheta|^2)$, this gives exactly (11.3). These expansions use only the analytic bounded-variance Gaussian length tilt, adjusting the projection tilt by the corresponding three-dimensional linear solve. Length coefficients are $O(q|\vartheta|)$, and scaled Taylor derivative estimates are unchanged. The calculation for o replaces the variance profile by constant b , giving the same terms. The Gaussian central prefactors for all three priors have uniformly bounded log derivatives in ϑ by the nonsingular saddle Hessians and the profile bounds. This proves (11.2).

The signed pair comparison. We next form the pair saddle functions on $|\vartheta|, |\vartheta'| \leq \epsilon_0$ and $|t|/q \leq \epsilon_1$, where ϵ_1 is fixed small and ϵ_0 may be much smaller. For types $i, j \in \{p, g_0, o\}$, put

$$\begin{aligned} D_\ell^{ij}(z, z') &= (D_{i,\ell}(z), D_{j,\ell}(z')), \\ c^{ij} &= (c_i(\delta), c_j(\delta')), \\ h_\ell(z, z') &= (z - p_\ell^T \delta)(z' - p_\ell^T \delta'), \end{aligned}$$

so $n^{-1} \sum_\ell h_\ell = t$ at the projection constraints. Define

$$\Lambda_{ij}(\theta, \beta) = \frac{1}{n} \sum_\ell \log \iint e^{\theta^T D_\ell^{ij} + \beta h_\ell} \gamma_{i,\ell}(dz) \gamma_{j,\ell}(dz').$$

Keep the targets and the δ 's inside h_ℓ fixed when differentiating. On the local interior branch let $\nabla_\theta \Lambda_{ij}(\theta_\beta, \beta) = c^{ij}$, and put

$$F_{ij}(\beta) = \Lambda_{ij}(\theta_\beta, \beta) - \theta_\beta^T c^{ij} - [\Lambda_{ij}(\theta_0, 0) - \theta_0^T c^{ij}].$$

Thus $F_{ij}(0) = 0$; it is the partial infimum over the single-constraint coefficients, minus its value at zero cross tilt. It is a formal optimized log moment, not the conditional log moment of an unsmoothed atomic event.

At the optimizing law write

$$D_2 = \langle \text{Cov}(D_\ell^{ij}, D_\ell^{ij}) \rangle, \quad v_\beta = \langle \text{Cov}(D_\ell^{ij}, h_\ell) \rangle, \quad w_\beta = \langle \text{Var}(h_\ell) \rangle.$$

These are averages of sitewise covariances. Implicit differentiation and the envelope identity give

$$\theta'_\beta = -D_2^{-1} v_\beta, \quad F'_{ij}(\beta) = \langle \mathbb{E}_\beta h_\ell \rangle, \quad F''_{ij}(\beta) = w_\beta - v_\beta^T D_2^{-1} v_\beta.$$

In particular the last expression is the variance after regressing the cross statistic on the single constraints.

Let $\beta_{ij}(t)$ solve $F'_{ij}(\beta_{ij}) = t$, with $t = LU/n$. The formal joint function in ordinary $d^3 a d^3 a' dU$, retaining every raw Gaussian length density, is

$$\begin{aligned} \hat{f}_{ij}^{a,a',U} &= \mathbf{c}_i \mathbf{c}_j \mathbf{r}_i(a) \mathbf{r}_j(a') n^3 L (2\pi n)^{-(d_i+d_j+1)/2} \\ &\quad \times [\det D_2^{ij}(\beta_{ij}) F''_{ij}(\beta_{ij})]^{-1/2} e^{n[\Lambda_{ij}(\theta_{\beta_{ij}}, \beta_{ij}) - \theta_{\beta_{ij}}^T c^{ij} - \beta_{ij} t]}. \end{aligned}$$

The factor $n^3 L$ converts the six raw projection sums and the raw cross sum $nt = LU$. Dividing this formula by its two single saddle functions, and converting U back to t , gives

$$\hat{h}_{ij}^t(t \mid a, a') = \sqrt{\frac{n}{2\pi}} \left[\frac{\det D_2^{ij}(0)}{\det D_2^{ij}(\beta_{ij}) F''_{ij}(\beta_{ij})} \right]^{1/2} e^{n[F_{ij}(\beta_{ij}) - \beta_{ij} t]}.$$

The determinant quotient uses each pair's own constraint dimension: 6, 7, 8 for pp , mixed pairs, and two Gaussian replicas. The Gaussian shell units and all explicit rank factors cancel in this quotient. Define the dimensionless formal ratio

$$\mathcal{R}_{ij} = \hat{h}_{ij}^t / \hat{h}_{oo}^t.$$

It is equally the ratio in U units, since both densities then acquire the factor L/n . The full formal pair ratio is

$$\frac{\hat{f}_{ij}^{a,a',U}}{\hat{f}_{oo}^{a,a',U}} = \frac{f_i^a(a)f_j^a(a')}{f_o^a(a)f_o^a(a')} \mathcal{R}_{ij}.$$

This is an algebraic quotient of saddle formulas. The local-limit argument below relates it to the actual mixed statistic measures. After the multiplier defining g , both effective single factors are f_p^a ; the projection cut does not change the conditional calculation.

The signed comparison and its verification below use the four ordered pairs $i, j \in \{p, g_0\}$. With signs $s_p = 1, s_{g_0} = -1$ and $E_* = 1 + |W| + |W'| + |U|$, the full signed pair-density comparison needed is

$$\frac{f_p^a(a)f_p^a(a')}{f_o^a(a)f_o^a(a')} \sum_{i,j \in \{p, g_0\}} s_i s_j \mathcal{R}_{ij} = \frac{q}{\sqrt{N}} e^{c_2 q U^2} (c_1 U + c_3' U^3) + O\left(\frac{q}{N} E_*^C e^{\zeta E_*^2}\right). \quad (11.4)$$

Here cuts can multiply both sides; the coefficients are bounded and independent of the statistics. The constant ζ can be made as small as needed by reducing the projection cut and the upper r . Central-prefactor relative errors $O(q/N)$ are allowed.

Here is a verification. Abbreviate F_{ij} by F for the pair under discussion. At $\beta = 0$ the canonical laws are independent single-replica laws. Write their site means and variances as λ_i, σ_i , where σ_i denotes variance rather than its square root, and put $d_i^0 = \lambda_i - P\delta$, with the primed target for the second replica.

Uniformly with scaled ϑ derivatives,

$$\begin{aligned} \|d_i^0 - (vPS_P^{-1}\delta - P\delta)\|_{2,\text{emp}} &\leq Cq^{3/2}|\vartheta|^2, & \|d_i^0\|_{2,\text{emp}} &\leq Cq^{3/2}|\vartheta|, \\ \|\sigma_i - \langle\sigma_i\rangle\|_{2,\text{emp}} &\leq Cq, & \langle\sigma_i\rangle &= B(\vartheta) - \|d_i^0\|_{2,\text{emp}}^2, & B(\vartheta) &= b - 2\sqrt{q}\delta_1 - |\delta|^2. \end{aligned} \quad (11.5)$$

Indeed the common linear response and nonlinear bounds were just obtained (for Gaussian the shifted length coefficient changes the response only nonlinearly).

The variance assertions use tanh or Gaussian differentiation as above; the last identity uses the exact target, automatic also for spins. The first bounds hold in any needed fixed typical-site empirical moment; on the tail d_i^0 and its scaled derivatives are bounded by $C|y|$.

Consequently

$$F'(0) = \langle d_i^0 d_j^0 \rangle, \quad F''(0) = \langle \sigma_i \sigma_j \rangle + O(q^3(|\vartheta| + |\vartheta'|)^2).$$

In the conditional curvature the centered-fluctuation product is orthogonal to the constraints, and the two linear remainder terms are regressed on them, proving the second formula, also to differentiated orders. For the size bounds with scaled projection derivatives used here, $\sigma_i = v$ at zero, and σ_i and its scaled derivatives after empirical mean subtraction have L^2 size $O(q)$: the logistic first derivative of variance brings a factor of the magnetization, and higher Taylor terms in the site shift or the Gaussian length coefficient already pay order q (use the same parameter estimates as for the single saddles).

On the tail all changes of the site fields and their scaled derivatives cost $O(1 + |y|)$ besides bounded quadratic coefficients, against fluctuation moments damped exponentially. Constraint

covariance matrices and their needed scaled derivatives at the independent saddles are bounded, with eigenvalues bounded below using just bounded-profile sites. Regression coefficients for the two linear remainders and their derivatives therefore use ordinary unscaled constraint units (profiles P , squares for Gaussian), with size $O(q^{3/2})$ by (11.5).

Thus $F'(0)$ to quadratic order, of size $O(q^3(|\vartheta| + |\vartheta'|)^2)$, is common to all four pairs. The curvature discrepancy from $B(\vartheta)B(\vartheta')$ and its derivatives are $O(q^2)$; it is common at zero projections, and to linear order varies by an *additive* sum of single-type terms. The third derivative is $O(q)$ at $\beta = 0$, with $O(q)$ projection derivatives. Indeed it is the third cumulant averaged over sites of the regression-adjusted cross observable.

At zero cross tilt the product of centered site fluctuations contributes the average product of individual third moments, $O(q)$ with derivatives. The linear corrections have coefficients d_i^0, d_j^0 , and regression coefficients on site constraints $D = (Pz, Pz', \text{applicable squares})$ of size $O(q^{3/2}(|\vartheta| + |\vartheta'|))$. Their cumulant contributions and derivatives satisfy the stated bound by typical moments in (11.5); on tails they use at most exponentially damped polynomials in $|y| + 1$, of tail average $O(q^3)$.

For clarity, fourth cross derivatives stay bounded up to $|\beta| \leq C\epsilon_1 q$. The constraint-coefficient shifts have first derivatives of norm $O(q^{3/2}(|\vartheta| + |\vartheta'|) + |\beta|)$, second derivatives bounded. Indeed bootstrap their first derivatives at $O(q)$, so shifts are $O(q^2)$, giving only small parameter changes even in Q^{-1} units. The constraint Hessian D_2 (averaged covariance of D) stays well-conditioned. Centered site fluctuations of z, z' remain exponentially damped in $|y|$ on tails, with bounded means: for a coupled Gaussian integrate the small quadratic tilt exactly, and for spins the conditional odds or Gaussian-integrated odds still have fields $y + O(1) + o(1)|y|$.

Thus the regression-adjusted observable $H = (z - P\delta)(z' - P\delta') + \theta'_\beta D$, with θ_β the optimizing single coefficients, has bounded empirical centered fourth moment, using the $O(q)$ coefficient times the worst P_3 profile. Implicit differentiation then bounds θ''_β by $C\|\langle \kappa(D, H, H) \rangle\| \leq C'$, closing the bootstrap (or use continuation at a larger threshold and small ϵ_1); κ denotes joint cumulant.

The fourth derivative of F uses only the fourth cumulant of H and the constraint regression of its square. Similarly $\partial_\beta D_2$ is bounded (using at most two constraint factors and one H), and $\partial_\beta^2 D_2 = O(1 + q^{-1/2})$: with two H 's the worst four-profile tail costs $Cq^2 q^{4-6}$, while the θ''_β term costs a third constraint moment. At zero cross tilt $\partial_\beta D_2$ varies with bounded scaled projection derivatives by the same estimates. Hessian lower bounds throughout use just typical sites; in particular F'' stays bounded below. All these statements apply to mixed constraint dimensions without dummy square variables for spins.

The pair conditional exponent is therefore

$$n \left[-\frac{(t - F'(0))^2}{2F''(0)} + \frac{F'''(0)(t - F'(0))^3}{6F''(0)^3} + O((t - F'(0))^4) \right].$$

The preceding calculations give the cross mean, curvature, and third cumulant at the independent saddle, with their projection derivatives. We now convert them into a conditional density. Both the prefactor and the Legendre exponent must be retained, because each contributes to the odd term in the signed comparison.

For the unweighted reference, the exact conditional law uses two independent uniform directions in dimension $n - 3$. Writing $B = B(\vartheta)$, $B' = B(\vartheta')$, its density is

$$h_{oo}^{t, \text{ex}}(t | a, a') = \frac{c_{n-3}}{\sqrt{BB'}} \left(1 - \frac{t^2}{BB'} \right)^{(n-6)/2}, \quad c_d = \frac{\Gamma(d/2)}{\sqrt{\pi} \Gamma((d-1)/2)}$$

on $|t| < \sqrt{BB'}$. The formal $\hat{h}_{oo}^t$ has the same power and replaces c_{n-3} by $\sqrt{n/(2\pi)}$; their ratio is $1 + O(1/n)$, a permitted relative error. The reference Legendre rate exponent is $\frac{n}{2} \log(1 - t^2/(BB')) =$

$-nt^2/(2BB') + O(nt^4)$. The remaining factor $(1 - t^2/(BB'))^{-3}$ belongs to the reference saddle prefactor. The pair conditional prefactor is $(F''(\beta) \det D_2(\beta) / \det D_2(0))^{-1/2}$ at the dual β , while the reference prefactor is $(BB')^{-1/2}$ up to $O(t^2 + 1/n)$ relative error. The curvature part at $\beta = 0$ thus has a common constant, additive linear log terms of size $O(q^2(|\vartheta| + |\vartheta'|))$, and error $O(q^2(|\vartheta| + |\vartheta'|)^2)$. The log change at β is a projection-independent linear term in t , up to $O(qE_*^C/N)$, by the preceding derivative bounds and dual inversion.

We organize the exponent difference by its dependence on the pair types. Common terms and terms additive in one type cancel after the signed sum. The remaining leading terms are odd in the normalized overlap; quadratic products of the smaller terms fit the remainder.

In the exponent difference, the t^2 terms give a common c_2qU^2 plus additive single-type terms $O(qU^2(|W| + |W'|)/\sqrt{N})$ and error $O(qE_*^C/N)$. The term linear in $F'(0)$ is common up to that error, of size $O(qE_*^3/\sqrt{N})$; its square contribution to the rate is negligible. The cubic term at leading order is odd, of order $qU^3/\sqrt{N}$, with coefficient at zero projections. All fourth-order remainders cost CqE_*^C/N . Indeed $nt^2 = q^{-1}U^2$, $nt^3 = U^3/\sqrt{N}$, $nt^4 = qU^4/N$.

In $F''(0) - BB'$ the terms through projection degree one are just from the empirical covariance of the two site-variance profiles (both v at zero). Hence the inverse-curvature discrepancy has a common constant and single-replica-summed linear terms, with remainder $O(q^2E_*^2/N)$. In $ntF'(0)$, the response product has a common bilinear part, leaving error $O(qE_*^C/N)$ after multiplication. The log-prefactor shift uses $\beta = (t - F'(0))/F''(0) + O((t - F'(0))^2)$; even its $O((1 + q^{-1/2})\beta^2)$ Taylor remainder fits.

Products of two of the small terms cost $O(qE_*^C/N)$. Exponentiating (with the small Gaussian-growth envelope stated, using small cutoffs), the common and additive terms cancel from the signed sum, leaving just the claimed odd terms. Finally (11.2) allows the outside projection factor to be replaced by one on those odd terms to the stated remainder. We check the remainder sizes term by term, including at the ends of the cut window. Writing $\mathcal{W} = |W| + |W'|$, the linear-mean term $ntF'(0)/F''(0)$ has common leading polynomial of size $Cq|U|\mathcal{W}^2/\sqrt{N}$ and remainder $Cq|U|\mathcal{W}^3/N$, while $nF'(0)^2 \leq Cq^3\mathcal{W}^4/N$.

The inverse curvature difference from $1/(B(\vartheta)B(\vartheta'))$ has quadratic-and-higher Taylor error $Cq^2\mathcal{W}^2/N$. The cubic coefficient discrepancy from its value at zero projections is $Cq\mathcal{W}/\sqrt{N}$, and replacing $t - F'(0)$ by t inside that cube also fits the stated error. Hence besides the common c_2qU^2 and projection-independent log-prefactor constant (common because both variances are v at zero), log contributions retained are of size $CqE_*^3/\sqrt{N}$, common, single-type summed, or odd as indicated.

In exponentiating, both the leading polynomials in the log and the full log deviations (using $n|t - F'(0)|^4$ for the Legendre remainder) have absolute upper bounds by constants plus CqE_*^2 on these cuts, before any outside projection log which adds $O(|\vartheta| + |\vartheta'| + (|\vartheta| + |\vartheta'|)E_*^2)$. This yields the stated Gaussian-growth envelope even at window ends, and the common and single-type-summed terms indeed drop out after multiplying by the common projection weights and summing both signs.

The zero-projection coefficients and the exact common/additive cancellation are also written explicitly in Lemma 11.5. This proves the smooth expansion (11.4). Individually the full ratios have at most the same Gaussian-growth envelope, and differ from one on bounded normalized statistics by $O(q) + o(1)$.

Density inversion, tails, and the common angular kernel. The four pair laws have different statistic measures. We keep their lattice coordinates and translated lengths exact, and smooth only the other coordinates. Each remainder is estimated on its own measure. The smooth leading terms are transferred to common volume before we perform the signed cancellation.

- On the central region just used, one can evaluate orbit integrals as lattice sums and smooth densities with the saddle formulas, each to relative remainder $O(q/N)$. Use lattice variables

$\sum x$ for each sign replica and $\sum xx'$ for a pair of signs, on the translated lattice generated by differences of sign patterns. One can use a_1, a_3 as the remaining continuous-approximation projection coordinates for sign replicas. Smooth all nonlattice variables at inverse arbitrarily high polynomial scale with compactly supported smooth kernels, *except* exact Gaussian length constraints $\sum(z^2 + 2mz) = nb$.

Changing an angular test by this smoothing costs an arbitrarily small inverse polynomial absolute error times the corresponding positive partition integrals (on slightly enlarged regions). Indeed projections and residual Gram factor parametrize the complement rotations, with Gram nonsingular here, so energies, pre-factors (in log), smooth cutoffs and the bounded-degree polynomial tests have polynomial variation. Only the Gaussian length itself needs remain imposed on the original site integral; inverse site variances need not be bounded.

Projection multipliers also have polynomial log derivatives. When lattice variables are present keep them exact in this procedure. For instance for two signs one may hold $(\sum x, \sum x', \sum xx')$ and smooth the two nonconstant projection coordinates for each sign; a_2, a'_2, U are recovered using those constant-coordinate sums, the other a 's and known site data. These are nonsingular affine or triangular coordinate changes (equivalently use the three raw statistics in site units together with the nonconstant profiles).

For a mixed pair the raw lattice variable is just $\sum x$ for the sign, and one also smooths the nonlattice overlap. Deterministic statistic Jacobians are understood in saddle densities and in converting lattice mass units. The tilt optimization using projected deviations rather than raw cross products is an affine change at the fixed projection constraints, with the same partial Legendre rate.

For the density estimation, tilt to the canonical product saddle at the evaluation point; variation of tilting factors across the mollifier costs negligible relative error. Fourier inversion there gives the Gaussian prefactor, including lattice covolume, to relative $O((1 + q^{-2})/n)$. Indeed covariance in site-sum units (normalized by $\sqrt{n}$) is bounded positive definite. We can use the well-conditioned equivalent raw coordinates (including zz' or xx' rather than the projection-subtracted statistic).

Maximum centered site variance in any such direction is $O(q^{-3})$, average third absolute and fourth moments $O(1 + q^{-1/2})$, $O(1 + q^{-2})$, by the bounds above. In the central Fourier expansion up to small times $\sqrt{N}$ in normalized frequency units, each site factor stays close to one. The cubic log term integrates to zero at first order by oddness, and the fourth remainder and squared cubic term, with Gaussian damping, give the indicated error. One can evaluate at the exact mean here even with smoothing (or keep its transform, which changes central frequencies negligibly).

For more detail about Fourier tails, bounded typical sites damp through small constant frequencies in unnormalized (single-site) units by nonsingular covariance on that subset. Up to arbitrarily high polynomial frequencies thereafter there is exponential-in-size damping off the central region of the reciprocal-lattice fundamental cell. For a sign pair, phase comparisons on the four patterns test the two nonconstant profile combinations plus constants by single-spin flips (with both signs for the other spin).

Thus if nonconstant coefficients for either replica are not tiny, the nonlattice diagnostics give damping; with tiny coefficients on the bounded typical sites, only the reciprocal lattice of the constant three-pattern statistics can escape the test. More explicitly these statistics are the coefficients of x, x', xx' , and equality of the four phases characterizes exactly that dual lattice. Typical-site pattern probabilities are bounded below. The single sign test works identically. For Gaussian variables on typical sites condition on any site signs first.

Quadratic Gaussian phases separated from zero give contraction irrespective of the linear part. With tiny quadratic coefficients, Gaussian linear phases separated from zero on a fixed fraction also give contraction; for a mixed sign pair one tests both sign choices to detect the cross coefficient. The Gaussian profile columns are well-conditioned on the bounded typical sites. Their conditional real means are bounded there. Thus failure of contraction from these tests requires all Gaussian-related coefficients tiny, when on this subset they only perturb the sign phase test by a small amount.

This argument can take successively small fixed cutoffs, or pass to sequences where contraction vanishes (quadratic phases must tend to zero, then Gaussian linears for each sign choice, then the empirical sign-phase dispersions). Beyond polynomial frequencies the mollifiers suffice; in unsmoothed length directions the Gaussian quadratic factors on typical sites themselves give a bound $(1 + c|\omega_{\text{lengths}}|^2)^{-c'n}$, also in the joint inversion. Here length direction means the single-site quadratic coefficients; the linear parts do not affect this bound.

Thus one is approximating ordinary joint constraint densities in the Gaussian length dimensions, and possibly smoothed/lattice joint masses elsewhere. Factoring a result into single prefactors and a conditional saddle expression (11.4) just uses determinant Schur factorization and rate subtraction at the level of these formulas; no division by tiny unsmoothed atomic projection probabilities is required. These estimates justify inversions in the different constraint dimensions and also the denominator calculations for conditional densities.

- Overlap tails outside $|t| \leq \epsilon_1 q$, with the projection cuts, are negligible (e^{-cN} times polynomial) in each pair orbit integral relative to Z_*^2 , even for polynomial tests. First take $\epsilon_1 q < |t| < c_*$ with c_* small fixed. Conditional density ratios can here be estimated as upper bounds on polynomially small boxes. At independent single canonical laws use exponential Markov on $(z - P\delta)(z' - P\delta')$ up to tilts Cc_* , without having to keep its constraint regressions optimized. Its mean is $O(q^3)$; unconditional curvature differs from BB' by $O(q^2)$ by (11.5); third cumulant is $O(q)$, fourth derivatives bounded. Indeed single saddle parameters remain fixed in this tilt and the damping on tails is preserved (no large new profile coefficients, $|P\delta| \leq C\epsilon_0|y|$ there, and site constants do not enter higher cumulants). Thus relative to the reference conditional overlap density the log cost, besides polynomial losses, is at most

$$Cn(q^3|t| + q^2t^2 + q|t|^3 + t^4).$$

Imposing exact Gaussian length densities costs at most polynomial under these tilts by bounded typical-site Gaussian smoothing as in the inversion bound.

Projection marginals relative to reference cost at most $\exp(C\epsilon_0^3 N)$ besides constants by (11.2), including our modification. Boxes can straddle lattice coordinates without needing any matching lower bound there for the numerator. In the physically tilted reference C_o , the length deficit of one of $(z \pm z')/\sqrt{2}$ is at least $n|t|/2$ from trace, on taking $\epsilon_0 \ll \epsilon_1$. This costs $\exp(-cn|t|^3)$ by negative square tilt of strength small-times t^2 (and conditioning at polynomial cost).

Indeed the covariance loses normalized trace at most $C\sqrt{\xi} + o(q)$ at added precision $\xi \geq 0$ small by (5.1) and finite-rank comparison; lengths have central denominators and upper density bounds polynomial under the tests by Gaussian Fourier damping. Equivalently exact shells here can use buffered radial conditioning on the spheres translated by m . The absorption is uniform down to the moving endpoint $|t| = \epsilon_1 q$, since

$$C \left(\frac{q^3}{t^2} + \frac{q^2}{|t|} + q + |t| \right) \leq C \left(\frac{q}{\epsilon_1^2} + \frac{q}{\epsilon_1} + q + c_* \right)$$

is the scalar cost divided by $n|t|^3$, while the projection cost has relative size at most $C(\epsilon_0/\epsilon_1)^3$. Hold ϵ_1 fixed, choose c_* and then the upper bound on q small, and finally choose ϵ_0/ϵ_1 small. The

reference penalty then leaves a fixed multiple of $n|t|^3$, which is at least a fixed multiple of N . Angular comparisons over these boxes have polynomially controlled variation.

For larger $|t|$, use directly the angular semicircle pair upper bound from (6.3) in the complement of $\mathcal{P}$, whose Gram eigenvalues here are $1 \pm t + o_{\text{small}}(1)$, and ignore projected components at $o_{\text{small}}(n)$ cost. The log gain besides independent baselines is at most $nt^2/2 + o_{\text{small}}(n)$. Indeed $\log Z_*/n \geq 1/4 - 1/2 - o_{\text{small}}(1)$ from its C_o formula by central length density, the semicircle log determinant upper bound and (9.1); finite-rank and prior normalization changes do not increase the log determinant by a nonvanishing order per spin.

The sign-sign prior pays the strict Cramér slack already used, with biased signs and translations changing bounds by $o_{\text{small}}(n)$. For Gaussian pairs, simultaneous length/overlap Chernoff gives at least the rate $-\frac{1}{2} \log(1 - t^2)$ in the small- q limit, strictly larger than $t^2/2$ throughout this range. For two Gaussian types this uses fixed tilts of the Gram matrix, with log moments tending to the iid unit-variance values ($v \leq 1$); for mixed types condition on the sign vector, of bounded entries after translation and near-unit empirical length, and tilt the one Gaussian length and its projection on that vector, with the same limiting rate.

This follows immediately from the normal moment formulas since $\langle |v - 1| \rangle = q$; the mixed vector entries are bounded. Interior finite tilts suffice including by approaching the endpoints, using compactness and only requesting strict slack. Length targets differ from 1 negligibly, and their exact Gaussian densities again cost polynomial under the tilts. Projection/pre-rank factors cost $o_{\text{small}}(n)$. These estimates also handle singular pair Grams directly, by splitting the fixed overlap range into sufficiently small fixed bins. This proves the tail assertion.

- Here is the accuracy of kernel integration and of converting lattice sums to ordinary integrals. Under the tilted reference, condition two C_o -normals independently on their length shells, whose density denominators in L units are bounded above and below as in (5.6). Conditional on projections a, a' before length conditioning, their white coordinates η_i have fixed projections described by standard-normal coordinates α, α' (exactly whitened a, a'), and fresh standard orthogonal complements.

In particular $|W| \leq C|\alpha|$, and raw projection covariance eigenvalues are bounded below since K_o has bounded norm. Lengths and U are linear-quadratic forms in the remaining normals. The quadratic parts in a Fourier direction of norm one have operator and Hilbert-Schmidt norms at most $C/\sqrt{N}, C$, with spread giving determinant decay $(1 + c|v_f|^2/N)^{-c'N}$ at frequency v_f . Indeed they use C_o/L tensored with a two-by-two symmetric matrix (whose norm in such a direction is bounded below), up to the finite-rank compressions/subtractions, and (9.1) applies.

The length linear part uses $\|C_o^{1/2}m\|/L \leq C$. Consequently central densities with any fixed derivatives in a, a', U , including the projection density, are bounded by

$$C(1 + |\alpha| + |\alpha'| + |U|)^C e^{-c(|\alpha|^2 + |\alpha'|^2 + U^2)} |\det(\partial\alpha/\partial a)|^2. \quad (11.6)$$

To see the U -tail factor before inversion, tilt in U by parameter small-times the target U ; up to $\epsilon_1\sqrt{N}$ the quadratic norm shift is small. Its centered determinant cost is $\exp(C \text{ parameter}^2)$ (off-diagonal tilt), and the projected translations cost at most $\exp(o_{\text{small}}(1)(|\alpha|^2 + |\alpha'|^2))$. The tilted spread is unchanged up to constants.

For the weighted matrix test, subtract C_o times the mass before forming the Hilbert-Schmidt square. The resulting insertion must be averaged with its sign: its absolute value would lose the Gaussian cancellation at the scale required here. Its insertion is

$$H = (\eta_1^T D \eta_2)^2 - \eta_1^T D^2 \eta_1 - \eta_2^T D^2 \eta_2 + \text{Tr } D^2, \quad D = C_o^{1/2} A_o^2 C_o^{1/2}, \quad \|D\| \leq C.$$

The next kernel estimate controls this signed average and, after the projection variables are integrated out, its smaller odd part.

Lemma 11.4, applied with $C = C_o$, $A = A_0$ and $E = \mathbf{P}/\sqrt{n}$, proves the signed-density bound (11.6) times Cnq , including its fixed derivatives. We verify its hypotheses explicitly. The formula for K_o , $S_P = I - O(q)$, and $\|J\| \leq 3$ give $\|K_o\| \leq C$; therefore $C_o \succeq C^{-1}I$. The lower precision bound, trace-square bound, and edge spread follow from (9.1). Its first projection estimate also gives

$$\|C_o^{1/2}m\|^2 = nq u^T C_o u \leq Cn/q = CL^2.$$

Thus

$$\text{Var}(\|z\|^2 + 2m^T z) = 2 \text{Tr } C_o^2 + 4m^T C_o m \asymp L^2, \quad \text{Tr } C_o = nb + o(L).$$

The lower variance bound uses the edge spread. The density estimate (5.6), with its Fourier upper bound, now puts both shell density denominators between fixed positive constants in L units. After integrating out a, a' , the odd parts of both the mass and matrix densities gain $N^{-1/2}$.

The lemma proves this by comparing the cross-frequency signs before projection conditioning; hence it does not assume an artificial symmetry of the translated shells. Projection cuts multiplying the leading odd term can be removed at exponentially small cost, since (11.6) gives Gaussian tails in the whitened projection variables.

Lattice sums after the density approximation use steps bounded by $C/\sqrt{n}$ in the raw a 's and C/L in full-overlap L units (fixing the other projection coordinates first, with affine shifts). Their covolumes are exactly the prefactor ones from Fourier inversion. Passing to U has bounded derivatives on the cut: besides subtracting $a \cdot a'/L$, converting a full x -overlap to a z -overlap subtracts the m -linears with derivative $\sqrt{nq}/L = q$. Converting constant-axis sums to a_2 's uses P_2 's formula with bounded coefficients.

Thus the centered-cell integration error for smooth leading terms is $O(1/n)$ times their respective polynomial-moment envelope scale. More explicitly use the individual expansions leading to (11.4) with second derivatives of the explicit constant/common/additive and odd terms, keeping the projection factors via (11.2); in raw a, a', U units these derivatives cost just polynomials times $e^{\zeta E_*^2}$. Remainders $O(qE_*^C e^{\zeta E_*^2}/N)$ require no derivatives.

The type of error matters here. Density-ratio errors, including relative saddle errors, are scalar functions of the statistics and multiply the *same angular kernel* as the leading terms. We make the exact denominator in this assertion explicit. Use the chart Φ , Jacobian J_Φ , mixed measures dm_i , and exact unweighted reference density f_{oo}^{unw} from Lemma 11.2. For the reference single projection, isotropy gives the exact formula

$$f_o^{a,\text{ex}}(a) = \mathbf{c}_o \mathbf{r}_o(a) \varphi_{bI_3}(a) f_{b\chi_{n-3}^2}(\rho(a)^2), \quad \rho(a)^2 = nb - 2\sqrt{nq}a_1 - |a|^2.$$

Its ratio to f_o^a is a dimension-only factor $1 + O(1/n)$ by Stirling's formula. Together with the exact conditional overlap density above, this proves, uniformly on the regular window,

$$f_{oo}^{\text{unw}} = \hat{f}_{oo}^{a,a',U}(1 + \varepsilon_{oo}), \quad |\varepsilon_{oo}| \leq C/n.$$

No raw shell density, rank factor, or normalizing constant has been removed from either side.

For the numerator inversion, first smooth the uncut statistic laws only in their nonlattice coordinates on a slightly enlarged window, and then apply the smooth projection cuts and the multiplier defining g . Let $\hat{f}_i^{\text{mod}}$, for $i = pp, pg, gp, gg$, denote the corresponding formal pair function with these factors included. Uniform inversion gives at each pair's own mixed point

$$f_i^{\text{sm}}(w) = J_\Phi \hat{f}_i^{\text{mod}}(\Phi(w))(1 + \varepsilon_i(w)), \quad |\varepsilon_i(w)| \leq Cq/N.$$

The factors are applied after smoothing, so this identity makes no relative-error assertion at a zero of a convolved cutoff. Commuting those factors through a mollifier of radius δ instead changes the tested integral by at most δn^C times the positive mass on the enlarged window. The multiplier's logarithmic derivatives and the cutoff derivatives have polynomial bounds there. The positive-mass estimate below and a sufficiently high inverse-polynomial choice of δ make this separate error negligible.

Write $\hat{R}_i = \hat{f}_i^{\text{mod}}/\hat{f}_{oo}$ in this chart, and let L_i be the finite smooth individual expansion used above for quadrature. Thus $\hat{R}_i = L_i + \hat{e}_i$, where $|\hat{e}_i| \leq C(q/N)E_*^C e^{\zeta E_*^2}$. Consequently the exact ratios used by the angular disintegration are

$$r_i(w) = \frac{f_i^{\text{sm}}(w)}{J_\Phi f_{oo}^{\text{unw}}(\Phi(w))} = L_i(w) + e_i(w),$$

$$e_i(w) = \hat{e}_i(w) + \hat{R}_i(w) \frac{\varepsilon_i(w) - \varepsilon_{oo}(\Phi(w))}{1 + \varepsilon_{oo}(\Phi(w))}, \quad |e_i(w)| \leq C \frac{q}{N} E_*^C e^{\zeta E_*^2}.$$

Here $1/n \leq q/N$, and the individual formal ratios have the Gaussian-growth envelope already proved. Only the smooth L_i undergo quadrature. Their signed sum has the leading odd term in (11.4), with its stated envelope remainder. These are absolute remainder bounds on each own mixed measure, including cutoff boundaries; no regularity of e_i is needed. Bounds (11.6) integrate or sum them against the absolute value of the signed angular average, not against the angular average of the absolute insertion. Thus the envelope in Lemma 11.3 is $\mathcal{E} = E_*^C e^{\zeta E_*^2}$.

Holding the length constraints exact ensures that at those points both angular averages use the same residual norms and Gram. Thus a scalar saddle error against an angular kernel is integrated using that kernel's reference signed density at the points, not by integrating absolute Wick polynomials. Mollification changes the angular test itself and therefore requires a separate estimate. For that change we use absolute bounds: comparison of positive angular integrands at neighboring regular Grams, with polynomial derivative costs relative to base exponential weights.

Such costs times an arbitrarily small inverse polynomial are negligible here. Indeed the positive partition integrals with smooth cut to this region are polynomially bounded in reference units already by the scalar upper saddle bounds against the mass version of (11.6), summing also on the meshes; neighboring unsmoothed mass needed on a smaller region obeys the same type of bound by the angular energy comparison itself and unit mollifier mass. One can slightly enlarge regular windows in this justification and use smooth boundaries, without moving length or lattice statistics.

Cut boundaries can all use smooth tapers, neglecting the t -boundary by the preceding tails or (11.6). Lemmas 11.2 and 11.3 below state this disintegration and quadrature step in a single explicit chart, including its Jacobian and primitive lattice covolumes. They justify (11.4) inside both mass and signed-kernel tests without identifying the distinct atomic and continuous statistic measures.

Transfer of moments and final parameter choices. We now finish (11.1). All four pair laws have been related to the same angular kernel, with their individual remainders controlled. We can therefore square differences of the single-replica integrals.

Under Haar averaging, the prior density ratios multiply the common angular integral at fixed statistics. When the reference prior density and the two energy factors in that integral are combined and divided by Z_*^2 , they give the tilted reference shell kernel. This is the mixed-measure identity made explicit below. The mass of modified g , divided by Z_* , tends to one in probability: its first and second moments follow from (11.2), the individual pair comparisons, and the tails.

The Hilbert-Schmidt second moments for the *difference* of the cut p and g integrals, still with

common denominator, are bounded as follows:

integrand	$\mathbb{E}_{\text{Haar}} \ \text{difference}\ _{\text{HS}}^2$
1	$O(q/N)$
z	$O(qL/\sqrt{N})$
zz^T	$O(qL^2/N)$
$A_0(zz^T - C_o)A_0$	$O(nq^2/N)$.

(11.7)

Each row follows by expressing the square as a pair integral and using (11.4), (11.6).

The insertion determines whether an odd-part gain is needed. For the mean the inner product kernel $LU + a \cdot a'$ needs no odd gain. For the raw second moment its square differs from L^2U^2 by at most $L^2E_*^4C/\sqrt{N}$, so the leading odd term is harmless. For the last test use exactly the signed H bounds. All constants here may use a fixed small upper bound on $\epsilon_1, \zeta, \epsilon_0$ satisfying the preceding absorptions.

On orientations obeying (10.1), the uncut g_0 physical law has centered covariance at most $(1 + o(1))K^{-1}$ and raw second moment at most CK^{-1} , exactly by the shell argument of (5.6). Here displacement of the standardized length target is bounded rather than vanishing, so use centered conditional Gaussian variance in that argument (joint Gaussian limit with any white projection, variance can only decrease). Projection and length density bounds needed for uniform moments follow after any bounded codimension compression by (9.1).

The modification defining g changes these estimates by $o(1)$ in covariance units. Indeed W has uniform sub-Gaussian tails, (11.2) gives a factor tending to one with arbitrarily small Gaussian growth, and the removed tail begins at order $\sqrt{N}$. Cauchy–Schwarz applies uniformly to each projection moment test.

We spell out the normalization and centering. Write

$$I_i(F) = Z_*^{-1} \int F(z) e^{z^T(J-bI)z/2} i(dz), \quad M_i = I_i(1), \quad i = p, g,$$

where both priors include their cuts and g includes its projection multiplier. Since $L/\sqrt{N} = q^{-2}$, the four root-mean-square errors in (11.7) are, in order,

$$O(\sqrt{q/N}), \quad O(q^{-1/2}), \quad O(q^{-3/2}), \quad O(q^{-1/2}).$$

Choose the failure probabilities in Markov’s inequality in advance, with their sum and the preceding orientation losses below $1/4$. For small q and large N , the resulting orientations have M_p, M_g bounded away from zero and close to one. For any matrix or vector test F ,

$$\frac{I_p(F)}{M_p} - \frac{I_g(F)}{M_g} = \frac{I_p(F) - I_g(F)}{M_p} + \frac{M_g - M_p}{M_p} \frac{I_g(F)}{M_g}.$$

The normalized modified Gaussian g has raw second moment $O(q^{-2})$, so its mean is $O(q^{-1})$. For $F = z$ and $F = zz^T$, the last display and the first three errors show that the normalized mean difference is $O(q^{-1/2}) + o(q^{-1})$, and the normalized raw second-moment difference is $O(q^{-3/2}) + o(q^{-2})$. Subtracting the two mean outer products costs at most $O(q^{-3/2}) + o(q^{-2})$. Consequently the cut spin covariance is bounded above by $(1 + o(1))K^{-1} + o(q^{-2})I$.

For the weighted estimates choose the loss in (9.1) separately, denoting it by η_{mat} , with $2\eta_{\text{mat}} < \gamma$. The normalized modified Gaussian g has raw weighted second moment $O(q^{-2\eta_{\text{mat}}})$. Use the last row of (11.7) in the normalization identity with $F = A_0(zz^T - C_o)A_0$. The subtraction is a constant under each normalized law, so the difference of raw weighted second moments is

$$O(q^{-1/2}) + o(q^{-2\eta_{\text{mat}}} + 1) = o(q^{-\gamma}).$$

Here $\gamma > 1/2$ is exactly what makes the first error smaller than the target. The cut spin raw weighted second moment is therefore $o(q^{-\gamma})$. It bounds the weighted covariance; Jensen's inequality bounds the squared weighted mean by the same quantity. In the unweighted case the raw second moment is $O(q^{-2})$, so the mean is $O(q^{-1}) = o(q^{-1-\eta})$ for the independently prescribed $\eta > 0$.

Finally restore the removed spin mass. If it is at most n^{-D} , then $\|X - m\| \leq 2\sqrt{n}$ bounds the changes in normalized means and raw second moments by $Cn^{1/2-D}$ and Cn^{1-D} , respectively; the same bounds, up to constants, hold after A_0 since $\|A_0\| \leq 5$. Choose D large. The sharp unweighted coefficient follows by choosing the precision, shell, transfer, and q/r tolerances so that

$$(1 + o(1))(1 - \epsilon_{\text{mat}})^{-1}s/q^2 \leq 1 + \epsilon.$$

This proves (11.1) with all the prescribed small constants.

The same event also provides the mass anchor needed in the next section. We may require $M_p \geq 1/2$, and positivity of the omitted integral gives

$$\int e^{z^T(J-bI)z/2} p_{\text{full}}(dz) \geq Z_*/2. \quad (11.8)$$

It holds simultaneously with the norm bounds on the retained orientation set. On any fixed compact observation-time interval bounded away from zero, contained in the established small-time regime, and on which r is bounded below, the translated posterior raw second moment $\mathbb{E}[(X - m)(X - m)^T \mid J, h]$ is bounded by a fixed constant in operator norm on these orientations, using the covariance and mean estimates.

All choices in this static assertion begin with coarse scalar tolerances and matrix budgets giving fixed constants in (8.2)–(10.1), and with the orientation failure allocations just described. Choose the overlap cutoff and then the projection cutoff small, including for the Gaussian envelopes, and use a sufficiently small scalar and overlap rate to put q/r close to one and control the projection-cut loss.

Stricter diagnostics do not require increasing the coarse constants. Typical-site moment and tail cutoff exponents can be chosen for the finite derivative orders used. The upper bound on r is then sufficiently small in terms of these choices, which have not used dynamical probability/moment inputs.

□

11.1 Exact statistic measures and signed cancellation

The following statements make the measure-theoretic and Gaussian parts of the preceding comparison explicit. The smooth saddle expansions (11.2)–(11.5) have already supplied the scalar coefficients and their errors; the first two lemmas specify exactly how these scalar quantities act on atomic and continuous priors.

Lemma 11.2 (Common angular kernel and mixed statistic measures). *Let $0 < q < 1$, $b = 1 - q$, $L = \sqrt{n/q}$, and $P^T P = nI_3$, $m = \sqrt{q}P_1$, $M = n^{-1} \sum m_i$, and $c = (1 - M^2/q)^{1/2} > 0$, with $P_2 = c^{-1}(1 - (M/q)m)$. For $z = x - m$ on $\|z\|^2 + 2m^T z = nb$, set $a = P^T z / \sqrt{n}$ and $z_{\perp} = z - Pa / \sqrt{n}$. Define primed variables likewise, and set $U = z_{\perp}^T z'_{\perp} / L$. Then*

$$\rho(a)^2 = nb - 2\sqrt{nq}a_1 - |a|^2, \quad \Gamma(a, a', U) = \begin{pmatrix} \rho(a)^2 & LU \\ LU & \rho(a')^2 \end{pmatrix}.$$

On $\Gamma \succ 0$, the statistic (a, a', U) is a complete invariant for simultaneous proper rotations of $\text{ran}(P)^{\perp}$, provided $n - 3 \geq 3$. Thus every integrable physical pair insertion has one Haar angular average,

independent of the choice of prior. To specify the reference used in the application, put $\mathbf{s} = (a, a', U)$, let $T(z, z') = \mathbf{s}$, and define the exact unweighted density

$$f_{oo}^{\text{unw}} = \frac{dT_{\#}(o \otimes o)}{d^3a d^3a' dU}$$

on the positive Gram region. It includes both rank factors, both raw shell densities, and $\mathfrak{c}_o^2$. It is distinct from the formal product of single saddle functions and a conditional saddle function.

For a pair insertion F , write

$$\mathcal{A}_F(\mathbf{s}) = \int_{\text{Haar}} e^{\mathcal{E}(gz) + \mathcal{E}(gz')} F(gz, gz') dg, \quad \mathcal{E}(z) = z^T(J - bI)z/2,$$

using any representative of $\mathbf{s}$. All physical matrices in F are fixed together in this vector-rotation representation. Let k_F be the signed statistic density under two independent normalized physical reference shells, with insertion F . Then

$$Z_*^{-2} f_{oo}^{\text{unw}}(\mathbf{s}) \mathcal{A}_F(\mathbf{s}) = k_F(\mathbf{s})$$

for almost every regular $\mathbf{s}$. This includes the mass insertion and the centered matrix insertion of Lemma 11.4.

There is one smooth chart for the four pair priors. Put $S = \sum x_i$, $S' = \sum x'_i$, and $R = \sum x_i x'_i$, and use $w = (a_1, a_3, a'_1, a'_3, S, S', R)$. Its map $\Phi : w \mapsto (a, a', U)$ is

$$a_2 = \frac{S - nM}{\sqrt{nc}} - \frac{M}{\sqrt{qc}} a_1, \quad a'_2 = \frac{S' - nM}{\sqrt{nc}} - \frac{M}{\sqrt{qc}} a'_1,$$

$$U = \frac{R - nq - \sqrt{nq}(a_1 + a'_1) - a \cdot a'}{L}, \quad |\det D\Phi| = \frac{1}{nc^2 L} =: J_\Phi.$$

For pp , the raw triple belongs to the affine lattice $n(1, 1, 1) + \Lambda$, where

$$\Lambda = \text{span}_{\mathbb{Z}}\{(2, 0, 2), (0, 2, 2), (0, 0, 4)\}, \quad \text{covol}(\Lambda) = 16.$$

For pg only $S \in n + 2\mathbb{Z}$ is retained as a lattice coordinate, and for gp only $S' \in n + 2\mathbb{Z}$ is retained; these covolumes are 2. The gg measure has no lattice coordinates.

Let dm_{pp} be 16 times counting measure in its lattice coordinates times Lebesgue measure in the remaining coordinates; define dm_{pg}, dm_{gp} with factor 2, and $dm_{gg} = dw$. Smooth only the nonlattice coordinates, preserving the exact translated shell constraints. For cut or multiplied priors apply their smooth factors after this convolution; the preceding commutator estimate charges the change from the original prior. If f_i^{sm} denotes the resulting density with respect to dm_i , set

$$r_i(w) = \frac{f_i^{\text{sm}}(w)}{J_\Phi f_{oo}^{\text{unw}}(\Phi(w))}, \quad \tilde{k}_F(w) = J_\Phi f_{oo}^{\text{unw}}(\Phi(w)) \mathcal{A}_F(\Phi(w)).$$

Then the smoothed pair integral is exactly $I_i^{\text{sm}} = \int r_i \tilde{k}_F dm_i$, and

$$Z_*^{-2} \tilde{k}_F(w) = J_\Phi k_F(\Phi(w)).$$

For a fixed insertion we abbreviate $\tilde{k}_F$ by $\tilde{k}$ below. In particular, a scalar density remainder multiplies the same signed reference kernel at each actual mixed mesh point.

Proof. Two ordered residual pairs with the same positive Gram matrix are related by an orthogonal map. If its determinant is negative, compose with a reflection on the orthogonal complement of the target pair; this fixes the pair and corrects the determinant. Haar invariance and Fubini therefore give, for every finite pair measure μ , atomic or continuous, and every integrable B ,

$$\int_G \int B(gz, gz') d\mu dg = \int \mathcal{A}_B(s) d(T_{\#}\mu)(s),$$

where $\mathcal{A}_B$ is its angular average. For signed insertions use absolute integrability. Moving the rotation from the interaction matrix to the pair vectors gives precisely this identity for orbit averages in the application; all matrix factors in the insertion move together. For the reference prior, o is invariant under the rotations: its isotropic Gaussian density, translated shell, and projection rank factor are all invariant. Applying this disintegration to $e^{\mathcal{E}(z)+\mathcal{E}(z')}F(z, z')o(dz)o(dz')$, and then using $Z_*^{-1}e^{\mathcal{E}}o = \text{Law}(Z \mid T_o = 0)$, proves $f_{oo}^{\text{unw}}\mathcal{A}_F = Z_*^2 k_F$. For a signed insertion absolute integrability justifies Fubini; the insertion keeps its sign inside the angular average.

The chart formulas follow by expanding $\sum x_i$ and $x^T x'$. Its Jacobian is triangular, with the three new diagonal factors $1/(\sqrt{nc}), 1/(\sqrt{nc}), 1/L$. For pp , the four sign-pattern counts are

$$\frac{n + S + S' + R}{4}, \quad \frac{n + S - S' - R}{4}, \quad \frac{n - S + S' - R}{4}, \quad \frac{n - S - S' + R}{4}.$$

Their integrality gives the stated lattice and determinant 16. The nonnegativity boundaries lie outside the central regular window. In particular the conditional R step is 4, so three independent step-2 lattices would give the wrong normalization.

The mixed lattices have only the one sign-sum coordinate. Finally the asserted integral formula follows by multiplying and dividing the smoothed scalar density by the positive unweighted reference density evaluated at the same point. This is an identity in the chosen mixed measures, not a Radon–Nikodym derivative of an atomic statistic law with respect to a continuous law. $\square$

Lemma 11.3 (Separate remainders and common-volume cancellation). *Suppose $r_i = L_i + e_i$, where L_i extends smoothly to the common chart, while e_i need only be defined at its own mixed points. Let $\mathcal{E}$ be a nonnegative envelope on the chart. Assume $|e_i| \leq \varepsilon \mathcal{E}$ and*

$$\int \mathcal{E}|\tilde{k}| dm_i \leq B, \quad \left| \int L_i \tilde{k} dm_i - \int L_i \tilde{k} dw \right| \leq \delta_i.$$

Then, with pair signs $+, -, -, +$,

$$\left| \sum_i s_i I_i^{\text{sm}} - \int \left(\sum_i s_i L_i \right) \tilde{k} dw \right| \leq 4\varepsilon B + \sum_i \delta_i.$$

Proof. Bound the four errors separately on their own meshes, perform quadrature on the four smooth terms, and only then add their ordinary-volume integrals. No matching support or regularity of the residual functions is needed. The relevant absolute value is $|\tilde{k}|$, the absolute value of the signed angular average; the insertion is not replaced by its absolute value. $\square$

For clarity, centered-cell quadrature supplies the displayed δ_i from two derivatives. If C_Λ is a centered fundamental parallelepiped of a fixed-dimensional lattice, Taylor's formula gives

$$\left| \text{covol}(\Lambda) \sum_\ell F(\ell) - \int F \right| \leq \frac{1}{2} \sum_\ell \int_{C_\Lambda} |u|^2 \sup_{0 \leq t \leq 1} \|\nabla^2 F(\ell + tu)\| du.$$

The linear term integrates to zero. In the application, $c^2 = 1 - M^2/q = 1 - O(q)$ is bounded below. Raw lattice steps then change a, a' by $O(n^{-1/2})$ and U by $O(L^{-1})$ on the projection cut. Hold the nonlattice coordinates (a_1, a_3, a'_1, a'_3) fixed when differentiating in the raw lattice coordinates. The only quadratic term of Φ in those directions has $\partial_S \partial_{S'} U = -1/(nc^2 L)$. Thus a summable polynomial–Gaussian envelope for two derivatives in (a, a', U) costs $O(1/n)$ times its envelope scale. The covolumes in this estimate are exactly those already present in the local Fourier inversion.

Mollification is a different error and is estimated before using signed scalar remainders. For a probability mollifier of radius δ in the nonlattice chart coordinates,

$$\int \mathcal{A} d(\nu * \kappa_\delta) - \int \mathcal{A} d\nu = \int \int [\mathcal{A}(w+h) - \mathcal{A}(w)] \kappa_\delta(h) dh d\nu(w).$$

On the enlarged regular Gram region choose a smooth Cholesky representative and impose $|\nabla \log \mathcal{A}_1| \leq n^C$, $|\nabla \mathcal{A}_H| \leq n^{C'} \mathcal{A}_1$, where $\mathcal{A}_1 > 0$ is the positive angular energy integral.

These inequalities follow by differentiating the representative: Gram eigenvalues are of order n , all physical parameters and the polynomial insertion are polynomially bounded, and the exponential derivative is its value times a polynomial. They preserve the translated lengths by recomputing $\rho(a)$. A smoothed positive-mass bound then bounds the unsmoothed mass within $e^{n^C \delta}$, so the signed smoothing error is at most $\delta n^{C'} e^{n^C \delta}$ times that positive mass. Taking an arbitrarily high inverse-polynomial smoothing scale makes this negligible. Smooth enlarged cuts handle boundaries. No Gaussian length coordinate and no retained lattice coordinate is convolved in this argument.

11.2 The centered Gaussian matrix kernel

Lemma 11.4 (Centered Gaussian kernel and its odd part). *Fix a derivative order $h \geq 0$. There is a threshold $N_0 = N_0(h)$ for the following statement. Let $0 < q \leq q_0 < 1$, $N = nq^3 \geq N_0$, and $L = \sqrt{n/q}$. Let C be positive definite, and let E have a fixed number d of orthonormal columns. Put $\Pi = I - EE^T$. Assume*

$$c_0 I \preceq C, \quad \|C\| \leq C_0 q^{-2}, \quad \text{Tr } C \leq C_0 n, \quad \text{Tr } C^2 \leq C_0 n/q,$$

and that at least $c_0 N$ eigenvalues belong to $[c_0 q^{-2}, C_0 q^{-2}]$. Suppose $m \in \text{ran } E$ and $\|C^{1/2} m\|/L \leq C_0$. Let A be symmetric, and assume $D = C^{1/2} A^2 C^{1/2}$ has norm at most C_0 . For independent standard normal vectors η_1, η_2 , set

$$z_i = C^{1/2} \eta_i, \quad T_i = (\|z_i\|^2 + 2m^T z_i - nb)/L, \quad a_i = E^T z_i, \quad U = z_1^T \Pi z_2 / L.$$

Assume the density of each T_i at zero belongs to $[c_0, C_0]$. Write $k_1(a_1, a_2, U)$ for the density conditional on $T_1 = T_2 = 0$, and k_H for the signed density with insertion

$$H = (\eta_1^T D \eta_2)^2 - \eta_1^T D^2 \eta_1 - \eta_2^T D^2 \eta_2 + \text{Tr } D^2.$$

Put $\alpha_i = (E^T C E)^{-1/2} a_i$ and $J_E = \det(E^T C E)^{-1}$. For every multi-index β with $|\beta| \leq h$, on $|U| \leq \epsilon \sqrt{N}$, with sufficiently small fixed ϵ , one has

$$|\partial^\beta k_1| \leq C_\beta P_\beta e^{-c(|\alpha_1|^2 + |\alpha_2|^2 + U^2)} J_E, \quad |\partial^\beta k_H| \leq C_\beta n q P_\beta e^{-c(|\alpha_1|^2 + |\alpha_2|^2 + U^2)} J_E,$$

where P_β is a fixed polynomial in $1 + |\alpha_1| + |\alpha_2| + |U|$. After integrating both projection variables, the resulting densities $\bar{k}_1, \bar{k}_H$ obey

$$|\bar{k}_1(U) - \bar{k}_1(-U)| \leq C N^{-1/2} P(U) e^{-cU^2}, \quad |\bar{k}_H(U) - \bar{k}_H(-U)| \leq C n q N^{-1/2} P(U) e^{-cU^2}.$$

The same bounds hold for derivatives of order at most h .

Proof. Both $B = C/L$ and $B_\Pi = C^{1/2}\Pi C^{1/2}/L$ have operator norm $O(N^{-1/2})$, Hilbert–Schmidt norm $O(1)$, and nuclear norm $O(\sqrt{nq})$. Their difference has rank at most d . Conditioning on a_i amounts to writing

$$\eta_i = S\alpha_i + \xi_i, \quad S = C^{1/2}E(E^TCE)^{-1/2}, \quad S^TS = I,$$

where ξ_i is standard Gaussian on $S^\perp$.

Take Fourier transforms in the two lengths and in U , allowing a real cross tilt u with $|u| \leq c\sqrt{N}$. The precision then has real part bounded above and below by fixed positive constants. Let Σ, μ denote the complex Gaussian covariance and mean. The resolvent identity, the preceding three norm bounds, and the fixed-rank projection correction give

$$\|\Sigma - I\|_{\text{HS}} \leq CP, \quad \|\Sigma - I\|_1 \leq C\sqrt{nq}P, \quad \|\mu\| \leq CP,$$

where P is a polynomial in the frequencies, $|u|$, and $1 + |\alpha_1| + |\alpha_2|$. The length linear coefficient is $C^{1/2}m/L$; projection conditioning adds coefficients of norm at most $C(|\alpha_1| + |\alpha_2|)/\sqrt{N}$.

Write $\Sigma_{11} = I + \Delta_1$, $\Sigma_{22} = I + \Delta_2$, $R = \Sigma_{12}$, and $s = \mu_1^T D\mu_2$. Wick’s formula is exactly

$$\begin{aligned} \mathbb{E}_{\Sigma, \mu} H = & \text{Tr}(D\Delta_2 D\Delta_1) + \text{Tr}(DR^T DR^T) + (\text{Tr}(DR^T))^2 \\ & + 2s \text{Tr}(DR^T) + s^2 + \mu_1^T D\Delta_2 D\mu_1 + \mu_2^T D\Delta_1 D\mu_2 + 2\mu_1^T DR^T D\mu_2. \end{aligned}$$

All zero-mean terms except the trace square cost a polynomial by Hilbert–Schmidt bounds. The exceptional term satisfies

$$|\text{Tr}(DR^T)|^2 \leq \|D\|^2 \|R\|_1^2 \leq CnqP.$$

The mean terms cost $C\sqrt{nq}P$. In particular the terms $\text{Tr} D^2$ and $\text{Tr}(D^2\Delta_i)$ have canceled identically. This is an estimate for the signed Gaussian expectation, not for the absolute insertion.

For real Fourier variable $v \in \mathbb{R}^3$, determinant damping is

$$|\det(I - 2iQ)|^{-1/2} \leq C(1 + c|v|^2/N)^{-c'N}.$$

Before fixed-rank changes, Q is the tensor product of C/L with the symmetric two-by-two frequency matrix. The latter has an eigenvalue of magnitude at least a constant times $|v|$. The spread assumption supplies cN singular values of size $c|v|/\sqrt{N}$; interlacing loses only a fixed number. A real cross tilt conjugates the Fourier quadratic matrix by bounded invertible operators, so the same bound holds with changed constants.

Every fixed frequency polynomial is integrable against this bound, uniformly for large N . Fourier inversion therefore proves the density bounds and their fixed derivatives. To obtain the Gaussian tail in U , shift the cross contour by $u = c_1 U$. Its centered determinant cost is $O(u^2)$, since the cross tilt has trace zero and bounded Hilbert–Schmidt norm. The projected mean cost is at most $C\epsilon(|\alpha_1|^2 + |\alpha_2|^2)$ on the stated window. Choose c_1 and then ϵ small. The factor $e^{-c_1 U^2}$ and the projection Gaussian density absorb these costs. Division by the bounded shell denominators completes these bounds.

The parity estimate uses the unconditioned projection variables. Integrate them out before Fourier inversion, so that reversing the cross frequency can be compared at the level of the Gaussian formula. Reversing U reverses both the cross Fourier frequency and the real cross tilt. Conjugation by $\text{diag}(I, -I)$ preserves the determinant and all zero-mean contractions of H . The inverse cross block satisfies the sharper operator estimate

$$\|\Sigma_{12}\| \leq CN^{-1/2}(|v| + |u|),$$

by the block inverse formula.

The length linear coefficients have bounded norms times $|v|$. Their Gaussian exponential factors therefore differ by at most $N^{-1/2}$ times a frequency polynomial. This estimate keeps the determinant damping: the two exponents and the segment between them have nonpositive real part, because the linear coefficients are imaginary and the real precision is positive. For the mass this proves the extra $N^{-1/2}$. For H , its zero-mean Wick expression costs Cnq times this factor. The mean terms can be bounded directly by $C\sqrt{nq}P$, which suffices since $\sqrt{nq} \leq nq/\sqrt{N}$ for $q \leq 1$. Fourier inversion with the same real tilt proves the odd-part bounds and their derivatives. $\square$

Lemma 11.5 (The surviving scalar coefficients). *Put $v_i = 1 - m_i^2$ and $\kappa_2 = \langle v^2 \rangle$, $\kappa_3 = 4\langle m^2 v^2 \rangle$. For $i, j \in \{p, g_0\}$, let $F_{ij}(\beta)$ be the per-site cross log moment after optimizing all single-replica constraints. At zero projection,*

$$F'_{ij}(0) = 0, \quad F''_{ij}(0) = \kappa_2, \quad F'''_{ij}(0) = \kappa_3 \mathbf{1}_{\{i=j=p\}}.$$

The common quadratic coefficient relative to the reference is $c_2 = (\kappa_2 - b^2)/(2q^2\kappa_2b^2)$. Up to the common prefactor $b/\sqrt{\kappa_2}$ and an error $O(qE_^C e^{\zeta E_*^2}/N)$, the signed zero-projection sum is*

$$\frac{q}{\sqrt{N}} e^{c_2 q U^2} \left(-\frac{\kappa_3}{2\kappa_2^2} U + \frac{\kappa_3}{6q\kappa_2^3} U^3 \right).$$

Proof. At zero constraints both canonical site laws have mean zero and variance v_i .

Their third centered moments are $-2m_i v_i$ for signs and zero for Gaussians. The cross observable is orthogonal to all single-replica constraints. The first derivative of the optimizing coefficients thus vanishes, so the first three optimized cross derivatives are its ordinary first three cumulants. This proves the displayed identities.

Let $D_2(\beta)$ be the constraint covariance. At zero cross tilt it is block diagonal. Each diagonal block of $D'_2(0)$ vanishes: the corresponding third cumulant contains a centered factor from the other replica. Hence $\partial_\beta \log \det D_2(\beta)|_0 = 0$. The conditional saddle prefactor is proportional to $[F''(\beta) \det D_2(\beta) / \det D_2(0)]^{-1/2}$, whose logarithmic t derivative is therefore $-F'''(0)/(2\kappa_2^2)$. Its cubic Legendre term is $nF'''(0)t^3/(6\kappa_2^3)$. Now use $t = qU/\sqrt{N}$, $nt^2 = U^2/q$, and $nt^3 = U^3/\sqrt{N}$. Reference finite-codimension corrections $O(t^2 + 1/n)$ lie within $O(q/N)$ on bounded normalized statistics.

Away from zero the Taylor estimates following (11.5) organize the logarithms as $\log R_{ij} = L_0 + C_* + A_i + A'_j + B_{ij} + r_{ij}$. Here L_0 is common constant/quadratic, C_* is common, A_i, A'_j are additive first projection terms, B_{ij} is the odd term just computed, and the retained small terms have size $CqE_*^C/\sqrt{N}$. The remainder has size $CqE_*^C e^{\zeta E_*^2}/N$. With $s_p = 1, s_{g_0} = -1$, $\sum s_i s_j = \sum s_i s_j (C_* + A_i + A'_j) = 0$. Quadratic products from exponentiation cost $q^2/N \leq q/N$.

The common projection multiplier differs from one by $N^{-1/2} E_*^C e^{\zeta E_*^2}$, so its effect on the surviving odd term also fits the stated remainder. The projection estimate (11.2), the derivative bounds following (11.5), and the mixed local-limit calculation therefore give precisely this cancellation for the smooth saddle ratios. Lemma 11.3 passes from these ratios to the actual mixed statistic measures. $\square$

Corollary 11.6 (Weighted moment comparison). *Use the priors, cuts, and shell-constrained reference mass Z_* of the preceding mixed comparison, whose physical reference covariance is $C_o = K_o^{-1}$. Let A be symmetric and assume the hypotheses of Lemma 11.4 with $C = C_o$. Suppose the mixed-coordinate disintegration and local saddle estimates give the preceding expansion, with scalar remainder $O(qE_*^C e^{\zeta E_*^2}/N)$ at each actual mixed mesh point. Suppose centered-cell quadrature for*

the smooth leading terms costs $O(1/n)$ times the signed kernel envelope. Then, with the common normalization by Z_* ,

$$\mathbb{E}_{\text{Haar}} \left\| Z_*^{-1} \int A(zz^T - C_o) A d(p - g)_{\text{tilted}} \right\|_{\text{HS}}^2 \leq Cnq^2/N.$$

Proof. There are three errors to account for: scalar remainders, the surviving odd term, and quadrature.

The scalar remainders cost $(q/N)(nq)$ using the signed density bound on each mesh separately. After quadrature, the odd leading term costs $(q/\sqrt{N})(nq/\sqrt{N})$. Quadrature itself costs $(nq)/n = q \leq nq^2/N$. Removing smooth projection cuts is exponentially negligible since $|W| \leq C|\alpha|$ and the cuts start at $|W|$ of order $\sqrt{N}$. Mollification is separately controlled by an absolute angular derivative bound and an arbitrarily high inverse-polynomial smoothing scale. The scalar saddle remainders never require placing $|H|$ inside the angular integral. $\square$

12 Amplification along observations

12.1 A compact-interval improvement of overlap tails

We next improve the probability of the static covariance bounds. The first ingredient concerns overlaps about the exact field root on a compact interval of small positive observation times. Throughout this section, $\mathcal{G}$ denotes the spectral restriction already imposed, $r = \sqrt{s}$, and $k = nr^3$. Probabilities with the indicator of $\mathcal{G}$ are integrated probabilities under the planted law; no claim about the planted probability of $\mathcal{G}$ is needed. We write $\Sigma_s = \text{Cov}_{\mu_{h_s}} X$ and $m_s^{\text{post}} = \mu_{h_s}(X)$. Thus the subscripted vector m_s^{post} is distinct from the posterior probability measure μ_{h_s} , and from the exact root m .

Lemma 12.1 (Overlap improvement on compact intervals). *There are $r_0 > 0$ and $a_* > 0$, with a_* independent of a lower endpoint r_1 , such that the following holds. For every fixed $0 < r_1 \leq r_0$ and $\chi > 0$, there is $c = c(r_1, r_0, \chi) > 0$ for which, at each $r \in [r_1, r_0]$,*

$$\mu_{h_s}^{\otimes 2}(|(X - m)^T(X' - m)| > \chi nr) \leq e^{-cn}$$

*outside an integrated planted exceptional set of mass at most e^{-a_*k} , with $\mathcal{G}$ imposed. The root is the unique regular root on the retained event.*

Proof. Use the root orbits, their three projection columns P , and the reference partition integral Z_* from the local comparison. We write ϑ for the normalized projection coordinate, so that $\delta = \text{diag}(\sqrt{q}, q, q^{3/2})\vartheta$, and put $z = X - m$, $N = nq^3$. For different replicas let

$$t_{ij} = n^{-1}(z_i - P\delta_i)^T(z_j - P\delta_j).$$

On this compact interval all polynomial bounds may depend on r_1 .

First, outside Haar mass e^{-Bk} , with any prescribed fixed B , the full tilted spin integral is at least $Z_*e^{-o(n)}$. Indeed (11.8) supplies a fixed positive fraction of orientations with full mass at least $Z_*/2$. The simultaneous covariance and mean bounds in (11.1) give a bounded raw operator second moment of z on this compact interval. Jensen's inequality, evaluated at such an anchor, bounds the loss under a rotation at fixed Frobenius distance by $C\sqrt{n} + O(1)$: the linear energy increment pairs with a raw second-moment matrix of Hilbert–Schmidt norm $O(\sqrt{n})$, and the quadratic increment costs $O(1)$.

Concentration of distance to the anchor set excludes distances greater than a sufficiently large fixed constant with probability e^{-Bk} . The determinant weight changes this estimate only by the already established arbitrarily small loss in its rate.

Fix a small absolute $D_* > 0$. For any fixed replica number l , consider the unnormalized tilted spin integral on

$$D_* \leq \sum_{i=1}^l |\vartheta_i|^2 + \sum_{i < j} |t_{ij}/q|^2 \leq 4D_*. \quad (12.1)$$

Its Haar mean, divided by Z_*^l , is at most e^{-cD_*N} , where $c > 0$ is independent of l , although the threshold in n may depend on l . We verify this uniformity because the replica number will depend on χ .

Under the tilted reference, the projection coordinates $\sqrt{N} \vartheta_i$ have bounded Gaussian covariance before length conditioning. Every linear combination with unit Euclidean coefficients of the off-diagonal coordinates $nt_{ij}/\sqrt{n/q}$ is a centered Gaussian quadratic form whose matrix has bounded Hilbert–Schmidt norm and operator norm at most $C/\sqrt{N}$. These constants are independent of l . Gaussian exponential Markov bounds, followed by a fixed-mesh net, therefore give cost at least $c_0 D_* N$ for (12.1). The net and conditioning on the finitely many lengths cost only polynomial factors for fixed l ; the latter follows either from the length-density bounds or from buffered shells.

The annulus penalty just obtained is under the *physically tilted* reference shell, with covariance C_o before length conditioning. The change of prior uses a different density: the *unweighted* o prior has uniform complement directions after its projections and lengths are fixed. The common angular energy integral is multiplied by the spin Gram density divided by this reference Gram density. We estimate this quotient using polynomial boxes and Chernoff upper bounds. The single-projection density ratios in (11.2) cost at most $CND_*^{3/2} + o(n)$. At fixed projections the unweighted reference conditional Gram density has exponent

$$-\frac{n}{2} \sum_{i < j} \frac{t_{ij}^2}{B_i B_j} + O(ND_*^{3/2}), \quad B_i = b - 2\sqrt{q}(\delta_i)_1 - |\delta_i|^2,$$

up to polynomial factors. At the independent spin projection saddles write the site deviations as $\xi_i + d_i^0$, with independent centered bounded ξ_i .

Set $R_{ij} = t_{ij}/(B_i B_j)$ and $R_{ii} = 0$. The quadratic log moment satisfies

$$\log \mathbb{E} \exp(\xi^T R \xi / 2) = \frac{1}{2} \text{Var}(\xi^T R \xi / 2) + O(\|R\|_{\text{HS}}^3),$$

with an absolute constant when $\|R\|_{\text{HS}}$ is small. For completeness, decouple the off-diagonal quadratic form using a random bipartition. Jensen’s inequality bounds its exponential moment by that of four times the corresponding directed cross sum. Integrating one side by the bounded-variable Gaussian moment bound produces a squared linear-image penalty; introducing a white Gaussian and integrating the other side gives a uniform exponential moment for a small multiple of $|\xi^T R \xi|/\|R\|_{\text{HS}}$. Taylor’s formula then gives the displayed third-order remainder.

The linear terms involving d_i^0 are treated by Hölder’s inequality with relative inflation $O(q\sqrt{D_*})$ in the centered quadratic moment. The remaining averaged log cost is at most

$$C(1 + (q\sqrt{D_*})^{-1}\|R\|)\|R\|q^3 D_*.$$

Here we used the response bounds (11.5) for d_i^0 and linear sub-Gaussianity. The site-averaged quadratic coefficient for each pair differs from $B_i B_j$ by $O(q^2)$. To see that the constant is independent

of l , note on the annulus that

$$\|R\|_{\text{HS}} \leq Cq\sqrt{D_*}, \quad \frac{1}{n} \sum_{\text{sites}} \sum_{i=1}^l |d_i^0|^2 \leq Cq^3 D_*.$$

The latter inequality is (11.5) summed over replicas. The displayed log-moment bound is first applied at each site and then averaged. Its cubic remainder costs $n\|R\|_{\text{HS}}^3 \leq CND_*^{3/2}$, the variance-profile discrepancy costs $nq^2\|R\|_{\text{HS}}^2 \leq CNqD_*$, and the displayed Hölder cost, after multiplication by n , is at most $CNqD_*^{3/2}$. These estimates use only sums of squares and the Hilbert–Schmidt norm. Their constants therefore do not grow with the fixed replica number. After the Chernoff subtraction, the total relative cost is bounded by $CND_*(q + \sqrt{D_*})$, besides polynomial factors. Choosing first D_* , then the upper bound on q , absorbs this cost into $c_0 D_* N$. Angular integrals on these boxes have controlled variation; their constants may depend on l, r_1 .

The annulus rate is independent of the fixed replica count. This lets us turn a requested overlap accuracy into a sufficiently large, but still fixed, number of replicas. A first-crossing selection will then find an annulus subset without allowing a single large statistic to account for the whole crossing.

We need an exponential posterior bound for each individually large statistic, not just the polynomial cutoff loss used in the local comparison. Let $\mathcal{R}_s$ be the domain where all root/scalar diagnostics retained for the local comparison hold. On this domain the root is unique and regular, $q > 0$, and the three axes are nondegenerate. Its excluded complement, including nonunique-root fields, has separate charge

$$\Pr_{\text{pl}}(\mathcal{G} \cap \mathcal{R}_s^c) \leq n^C e^{-a_0 k},$$

after taking the fixed $a_0 > 0$ below the preceding exclusion rates.

Only on $\mathcal{R}_s$, define $E_1 = \{|\vartheta_1|^2 > D_*/4\}$ for one fresh posterior replica and $E_2 = \{|t_{12}/q|^2 > D_*/4\}$ for two. The plant and retained root/scalar data remain fixed inside $\mu_{h_s}^{\otimes j}$. The scalar-root estimates after equivariant plant replacement, and the plant–replica and replica–replica overlap estimates, give

$$\int_{\mathcal{G} \cap \mathcal{R}_s} \mu_{h_s}^{\otimes j}(E_j) d\Pr_{\text{pl}} \leq n^C e^{-a_0 k}, \quad j = 1, 2,$$

after decreasing a_0 if necessary. Conditional Markov on this retained domain gives

$$\Pr_{\text{pl}}\left\{\omega \in \mathcal{G} \cap \mathcal{R}_s : \mu_{h_s}^{\otimes j}(E_j) > e^{-a_0 k/2}\right\} \leq n^C e^{-a_0 k/2}.$$

Relabeling E_1, E_2 and taking a finite union covers every individual statistic in any fixed replica family. The complement of $\mathcal{R}_s$ remains the separate charge displayed above.

Suppose the conclusion of the lemma fails. Draw a sufficiently large fixed number l_{pair} of disjoint violating pairs, depending on χ . It uses $2l_{\text{pair}}$ replicas; the preceding estimates apply to every subset with l equal to that subset’s replica count. Since

$$n^{-1} z_i^T z_j = t_{ij} + \delta_i \cdot \delta_j,$$

their statistics force the total in (12.1) to cross D_* . Requiring all l_{pair} pairs costs at most $l_{\text{pair}} cn$ in log likelihood. Choose c so that $l_{\text{pair}} cn \leq a_0 k/4$, possible uniformly on the compact interval because $k \geq nr_1^3$. The preceding exponential conditional bounds then make all individually large statistics, including those across pairs, negligible even after this conditioning. This choice explains the dependence of c on r_1, χ , while the field-rate constant remains fixed.

A subset of the selected replicas then satisfies (12.1). Indeed take the first prefix crossing D_* . If its new row contributes too much, retain enough earlier indices to make that row contribution

comparable to D_* ; each of its terms is at most $D_*/4$, while the earlier internal sum is below D_* . A finite union over subsets, the denominator lower bound, and the preceding Haar estimate exclude the violating orientations at a rate bounded below independently of the number of replicas. Returning to the counting weight costs $o(k)$ on this compact interval. The initially excluded orbits had a fixed small rate as well. This proves the lemma. $\square$

12.2 A conditional weak Poincaré inequality

The next lemma transports variance one short step backwards in the observation filtration. This Gaussian posterior interpolation belongs to the stochastic-localization framework of Chen and Eldan [12], Eldan [19]; the exceptional-set variance estimate needed here is proved below. Given (J, h_s) , let $u = (1 + d)s$, $\tau = u - s$, with $d > 0$ fixed and small, and write

$$g = \frac{h_u - h_s - \tau m_{\text{sh}}}{\sqrt{\tau}}.$$

The shift m_{sh} is measurable at time s and has polynomially bounded norm. Let P be the conditional law of g , G standard Gaussian measure, and $L_P = \log(dP/dG)$. The Gaussian-mixture formula gives convexity of L_P and

$$\nabla^2 \log \frac{dP}{dG} = -I + \tau \Sigma_u,$$

where dP/dg is the Lebesgue density.

Lemma 12.2 (Conditional variance estimate). *Fix an observation bound $T < \infty$ and positive constants α, β . Suppose $0 < s < u = (1 + d)s \leq T$, $k = ns^{3/2}$, and*

$$P\{\tau \|\Sigma_u\| > 1/4\} \leq e^{-\beta k}, \quad \log \int e^{2L_P} dG \leq \alpha k, \quad (12.2)$$

where α/β is less than a sufficiently small absolute constant. For every fixed $D > 0$ and smooth test ϕ whose values and required derivatives have fixed polynomial bounds,

$$\text{Var}_P \phi \leq 3\mathbb{E}_P \|\nabla_g \phi\|^2 + n^{-D}, \quad (12.3)$$

provided $\beta k / \log n$ is sufficiently large in terms of those bounds and D . Bounded tests with polynomial cutoffs are included.

Proof. We first transfer small exceptional sets from P to G . Since $\int e^{L_P} dG = 1$, the second moment in (12.2) and Paley–Zygmund give

$$G\{L_P \geq -1\} \geq (1 - e^{-1})^2 e^{-\alpha k} \geq e^{-2\alpha k}$$

for sufficiently large αk ; enlarging harmless constants covers the other case. Also $G\{L_P > \beta k/8\} \leq e^{-\beta k/8}$. Convexity implies that

$$G\{L_P < -\beta k/4\} \leq e^{-c\beta k} \quad (12.4)$$

for an absolute $c > 0$.

Here is a proof that does not presume regular boundaries of the two level sets. Use Gaussian endpoint conditioning to drive two Brownian motions with the same noise into $\{L_P \geq -1\}$ and a proposed low set of mass at least $e^{-c\beta k}$, with joint success probability greater than 0.8. Positive smooth approximations to the conditioning densities and energy stopping, in the entropy–drift

representation of Lehec [25, Proposition 1 and Section 2.5], give total expected squared drift energy at most $C(\alpha + c\beta)k$.

Reflect the second control about the first. The third endpoint is the reflected point, so convexity forces its likelihood above $\beta k/8$ on the joint hit, for large k . Its relative entropy with respect to G is at most $C'(\alpha + c\beta)k$, by Girsanov and data processing. Binary entropy applied to $\{L_P > \beta k/8\}$ contradicts this bound if c and α/β are small enough. This proves (12.4). Consequently, if $P(A) \leq e^{-\beta k/2}$, then

$$G(A) \leq e^{-\beta k/4} + e^{-c\beta k}.$$

The likelihood comparison has transferred rare starting sets to the Gaussian reference. We now obtain contraction along typical Langevin paths and telescope short-time variances to get the gradient term. A final comparison through the Gaussian reference will remove the remaining terminal variance.

Run stationary Langevin diffusion for P , with generator $\Delta + \nabla \log(dP/dg) \cdot \nabla$, for $T' = C_D \log n$. Apart from probability $\text{poly}(n)e^{-\beta k} + e^{-n^2}$, its log-density Hessian is at most $-2I/3$ all along the path. Here are the buffers needed for this path statement. Let M_n be a polynomial bound for the Lipschitz constant of the Hessian, supplied by the bounded-spin formulas for posterior derivatives, and take $\rho = (100M_n)^{-1}$. At every point of a polynomial time grid, stationarity and (12.2) give the upper bound $-3I/4$ outside total probability $\text{poly}(n)e^{-\beta k}$. The drift has linear growth with a polynomial additive bound. Choose the grid mesh polynomially small enough that, on a polynomial ball, its drift and Brownian increments are less than $\rho/4$ between successive grid points. Gaussian increment bounds and the tails of the Gaussian mixture P make the remaining probability at most e^{-n^2} . The Hessian Lipschitz bound then preserves a strict upper bound below $-2I/3$ throughout the tube of radius ρ about the path. The fixed observation bound and fixed β give $\beta k = O(n)$. Thus these Gaussian-tail errors remain negligible after the exponential reweightings below. The lemma is only asserted on this compact observation domain.

This tube also controls neighboring flows for that same future noise. Start a synchronous flow within $\rho/4$ of the path's initial point and stop at the first time its distance from the path reaches $\rho/2$. Before that time, the mean Hessian along the line segment between the flows is at most $-2I/3$, so their distance contracts. It cannot reach the stopping boundary. Differentiating this stopped comparison and keeping a little slack gives Jacobian norm at most $e^{-v/2}$ for every $v \leq T'$. This is a neighborhood of a good random path, not a claim about every initial point.

Let K_v be the Langevin semigroup. Choose a polynomially small Δt and an integer N_t with $T' = N_t \Delta t$, adjusting T' negligibly. Stationarity gives the exact telescope

$$\text{Var}_P \phi - \text{Var}_P K_{T'} \phi = \sum_{j=0}^{N_t-1} \mathbb{E}_{x \sim P} \text{Var}_{K_{\Delta t}(x, \cdot)}(K_{j\Delta t} \phi).$$

Every short transition $K_{\Delta t}(x, \cdot)$ has Poincaré constant at most $(2 + o(1))\Delta t$, uniformly in x . Indeed the global upper Hessian bound is polynomial, hence the flow Jacobian is $1 + o(1)$ up to this time; the conditional heat variance identity integrated over the short interval gives the assertion.

For a variance term involving $K_v \phi$, Jensen in the common future noise bounds $\text{Var}_{K_{\Delta t}(x, \cdot)} K_v \phi$ by the average variance of $y \mapsto \phi(Z_v(y))$, where Z_v is that future Langevin flow. On a good path from x , cut this test off on the inverse-polynomial ball just described. Its gradient there is bounded by $e^{-v/2} \|\nabla \phi(Z_v(y))\|$. Crossing and cutoff-annulus errors are negligible by short-time Gaussian tails after making Δt small enough.

The good-path event is determined from the future path started at x , and the cutoff test has polynomial bounds on this ball. Bad paths, averaged over stationary x , cost only a polynomial

times their probability. On good paths, drop the event indicator from the resulting nonnegative gradient integral. Then $x \sim P$, $y \sim K_{\Delta t}(x, \cdot)$, and independent future noise make $Z_v(y)$ stationary. Each variance term is therefore at most

$$(2 + o(1))\Delta t e^{-v} \int \|\nabla \phi\|^2 dP$$

plus an arbitrarily small inverse-polynomial error. Summing over the polynomial number of time steps preserves the negligible error. The gradient contributions sum to at most $(2 + o(1)) \int \|\nabla \phi\|^2 dP$.

It remains to bound $\text{Var}_P K_{T'}\phi$. Let $b(x)$ be the conditional probability that the future path from x fails the preceding tube test. Its P mean is at most $\text{poly}(n)e^{-\beta k} + e^{-n^2}$. Markov at $e^{-\beta k/3}$, using $\beta k \gg \log n$, leaves a set of P -mass at most $e^{-\beta k/2}$. Off that set, averaging the common-noise coupling gives local differences bounded by $e^{-T'/2} \text{Lip}(\phi)$ times distance, plus an arbitrarily small inverse-polynomial error. Transfer the exceptional starting set to G using (12.4). For independent $g_0, g_1 \sim G$, use a polynomial mesh on $\cos \theta g_0 + \sin \theta g_1$, $0 \leq \theta \leq \pi/2$. Every vertex has law G , and Gaussian norm tails put adjacent vertices within the local ball. A union bound over the vertices leaves Gaussian exceptional probability $\text{poly}(n)e^{-c\beta k}$.

The local estimates telescope along this polygon. To reweight its endpoints to two independent P -points, Cauchy–Schwarz and (12.2) bound the exceptional probability by

$$e^{\alpha k} (\text{poly}(n)e^{-c\beta k})^{1/2}.$$

It is smaller than any required inverse polynomial when α/β is sufficiently small and $\beta k/\log n$ is sufficiently large. Taking C_D large gives the desired residual variance bound. Absorb the remaining errors in the slack between $2 + o(1)$ and 3. $\square$

12.3 Backward amplification to logarithmic observation scale

We use two positive-time inputs: for every compact $[s_1, T] \subset (0, \infty)$, posterior covariance is bounded by a constant except at integrated planted probability $e^{-c(s_1, T)n}$; and at a sufficiently large fixed T , the posterior unscaled gap is bounded below with the same type of exception. The critical endpoint verification of these inputs is given in the positive-time argument. Their use here is only with the rotation-invariant restriction $\mathcal{G}$, and only on fixed positive intervals.

Proposition 12.3 (Amplified covariance estimate). *For every sufficiently small fixed $\epsilon > 0$, there are $s_0, a > 0$ and $K_{\log} < \infty$ such that, at every deterministic time in the indicated range,*

$$\|\Sigma_s\| \leq \frac{1 + \epsilon}{s}, \quad 0 < s \leq s_0, \quad ns^{3/2} \geq K_{\log} \log n, \quad (12.5)$$

outside an integrated planted exceptional set, intersected with $\mathcal{G}$, of mass at most $e^{-ans^{3/2}}$. For fixed $1/2 < \gamma < \gamma' < 1$, the same exceptional set may be used for

$$\begin{aligned} \|A_0 \Sigma_s A_0\| &\leq r^{-\gamma}, \quad A_0 = 2I - J, \\ v_s &:= \sup_{\substack{H^T = -H \\ \|H\|_{\text{HS}} \leq 1}} \text{Var}_{\mu_{h_s}}((X - m_s^{\text{post}})^T [H, J] (X - m_s^{\text{post}})) \leq C_v s^{-1-\gamma'/2}. \end{aligned} \quad (12.6)$$

Proof. Each backward step first uses later observations to obtain coarse covariance and moment bounds. These control derivatives along rotation orbits, where the static majority estimates reset covariance to its sharp threshold. The initial positive-time band uses the same reset; its coarse

bounds come from the separate fixed-positive-time and terminal inputs. We carry all three estimates together. First fix $0 < \iota < \gamma'/2$, with the target covariance slack ϵ small enough for the variance calculation below. That calculation gives a constant C'_ι independent of L_0 and of the relative step size. Choose L_0 large enough that $C'_\iota L_0^{\iota-\gamma'/2} < 1/4$, and then choose s_0 small enough for the static medians and Lemma 12.1.

The quadratic-variance constant C_v and the auxiliary fourth-moment constants below may depend on these choices, but will not depend on the eventual relative step size. The initial band is $[s_0/L_0, s_0]$. Let F_b^+ denote the positive-time covariance failure, including the terminal gap failure when $b = T$, and define on this band

$$Q_s^+ = \mathbf{1}_{F_s^+} + \Pr_s(F_T^+) + \int_s^T (1 + b^{-C}) \Pr_s(F_b^+) db.$$

Here and below $\Pr_s$ conditions on (J, h_s) , and the fixed C dominates the weights in the moment transfers. Tonelli and Markov on the fixed interval $[s_0/L_0, T]$ give a fixed $c_{\text{init}} > 0$ such that $Q_s^+ \leq e^{-c_{\text{init}}n}$ outside integrated mass $e^{-c_{\text{init}}n}$. On this one event, the bounded-covariance martingale transfer to T and its gap bound give all the initial quadratic and fourth-moment bounds proved below. No conditional variance lemma is needed for this initial transfer. Orientation concentration will reset the covariance thresholds on the same band.

Let $c_+ > 0$ be a positive-time failure rate on $[s_0, T]$, and let $a_{\text{stat}} > 0$ be smaller than the fixed root, orbit, diagnostic, and spherical-width exclusion rates at the accuracies used below. The compact bridge has its endpoint-independent rate a_* . Choose $a > 0$ once, with

$$8a < \min\{a_{\text{stat}}, a_*, c_{\text{init}}/s_0^{3/2}, c_+/s_0^{3/2}, 1\},$$

and small enough for the final orbit-gradient condition whenever its concentration exponent is at most $2a$. That condition uses only the moment constants and the already fixed s_0, L_0 .

A conditional event with room in its probability exponent. For $s < s_0/L_0$, put $u = (1+d)s$ and $t = L_0s$, and suppose the induction estimates hold at all required times at least u . The active d will be one of two fixed values specified below; the rate a will not be decreased after those choices. Let F_b be the union of the three inductive failures at $b \leq s_0$, and the positive-time covariance failure at $s_0 < b \leq T$. Include the terminal gap failure in F_T . Define

$$Q_s = \Pr_s(F_u) + \Pr_s(F_t) + \Pr_s(F_T) + \int_u^T (1 + b^{-C}) \Pr_s(F_b) db.$$

Choose the fixed exponent C large enough to dominate the polynomial weights in the variance and fourth-moment estimates below. All these weights are polynomial in n on the stated time range. Tonelli's theorem and the deterministic-time induction give

$$\mathbb{E}_{\text{pl}}[\mathbf{1}_{\mathcal{G}} Q_s] \leq \text{poly}(n) e^{-ak(u)} + \text{poly}(n) e^{-c_+n}.$$

Here $\mathcal{G}$ is measurable at time s ; the equality behind Tonelli therefore retains its indicator. Above s_0 we used the positive-time inputs on the fixed interval $[s_0, T]$.

Set $\beta = c'da$, with a sufficiently small absolute $c' > 0$. Markov at $Q_s = e^{-\beta k}$ gives, before polynomial costs, the exponent

$$(a(1+d)^{3/2} - \beta)k.$$

For small fixed d this strictly exceeds ak . The other Markov term is $\text{poly}(n) e^{-c_+n+\beta k}$; the prior choice of a , and $k \leq ns_0^{3/2}$, also place it strictly above the required rate. We next intersect this test with the likelihood and diagnostic events needed for the derivative bounds.

For sufficiently small d , the covariance hypothesis in (12.2) follows from $Q_s \leq e^{-\beta k}$. The likelihood hypothesis follows from the exact Gaussian identity

$$\int e^{2L_P} dG = \mathbb{E}_s \exp\{\tau(X^1 - m_{\text{sh}})^T(X^2 - m_{\text{sh}})\}.$$

The initial band is already fixed, so the back-step must be chosen without shrinking that band again. Two likelihood estimates provide this separation of choices: the spherical log-moment estimate near zero, and the compact overlap bridge on the remaining interval.

There are two parameter ranges. At sufficiently small r , use the spherical shift $m_{\text{sh}} = m_o$. Choose $d_{\text{near}} > 0$ so that Cd_{near}^2 is a sufficiently small multiple of $d_{\text{near}}a$. Apply the off-diagonal replica log-moment bound (6.2) to this fixed two-replica test, with requested failure rate $4a$. Given $\varepsilon_{\text{like}} > 0$, it gives

$$\log \int e^{2L_P} dG \leq Cd_{\text{near}}^2 k + \varepsilon_{\text{like}} k + C_2 \log n$$

on a sufficiently small upper range. Choose $\varepsilon_{\text{like}} \ll d_{\text{near}}a$, then its upper endpoint $r_c > 0$, and finally take $K_{\log}$ large enough that $C_2 \log n \ll d_{\text{near}}ak$. Shrinking r_c alone does not absorb this logarithmic term. Only this selected test is needed, not simultaneous control for all prospective step sizes. The width exclusion has the larger rate fixed above. On the remaining compact interval use $m_{\text{sh}} = m$ and Lemma 12.1, with $\chi \ll a$.

Choose a second step $d_{\text{comp}} > 0$ on this compact interval. The good-pair contribution to the second moment is at most $e^{d_{\text{comp}}\chi k}$, while the exceptional-pair contribution is at most

$$e^{-cn+4d_{\text{comp}}sn},$$

using $\|X - m\| \leq 2\sqrt{n}$. Choose d_{comp} after the compact-tail constant c so that this term is negligible. Then $\alpha \leq 2d_{\text{comp}}\chi$ for large n , while $\beta = c'd_{\text{comp}}a$. In either range, α/β in (12.2) is as small as required. Thus no choice of s_0 depends on a or on these small step sizes. Choose $K_{\log}$ last to absorb the polynomial costs of Lemma 12.2.

For each active step let $\mathcal{L}_s$ be its likelihood event just constructed, and let $\mathcal{D}_s$ retain the diagnostics and root or spherical-width events used in its orbit comparison. The full event used from now on is

$$\mathcal{C}_s = \{Q_s \leq e^{-\beta k}\} \cap \mathcal{L}_s \cap \mathcal{D}_s.$$

The first two factors are tests of (J, h_s) , and are the only ones used to apply the conditional variance lemma under Pr_s . The diagnostic factor may also use the plant and root-orbit data; it restricts the subsequent orbit integration without further conditioning the posterior spin law. Put

$$\Delta = \min_{d \in \{d_{\text{near}}, d_{\text{comp}}\}} \{a((1+d)^{3/2} - 1) - c'da\} > 0.$$

The preceding estimates give

$$\Pr_{\text{pl}}(\mathcal{G} \cap \mathcal{C}_s^c) \leq \text{poly}(n)e^{-(a+\Delta)k} + \text{poly}(n)e^{-4ak}.$$

All assertions remain integrated with $\mathcal{G}$ imposed; none is a bound conditional on every spectrum. On the initial band define $\mathcal{C}_s$ instead as $\{Q_s^+ \leq e^{-c_{\text{init}}n}\} \cap \mathcal{D}_s$, whose excluded mass has a fixed rate larger than $4a$.

Coarse covariance and a longer variance transfer. On $\mathcal{C}_s$, total covariance gives

$$\Sigma_s = \mathbb{E}_s \Sigma_u + \text{Cov}_s(m_u^{\text{post}}).$$

Apply (12.3) to every linear projection of m_u^{post} . Its derivative in g is $\sqrt{\tau} \Sigma_u$, so the second term is at most $3\tau \mathbb{E}_s \Sigma_u^2$, up to an arbitrarily small inverse-polynomial operator error. Apply the same argument after multiplication by A_0 . This gives both covariance bounds with fixed coarse factors.

For any smooth polynomially bounded test of the later field,

$$\text{Var}_s(\phi(h_t)) \leq C_\iota t L_0^{1+\iota} \mathbb{E}_s \|\nabla \phi(h_t)\|^2 + n^{-D}. \quad (12.7)$$

Here $\iota > 0$ is arbitrarily small if ϵ is chosen sufficiently small, and C_ι is independent of L_0, d . To prove this, the posterior-expectation martingale on $[u, t]$ has diffusion coefficient

$$\nabla \mathbb{E}_b \phi(h_t) = \mathbb{E}_b \nabla \phi(h_t) + \text{Cov}_b(X, \phi(h_t)).$$

This follows directly from the Gaussian likelihood of the future observation increment given X .

The observations have unit diffusion and posterior-mean drift in their own filtration. If $R_b = \mathbb{E}_s \text{Var}_b(\phi(h_t))$, their martingale variance identity and Cauchy–Schwarz yield

$$-R'_b \leq \frac{1+\iota}{b} R_b + C_\iota \mathbb{E}_s \|\nabla \phi(h_t)\|^2 + \text{poly}(n) \Pr_s(\text{failure at } b).$$

In the covariance term use $\|\text{Cov}_b(X, Y)\|^2 \leq \|\Sigma_b\| \text{Var}_b Y$, and absorb the mixed term by the slack in ι .

Integrating backwards from $R_t = 0$ gives explicitly

$$R_u \leq C_\iota t (t/u)^{1+\iota} \mathbb{E}_s \|\nabla \phi(h_t)\|^2 + \int_u^t (b/u)^{1+\iota} \text{poly}(n) \Pr_s(F_b) db.$$

The last term is negligible on the conditional event. For the first interval $[s, u]$, apply (12.3) to $\mathbb{E}_u \phi(h_t)$ and use the same gradient estimate. It adds at most a fixed multiple of $R_u + \tau \mathbb{E}_s \|\nabla \phi(h_t)\|^2$, since $\tau/u \leq d$. As $t/u \leq L_0$, this proves (12.7) with a constant independent of L_0, d . When covariance is bounded on a fixed positive band beginning at s , this same martingale argument works directly from s up to T , with fixed constants in place of the displayed powers.

Quadratic and fourth moments. Let $M = [H, J] = -[H, A_0]$, with $\|H\|_{\text{HS}} \leq 1$, and put $D_t = m_t^{\text{post}} - m_s^{\text{post}}$. Total covariance gives $\mathbb{E}_s D_t D_t^T \preceq \Sigma_s$. At time t , the centered quadratic from time s splits into a centered quadratic there and the linear term $2D_t^T M(X - m_t^{\text{post}})$, plus its conditional mean. The averaged variance of this linear term is $O_{L_0}(s^{-1-\gamma/2})$. Indeed expand $M = -HA_0 + A_0H$. On successful descendants the two terms obey

$$\begin{aligned} \mathbb{E}_s[\mathbf{1}_{\text{good}} D_t^T H A_0 \Sigma_t A_0 H^T D_t] &\leq C t^{-\gamma/2} \text{Tr}(H^T \Sigma_s H) \leq C t^{-\gamma/2} s^{-1}, \\ \mathbb{E}_s[\mathbf{1}_{\text{good}} D_t^T A_0 H \Sigma_t H^T A_0 D_t] &\leq C t^{-1} \text{Tr}(H^T A_0 \Sigma_s A_0 H) \leq C t^{-1} s^{-\gamma/2}. \end{aligned}$$

The norm bound at t is applied before averaging, and $\mathbb{E}_s D_t D_t^T \preceq \Sigma_s$ is then used in the trace. No independence between D_t and Σ_t is needed. Polynomially bounded errors on failed descendants are negligible.

The conditional mean of the whole quadratic is $\text{Tr}(M \Sigma_t) + D_t^T M D_t$. As a function of the field at t , its gradient is its conditional covariance, as a spin test, with X . On successful descendants its

squared norm is thus at most $(1 + \epsilon)/t$ times its conditional variance. Elsewhere use the polynomial deterministic bound. Total variance, (12.7), and splitting sums with a factor two give

$$v_s \leq C'_\ell L_0^{1+\iota} C_v t^{-1-\gamma'/2} + C_{L_0} s^{-1-\gamma/2} + o(1).$$

The coefficient of the first term relative to $C_v s^{-1-\gamma'/2}$ is $C'_\ell L_0^{\iota-\gamma'/2} < 1/4$ by our earlier choice. Choose C_v large. The strict inequality $\gamma < \gamma'$ absorbs the second term. This closes the quadratic-variance induction.

On the initial band, apply the bounded-covariance version of the transfer to T . At T , the gap input bounds the centered quadratic variance because its single-site heat-bath energy is at most a constant times $\text{Tr}(M\Sigma_T M) + \|M\|_{\text{HS}}^2$, as follows by expanding a single spin flip. Coarse covariance on the initial band comes directly from the positive-time input.

We also obtain uniformly in unit e

$$\mathbb{E}_s |e^T (X - m_s^{\text{post}})|^4 \leq Cr^{-4}, \quad \mathbb{E}_s |e^T A_0 (X - m_s^{\text{post}})|^4 \leq Cr^{-2\gamma}.$$

Integrate the posterior-mean martingale from u to T , whose diffusion coefficient is Σ_b . The respective directional quadratic variations have second moments bounded by the squares of the integrals of Cb^{-2} and $Cb^{-1-\gamma/2}$ on small scales; use bounded covariance later. These estimates follow from $\|\Sigma_b\|^2$ and $\|\Sigma_b A_0\|^2 \leq \|\Sigma_b\| \|A_0 \Sigma_b A_0\|$, with deterministic polynomial bounds on exceptions.

Martingale fourth-moment inequalities apply. Terminal fourth moments follow by applying the gap first to a linear test and then to its centered square. Finally apply (12.3) to a centered linear projection of m_u^{post} and to its square; their gradients involve Σ_u . This treats $m_u^{\text{post}} - m_s^{\text{post}}$ without constants depending on d . On the initial band start the mean martingale directly at s .

Restoring the sharp covariance thresholds. On each root orbit, monitor the two covariance norms and the two mean deviations in the static comparison, in their own normalized units. Use tolerance $\epsilon/2$ for the sharp covariance median and ϵ for its failure threshold, with fixed slack also for the other quantities. In particular the mean thresholds are $r^{-1-\eta}$ and $r^{-\gamma/2}$, with an arbitrarily small $\eta > 0$. Let a Lipschitz ramp be zero below the median thresholds and one when any failure threshold is reached.

On the conditional events already constructed, its gradient in the transition region is bounded by

$$D_{\text{grad}} = Cr^{-1-\gamma''/2}, \quad \gamma' < \gamma'' < 1.$$

To check this, a root-preserving rotation has score $z^T M z/2$, with m fixed. Its centered quadratic variance is bounded by (12.6). The additional linear term about m_s^{post} has variance at most $Cr^{-2-\gamma-2\eta}$, by the coarse covariance bounds and the transition thresholds.

Choose η sufficiently small. Cauchy–Schwarz with the fourth and second moments just proved then controls derivatives of directional centered squares and means. The derivative of A_0 itself costs at most $Cr^{-1-\eta}$ in these units. Outside these conditional events the ramp has a polynomial global gradient bound. This calculation does not differentiate the root.

When $K_{\log} \log n \leq k < \log^{10} n$, use instead unconstrained orbits that co-rotate the field and matrix, centered at m_o . The polylogarithmic edge comparison and the spherical shell estimate give the required medians, after excluding the small-rate event on which the spherical width squared differs from s by more than a fixed relative tolerance $\epsilon' \ll \epsilon$. Here is the conversion from raw moments. In this band $r = n^{-1/3} \log^{O(1)} n$, so the unweighted error $n^{1/2+o(1)}$ in Theorem 7.1 is $o(s^{-1})$. On the width event, (5.6) bounds the spherical raw moment about m_o by $(1 + O(\epsilon') + o(1))s^{-1}I$. The cube covariance is bounded by its raw moment about m_o , giving the sharp covariance median. The

weighted raw moment is at most $n^{1/6+o(1)}$, which is $o(r^{-\gamma})$ because $\gamma/3 > 1/6$. Jensen bounds the squares of the ordinary and weighted mean deviations by their respective raw moment operators. These give the two looser mean medians $r^{-1-\eta}$ and $r^{-\gamma/2}$. All four statements concern a Haar majority on the orbit that co-rotates this field.

The extra linear score coefficient is

$$Mm_o + Hh = (\lambda - J)Hm_o.$$

Since $\|m_o\| = O(\sqrt{k}/r)$, its standard deviation has an additional multiplier at most $C\sqrt{k}$ in the preceding bounds. Differentiating the center costs the same allowance. Thus D_{grad} may be enlarged by a fixed power of $\log n$ in this range. The unconstrained partition-ratio comparison (6.1) transfers the orbit weights.

The conditional derivative bound has exceptions whose masses depend on the orbit. We need a concentration estimate that is linear in those masses, so that it can be averaged over the orbit parameters. A Haar ramp zero on a fixed majority, with gradient at most D_{grad} off a set of mass p , and polynomially bounded gradient everywhere, has plateau mass at most

$$e^{-bk} + \text{poly}(n)p,$$

provided $D_{\text{grad}}\sqrt{bk/n}$ is sufficiently small. Indeed split the upper ramp levels at tail probabilities successively halving from $1/2$ to e^{-bk} .

If the plateau exceeds both this last probability and a large polynomial times p , truncate and rescale the ramp across one such quantile interval of width w and probability level p' . The entropy of its square is at least $cp' \log(1/p')$, whereas its gradient-square integral is at most

$$w^{-2}(2D_{\text{grad}}^2 p' + \text{poly}(n)p).$$

The rotation-group heat log-Sobolev inequality gives $w \leq CD_{\text{grad}}/\sqrt{n \log(1/p')}$.

Strict supports and inclusive plateaux handle atoms or zero-width intervals. Summing these widths bounds the full rise by $CD_{\text{grad}}\sqrt{bk/n}$, a contradiction. In particular the orbit-dependent exceptional masses enter linearly and can be averaged without taking a fractional power.

Take $b = a + \Delta/2 < 2a$. The gradient condition uses only L_0, s_0, C_v , the moment constants and $\gamma'' < 1$, and holds by the earlier choice of a . It also holds throughout the polylogarithmic band despite the extra logarithms. On regular root anchors, the determinant-weight comparison at the concentration radius costs

$$O(\sqrt{k} + k/(nq^2)) = o(k),$$

so it does not require shrinking r after the rate has been chosen.

We display the two measure changes. Let $p(\omega)$ be the orbit mass of $\mathcal{C}_s^c$, including its likelihood and diagnostic exclusions, and let $h(\omega)$ be its ramp plateau mass. The quantile estimate and $\int w_0 d\pi \leq n^C$ give

$$\int w_0 h d\pi \leq e^{-bk} \int w_0 d\pi + \text{poly}(n) \int w_0 p d\pi.$$

The first comparison in Section 10 bounds $\int w_0 p d\pi$ by the counting derivative-failure mass times $e^{\xi k}$, plus larger-rate exclusions. Returning the last display to the counting law by the reverse comparison costs a second $e^{\xi k}$. Thus the derivative term retains exponent at least $a + \Delta - 2\xi$, and the direct plateau term retains exponent at least $b - \xi$, before polynomial factors. Choose $0 < \xi < \Delta/8$. The excluded diagnostic and base-orbit terms have already been assigned larger rates. Finally the degree identity charges excess root multiplicity to the excluded negative-Jacobian roots when returning

from counts to planted fields. For unconstrained orbits choose partition-ratio accuracies relative to the small-range step size, then enlarge $K_{\log}$. Every reset cost and discarded event therefore fits within e^{-ak} . The same reset works on the initial band with its fixed constants and the larger initial-event rate already reserved; a is unchanged.

For clarity, the order of choices is: the target accuracies, $\iota < \gamma'/2$, and then large L_0 ; small s_0 for the static medians, the fixed outer cuts, and the compact bridge; then C_v and all moment constants, including those on the initial band; then a ; then the two positive relative step sizes $d_{\text{near}}, d_{\text{comp}}$; and finally $K_{\log}$. Put $d_{\min} = \min(d_{\text{near}}, d_{\text{comp}}) > 0$. On successive backward bands with ratio $1 + d_{\min}$, every active $u = (1 + d(s))s$ lies in an earlier band, as does t . Induction on these bands proves (12.5)–(12.6) for each deterministic s . Tonelli supplies the time integrals used above from these pointwise statements; no union over uncountably many observation times is taken. No positive-time input is required uniformly down to an endpoint depending on a . $\square$

12.4 Covariance below the logarithmic threshold

The backward induction stops at logarithmic k . Below that threshold we prove a separate, coarser covariance bound with an arbitrarily large polynomial probability exponent. The required probability cannot be obtained by merely repeating the second-moment comparison. The Haar Gram comparison and the mixed cube–sphere tensor cancellation are closely related to Du and Huang [16, Section 3.3] and Du and Huang [17, Lemma 3.5]. Here the full finite-replica expansion keeps track of every normalization factor to obtain the required probability exponent along the observation fields.

Proposition 12.4 (Bottom-scale covariance). *For every fixed $B > 0$ and $K_{\log} < \infty$, at each deterministic time satisfying $ns^{3/2} \leq K_{\log} \log n$,*

$$\|\Sigma_s\| \leq Cn^{2/3}\ell^C, \quad \ell = \log \log(n + 1000), \quad (12.8)$$

except on an integrated planted event, intersected with $\mathcal{G}$, of mass at most n^{-B} .

Proof. Set $L = n^{2/3}$. The spherical raw second moment about m_o already satisfies the bound in (12.8). The hypotheses of the polylogarithmic edge comparison, namely

$$\lambda - \lambda^0 \leq n^{-2/3} \log^C n, \quad \|h\| \leq \sqrt{n} r \log n,$$

may be imposed with arbitrarily small inverse-polynomial failure by testing the spherical observation at a larger polylogarithmic edge width. We prove a higher-moment version of that comparison. Only unweighted raw second moments are needed here.

On an unconstrained Haar orbit divide every integral by $Z_o(h)$. Let Δ be either the signed cube-minus-sphere mass, or the signed matrix integral of $(x - m_o)(x - m_o)^T/L$, with this common denominator. For a fixed even integer p , expand $\|\Delta\|_{\text{HS}}^p$ (the mass case means $|\Delta|^p$). A subset $B' \subseteq [p]$ specifies which of the p draws have sign priors; the other draws have spherical priors. The common kernel is respectively 1, or

$$\prod_{a=1}^{p/2} \left(\frac{(x_{2a-1} - m_o)^T (x_{2a} - m_o)}{L} \right)^2.$$

The sign of that term is $(-1)^{p-|B'|}$.

Truncating the moment before annealing. In each term first restrict the internal sign overlaps $Q_{ij} = x_i^T x_j / n$, $i, j \in B'$, to $|Q_{ij}| \leq \eta_p$, where $\eta_p > 0$ is fixed and small. This avoids assuming an unrestricted high moment of the partition function. Outside exponentially exceptional orientations the truncated alternating sum differs negligibly from the desired power: the unnormalized large-overlap sign-pair integral is exponentially small by the global pair bound, while the total normalized cube mass is at most $e^{c_{\text{tiny}} n}$ by its first moment. Choose c_{tiny} sufficiently small after p, η_p . The polynomial kernel costs can also be absorbed.

To discard the exceptional orientations inside an alternating sum, we need a bound on each term separately. Let $T_{B'}(O) \geq 0$ be the normalized, individually truncated term at orientation O , including its mass or product-of-squares kernel. We claim

$$\mathbb{E}_{\text{Haar}} T_{B'}^2 \leq e^{o(n)}.$$

To verify this, bound the polynomial insertions crudely and integrate out the spherical priors. Two groups of sign vectors remain, with nearly identity internal Grams. If R is their matrix of cross overlaps, the empirical-site-type entropy cost is at least

$$\frac{\|R\|_{\text{HS}}^2}{2(1 + C_p \eta_p)} + c_p \|R\|_{\text{HS}}^4.$$

Indeed let E be the first-group sign vector in the empirical site-type law, and let $b_j(E)$ be the conditional bias of the j th sign in the second group. Then

$$R_{ij} = \mathbb{E}[E_i b_j(E)], \quad \sum_j \mathbb{E} b_j(E)^2 \geq \frac{\|R\|_{\text{HS}}^2}{1 + C_p \eta_p},$$

by projection onto the first-group coordinates and their Gram bound. Conditional relative entropy dominates the sum of marginal conditional entropies. The binary entropy inequality $I(b) \geq b^2/2 + cb^4$, followed by Jensen and the fixed number of coordinates, gives the displayed quadratic and quartic costs. On the other hand, diagonalizing the full Gram and applying the semicircle angular bound successively shows that the angular energy gain above the independent baseline is at most $n/4$ times the squared Hilbert–Schmidt norm of the full off-diagonal Gram, plus $o(n)$. The cross block R occurs twice in that norm, so the angular gain is bounded above by

$$\frac{n}{2} \|R\|_{\text{HS}}^2 + O_p(n \eta_p^2) + o(n).$$

The leading quadratic halves cancel against the entropy cost. The remaining positive terms are $C_p n \eta_p \|R\|_{\text{HS}}^2 + O_p(n \eta_p^2)$, whereas the quartic term is $-c_p n \|R\|_{\text{HS}}^4$. Thus cross overlaps outside any sufficiently small fixed neighborhood of zero have a strict negative rate after choosing η_p sufficiently small. Within that neighborhood the small-Gram comparison and the spherical length-deficit tails give $e^{o(n)}$. This proves the second-moment bound. Cauchy–Schwarz applied to each $T_{B'}$ removes an exponentially exceptional orientation set at cost $e^{o(n)} e^{-cn/2}$. It may therefore be removed before the signed sum, even though that sum need not be nonnegative on the exceptional set.

The same estimates restrict each term’s Haar integral, to arbitrarily high inverse-polynomial accuracy, to

$$|Q_{ij}| \leq (L/n) \log^{C_p} n \quad (i \neq j). \quad (12.9)$$

For internal sign overlaps, use the small-Gram comparison until this window and choose its logarithmic power large enough that the length-deficit decay absorbs all losses. Once those overlaps are in the window, all overlaps involving spherical priors have the same tails: conditional on the

internal sign Gram, the rotated prior is exactly the spherical prior conditioned on that Gram. Polynomially fine boxes have controlled angular variation on these nonsingular internal Grams. The central cutoffs may be smooth. The large-overlap truncation has therefore disappeared before the alternating cancellation below.

The finite-label expansion. The overlap window is now common to all terms. The only remaining prior ratio is the internal sign Gram mass, in lattice density units, divided by the spherical internal Gram density on the label set B' . We expand this ratio with every normalization factor included. Each coefficient must have an explicit finite set of replica labels: the alternating sum cancels exactly those terms that omit a label.

For $B' \subseteq [p]$, put $v = |B'|$, $d_v = \binom{v}{2}$, and let $Q_{B'}$ be the Gram with diagonal one. Zero-dimensional densities and empty products are one. Let $\rho_{B'}$ be the preceding density ratio. To every fixed expansion order it has the form

$$\rho_{B'}(Q, n) = \exp \left\{ n \sum_{b \geq 4} F_{B', b}(Q) \right\} \sum_{b, j \geq 0} n^{-j/2} A_{B', b, j}(Q), \quad A_{B', 0, 0} = 1. \quad (12.10)$$

Here the displayed coefficients are homogeneous of degree b in the off-diagonal overlaps.

Every degree- b rate coefficient is a sum of terms involving at most $6b$ replica labels; every amplitude coefficient of degree b and order $n^{-j/2}$ is such a sum involving at most $6(b+j)$ labels. A term involving a label set S means a universal summand indexed by labels in S , including all internally summed indices, whose value is independent of whether further labels belong to B' . This qualification includes the normalized central density prefactors, rather than merely the visible overlap monomial. On (12.9), the expansion and its exponential can be truncated with uniform relative error $O(n^{-A})$, for any prescribed fixed A .

We prove these assertions, starting with the normalizations. For one sign site let $Y = (\varepsilon_a \varepsilon_b)_{a < b, a, b \in B'}$. If $\delta(S)$ denotes the cut vector of a subset of labels, then $Y = \mathbf{1} - 2\delta(S)$. Its difference lattice is $\Lambda_v = \text{span}_{\mathbb{Z}}\{Y - \mathbf{1}\}$. Distinguish one label, denoted $i_\bullet$ in this lattice calculation. A basis consists of

$$\{2\delta(\{i\}) : i \neq i_\bullet\} \cup \{4e_{ij} : i < j, i, j \neq i_\bullet\}.$$

Indeed

$$\delta(S) = \sum_{i \in S} \delta(\{i\}) - 2 \sum_{\{i, j\} \subseteq S} e_{ij}, \quad \delta(\{i\}) + \delta(\{j\}) - \delta(\{i, j\}) = 2e_{ij}.$$

Ordering the edges incident to $i_\bullet$ first makes the basis block triangular, so its covolume is

$$c_v = 2^{v-1} 4^{\binom{v-1}{2}} = 2^{(v-1)^2} \quad (v \geq 1), \quad c_0 = 1.$$

The sample overlap belongs to $\mathbf{1} + n^{-1}\Lambda_v$, whose cell volume is $c_v n^{-d_v}$.

Thus the sign density in lattice units is

$$f_{n, B'}^S(Q) = \frac{n^{d_v}}{c_v} \Pr \left\{ n^{-1} \sum_{i=1}^n Y_i = (Q_{ab})_{a < b} \right\}.$$

The reciprocal lattice, using the 2π convention, is

$$\Lambda_v^\vee = \frac{\pi}{2} \left\{ z \in \mathbb{Z}^{d_v} : \sum_{b \neq a} z_{ab} \equiv 0 \pmod{2} \text{ for every } a \right\}.$$

Equality in the triangle inequality shows that the characteristic function of Y has modulus one precisely on this reciprocal lattice. The same is true under every sufficiently small real exponential tilt, whose support is unchanged. There is consequently a single central saddle on the reciprocal torus, and uniform contraction away from it. Boundary representatives of that one saddle are identified on the torus.

For the spherical density, use independent Gaussian coordinates and the one-site statistics

$$\left((g_a g_b)_{a < b}, ((g_a^2 - 1)/\sqrt{2})_{a \in B'} \right).$$

Their covariance is the identity in dimension $d_v + v$; the sign statistics also have identity covariance in dimension d_v . Denote the two log transforms by Ψ_G, Ψ_S , and their Legendre transforms by I_G, I_S . At target $(Q_{ab}, 0)$, $I_G = -\frac{1}{2} \log \det Q_{B'}$. If f^G is the full Gaussian statistic density, conditioning on the zero normalized-length targets gives exactly

$$f_{n,B'}^O(Q) = f_{n,B'}^G(Q, 0) / f_{n,B'}^{\text{len}}(0).$$

There is no extra Jacobian because both densities use the same normalized length coordinates.

With $z_n = n/2$, their exact length-only factor is

$$f_{n,B'}^{\text{len}}(0) = \left(\frac{n}{2\pi} \right)^{v/2} b_n^v, \quad b_n = \frac{\sqrt{2\pi} z_n^{z_n-1/2} e^{-z_n}}{\Gamma(z_n)}, \quad \log b_n = -\frac{1}{6n} + O(n^{-3}).$$

The saddle densities have forms

$$\begin{aligned} f_{n,B'}^S(Q) &= \left(\frac{n}{2\pi} \right)^{d_v/2} e^{-n I_S(Q)} \mathcal{A}_{B'}^S(Q, n), \\ f_{n,B'}^G(Q, 0) &= \left(\frac{n}{2\pi} \right)^{(d_v+v)/2} e^{-n I_G(Q, 0)} \mathcal{A}_{B'}^G(Q, n). \end{aligned}$$

Their amplitudes at zero target and leading order equal one.

Hence

$$\rho_{B'} = e^{n(I_G - I_S)} \mathcal{A}_{B'}^S b_n^v / \mathcal{A}_{B'}^G.$$

Every power of $n/(2\pi)$, and the entire lattice covolume, have cancelled before expanding. As a check, the exact spherical Gram density is

$$\frac{\Gamma(n/2)^v}{\pi^{d_v/2} \prod_{j=0}^{v-1} \Gamma((n-j)/2)} (\det Q_{B'})^{(n-v-1)/2},$$

whose central factor has the same leading value $(n/(2\pi))^{d_v/2}$.

The sign transform through degree three is

$$\Psi_S(t) = \frac{1}{2} \sum_{a < b} t_{ab}^2 + \sum_{a < b < c} t_{ab} t_{ac} t_{bc} + O(|t|^4).$$

Indeed the only nonzero third products are triangles, each with value one. Its Legendre transform therefore has quadratic term $\frac{1}{2} \sum Q_{ab}^2$ and cubic term $-\sum Q_{ab} Q_{ac} Q_{bc}$. These agree with the expansion of $-\frac{1}{2} \log \det Q_{B'}$. Thus the rate difference starts in degree four, as in (12.10).

Here is the label count for all subsequent terms. Each statistic coordinate carries at most two labels, including the single-label Gaussian length coordinates. A zero-tilt cumulant tensor with M coordinate slots therefore involves at most $2M$ labels, and its value is unaffected by additional

replicas. Formal inversion of $\nabla\Psi(t) = Q$ produces rooted trees with nonlinear vertex valences $m \geq 3$. A degree- b Legendre coefficient satisfies $\sum(m-2) = b-2$. Since $m \leq 3(m-2)$, it uses at most $3(b-2)$ cumulant slots, hence at most $6b$ labels. Setting the Gaussian diagonal targets to zero preserves this bound.

For the central amplitude, perform saddle Fourier inversion on the $n^{-1/2}$ frequency scale against an identity-covariance Gaussian. This expands the determinant instead of leaving an implicit factor depending on dimension. A higher Fourier vertex of degree $m \geq 3$ carries weight $n^{-(m-2)/2}$; at order $n^{-j/2}$, their total Fourier-slot count is at most $3j$. A vertex from the perturbed Hessian has two Fourier slots and at least one tilt slot. If the initial Taylor expansion has T tilt slots, all its slots therefore number at most $3j + 3T$.

Substitute the rooted-tree expansion for the saddle, with a total of b target leaves. This adds at most $3(b-T)$ slots. There are thus at most $3(j+b)$ slots, or $6(j+b)$ labels. Gaussian pairings identify slots and introduce no further labels; unused coordinates integrate to one. This counts the determinant, inverse-Hessian contractions, and every higher correction. Odd j vanish by Fourier parity, although allowing them in the bookkeeping is harmless. The factor b_n^v obeys the same rule by multiplying its single-label expansions. Products and reciprocals of series with leading constant one preserve the bound, because the weights $b+j$ add. This proves the asserted support rule even for coefficients that depend on $|B'|$ after their label sums are evaluated.

For fixed p , the transforms are analytic near zero with uniformly positive Hessian there. The saddles on (12.9) lie in this neighborhood. Taylor expansion under Gaussian damping on logarithmic frequency balls gives any required inverse-polynomial relative error. The sign Fourier tails are negligible by the reciprocal-lattice contraction just proved; the Gaussian quadratic determinant formula controls its remaining Fourier tails. This proves the uniform saddle remainders. Moreover $nQ^4 = O(n^{-1/3} \log^{C_p} n) = o(1)$ on the window, so the exponential in (12.10) can also be expanded to arbitrary inverse-polynomial precision.

A common integration measure and high-order quadrature. Fix the orbit and let μ_o be its normalized spherical Gibbs law, equivalently the Gaussian law below conditioned on length. Let $K(x_1, \dots, x_p)$ be either 1 or the product of squares displayed above, and write $\mathcal{U}(x) = (x_a^T x_b / L)_{a < b}$. Define the nonnegative common kernel density G_p by

$$\int_A G_p(U) dU = \mathbb{E}_{\mu_o^{\otimes p}}[K(x_1, \dots, x_p) \mathbf{1}_{\{\mathcal{U}(x) \in A\}}]$$

for Borel A in the interior Gram region. This definition includes the common normalization $Z_o(h)^{-p}$. We now bound the density and its derivatives.

Use p independent Gaussian representatives $x_a = m_o + D^{1/2} \eta_a$, with $D = (\lambda - J)^{-1}$, and impose all their lengths exactly. Fourier-invert the length statistics $(\|x_a\|^2 - n)/L$ and overlaps $U_{ab} = x_a^T x_b / L$. For a symmetric Fourier matrix T , the tilted white covariance is $(I - 2iT \otimes D/L)^{-1}$, up to fixed conventions on off-diagonal coefficients, and has norm at most one. The tilted mean is bounded by $C_p \|T\| \|D^{1/2} m_o\| / L$.

The polylogarithmic edge inputs give

$$\|D/L\| + \text{Tr}((D/L)^2) + \|D^{1/2} m_o\| / L \leq \log^C n,$$

with logarithmically bounded reciprocals of the exact length densities in these units. The spectral spread supplies arbitrarily many covariance eigenvalues of inverse-polylogarithmic size. As a nonzero T has an eigenvalue at least $\|T\|/\sqrt{p}$, the quadratic determinant is integrable against every fixed frequency power with only a logarithmic loss; the noncentral characteristic factor has modulus at most one.

The matrix-moment kernel is a product of squares of $\eta_a^T(D/L)\eta_b$ with $a \neq b$. Its Wick expansion under this complex tilt also costs only powers of $\log n$ and T . A contraction loop of length at least two is bounded by powers of $\|D/L\|$ times $\text{Tr}((D/L)^2)$. A loop of length one uses an off-diagonal replica block of the tilted covariance, whose identity term vanishes; the resolvent identity supplies an additional D/L , giving the same bound with a frequency factor.

Open contractions end in the already bounded means. Differentiation of the resulting density adds only frequency powers. Consequently $G_p(U)$ has all fixed derivatives bounded by powers of $\log n$; on the common smooth window their L^1 norms have the same bounds. This holds for both the mass and matrix kernels.

The density and all required derivatives are now controlled. This is what allows high-order quadrature on each subset's actual measure. For the subset B' , only its internal coordinates initially lie on a lattice. In U units that lattice has covolume c_v/L^{d_v} and a fixed-shape basis of size $O_p(L^{-1})$. Its mixed lattice–continuous integral is exactly the quadrature of $\rho_{B'}((L/n)U, n)G_p(U)$, with that cell-volume factor. For every finite smooth expansion term, repeated quadrature after the fixed basis change, or Poisson summation, gives error $O_{p,A}(L^{-A} \log^C n)$, uniformly over affine lattice shifts.

The relative expansion remainder also has a uniform absolute bound on the common window. Indeed the rate transforms agree through degree three, and their analytic amplitudes are one at zero, so for fixed p

$$\rho_{B'}(Q, n) = \exp\{O_p(n \max_{a < b} |Q_{ab}|^4)\}[1 + O_p(\max_{a < b} |Q_{ab}| + n^{-1})] = 1 + o(1).$$

Here $\max |Q_{ab}| = O(n^{-1/3} \log^{C_p} n)$, so the exponent tends to zero uniformly over all subsets B' . Thus a relative remainder $O_{p,A}(n^{-A})$ is also an absolute remainder of that order. It requires no derivatives, but is still evaluated on its own mixed lattice–continuous measure. Apply the same high-order quadrature to the smooth windowed G_p itself. Its derivative bounds show that its mixed sum is at most $\log^C n + O_{p,A}(L^{-A} \log^C n)$, uniformly in the affine shift, because its continuous mass has that bound. Multiplying this mixed-sum bound by the uniform absolute remainder controls the remainder on its actual measure. We may therefore convert all subset terms to one continuous integral, with arbitrarily high inverse-polynomial error and with all covolumes included. The different subset lattices need not have matching supports before conversion.

Cancellation and the probability bound. Quadrature has put every subset term on one continuous integration measure. We can now take the alternating sum without identifying the original lattice supports. A surviving term must use every replica label, and the coefficient-count rule turns that requirement into a power of the dimension that grows with the moment order. A term in the fully expanded expression has rate factors of degrees $b_1, \dots, b_t \geq 4$, and an amplitude of degree b and order $n^{-j/2}$. Put $W = b_1 + \dots + b_t + b + j$.

Its coefficient involves at most $6W$ labels. On the window,

$$nQ^{b_i} = O(n^{-b_i/12} \log^{C_p} n), \quad n^{-j/2}Q^b = O(n^{-b/3-j/2} \log^{C_p} n),$$

where the first estimate uses $b_i/3 - 1 \geq b_i/12$ for $b_i \geq 4$. The whole term thus costs at least $n^{-W/12}$, up to logarithms. A diagram whose label set S is a proper subset of $[p]$ cancels exactly, since

$$\sum_{B': S \subseteq B' \subseteq [p]} (-1)^{p-|B'|} = 0.$$

Every surviving term has $6W \geq p$, and is bounded by $n^{-p/72}$ before logarithmic losses.

Choose the expansion order sufficiently large in terms of p and the desired error power. The logarithmic kernel bounds and integration, followed by an enlargement of the threshold in n , give the safe absolute bound n^{-cp} , for example with $c = 1/200$.

We now return from the expansion to probability. The calculation bounds the Haar expectation of the truncated alternating power. The separate second-moment bounds remove the exponentially exceptional orientations even though the truncated sum may be signed there. On their complement it agrees with the nonnegative quantity $\|\Delta\|_{\text{HS}}^p$ up to negligible error. For the mass choose threshold $n^{-c/2}$; Markov then gives Haar failure at most $n^{-cp/2}$. For the matrix choose threshold one in Hilbert–Schmidt norm; its Haar failure is at most n^{-cp} . The planted likelihood $R = Z(h)/Z_o(h)$ has bounded second moment by the two-replica comparison and pair tails. Cauchy–Schwarz therefore leaves planted failure exponents at least $cp/4$ and $cp/2$, respectively. A sufficiently large fixed even p gives any requested B , with room for the earlier width exclusions.

Let M_{cube} and M_{sphere} be the normalized raw second moments about m_o . On the retained event,

$$R = 1 + O(n^{-c/2}), \quad RM_{\text{cube}} - M_{\text{sphere}} = L\Delta_{\text{matrix}}, \quad \|\Delta_{\text{matrix}}\| \leq \|\Delta_{\text{matrix}}\|_{\text{HS}} \leq 1.$$

The spherical bound is $\|M_{\text{sphere}}\| \leq CL\ell^C$. Thus $\|M_{\text{cube}}\| \leq CL\ell^C$, and $\Sigma_s \preceq M_{\text{cube}}$. This proves (12.8). $\square$

13 Positive-time covariance and macroscopic entropy inputs

We establish the fixed-positive-time estimates used to start the backward induction, and the estimates for changes of mean needed at macroscopic entropy. The matrix in a heat-bath conditional mean is $J^\circ = J - \text{diag}(J_{11}, \dots, J_{nn})$. Thus, in this section, put

$$\begin{aligned} m(y) &= \tanh y, \quad q(y) = n^{-1}\|m(y)\|^2, \quad b(y) = 1 - q(y), \\ V_y &= \text{diag}(1 - m_i(y)^2), \quad F_h(y) = y - h + (b(y)I - J^\circ)m(y). \end{aligned}$$

For a nonnegative diagonal matrix A define

$$\mathcal{H}(y, A) = I + A^{1/2} \left(b(y)I - J^\circ - \frac{2m(y)m(y)^\top}{n} \right) A^{1/2}.$$

The relevant stability property, with fixed positive constants, is

$$\left. \begin{aligned} \|F_h(y)\| &\leq \rho_0\sqrt{n}, & 0 \leq A \leq (1 + \eta_0)I, \\ n^{-1/2}\|A - V_y\|_{\text{F}} &\leq \epsilon_A \end{aligned} \right\} \implies \mathcal{H}(y, A) \succeq c_0 I. \quad (13.1)$$

Only diagonal A are quantified in this implication.

The companion high-temperature article [28] proves the requisite deterministic residual identities and posterior consequences. Its probabilistic matrix and stable-field statements are stated for $j = \beta^2 < 1$; they do not directly supply (13.1) at $j = 1$. We give the endpoint replacements below. In particular, none of the following arguments uses its spectral-gap theorem at the critical temperature.

13.1 Inverse words with a strict norm–trace margin

For a diagonal $A \geq 0$ write $\chi_A = n^{-1} \text{Tr } A$ and set

$$S_A(z; M) = I + z^2 \chi_A A - z A^{1/2} M A^{1/2}, \quad K_A(z; M) = (I - z A M + z^2 \chi_A A)^{-1}.$$

Fix $a_+ < \infty$ and $\kappa \in (0, 1)$, and define the closed parameter set

$$\mathcal{A}_\kappa = \{A \text{ diagonal} : 0 \leq A \leq a_+ I, \quad \|A\|_{\chi_A} \leq 1 - \kappa\}.$$

An ordinary word is a finite product of bounded diagonal matrices and copies of M . An inverse word has, in addition, at most one factor $K_A(1; M)$. Its diagonal prediction is obtained by noncrossing contractions $MDM \mapsto n^{-1} \text{Tr}(D)I$. For an inverse factor use its formal series in z before contracting; the prediction at $z = 1$ is well defined by the polynomial assertion in the next lemma. The associated normalized trace prediction is cyclic: rotating the factors of a word does not change $n^{-1} \text{Tr} P(Q)$, coefficient by coefficient in the formal variable z .

Lemma 13.1 (Restricted endpoint word estimates). *Let W be GOE with off-diagonal variance $1/n$ and diagonal variance $2/n$, and let W° be its zero-diagonal version. Fix a word length L , a bound M_0 on ordinary diagonal factors, and an inverse margin $c > 0$. Except on an event of probability $Ce^{-c_1 n}$, simultaneously for $M = W, W^\circ$ and all such words Q ,*

$$\|Q\| + \left(\sum_{i \neq j} |Q_{ij}|^4 \right)^{1/4} + \|\text{diag}(Q - P(Q))\|_2 \leq C.$$

For inverse words this assertion is restricted to $A \in \mathcal{A}_\kappa$ and $S_A(1; M) \succeq cI$. All diagonal parameters and permitted word choices within the fixed bounds may be selected after observing M ; the bounds L, M_0, c, a_+, κ are fixed beforehand. Constants depend on the displayed fixed bounds, including κ . The prediction for a word with r ordinary matrix letters outside its inverse is a polynomial in z of degree at most r .

For each deterministic $A \in \mathcal{A}_\kappa$ there is a constant $c_\kappa > 0$ such that $S_A(z; W) \succeq c_\kappa I$ for all $z \in [0, 1]$ except with probability $Ce^{-c_2 n}$. For any fixed finite family of deterministic bounded vectors, the bilinear deterministic equivalents are, to any fixed positive accuracy,

$$G_A \simeq A, \quad G_A W \simeq \chi_A A, \quad W G_A W \simeq \chi_A I + \chi_A^2 A, \quad G_A = A^{1/2} S_A(1; W)^{-1} A^{1/2},$$

with the same exponential form of exceptional probability. Here each comparison concerns only the specified bilinear forms.

Proof. We give the endpoint changes to Lemmas 3.1–3.4 of [28], including the uniformity needed below. The proof separates fixed parameters from adaptive ones. A strict norm–trace bound gives the deterministic-diagonal path margin and controls expectations. Chaining makes the word errors simultaneous over the diagonal parameters. An inverse chosen after observing the matrix is used only where its own stated positive margin holds. For $\|p\| = 1$ and $w = A^{1/2}p$, Gaussian comparison bounds the expected supremum of $zw^\top Ww - z^2 \chi_A \|w\|^2$ by that of

$$2zT_g v - z^2 \chi_A v^2, \quad 0 \leq v \leq \|A\|^{1/2}, \quad T_g = n^{-1/2} \|A^{1/2} g\|.$$

Indeed the increment difference between the comparison process $2n^{-1/2} \|w\| g^\top w$ and $w^\top Ww$ is

$$\frac{2}{n} \left[(\|w\|^2 - \|w'\|^2)^2 + 2(\|w\| \|w'\| - w^\top w')^2 \right] \geq 0.$$

Also $\mathbb{E} T_g^2 = \chi_A$ and $\text{Var}(T_g^2) \leq 2a_+^2/n$, so replacing T_g by $\sqrt{\chi_A}$ costs $o(1)$ uniformly in A .

The supremum then has integrand $2d - d^2$, where

$$0 \leq d = z\sqrt{\chi_A} v \leq \sqrt{\|A\|_{\chi_A}} \leq \sqrt{1 - \kappa}.$$

Its distance below one is at least $\delta_\kappa = (1 - \sqrt{1 - \kappa})^2$. The supremum is a_+ -Lipschitz in W for the Frobenius metric. Gaussian concentration, the GOE norm tail, and a fixed mesh in z give the asserted path margin, for example with $c_\kappa = \delta_\kappa/4$.

To prove expectations without conditioning on a matrix event, project W onto a fixed operator-norm ball and replace the symmetric inverse by a smooth bounded functional-calculus function agreeing with x^{-1} on an interval containing all spectra with either required margin. Use

$$K_A = I + A^{1/2} S_A^{-1} A^{1/2} (zW - z^2 \chi_A I), \quad K_A A = A^{1/2} S_A^{-1} A^{1/2}.$$

This defines bounded truncated words. For fixed length their Frobenius Lipschitz constants in W are bounded, and the increment between parameter tuples at sup-norm distance Δ has Frobenius Lipschitz constant at most $C\sqrt{\Delta}$.

These follow by the product rule, $\|A^{1/2} - B^{1/2}\| \leq \|A - B\|^{1/2}$, and the Fourier formula for differentiating a smooth matrix function. Thus entry variances are $O(n^{-1})$, normalized-trace variances are $O(n^{-2})$, and fourth centered entry moments are $O(n^{-2})$. These bounds gain respectively Δ, Δ, Δ^2 for increments.

The fixed-parameter agreement event has exponentially small complement. Gaussian integration by parts therefore gives, with errors $O(n^{-1})$ uniformly in i, z, A , the same loop equations as in Lemma 3.3 of the companion article. Writing $a_i = A_{ii}$, $k_i = \mathbb{E}(K_A)_{ii}$, $r = n^{-1} \sum_i a_i k_i$, and $h = r - \chi_A$, these equations are

$$\mathbb{E}(WK_A)_{ii} = zr k_i + O(n^{-1}), \quad k_i = 1 + z^2 a_i h k_i + O(n^{-1}).$$

All expectations here may use the bounded truncations. Their integration-by-parts error is exponentially small times a fixed polynomial in n , because both true and formal differentials are bounded and agree on the path event.

If $|h| \leq \delta$, the second equation gives $k_i = 1 + O(\delta + n^{-1})$. Consequently

$$\left| z^2 n^{-1} \sum_i a_i^2 k_i \right| \leq (1 + C\delta + C/n) n^{-1} \sum_i a_i^2 \leq (1 + C\delta + C/n)(1 - \kappa) \leq 1 - \kappa/2$$

for fixed sufficiently small δ . Averaging the loop equation with weights a_i gives $|h| \leq C/n$. Continuity from $h(0) = 0$ prevents an exit from $|h| \leq \delta$. It follows that $k_i = 1 + O(n^{-1})$ and $\mathbb{E}(WK_A)_{ii} = z\chi_A + O(n^{-1})$.

For completeness, the higher-word recursion has no new stability requirement. Mark an ordinary matrix letter, integrate it by parts, and let its differentiated partner split the word as $U \boxed{W} V \boxed{W} Q$. The leading term is

$$\mathbb{E}(UQ)_{ii} n^{-1} \mathbb{E} \text{Tr } V;$$

the reversed contraction is $n^{-1} \mathbb{E}(UV^\top Q)_{ii}$ and is $O(n^{-1})$. Covariance errors are $O(n^{-1})$ by the preceding variance bounds. If the derivative meets the inverse, use $dK_A[E] = zK_A A E K_A$.

Splitting then puts one copy of K_A in each smaller word, whose total number of ordinary letters is reduced by one. Otherwise it is reduced by two. The base formal prediction is $P(K_A) = I$: for each nonempty noncrossing-paired chain, summing over subsets of its originally adjacent pairs gives $\sum_T (-1)^{|T|} = 0$. Induction proves the polynomial assertion and entrywise expectation error $O(n^{-1})$. Every inverse in this calculation has the original coefficient A ; no enlarged diagonal parameter set is used. For a normalized trace, the noncrossing face formula has no marked starting position on the trace cycle. Rotating the word permutes its faces and contractions, proving the stated cyclic invariance coefficient by coefficient, also after the formal inverse expansion.

To make the result simultaneous, take sup-norm meshes whose points lie in $\mathcal{A}_\kappa$ and in the bounded boxes for the other diagonal parameters. At scale 2^{-l} their cardinalities are at most $\exp(C_L(l+1)n)$. The truncated estimates hold in the ambient box, so mesh increments are legitimate even though no convexity of $\mathcal{A}_\kappa$ is assumed. The diagonal ℓ^2 and off-diagonal ℓ^4 seminorms of centered increments have expectations at most $C2^{-l/2}$ and concentration scale $C\sqrt{2^{-l}/n}$.

A union bound bounds every increment by $C\sqrt{l+1}2^{-l/2}$, with summable failure probabilities $Ce^{-c(l+1)n}$. Sum the increments and the fixed-parameter biases. Sign conjugation makes all off-diagonal expectations zero. This proves the asserted simultaneous estimates for W . Since $\|\text{diag } W\|_F$ is bounded except with exponentially small probability, the truncated Frobenius Lipschitz estimate proves the result for W° as well. Restriction to the stated positive margin restores the exact inverse factors.

Finally sign conjugation and the loop equations give the diagonal expectations of G_A , $G_A W$ and $W G_A W$. They are respectively A , $\chi_A A$ and $\chi_A I + \chi_A^2 A$, to $O(n^{-1})$ in each entry. Their truncated bilinear forms are uniformly Lipschitz. Gaussian concentration proves the last assertion for each fixed finite list of vectors. $\square$

Lemma 13.2 (Restricted determinant bound). *Fix $\kappa, \varepsilon > 0$, a rank bound r_0 , and a norm bound M_2 . There is $\gamma_0 > 0$ such that for each fixed $0 < \gamma \leq \gamma_0$ and each fixed $B > 0$, with probability at least $1 - e^{-Bn}$ for sufficiently large n , simultaneously for*

$$V \text{ diagonal}, \quad 0 \leq V \leq I, \quad b = n^{-1} \text{Tr } V \leq 1 - \kappa, \quad \text{rank } E \leq r_0, \quad \|E\| \leq M_2,$$

the matrix $L = I + (bI - W)V + E$ satisfies

$$n^{-1} \log \det(L^\top L + \gamma I) \leq b^2 + \varepsilon.$$

The choices of γ_0 and the threshold on n may depend on the fixed margins. The threshold on n may also depend on B .

Proof. For deterministic V use Lemma 13.1 along $L_z = I + z^2 b V - z W V$. On its path event,

$$\frac{d}{dz} \frac{1}{n} \log \det L_z = 2zb \frac{\text{Tr}(K_V(z)V)}{n} - \frac{\text{Tr}(W K_V(z)V)}{n}.$$

The expectation is $zb^2 + O(n^{-1})$, by the loop calculation. Using a fixed norm cutoff in the path event $\mathcal{E}_V$, integration from zero to one gives $\mathbb{E}[\mathbf{1}_{\mathcal{E}_V} 2n^{-1} \log \det L_1] = b^2 + O(n^{-1})$. The inverse norm there is bounded; regularization costs $C_\kappa \gamma$. Off this event, the regularized logarithm is bounded in absolute value by $|\log \gamma| + C + 2 \log(1 + \|W\|)$ and its expected contribution tends exponentially to zero. Hence the regularized mean is at most $b^2 + C_\kappa \gamma + o(1)$ uniformly in V .

The map $M \mapsto n^{-1} \log \det(M^\top M + \gamma I)$ has Frobenius Lipschitz constant $1/\sqrt{n\gamma}$. Gaussian concentration thus gives a tail $2 \exp(-\gamma n^2 u^2/4)$ at deviation u . Choose γ_0 so $C_\kappa \gamma_0 < \varepsilon/8$. A fixed sup-norm mesh of the restricted diagonal set has size $\exp(O(n))$. On $\|W\| \leq R$, a mesh spacing depending on R, γ, ε controls both the logarithm and b^2 to $\varepsilon/4$. For this last mesh step choose a possibly larger R so that its exceptional probability is at most $e^{-(B+1)n}$.

This choice does not change the preceding expectation estimate or its regularizer threshold. The n^2 concentration bound pays for this mesh for sufficiently large n . Finally a rank- r_0 perturbation changes the logarithm by at most $M_2 \sqrt{r_0/(n\gamma)}$. This proves the statement, including its arbitrarily large fixed exponential budget. The exponentially small path exception was used only in the mean estimate; its exponent does not limit this final budget. $\square$

The diagonality restriction is essential for this simultaneous statement. If arbitrary positive contractions chosen after W were allowed, take $V = \mathbf{1}_{\{W \leq 0\}}$ and $E = 0$. Then $b \rightarrow 1/2$, so the constraint holds, for example, at $\kappa = 1/4$, while every eigenvalue of L on the range of V is at least $1 + b$, and the other eigenvalues equal one. For every $\gamma > 0$,

$$\frac{1}{n} \log \det(L^T L + \gamma I) \geq 2b \log(1 + b) \longrightarrow \log(3/2) > 1/4.$$

Thus the asserted bound with arbitrarily small ε would fail in that larger class. The application below uses only the diagonal matrix V_y .

13.2 Scalar equality and stable fields at positive times

Lemma 13.3 (Endpoint scalar rate). *Let P have finite second moment, put $q = \mathbb{E}_P \tanh^2 y$, $b = 1 - q$, and $a = \mathbb{E}_P[(y - d) \tanh y]$. For $\nu > 0$ with $\nu \geq q$,*

$$D(P \| N(d, \nu)) \geq \frac{(a - \nu b)^2}{2\nu(\nu + q)}.$$

Equality holds exactly when $P = N(d, \nu)$. At every observation time $s > 0$ the consistent equality law, with $d = s + \mathbb{E}_P \tanh y$ and $\nu = s + q$, is uniquely $N(v_(s), v_*(s))$, where*

$$v_*(s) = s + \mathbb{E}_{N(v_*(s), v_*(s))} \tanh y.$$

On compact positive time intervals this curve is continuous and its q is bounded away from zero. The scalar integral and rate conclusions of Lemma 5.4(i), (ii), and (iv) of [28] therefore hold at $j = 1$ on such intervals. In particular, outside any fixed neighborhood of the equality curve, the smoothed scalar integral has strictly negative rate for sufficiently small smoothing variance.

Proof. Suppose the entropy is finite; then $0 < q < 1$. Set $k = q/\nu \in (0, 1]$ and $C = a/\nu$. The change of variable $T_u(y) = y - u \tanh y$, $u < 1$, gives

$$D(P \| N(d, \nu)) \geq u(C - b) - \frac{ku^2}{2} - b[-\log(1 - u) - u].$$

This identity is the entropy change-of-variables calculation in Lemma 4.1 of the companion article and uses no strict restriction on j . Put $h(u) = u^2/2 - b[-\log(1 - u) - u]$. Since $b = 1 - q < 1$,

$$h(u) > 0 \quad (u \leq q, u \neq 0), \quad h(q) \geq q^3/6.$$

For negative u use $-\log(1 - u) - u \leq u^2/2$; for positive $u \leq q$ use the increasing series ratio and $h(q) = \sum_{l \geq 3} q^l/[l(l - 1)]$.

If $(C - b)/(1 + k) \leq q$, substitution proves strict inequality unless $C = b$. Otherwise let $C_0 = 1 + kq$ and $C_1 = C_0 + \sqrt{(1 + k)/3} q^{3/2}$. The choice $u = q$ proves strictness for $C_0 < C < C_1$. For $C \geq C_1$, variance comparison and Cauchy–Schwarz give

$$2D(P \| N(d, \nu)) \geq C^2/k - 1 - \log(C^2/k).$$

Subtract the target and write $f(C) = C^2/k - 1 - \log(C^2/k) - (C - b)^2/(1 + k)$. Direct differentiation gives

$$f(C_0) \geq -2q^3/3, \quad f'(C_0) \geq \frac{2q}{1 + q}, \quad f''(C) \geq 1.$$

Consequently

$$f(C_1) \geq q^3 \left[-\frac{2}{3} + \frac{2}{\sqrt{3}(1+q)\sqrt{q}} + \frac{1}{6} \right] > 0.$$

Since f' is positive and increasing after C_0 , this finishes the strict cases. If $C = b$, the target is zero and equality is equivalent to vanishing entropy. Gaussian integration by parts verifies equality for the stated Gaussian law.

For a fixed variance $\nu > 0$, the function $d \mapsto d - \mathbb{E}_{N(d,\nu)} \tanh y$ is strictly increasing, onto $\mathbb{R}$, and zero at $d = 0$; its derivative is $\mathbb{E} \tanh^2 y > 0$. Thus its solution at $s \geq 0$ is nonnegative. Let $H_\nu(d) = \mathbb{E}_{N(d,\nu)}(\tanh y - \tanh^2 y)$. Gaussian reflection gives $H_\nu(\nu) = 0$ and $\mathbb{E}_{N(d,\nu)}[\tanh y \operatorname{sech}^2 y] \geq 0$ for $d \geq 0$. Hence $|H'_\nu(d)| < 1$ for every finite $d \geq 0$: the upper bound is strict since $\mathbb{E} \operatorname{sech}^2 y < 1$, and the pointwise lower bound of $(1 - m^2)(1 - 2m)$ is strictly greater than -1 . The strict derivative bound is uniform on the compact segment joining d and ν . Consistency gives $d - \nu = H_\nu(d)$ and therefore forces $d = \nu$.

Finally $F(v) = \mathbb{E}_{N(v,v)} \tanh y$ satisfies $F(0) = 0$ and

$$F'(v) = \mathbb{E}_{N(v,v)}[(1 - m^2)(1 - m)] \in [0, 1) \quad (v > 0).$$

The upper bound uses the same reflection identity. Thus $v - F(v)$ is strictly increasing from zero to infinity, since $0 \leq F(v) \leq 1$. It has a unique inverse at every $s > 0$, with $s \leq v_*(s) \leq s + 1$. This proves continuity and the positive lower bound for q on each compact positive interval. At $s = 0$ there is no equality law with $q > 0$: its mean equation forces $d = 0$, while $\mathbb{E}_{N(0,q)} \tanh^2 y < q$.

We specify why the integral consequences extend as stated. In the smoothed integral the parameters are $d = s + M$, $\nu = s + \sigma^2 + q$, and the exponent is $\psi = (a - \nu b)^2 / [2\nu(\nu + q)]$. With $\varphi_{d,\nu}$ the density of $N(d, \nu)$, the scalar weight meant here is

$$G_{s,\sigma}(y) = \prod_{i=1}^n \varphi_{d,\nu}(y_i) \exp\{n\psi\}.$$

When $\nu \geq \nu_{\min} > 0$, Cauchy–Schwarz gives the uniform envelope $G_{s,\sigma}(y) \leq \exp(Cn - c \sum_i y_i^2)$. On a fixed empirical second-moment ball, M, q, a, d, ν, ψ are continuous for weak convergence, since only bounded or at most linearly growing functions are averaged. The entropy variational formula and a finite covering by fixed product-Gaussian neighborhoods give

$$\limsup_{n \rightarrow \infty} n^{-1} \log \int_{P_y \in \mathcal{C}} G_{s,\sigma}(y) dy \leq \sup_{P \in \mathcal{C}} \{\psi - D(P \| N(d, \nu))\}$$

for each compact parameter set $\mathcal{C}$; Gaussian density ratios on the second-moment ball tend uniformly to one at the exponential rate when their parameters converge.

The rate is nonpositive, and its zero set at $\sigma = 0$ is exactly the curve just proved. Upper semicontinuity therefore makes the rate strictly negative on closed sets outside its prescribed neighborhood. Compactness preserves this negativity when σ is sufficiently small. The exterior of the ball has any required negative rate by the envelope. For the tail-square assertion insert $\exp(\lambda \sum_i y_i^2 \mathbf{1}_{|y_i| > R})$ into each of the finite Gaussian comparisons, with fixed sufficiently small $\lambda > 0$.

Its additional logarithmic moment tends uniformly to zero as $R \rightarrow \infty$. Removing the tilt on a positive tail-square level produces strict negative rate. These are precisely the three stated integral conclusions, now with no subcritical assumption. $\square$

Proposition 13.4 (Positive-time field inputs). *Fix $0 < s_1 < T < \infty$. There are positive constants $\rho_0, \eta_0, \epsilon_A, c_0, c_*, a$, depending on this interval, such that, at every deterministic $s \in [s_1, T]$, the*

probability under the planted law, restricted to the original spectral event, of either failure of (13.1) or existence of y with

$$\|F_{h_s}(y)\| \leq \rho_0 \sqrt{n}, \quad q(y) < c_*$$

is at most e^{-an} for sufficiently large n . The same assertion holds uniformly on each fixed compact interval after decreasing the residual threshold.

There are also absolute $c, C > 0$ with the following small-time refinement. For each fixed $A_* > 0$, at all sufficiently small $r = \sqrt{s} > 0$, failure of (13.1) has probability at most $e^{-A_* nr^3}$; its margins may depend on the scale. For $u \geq Cr$ in a sufficiently small fixed range, existence of an approximate zero with $q(y) > u$ has probability at most e^{-cnu^3} . The approximate-zero tolerance is fixed before taking $n \rightarrow \infty$ and may depend on s, u, A_* . Uniformity is asserted only on compact positive bands; finitely many or dyadic values of u may be imposed together. In the small-time assertions a vanishing maximum diagonal may be included in the external disorder restriction.

Proof. The scalar calculation separates the overlap from zero and places the retained parameters near the positive-time equality curve. This gives a norm–trace margin for nearby diagonals. We next check the conditional finite-rank matrix correction and make that check simultaneous over the diagonal neighborhood.

First retain the diagonal and gauge the planted spin to $\mathbf{1}$. Then $J = W + \mathbf{1}\mathbf{1}^\top/n$ and $h_s = s\mathbf{1} + \sqrt{s}g$. Until the final diagonal-removal paragraph, use the temporary full-matrix residual and Hessian

$$\tilde{F}_h(y) = y - h + (b(y)I - J)m(y), \quad \tilde{\mathcal{H}}(y, A) = I + A^{1/2} \left(b(y)I - J - \frac{2m(y)m(y)^\top}{n} \right) A^{1/2}.$$

Let $E \subseteq \mathbb{R}^n \times \text{Sym}_n$ be Borel and independent of the observation h_s . For $\sigma > 0$ set

$$I_{s,\sigma}(E) = (2\pi\sigma^2)^{-n/2} \int_{\mathbb{R}^n} \mathbf{1}_{\{(y,J) \in E\}} \exp \left[-\frac{\|\tilde{F}_{h_s}(y)\|^2}{2\sigma^2} + \frac{nb(y)^2}{2} \right] dy.$$

The exact Gaussian reduction, Lemma 5.3 of [28], is valid for every positive j and here reads

$$\mathbb{E} I_{s,\sigma}(E) = \int \sqrt{\frac{\nu}{\nu+q}} G_{s,\sigma}(y) \mathbb{P}((y, J) \in E \mid Wm + \sqrt{s+\sigma^2}g' = y - d\mathbf{1} + bm) dy,$$

where $d = s + M$, $\nu = s + \sigma^2 + q$ and $G_{s,\sigma}$ is the explicit scalar weight in Lemma 13.3. Here g' is a fresh standard normal independent of W . The identity follows directly by inverting $\nu I + mm^\top/n$; the determinant factor is $\sqrt{\nu/(\nu+q)}$.

On $[s_1, T]$, discard a large second-moment ball's exterior, a fixed neighborhood of the equality curve's complement, and the scalar square-tail event. Lemma 13.3 gives strictly negative rates on these discarded parts and an arbitrarily small positive rate for the total integral. On the remaining parameters $q \geq q_0 > 0$. Choose η_0 and the diagonal Frobenius tolerance so small that

$$\|A\|_{\chi_A} \leq (1 + \eta_0)(1 - q_0 + \epsilon_A) \leq 1 - q_0/2.$$

The restricted fixed-parameter and bilinear estimates now apply. We verify their conditional use and its positive margin.

Set $u = m/\sqrt{nq}$, $Z_1 = (y - d\mathbf{1})/\sqrt{n\nu}$ and $P_u = I - uu^\top$. Under the displayed conditioning, using a fresh GOE W_0 , put $\ell_0 = u^\top W_0 u$ and

$$\begin{aligned} P_u W P_u &= P_u W_0 P_u, \\ P_u W u &= \bar{w} + \tau P_u W_0 u, \\ \bar{w} &= \sqrt{q/\nu} Z_1 - (a/\nu)u, \quad \tau^2 = (s + \sigma^2)/\nu, \\ u^\top W u &= \alpha_c = 2(a + bq)/(\nu + q) + \sqrt{(s + \sigma^2)/(\nu + q)} \ell_0. \end{aligned}$$

Writing out the three blocks gives the exact finite-rank difference

$$\begin{aligned} W - W_0 &= u\bar{w}^\top + \bar{w}u^\top - (1 - \tau)\{u(W_0 u)^\top + (W_0 u)u^\top\} \\ &\quad + \{\alpha_c + (1 - 2\tau)\ell_0\}uu^\top. \end{aligned}$$

The scalar ℓ_0 has variance $2/n$ and is smaller than any fixed tolerance outside an exponentially small event.

Write $S_A = S_A(1; W_0)$ and define $\mathcal{Y}(p) = S_A^{-1/2} A^{1/2} p$. Set

$$U = \mathcal{Y}(u), \quad T_1 = \mathcal{Y}(W_0 u), \quad \bar{W} = \mathcal{Y}(\bar{w}), \quad X_1 = \mathcal{Y}(\mathbf{1}/\sqrt{n}).$$

The exact whitened Hessian is

$$\begin{aligned} S_A^{-1/2} \tilde{\mathcal{H}}(y, A) S_A^{-1/2} &= I + (b - \chi_A) S_A^{-1/2} A S_A^{-1/2} - X_1 X_1^\top - 2q U U^\top \\ &\quad - [U \bar{W}^\top + \bar{W} U^\top - (1 - \tau)(U T_1^\top + T_1 U^\top) \\ &\quad + \{\alpha_c + (1 - 2\tau)\ell_0\} U U^\top]. \end{aligned}$$

For each fixed y, A , the vectors $\mathbf{1}/\sqrt{n}, u, Z_1$ are deterministic relative to W_0 . The bilinear tests in [Theorem 13.1](#) give

$$\begin{aligned} \mathcal{Y}(p)^\top \mathcal{Y}(r) &\simeq p^\top A r, \\ \mathcal{Y}(p)^\top T_1 &\simeq \chi_A p^\top A u, \\ \|T_1\|^2 &\simeq \chi_A + \chi_A^2 u^\top A u \end{aligned}$$

for this fixed finite vector list. In particular $\|X_1\|^2 \simeq \chi_A$, rather than b for an arbitrary A . These are comparisons of actual Gram entries, with a chosen fixed accuracy.

The comparison Gram for T_1 is represented by $\chi_A U + \sqrt{\chi_A} N$, where N is an abstract unit direction orthogonal to the other three images. This is a model of Gram entries, not a coupling with a new random vector. Set aside the small full-rank term $b - \chi_A$ and the scalar ℓ_0 . Eliminating the N direction, whose Hessian pivot is the identity, gives the ideal coefficient

$$\begin{aligned} \beta_U &= 2q - \frac{2a}{\nu} + \frac{2(a + bq)}{\nu + q} - 2(1 - \tau)\chi_A + (1 - \tau)^2 \chi_A \\ &= q \left(2 - \frac{\chi_A}{\nu} - \frac{2(a - \nu b)}{\nu(\nu + q)} \right) \end{aligned}$$

on $U U^\top$ in the form being subtracted. The final term in the first line is the Schur correction.

At the equality-law limit, $\sigma = 0$, $d = \nu = v = v_*(s)$, $a = vb$, and the limiting diagonal is $A = V_y$, so $\chi_A = b$. Only in this limit is $\beta_U = q(2 - b/\nu)$. Write $b = \mathbb{E} \text{sech}^2 y$ and $e = \mathbb{E} [\tanh^2 y \text{sech}^2 y]$ under $N(v, v)$. Use the rescaled triple

$$M_1 = \sqrt{q} \mathcal{Y}(u), \quad G_1 = \mathcal{Y}(Z_1)/\sqrt{v}.$$

The ideal remaining matrix is identity minus the form on (X_1, M_1, G_1) with coefficient matrix C and limiting equality-law Gram matrix H :

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 - b/v & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} b & e & -2e \\ e & e & b - 3e \\ -2e & b - 3e & b/v - 2b + 6e \end{pmatrix}.$$

The Gram matrix here is the limit of the empirical $A = V_y$ comparison data, not the actual finite-sample Gram. Its entries follow by Gaussian integration by parts; for example $\mathbb{E}[m(1 - m^2)] = e$, $v^{-1}\mathbb{E}[(y - v)m(1 - m^2)] = b - 3e$, and the double integration by parts gives $b/v - 2b + 6e$. Elementary column operations yield

$$\det(I - HC) = (q + 2e)(q + e)^2 > 0.$$

Subtracting the first vector alone is positive because its squared length is $b < 1$.

The remaining two-vector coefficient block has only one positive eigenvalue. Hence the final matrix can have at most one nonpositive direction; its positive determinant rules that out. The preceding orthogonal-direction Schur complement is therefore positive as well. Continuity on the compact equality curve gives a fixed positive margin.

The fixed-parameter calculation is complete. To make it simultaneous, we first quantify its continuity. The margin survives small errors in all Gram entries, coefficients, and $\chi_A - b$. To check uniformity over nearby A , each tested coordinate product is bounded by $C(1 + y_i^2)/n$. On $|y_i| \leq R_1$, its change under $A - V_y$ is at most $C_{R_1}\|A - V_y\|_F/\sqrt{n}$; the remaining part costs the prescribed square-tail tolerance. The V_y -weighted coordinate products are bounded continuous functions, so their equality-law limits follow from weak closeness.

This proves the fixed- A conditional estimate with a uniform positive exponential rate. The path margin $S_A \succeq c_\kappa I$ returns the whitened margin to $\tilde{\mathcal{H}}$, after the small $(b - \chi_A)$ term is absorbed. We pay for simultaneity only on the coordinates where a diagonal can move appreciably. On the bounded conditional-matrix norm event choose a fixed sup-norm mesh spacing δ so the $1/2$ -Hölder error of $\tilde{\mathcal{H}}(y, A)$ is smaller than one quarter of the margin. Only $\epsilon_A^2 n/\delta^2$ sites can differ from V_y by more than δ . Mesh those sites and leave the others at V_y . The logarithm of the number of approximants is at most

$$n[\mathfrak{h}(\alpha) + \alpha \log(C/\delta) + o(1)], \quad \alpha = \epsilon_A^2/\delta^2,$$

where $\mathfrak{h}(\alpha) \rightarrow 0$ as $\alpha \downarrow 0$.

Use twice the Frobenius radius in the fixed-parameter estimate and then decrease ϵ_A to pay this mesh. Thus the conditional matrix failure, integrated against G , has strictly negative rate. This explicitly extends the positive-time part of Lemma 5.5 of the companion article; its subcritical small- q argument is not used.

The estimates so far control a smoothed integral of bad configurations. To exclude a bad approximate zero, a witness must force that integral to be large enough. The deterministic local-volume estimate supplies this implication, with determinant, smoothing, and residual tolerances chosen in that order.

Let $\mathcal{D}_{\eta_0}$ be the compact set of diagonal $0 \leq A \leq (1 + \eta_0)I$. Let $\epsilon_* > 0$ be the full diagonal radius retained by the conditional calculation, and choose c_E below half its matrix margin. Use the relaxed Borel set

$$E^{\text{mat}} = \left\{ (y, J) : \exists A \in \mathcal{D}_{\eta_0}, \|A - V_y\|_F \leq \epsilon_* \sqrt{n}, \lambda_{\min}(\tilde{\mathcal{H}}(y, A)) \leq c_E \right\},$$

intersected with the external spectral and norm restriction. Compactness of the diagonal range and continuity make this set Borel and independent of h_s . The scalar exclusions and the conditional

matrix rate just proved give, for sufficiently small $\sigma > 0$, uniformly on the fixed positive band,

$$\mathbb{E}I_{s,\sigma}(E^{\text{mat}}) \leq e^{-a_0 n}$$

for some $a_0 > 0$.

The stricter witness event uses

$$\|\tilde{F}_{h_s}(y)\| \leq \rho\sqrt{n}, \quad \|A - V_y\|_F \leq \frac{1}{2}\epsilon_*\sqrt{n}, \quad \lambda_{\min}(\tilde{\mathcal{H}}(y, A)) \leq \frac{1}{2}c_E,$$

with $A \in \mathcal{D}_{\eta_0}$. It depends on the observation, unlike E^{mat} . Keeping this same A , for $d_y = \|y' - y\|/\sqrt{n}$ we have

$$\|V_{y'} - V_y\|_F \leq 2\|y' - y\|, \quad |b(y') - b(y)| \leq 2d_y, \quad \left\| \frac{m(y')m(y')^T - m(y)m(y)^T}{n} \right\| \leq 2d_y.$$

Thus the Hessian changes by at most $6(1 + \eta_0)d_y$. Every point of the ball $\|y' - y\| \leq r_{\text{loc}}\sqrt{n}$ lies in the same E^{mat} fiber if

$$r_{\text{loc}} \leq \min\{\epsilon_*/4, c_E/[12(1 + \eta_0)]\}.$$

We next obtain the overlap lower bound needed for the determinant estimate. A uniform lower bound on q at any witness is available before using the determinant estimate. On the external norm event,

$$\|\tanh h_s\| \leq \|m(y)\| + \|h_s - y\| \leq C\sqrt{nq(y)} + \|\tilde{F}_{h_s}(y)\|.$$

Independence of the planted observation coordinates and a fixed Bernoulli deviation bound give $\|\tanh h_s\| \geq c \min(\sqrt{s}, 1)\sqrt{n}$ outside a fixed exponentially small event, uniformly on compact time intervals. For instance, at small s , a fixed positive fraction of the standard Gaussian coordinates lie in $[1/2, 1]$, on which the hyperbolic tangent has magnitude at least $c\sqrt{s}$. On the fixed band let $v_- > 0$ be a common observation lower bound, and let the external matrix norm be at most K . Taking $\rho \leq v_-/2$ forces

$$q(y) \geq q_{\text{vol}} := \left(\frac{v_-}{2(K+2)} \right)^2 > 0.$$

At such a witness the Jacobian is exactly

$$\tilde{F}'_{h_s}(y) = I + (bI - W)V_y - \frac{\mathbf{1}\mathbf{1}^T}{n}V_y - \frac{2mm^T}{n}V_y.$$

The last two terms have rank at most two and norm at most three. Since $b \leq 1 - q_{\text{vol}}$, Lemma 13.2 with $\kappa = q_{\text{vol}}$ gives its regularized determinant upper bound with any needed fixed failure budget. The deterministic local-volume lemma, Lemma 5.6 of [28], has hypotheses $j, K < \infty$ and hence applies at $j = 1$ without modification.

With the regularized Jacobian $Q = \tilde{F}'_{h_s}(y)^T \tilde{F}'_{h_s}(y) + \gamma I$, its change of variable $y' = y + \sigma Q^{-1/2}g$ proves a lower bound $e^{-\varepsilon n}$ for this bad integral whenever the residual is sufficiently small and $2\sigma/\sqrt{\gamma} \leq r_{\text{loc}}$. Its losses per spin are bounded by

$$\frac{\varepsilon_1}{2} + C \left[\frac{\rho}{\sigma} + \frac{\sigma}{\gamma} + \left(\frac{\rho}{\sigma} \right)^2 + \left(\frac{\sigma}{\gamma} \right)^2 + \frac{\sigma}{\sqrt{\gamma}} \right] + o(1).$$

Choose the target loss $\varepsilon < a_0/2$, then the determinant tolerance ε_1 , then the regularizer γ , then σ , and finally the residual tolerance ρ .

Markov's inequality now bounds the witness probability by $e^{-(a_0-\varepsilon)n}$, together with the assigned determinant and observation exceptions. For low overlap, choose q_{rel} below the equality curve's minimum q , use the relaxed set $E^{\text{low}} = \{q \leq q_{\text{rel}}\}$ with the same external restriction, and take the witness threshold $q \leq q_{\text{rel}}/2$. Since $|q(y') - q(y)| \leq 2d_y$, a radius $r_{\text{loc}} \leq q_{\text{rel}}/4$ gives the same inclusion. The scalar negative rate then replaces the matrix rate in the same volume argument.

For the stronger small-time budgets use [Theorem 8.3](#), without the determinant weight. Its scalar integral estimates permit any preliminary fixed exclusion budget $A_{\text{pre}}nr^3$, restrict $q \asymp_{A_{\text{pre}}} r$, and give the outer rate cnu^3 for $u \geq Cr$. Choose $A_{\text{pre}} > A_*$ to reserve rate for smoothing and local volume. For an upper-tail witness $q > u$, use the relaxed set $E_u^{\text{up}} = \{q \geq u/2\}$ and radius $r_{\text{loc}} \leq u/4$. Taking $u \geq 2Lr$, with L from the scalar proposition, gives a preliminary rate proportional to $n(u/2)^3$.

We compare the actual smoothed scalar weight with the root weight. On a fixed band $s \in [s_-, s_+]$, $s_- > 0$, put $\nu_0 = s + q$, $\nu_\sigma = \nu_0 + \sigma^2$, and $\mathbf{v} = \langle (y - d)^2 \rangle$. For fixed y ,

$$\frac{1}{n} \log \frac{G_{s,\sigma}(y)}{G_{s,0}(y)} = -\frac{1}{2} \log \frac{\nu_\sigma}{\nu_0} + \frac{\sigma^2 \mathbf{v}}{2\nu_0 \nu_\sigma} + \psi(\nu_\sigma) - \psi(\nu_0), \quad \psi(\nu) = \frac{(a - \nu b)^2}{2\nu(\nu + q)}.$$

On $\langle y^2 \rangle \leq R^2$, all numerators are bounded and $\nu \geq s_-$, so the absolute value is at most $C_{R,s_-,s_+} \sigma^2$. The uniform Gaussian envelope from [Theorem 13.3](#) makes the exterior integral have any prescribed fixed negative rate when R is large. Choose R first, then σ so this logarithmic loss is smaller than one quarter of the reserved gap $(A_{\text{pre}} - A_*)r_-^3$, or than the reserved fixed multiple of u_-^3 for a finite outer-threshold family. Polynomial factors are absorbed for large n . This preserves every scalar rate used here for the smoothed integral.

If a diagnostic itself uses ν_σ , give its thresholds fixed slack first. On the same ball its scalar parameters change by $O_R(\sigma^2)$. On the retained scalar region, $q \geq cr_- > 0$, and the conditional column coefficients have the same bound. The stricter smoothed bad set is then contained in the relaxed unsmoothed one. This supplies the required set comparison as well as the weight comparison. For the matrix estimate the spectral indicator must remain attached to the actual conditioned matrix. Denote that matrix by $J_\sigma = W_\sigma + \mathbf{1}\mathbf{1}^T/n$; the subscript records the conditioning, not a replacement of the external disorder. With $u = m/\sqrt{nq}$, $E = u^\perp$, and $e = P_E \mathbf{1}/\sqrt{n}$, its exact regression gives

$$J_\sigma|_E = W_E + ee^T, \quad P_E J_\sigma u = \frac{q}{\nu_\sigma} \frac{P_E(y - d\mathbf{1})}{\sqrt{nq}} + \frac{M}{\sqrt{q}} e + \sqrt{\frac{s + \sigma^2}{\nu_\sigma}} w,$$

where the GOE block W_E and $w \sim N(0, I_E/n)$ are independent. Its scalar block is the regression displayed earlier. In decreasing eigenvalue order, compression and rank-one interlacing give, on $\mathcal{G}_{\text{spec}}(J_\sigma)$,

$$\lambda_j(J_\sigma) \geq \lambda_j(W_E) \geq \lambda_{j+2}(J_\sigma), \quad 1 \leq j \leq n-2.$$

Thus every cumulative count changes by at most two, and the top edge of W_E is bounded by that of this same full matrix.

Here is the precise spectral information used by the stability calculation. Put $\Delta_{\text{sp}} = q^{8/5}$ and restrict $R = (2 + O(q^2) - W_E)^{-1}$ to depths greater than Δ_{sp} . The inherited edge counts and bulk semicircle limit give, uniformly for the $O(q^2)$ shifts in the stability Schur calculation,

$$\frac{\text{Tr } R}{n} = 1 + O(\sqrt{\Delta_{\text{sp}}}) + o_n(1), \quad \frac{\text{Tr } R^j}{n} \asymp_j \Delta_{\text{sp}}^{3/2-j} \quad (j \geq 2).$$

The first estimate follows by removing the edge contribution to the first trace; dyadic counts give the upper estimates for higher powers, and one band at depth comparable to Δ_{sp} gives their lower

estimates. The fixed-rank count error contributes only $O((n\Delta_{\text{sp}}^j)^{-1})$. Likewise the restriction to depths at most a fixed ζ has first trace divided by n at most $O(\sqrt{\zeta}) + o_n(1)$. In particular the first trace is normalized to one, as required for the cancellation in (9.2). The other needed margin uses the actual full spectrum:

$$(b + b^{-1})I - J_\sigma \succeq (1 - o(1))q^2I, \quad b + b^{-1} = 2 + q^2 + O(q^3).$$

Condition first on the spectrum of W_E . On every spectrum satisfying these inherited bounds, its Haar eigenframe and the independent Gaussian column obey the tests in (9.2) at any prescribed fixed rate in nr^3 , before imposing $\mathcal{G}_{\text{spec}}(J_\sigma)$. The scalar diagnostics bound every regression coefficient uniformly on the fixed positive band. Choosing their error tolerances first and then σ small preserves the column tests and their q^2 Schur margin. Consequently the estimate is for

$$\int_S G_{s,\sigma}(y) \mathbf{Q}_{y,\sigma}(\mathcal{G}_{\text{spec}}(J_\sigma) \cap \mathcal{F}_y) dy,$$

with the preliminary rate paying the scalar-weight and mesh losses. Here $\mathcal{F}_y$ is failure of the required Hessian margin. No event for the fresh comparison matrix W_0 replaces the displayed indicator, and no division by its conditional probability is used. This argument transfers the stability calculation, not the sharper inverse-trace accuracy for K_o , which is not needed here. All smoothing choices are made on a fixed positive band before taking n large.

We spell out the additional diagonal uniformity in this use. At the given positive scale truncate $A^{-1} - I$ above at the same level $q^{1-\eta}$ used for $V_y^{-1} - I$, assigning that truncation value at $A_{ii} = 0$. The truncated scalar function is Lipschitz on $[0, 1 + \eta_0]$; its negative part is at most η_0 . Choosing η_0 small preserves the norm bound for the truncated perturbation. Shrinking $\|A - V_y\|_F/\sqrt{n}$ makes its normalized first two moments and all the fixed-axis quadratic and column tests as close as required to those in (9.2).

For axes involving y , first discard their square tails by the scalar diagnostics, then use bounded site profiles on the remaining sites. For $u = m/\sqrt{nq}$ one can directly use $\|(B(A) - B(V_y))u\| \leq C\epsilon_A/\sqrt{q}$. All scales here are fixed before $n \rightarrow \infty$. Thus the Schur calculation retains its margin with the prescribed rate. The original precision is larger. Its positive lower bound also gives a positive symmetric-normalization margin: combine it with the bound $A^{-1} - CI$, then take limits if some entries vanish. The preceding sparse diagonal mesh costs an arbitrarily small exponential rate and makes the estimate simultaneous.

The observation lower bound just used has a fixed rate independent of sufficiently small r , so it pays any specified A_*r^3 after shrinking the upper scale. The determinant event can have an arbitrarily large fixed rate by Lemma 13.2. Apply the local-volume argument in the same order, now taking its loss below the available A_*r^3 or cu^3 rate. This proves the small-time assertions, including their order of tolerance choices.

Finally remove the diagonal. On $\max_i |J_{ii}| \leq \delta_D$ the normalized residual changes by at most δ_D , and the symmetric Hessian by at most $(1 + \eta_0)\delta_D$. On fixed positive bands take a sufficiently small fixed δ_D ; its failure has exponential probability. In the small-scale assertions keep the vanishing-diagonal event as an external restriction and then, for each fixed band, take n large enough. For example, take $\delta_D \leq \rho/2$ and $(1 + \eta_0)\delta_D \leq c_E/4$. Then a zero-diagonal witness with residual at most $\rho\sqrt{n}/2$, diagonal radius $\epsilon_*/2$, and Hessian threshold $c_E/4$ implies the full-matrix witness above.

For the compact path version write $h_t = t\mathbf{1} + B_t$ in the gauged planted process. On a mesh interval of length Δ , the coordinate Brownian suprema have a fixed exponential moment after division by Δ . Independence across coordinates gives

$$\Pr \left\{ \sup_{t_j \leq t \leq t_{j+1}} \frac{\|B_t - B_{t_j}\|}{\sqrt{n}} > \kappa_0 \right\} \leq \exp\{-n(c\kappa_0^2/\Delta - C)\}.$$

Choose κ_0, Δ so this bound has the required fixed rate and the Brownian increment plus the deterministic drift is at most $\rho/4$ in normalized norm. A finite union covers the fixed compact interval. A path witness with residual $\rho\sqrt{n}/8$ then has residual below $\rho\sqrt{n}/2$ at the left mesh endpoint. The Hessian, overlap, and diagonal predicate are independent of the field and persist there. This proves the compact assertion with its reduced residual threshold, also for a fixed finite family of scalar thresholds. All previous exclusions estimated failure intersected with the original spectral restriction, using interlacing rather than a new typicality assertion for a conditional largest eigenvalue. $\square$

13.3 Residual identities and the complete derivative buffer

Proposition 13.5 (Endpoint posterior consequences). *Fix a matrix norm bound. For the covariance conclusion, assume positive margins in (13.1) and suppose that every point in its small-residual region has $q(y) \geq c_* > 0$. On the fixed-length ordinary and restricted inverse-word events specified below,*

$$\|\text{Cov}_{\mu_h}(G, X)\| \leq C(\text{Var}_{\mu_h}(G) + \mathcal{D}_h(G))^{1/2} \quad \text{for every } G.$$

In particular $\|\Sigma_h\| \leq C$.

For the following mean conclusion, only the whole-region root stability implication

$$\|F_h(y)\| \leq \rho_0\sqrt{n} \implies \mathcal{H}(y, V_y) \succeq c_0 I$$

is required, together with ordinary word diagnostics. No lower bound on $q(y)$ is assumed in this mean conclusion. For $N = \mu_h(f^2) > 0$ and $\nu_h = f^2\mu_h/N$, and each sufficiently small fixed $\zeta > 0$,

$$\frac{\|\mathbb{E}_{\nu_h} X - \mathbb{E}_{\mu_h} X\|}{\sqrt{n}} \leq C\zeta + \frac{C_\zeta}{\sqrt{n}} \left(1 + \frac{\mathcal{D}_h(f)}{N}\right). \quad (13.2)$$

The leading constant C depends on the fixed norm and stability margins, but not on ζ . Without the stability hypothesis, put

$$Q_\delta(h) = \sup_{\|F_h(y)\| \leq \delta\sqrt{n}} q(y),$$

with an empty supremum equal to zero. The same ordinary diagnostics give

$$\frac{\|\mathbb{E}_{\nu_h} X - \mathbb{E}_{\mu_h} X\|}{\sqrt{n}} \leq 2\sqrt{Q_\delta(h)} + \frac{C_\delta}{\sqrt{n}} \left(1 + \frac{\mathcal{D}_h(f)}{N}\right). \quad (13.3)$$

For each fixed $\delta > 0$ this assertion holds for sufficiently large n on the corresponding ordinary diagnostics. All events are independent of f, G and the unit directions used to test these inequalities.

Proof. The root and inverse statement, Lemma 5.7 of [28], is deterministic for every finite positive j and applies here at $j = 1$. Its root conclusion uses only the whole-region implication at $A = V_y$ stated above; the overlap lower bound is used later only for the inverse recipe. Denote the unique zero by y_* , and set $m_* = \tanh y_*$, $q_* = n^{-1}\|m_*\|^2$. It gives $\|y - y_*\| \leq C\|F_h(y)\|$ when $\|F_h(y)\| < \rho_0\sqrt{n}$, with $C = 1 + (\|J^\circ\| + 3)/c_0$. The strict residual inequality will be preserved by the inner buffer below.

We verify the precise diagnostic hypotheses of the companion residual proof. Fix J, h throughout this deterministic subsection; all spin expectations below are under μ_h , unless a displayed square reweighting specifies otherwise. For a spin function define

$$d_i g(x) = \frac{g(x) - g(x^{(i)})}{2x_i}, \quad dg = \nabla g = (d_i g)_i.$$

For a vector function, $(\nabla g)_{ji} = d_i g_j$. These are discrete spin-flip derivatives, distinct from the formal site derivatives in the recipes. Put

$$m^0 = x, \quad b_0 = 0, \quad h^1 = h + J^\circ x, \quad m^l = \tanh h^l, \quad b_l = n^{-1} \sum_i (1 - (m_i^l)^2), \quad R_l = m^{l-1} - m^l,$$

$$h^{l+1} = h + J^\circ m^l - b_l m^{l-1}.$$

For each fixed depth the primary differentiation estimates of Lemma 6.1 of the companion article need only ordinary words, with contraction coefficient one. Explicitly their formal derivatives are

$$H_1 = J^\circ, \quad M_0 = I, \quad M_l = \text{diag}(1 - (m^l)^2) H_l, \quad H_{l+1} = J^\circ M_l - b_l M_{l-1}.$$

The diagonal predictions of H_l cancel by inclusion–exclusion over originally adjacent pairs of each noncrossing pairing.

The word estimates give bounded operator, diagonal ℓ^2 , and off-diagonal entrywise ℓ^4 norms. Taylor’s error in a coordinate difference is at most $C|d_i h_j^l|^2$; summing its square uses that ℓ^4 bound. Differentiating $b_l m^{l-1}$ adds a rank-one term of bounded Frobenius norm, since $\|db_l\| \leq C_l/\sqrt{n}$. Thus actual derivatives differ from H_l, M_l by bounded Frobenius errors. Every single flip changes a fixed primary vector by at most C_l in Euclidean norm. None of these finite recurrences uses $j < 1$ or convergence as $l \rightarrow \infty$.

We use the following finite recipe class, as in Definition 6.2 of [28]. Fix the primary depth and finitely many seeds $s^1, \dots, s^v \in \mathbb{R}^n$ with bounded Euclidean norms. The seeds may depend on the fixed (J, h) but are constant as functions of the spin configuration x . Let $\theta(x)$ be a finite tuple of scalar parameters with $\sup_x (|\theta_\alpha(x)| + \|d\theta_\alpha(x)\|) \leq C$ for every component; the averages b_l may be among them. Given auxiliary fields $y^1, \dots, y^{a-1}$, a source has the form

$$w_i^a = \sum_{j=1}^v \phi_i^{a,j}((h_i^l)_{l \in I_a}, \theta) s_i^j + \sum_{b < a} \psi_i^{a,b}((h_i^l)_{l \in I_a}, \theta) y_i^b.$$

All index sets are finite. The site coefficients and every fixed derivative needed by the recipe are uniformly bounded on their argument ranges and on the line segments joining evaluations. Deterministic site labels are allowed with the same bounds. Define

$$y^a = J^\circ w^a - \sum_l \left(n^{-1} \sum_i \partial_l w_i^a \right) m^{l-1} - \sum_{b < a} \left(n^{-1} \sum_i \partial_{y^b} w_i^a \right) w^b.$$

The partial ∂_l differentiates only the explicit h_i^l argument, keeping θ and all auxiliary arguments fixed; $\partial_{y^b} w_i^a = \psi_i^{a,b}$, and an absent partial is zero. A terminal vector has the same source form. When a source is reused in a later source or terminal vector, substitute its source formula without expanding the auxiliary fields.

A recipe is admissible from level p when every explicit primary argument in these source and terminal formulas has index at least p . This restriction does not apply to scalar parameters or to the correcting vectors m^{l-1} in the auxiliary formula. All constants may depend on the fixed recipe and its coefficient bounds. In particular, the finite residual inductions below may enlarge the recipe before its diagnostic length and the eventual dimension threshold are fixed.

Only the directional covariance argument uses an inverse recipe. Fix $k \geq 2$ and $\Omega_{a\rho} = \{\max(\|R_k\|, \|R_{k-1}\|) \leq a\rho\sqrt{n}\}$. When several indices occur we write this set as $\Omega_{a\rho}^{(k)}$; the superscript is suppressed while k is fixed. The exact recurrence gives

$$F_h(h^k) = J^\circ R_k - b_{k-1}(R_{k-1} + R_k) + (b_k - b_{k-1})m^k, \quad |b_k - b_{k-1}| \leq 2\|R_k\|/\sqrt{n}.$$

Choose ρ so the displayed recurrence puts $\Omega_{2\rho}$ strictly inside the root lemma's residual region. Consequently on $\Omega_{2\rho}$,

$$\|h^k - y_*\| + \|m^{k-1} - m_*\| \leq C\rho\sqrt{n}, \quad |1 - b_{k-1} - q_*| \leq C\rho, \quad q(h^k) \geq c_*.$$

These constants are independent of k . Define

$$\Phi(z; r, a) = \int_0^1 e^{a(1-u^2)/2} \frac{\cosh(u(z-r))}{\cosh z \cosh r} du, \quad a = 1 - b_{k-1} - q_*, \quad A_i = \Phi(h_i^k; (y_*)_i, a).$$

Differentiation in u proves the exact identity

$$[(z - r - a \tanh z) - a \partial_z] \Phi(z; r, a) = \tanh z - \tanh r.$$

At $a = 0$, Φ is the secant slope of $\tanh$.

Its positive integral and the bounded derivatives of its logarithmic factors give $0 < \Phi \leq e^{|a|/2}$ and bounded ratios of every fixed derivative to Φ . The derivative recipes evaluate their coefficients at nearby spin configurations as well as at the selected residual point. We therefore establish the diagonal estimates on the whole outer buffer. Once the finite recipe and its number of flips are fixed, the threshold in the dimension will ensure that every evaluation stays in that buffer.

The preceding root-distance estimates imply, on the whole outer buffer,

$$\|A - V_{h^k}\|_F \leq C\rho\sqrt{n}, \quad |\chi_A - b_k| \leq C\rho, \quad \|A\| \leq e^{C\rho}, \quad \|A\|_{\chi_A} \leq e^{C\rho(1-c_*+C\rho)} \leq 1 - c_*/2. \quad (13.4)$$

Choose ρ small enough for the last inequality and for all stable field tolerances. This choice precedes the depth and recipe choices. The inverse diagonal therefore belongs to $\mathcal{A}_\kappa$ with $\kappa = c_*/2$ everywhere it will be evaluated.

For either $w_{0,i} = A_i e_i$, $\|e\| \leq 1$, or $w_{0,i} = n^{-1/2} \partial_r \Phi(h_i^k; (y_*)_i, a)$, the implicit recipe is

$$w = w_0 + A(y^{\text{imp}} - c_1 m_*), \quad y^{\text{imp}} = J^\circ w - \chi_A w - c_1 m^{k-1}, \quad c_1 = n^{-1} \sum_i \partial_k w_i.$$

In this site partial, y^{imp} , c_1 and a are held fixed. After this one implicit solve, y^{imp} may be used as the first auxiliary in an ordinary recipe, with $\sqrt{n}c_1$ as an additional scalar parameter and $m_*/\sqrt{n}$ as an additional seed. The corresponding coefficient and parameter bounds are required on the flip buffers below; no such bounds are imposed on the arbitrary exterior extension of the implicit solution. Put $\ell_i = \partial_z \Phi / \Phi$, $u_0 = w_0 / A$ and $p_0 = A \partial_k u_0$. Then $\partial_k w = \text{diag}(\ell)w + p_0$ and elimination gives the actual finite-dimensional solve

$$\left[I - A(J^\circ - \chi_A I) + \frac{1}{n} A(m^{k-1} + m_*) \ell^\top \right] w = A \left[u_0 - (m^{k-1} + m_*) n^{-1} \sum_i (p_0)_i \right].$$

Put

$$\mathcal{L} = \chi_A I - J^\circ + (m^{k-1} + m_*) \ell^\top / n.$$

Because $\|\ell + m_*\| \leq C\rho\sqrt{n}$, the two-sided normalized coefficient $I + A^{1/2} \mathcal{L} A^{1/2}$, where the displayed solve is $(I + A\mathcal{L})w = \dots$, differs in operator norm by $O(\rho)$ from $\mathcal{H}(h^k, A)$. The normalized coefficient need not be symmetric. The latter matrix has a fixed positive margin by (13.1), so a perturbation smaller than half that margin still has smallest singular value bounded below and a bounded inverse. The unnormalized inverse is bounded by

$$(I + A\mathcal{L})^{-1} = I - A^{1/2}(I + A^{1/2} \mathcal{L} A^{1/2})^{-1} A^{1/2} \mathcal{L}.$$

This proves $\|w\| + \|y^{\text{imp}}\| + \sqrt{n}|c_1| \leq C$, uniformly in k and the unit seed. Removing the negative rank term in $\mathcal{H}(h^k, A)$ and replacing b_k by χ_A also gives

$$S_A(1; J^\circ) \succeq cI.$$

This separate adaptive stability condition is essential; the norm–trace restriction alone was only used for fixed-parameter expectations.

Every coordinate-difference solve uses the same displayed coefficient matrix evaluated at the flipped configuration. All coefficient differences multiply vectors at the common starting configuration. Their squared column norms sum to a constant: for a coefficient ψ and a fixed bounded-norm vector v ,

$$\sum_{i,j} v_i^2 |d_j \psi_i|^2 \leq \|v\|^2 \max_i \sum_j |d_j \psi_i|^2 \leq C.$$

The averaged coefficients have differences of order $n^{-1/2}$; the factors $m^{k-1} + m_*$ have norm at most $2\sqrt{n}$.

Columnwise inversion therefore gives $\|\nabla w\|_F + \|\nabla y^{\text{imp}}\|_F + \sqrt{n}\|dc_1\| \leq C_k$, as in Lemma 6.3 of the companion article. If the finished finite recipe uses at most N_f successive flips, the primary bounds put every required flip-neighborhood of $\Omega_{3\rho/2}$ inside $\Omega_{2\rho}$ once $n > (2N_f \max_l C_l/\rho)^2$. Thus all four bounds in (13.4) and the adaptive inverse margin hold at every derivative evaluation. The recipe and its depth are fixed before this threshold on n is chosen.

We also fix the whole-cube convention for these local recipes. Keep the actual implicit solution y^{imp}, c_1 on $\Omega_{2\rho}^{(k)}$, assign arbitrary finite values to them outside that set, and define w and all later ordinary auxiliaries by their stated formulas. The implicit equation is required only on the outer buffer. Every scalar test in a local residual statement is supported on $\Omega_\rho^{(k)}$. Its discrete differences use only the fixed finite flip neighborhoods already placed inside the buffers, so the exterior values never enter a tested estimate.

We record the remaining closure of the residual proof to ensure that no other inverse diagonal is being assumed. Freezing parameters in formal differentiation yields

$$\nabla w = A \nabla y^{\text{imp}} + \text{diag}(\partial_k w) H_k + E_1, \quad \nabla y^{\text{imp}} = (J^\circ - \chi_A I) \nabla w - c_1 M_{k-1} + E_2.$$

Solving this equation introduces exactly the one factor $K = (I - A J^\circ + \chi_A A)^{-1}$ multiplying $\text{diag}(\partial_k w) H_k - c_1 A M_{k-1} + E_1 + A E_2$. Later auxiliaries are ordinary, so error propagation multiplies by ordinary words only.

The Taylor errors have bounded Frobenius norm and bounded trace against the required words. For the only non-nuclear product error, the decisive bound is

$$\sum_{i \neq j} |P_{ji}| |d_j s_i| Q_{ij} \leq \|P_{\text{off}}\|_{\ell^4} \|\nabla s\|_F \|Q\|_{\ell^4} \leq C, \quad Q_{ij} = \sum_l |d_j h_i^l|.$$

The diagonal part uses the primary diagonal ℓ^2 estimate. All other dropped terms are bounded in nuclear norm, using $\|BC\|_{\text{nuc}} \leq \|B\|_F \|C\|_F$.

The formal trace cancellation in Lemma 6.4 of the companion article is algebraic. Its only cyclic node closes as

$$\mathfrak{S}_{\text{imp}} = K_A(z; J^\circ) \{ \text{diag}(\partial_k w) \mathfrak{H}_{h^k} - z^2 c_1 A \mathfrak{S}_{h^k} \}.$$

Both terms, including the averaged correction, must be retained. For each dependency path, inclusion–exclusion over its originally adjacent contracted pairs cancels every complete noncrossing pairing. There is no empty derivative path since seeds do not depend on x .

The contracted trace prediction vanishes as a formal series and is a polynomial by Lemma 13.1. Its evaluation at $z = 1$ is therefore zero; this does not require convergence of the inverse's formal series. Each resulting trace has either a bounded-norm small vector as a diagonal factor or a scalar $O(n^{-1/2})$. In the first case rotate the small diagonal to the beginning of the trace. Cyclic invariance of the normalized prediction preserves the canceled value, and Cauchy–Schwarz gives

$$|\text{Tr}(\text{diag}(u)(Q - P(Q)))| \leq \|u\|_2 \|\text{diag}(Q - P(Q))\|_2 \leq C.$$

In the scalar case the diagonal word error has trace at most $C\sqrt{n}$, canceled by its $O(n^{-1/2})$ coefficient. These bounds remain uniform after a norm-one diagonal test. Thus every small vector U in an ordinary recipe obeys the following bound on the whole cube, and every local recipe vector obeys it on $\Omega_\rho^{(k)}$ with the stated flip buffers, including the vector $\partial_k w$:

$$\|U\| + \|\nabla U\|_F + \|\text{diag } \nabla U\|_1 \leq C.$$

This proves the derivative assertion with only the single inverse whose full buffer has already been checked.

We state the residual interfaces that now apply, specializing Lemmas 6.5–6.6 of [28] to coefficient one. Put

$$\mathcal{E}_h(G) = (\mu_h(G^2) + \mathcal{D}_h(G))^{1/2}, \quad \mathcal{D}_h(G) = \mu_h \sum_i (1 - (m_i^1)^2)(d_i G)^2.$$

If U is a terminal vector of an ordinary recipe admissible from level p , then

$$|\mu_h[GR_p^T U]| \leq C\mathcal{E}_h(G)$$

for every scalar G . For a terminal vector of a local recipe admissible from level p , with implicit starting index k , the same inequality holds when $p \leq k$ and G is supported on $\Omega_\rho^{(k)}$, with the buffers active. For an ordinary recipe admissible from level p and a multiplier satisfying $\sup_x (|H_0(x)| + \|dH_0(x)\|) \leq C_0$, the square-density version is

$$|\mu_h[f^2 H_0 R_p^T U]| \leq C(\mu_h(f^2) + \mathcal{D}_h(f)).$$

Only ordinary recipes are used in this last version.

The induction can be checked directly. Its base identity is

$$\mu_h \sum_i (x_i - m_i^1) G U_i = \mu_h \sum_i (1 - (m_i^1)^2) d_i (G U_i).$$

The preceding diagonal ℓ^1 bound and Cauchy–Schwarz control it by $C\mathcal{E}_h(G)$. For the induction from p to $p+1$ multiply the terminal vector by $\Phi(h_i^p; h_i^{p+1}, b_p - b_{p-1})$, add one ordinary auxiliary, and use the displayed scalar identity for Φ .

Here $|b_p - b_{p-1}| \leq 1$, so all required coefficient bounds hold. The term $n(b_p - b_{p-1}) \text{av}(\partial_p U')$ cancels by

$$R_p^\top m^{p-1} = n(b_p - b_{p-1}) - R_p^\top m^p.$$

Only smaller residual indices remain; no further inverse is introduced. The square-density version follows from $d_i(f^2) = 2f d_i f - 2x_i(d_i f)^2$ and has right side $C(\mu_h(f^2) + \mathcal{D}_h(f))$. It uses ordinary recipes only. All these operations are finite, so one fixed diagnostic length covers the whole argument.

For the implicit recipe define the terminal residual

$$\mathcal{T}_k = \{h^k - h - (J^\circ - b_{k-1}I)m^k\}^T w - (b_k - b_{k-1})nc_1.$$

For every G supported on $\Omega_\rho^{(k)}$,

$$|\mu_h[G\mathcal{T}_k]| \leq C\mathcal{E}_h(G).$$

Indeed, on the buffer the exact expansion is

$$\mathcal{T}_k = R_k^T y^{\text{imp}} + (\chi_A - b_{k-1})R_k^T w - b_{k-1}R_{k-1}^T w - c_1 R_k^T m^k.$$

The local weak rule applies to the first three terms. In the last one use target $m^k/\sqrt{n}$ and multiplier $\sqrt{n}c_1$, whose size and discrete differences are bounded on the buffer.

We finish by spelling out selection and the uniformity in the tests. Put $T_l = R_l^\top m^l/\sqrt{n}$. The ordinary residual rule gives $\mu_h(T_l^2) \leq C_l$ and

$$\mathbb{E}_{\nu_h}(T_l/\sqrt{n})^2 \leq \frac{C_l}{\sqrt{n}} \left(1 + \frac{\mathcal{D}_h(f)}{N}\right).$$

The exact telescoping identity $\|R_l\|^2/n = b_l - b_{l-1} - 2T_l/\sqrt{n}$ supplies a pair of consecutive small residuals. More precisely, choose a fixed depth d_{sel} with $\lfloor d_{\text{sel}}/2 \rfloor \rho^2/4 > 2$, and then a scalar tolerance with $2d_{\text{sel}}\varepsilon < 1$. If $\max_{l \leq d_{\text{sel}}} |T_l|/\sqrt{n} \leq \varepsilon$, the telescoping sum is less than 2, forcing a consecutive pair whose two norms are at most $\rho\sqrt{n}/2$.

Take a smooth $\vartheta : [0, \infty) \rightarrow [0, 1]$, equal to one on $[0, 1/4]$ and zero on $[1, \infty)$, and set

$$C_k = \vartheta\left(\frac{\|R_k\|^2}{\rho^2 n}\right) \vartheta\left(\frac{\|R_{k-1}\|^2}{\rho^2 n}\right), \quad \widehat{C}_k = C_k \prod_{i=2}^{k-1} (1 - C_i), \quad 2 \leq k \leq d_{\text{sel}}.$$

Pointwise,

$$0 \leq \widehat{C}_k \leq C_k \leq 1, \quad \text{supp } C_k \subseteq \Omega_\rho^{(k)}, \quad \|dC_k\| + \|d\widehat{C}_k\| \leq C_\rho/\sqrt{n}.$$

The derivative bounds follow from the primary differences and the finite product rule. Also $C_k = 1$ when both residual norms are at most $\rho\sqrt{n}/2$, and

$$D := 1 - \sum_{k=2}^{d_{\text{sel}}} \widehat{C}_k = \prod_{k=2}^{d_{\text{sel}}} (1 - C_k) \in [0, 1].$$

The scalar moment bounds and a finite union bound give

$$\mu_h(D > 0) \leq C_\rho/n, \quad \nu_h(D > 0) \leq \frac{C_\rho}{\sqrt{n}} \left(1 + \frac{\mathcal{D}_h(f)}{N}\right).$$

These are the conclusions of the companion Lemma 7.1, with the preceding derivation of its endpoint hypotheses.

Residual selection now turns the finite recipe estimates into posterior bounds. The covariance argument uses its one controlled inverse; the square-reweighted mean argument below uses only ordinary recipes. This distinction also separates their diagnostic requirements.

We make the terminal implication explicit. On the selected buffer put $p_k = m^k - m_*$ and $L_* = \sqrt{n}(\chi_A - (1 - q_*))$. Subtracting the root equation gives

$$h^k - h - (J^\circ - b_{k-1}I)m^k = h^k - y_* - am^k - J^\circ p_k + (1 - q_*)p_k.$$

Using the implicit equations and the scalar identity for Φ then yields the exact identity

$$\begin{aligned} \mathcal{T}_k &= \sum_i [(h_i^k - (y_*)_i - am_i^k)w_{0,i} - a \partial_k w_{0,i}] \\ &\quad + (1 - q_* - \chi_A)p_k^T w - c_1 R_k^T p_k. \end{aligned}$$

The final term is controlled by the local weak rule with target $p_k/\sqrt{n}$ and multiplier $\sqrt{n}c_1$. The remaining scalar error satisfies pointwise

$$|(1 - q_* - \chi_A)p_k^T w| = \frac{|L_*|}{\sqrt{n}} |p_k^T w| \leq C_0 \rho |L_*|.$$

Here the root-distance and source-size bounds give C_0 depending only on the fixed norm and stability margins, not on k or the selection depth. The earlier choice of ρ includes $C_0 \rho < 1/2$.

For the source $w_{0,i} = n^{-1/2} \partial_r \Phi$, differentiation of the scalar identity in r , with a fixed, gives

$$[(z - r - a \tanh z) - a \partial_z] \partial_r \Phi = \Phi - \operatorname{sech}^2 r.$$

Thus the sum in the terminal identity is L_* . Globally, $|L_*| \leq C\sqrt{n}$ and $\|dL_*\| \leq C_k$. Use the local test $G = C_k^2 L_*$, whose support lies in $\Omega_\rho^{(k)}$. The cutoff difference bound gives

$$\mathcal{E}_h(C_k^2 L_*) \leq C_{\rho,k} + \|C_k L_*\|_{L^2(\mu_h)}.$$

The tested scalar error is at most $C_0 \rho \mu_h(C_k^2 L_*^2)$. Absorbing it into the left side of the terminal inequality proves

$$\|C_k L_*\|_{L^2(\mu_h)} \leq C_{\rho,k}.$$

For the source $w_0 = Ae$, $\|e\| \leq 1$, the same sum is $p_k^T e$. Use the local test $\hat{C}_k G$. Its support and difference bounds control its test norm by $C_\rho \mathcal{E}_h(G)$. For the scalar error use $\hat{C}_k \leq C_k$, the last L^2 bound, and Cauchy-Schwarz. The terminal and weak residual inequalities give

$$|\mu_h[\hat{C}_k G(m^k - m_*)^T e]| \leq C_{\rho,k} \mathcal{E}_h(G).$$

Summing the ordinary weak rule with target e from level one through k also controls the tested $(X - m^k)^T e$.

The deficit costs at most $2\sqrt{n}\|G\|_2 \mu_h(D > 0)^{1/2}$. This is the deterministic argument of Proposition 7.2, applied only after checking its hypotheses on the selected regions and verifying both of its actual source diagonals above. It yields the asserted covariance inequality after replacing G by $G - \mu_h(G)$ and taking the supremum over e . For $G = e^T X$, $\mathcal{D}_h(G) \leq 1$; taking a top covariance eigenvector gives $\|\Sigma_h\| \leq C(\|\Sigma_h\| + 1)^{1/2}$ and hence a constant bound.

For the mean estimate, the ordinary residual rule suffices; no implicit source is needed. On each selected region m^k is within $C\zeta\sqrt{n}$ of m_* if the residual threshold was chosen below a fixed multiple of ζ . The ordinary square-density rule, summed with the constant target e through k , bounds the tested difference between X and m^k by $C_\zeta(1 + \mathcal{D}_h(f)/N)$. The deficit has the same bound. This proves the conclusion of the companion Proposition 7.3 relative to m_* .

Apply it also with $f = 1$ and use the triangle inequality to obtain (13.2). If stability is absent, choose the iteration threshold so the exact formula for $F_h(h^k)$ is at most $\delta\sqrt{n}$ on each selected region. Then $\|m^k\| \leq \sqrt{nQ_\delta(h)}$, and the same argument, once for ν_h and once for μ_h , proves the coarser assertion. This also covers an empty approximate-zero set, since then every selected cutoff vanishes and the displayed deficit estimate controls the entire measure.

All bounds use only norm bounds on seeds, so they hold for the whole Euclidean unit ball simultaneously. The finite recipe choices may depend on the requested scalar tolerance, but the diagnostic event quantifies over every bounded diagonal parameter after J is observed. It therefore does not depend on f, G , or the directional seed. For covariance choose the stability margins and c_* , then ρ and κ , then the selection depth and finite recipes, then their word length, coefficient bounds and truncations, and finally the size threshold guaranteeing every flip buffer. For (13.2) and its coarser form, only the ordinary diagnostics are required throughout. $\square$

13.4 Large observation time and the product endpoint

Proposition 13.6 (Terminal extension, uniformly in the initial spin law). *There are fixed $T, C, c > 0$ and disorder events of ordinary probability at least $1 - e^{-cn}$ with the following property. For every matrix in these events, let X have any probability law on the cube, possibly depending on the matrix, and let the observation noises be independent of X . Set $h_T = TX + \sqrt{T}g$. The reference Gibbs law μ_{h_T} has unscaled gap at least $1/2$ except with probability e^{-cn} .*

On the original spectral event choose a fixed c_+ such that $R = J + c_+I \succeq 0$. For one further time unit add observations with drift RX and covariance rate R , obtaining $h_{T+\lambda}$. Define the reference laws

$$\mu_{h,\lambda}(x) \propto \exp\left(\frac{1-\lambda}{2}x^\top J^\circ x + h^\top x\right), \quad 0 \leq \lambda \leq 1.$$

The covariance and inverse unscaled gap of $\mu_{h_{T+\lambda},\lambda}$ are bounded by C , simultaneously in λ , outside an event of conditional probability e^{-cn} , uniformly over the law generating X . At $\lambda = 1$ the reference law is product. These reference laws are the posteriors of the original Gibbs prior evaluated at the observations; no assertion about the posterior of the arbitrary law generating X is made.

Proof. The signed curvature inequality, Lemma 8.1 of [28], is deterministic for arbitrary symmetric zero-diagonal interaction matrices with sufficiently small entries. For interaction B , write $U_{ij} = \tanh B_{ij}$, $Q_{ij} = U_{ij}^2$ (entrywise), $r_4(U) = \max_i \sum_j U_{ij}^4$, and $a_i = v_i d_i f$, where v_i is the conditional spin variance. It states

$$\mu_h[(-Lf)^2] \geq (1 - Cr_4(U))\mathcal{D}_h(f) - \mu_h[a^\top Ua + C|a|^\top Q|a|].$$

Its proof has no inverse-temperature parameter, so this is an exact companion input.

Here is the endpoint matrix event, with exponential rather than only asymptotic probability. Fix a curvature tolerance $\varepsilon > 0$ so small that $(1 + C)\varepsilon \leq 1/8$. Choose a fixed global bound M for the row-square sums and $\|J^\circ\|$, with exponentially small exception. Next choose an entry tolerance $\delta > 0$ small enough for the curvature lemma and for $M\delta/3 \leq \varepsilon/2$ and $CM\delta^2 \leq 1/8$. Gaussian tails give $\max |J_{ij}^\circ| \leq \delta$ outside probability $e^{-c_\delta n}$. For each fixed unit vector z ,

$$\mathbb{P}(|z^\top J^\circ z| > t) \leq 2e^{-nt^2/4}, \quad \mathbb{P}\left(\sum_{l \in S} (J_{il}^\circ)^2 > t\right) \leq e^{-nt/4 + |S| \log 2/2}.$$

A $1/4$ -net of a k -dimensional unit sphere has size at most 9^k .

Union these bounds over supports of size at most αn and the $9^{\alpha n}$ nets. Since $n^{-1} \log \binom{n}{\lfloor \alpha n \rfloor} \rightarrow \mathfrak{h}(\alpha)$, choosing α sufficiently small gives, with exponential probability,

$$\|(J^\circ)_{SS}\| \leq \varepsilon/2, \quad \max_i \sum_{l \in S} (J_{il}^\circ)^2 \leq \varepsilon \quad (|S| \leq \alpha n).$$

These are the net and row arguments of Lemma 8.2 of the companion article, now evaluated at variance $1/n$.

They work simultaneously for all $B = \theta J^\circ$, $0 \leq \theta \leq 1$. Indeed $|\tanh z - z| \leq |z|^3/3$ gives $\|\tanh(\theta J^\circ) - \theta J^\circ\| \leq M\delta/3$ by row sums; $Q_{ij} \leq (J_{ij}^\circ)^2$ and $r_4(U) \leq M\delta^2$. By the preceding choice of δ , these errors fit the ε and curvature budgets. We obtain uniformly in θ bounded global norms of B, U, Q , small principal submatrix norms of U, Q on $|S| \leq \alpha n$, and $Cr_4(U) \leq 1/8$. The bounds hold pointwise over spin configurations and require no union over configurations or initial laws.

The observation source law may differ from the Gibbs law whose gap is being estimated. Throughout this argument the posterior is formed from the Gibbs reference law. A norm estimate

makes most coordinates strongly biased for every source configuration; the small-submatrix bounds control the remaining coordinates uniformly.

At time T the observation field is $TX + \sqrt{T}g$. During the extension it is

$$h_{T+\lambda} = TX + \sqrt{T}g + \lambda RX + R^{1/2}B_\lambda,$$

where B is a fresh standard Brownian motion independent of X, g . Updating the original Gibbs prior by this likelihood subtracts $\lambda x^\top R x / 2$ from its interaction; the diagonal is constant on the cube. This identifies its posterior with the stated reference law. The probability estimates below are uniform over the possibly different law generating the observation source X .

On the norm event $\|R\| \leq C$, the perturbation to TX obeys

$$\sup_{0 \leq \lambda \leq 1} \|h_{T+\lambda} - TX\| \leq C(1 + \sqrt{T})\sqrt{n}$$

outside an event of probability e^{-cn} , independently of the law of X . The Gaussian norm tail controls g , and exponential Markov on the sum of the coordinate Brownian suprema squared controls B . Thus, for sufficiently large fixed T , fewer than $\alpha n / 2$ coordinates have $|h_{T+\lambda, i}| < T/2$, uniformly in λ . For every configuration x simultaneously, $\|(1 - \lambda)J^\circ x\| \leq M\sqrt{n}$, so at most $16M^2n/T^2 \leq \alpha n / 2$ coordinates have interaction field larger than $T/4$. Consequently the set where the full conditional field has magnitude below $T/4$ has size at most αn , for every x and λ .

Put $q_f(x) = \sum_i v_i(x)(d_i f(x))^2$ and $\eta = \text{sech}^2(T/4)$. On the complement of that small set, $\|a_{S^c}\|^2 \leq \eta q_f$, while $\|a\|^2 \leq q_f$. The small-submatrix and global bounds therefore give pointwise

$$a^\top U a + C|a|^\top Q|a| \leq (1 + C)\{\varepsilon + M(2\sqrt{\eta} + \eta)\}q_f.$$

With ε already fixed, choose T large enough both for the sparse-set bounds above and for $(1 + C)M(2\sqrt{\eta} + \eta) \leq 1/8$. The displayed expression is then at most $\mathcal{D}_h(f)/4$ after averaging. The curvature inequality yields $\mu_h[(-Lf)^2] \geq \mathcal{D}_h(f)/2$. We finish by converting curvature to the gap.

By the finite-dimensional spectral theorem each nonzero eigenvalue of $-L$ is at least $1/2$. The kernel consists of constants because the Ising law is positive on every cube edge. Thus $\text{Var}_{\mu_h} f \leq 2\mathcal{D}_h(f)$, uniformly in λ and f . Applying this to unit linear tests also gives $\|\Sigma_h\| \leq 2$. This proves the proposition. $\square$

We conclude by making explicit the probability interfaces used elsewhere. For each fixed positive compact interval, combine Propositions 13.4 and 13.5 with the finite diagnostic events of Lemma 13.1. Their ordinary exponentially small exceptions remain exponentially small after planting: if $L = Z(0)/\mathbb{E}Z(0)$, the critical overlap-binomial identity gives $\mathbb{E}L^2 \leq C(n + 1)$, for example by the pointwise Chernoff bound $\mathbb{P}(\sum_i \varepsilon_i = k) \leq 2e^{-k^2/(2n)}$ and summing its $n + 1$ possible values.

Cauchy–Schwarz therefore pays only a polynomial factor in transferring any disorder event. It follows that, at each deterministic positive time in that interval, $\|\Sigma_s\|$ is bounded except with probability e^{-cn} integrated under planting and the original spectral restriction. The terminal proposition gives the corresponding terminal gap input.

For the macroscopic entropy argument, only the ordinary diagnostics in (13.2) and its Q_δ version are needed. They are field-independent and may be imposed externally on typical ordinary disorder for each fixed tolerance. Their exponent need not be compared with the arbitrarily large multiplier A_* in Proposition 13.4. On any compact positive band with finitely many outer thresholds, use one residual tolerance smaller than every threshold’s permitted tolerance.

Existence of a large- q point at this tolerance implies existence at each corresponding looser tolerance, so the outer rates remain valid. Here is how the moving thresholds in the entropy

argument fit this finite construction. Let $u_* > 0$ be a fixed upper endpoint for the small-range outer estimate, and on a fixed compact band put $R_b = C' A_*^{1/3} \sqrt{b}$, with $A_* \geq 1$ and $C' \geq 2C$. Restrict the initial interval so that $R_b \leq u_*$. For every moving threshold $u = 2^j R_b \leq u_*$, choose a lower point v in a fixed dyadic grid on that band with $u/2 \leq v \leq u$. Then $v \geq C\sqrt{b}$, and

$$\{Q_\delta(h_b) > u\} \subseteq \{Q_\delta(h_b) > v\}, \quad cv^3 \geq (c/8)u^3.$$

Only finitely many grid points occur on a fixed positive band. For each point use the compact time subband on which $v \geq C\sqrt{b}$, and choose one residual tolerance for this finite collection. Thresholds above u_* are grouped into the single event $Q_\delta > u_*$, using its fixed rate $e^{-cnu_*^3}$ and the bound $Q_\delta \leq 1$. Countably many bands and tolerances are handled by the diagonal subsequence procedure in the entropy argument; no estimate is claimed uniformly as positive time or a chosen residual tolerance tends to zero.

14 Entropy localization and the support profile

We now convert the posterior estimates into an inequality for every test function of small support. All final typical-environment assertions concern the original, unplanted disorder. The passage from the planted estimates to these statements will be made before choosing a test function; this order is essential for the uniformity of the resulting profile.

Write $\mu = \mu_J$ for the critical zero-field Gibbs measure. For a probability measure π on the cube, use the unscaled heat-bath form

$$\mathcal{D}_\pi(f) = \sum_{i=1}^n \pi[\text{Var}_\pi(f \mid X_{-i})].$$

Thus $\mathcal{D}_J$ is the form at μ_J , and $\mathcal{D}_b$ will denote the same form for the posterior law at observation parameter b . When used for heat-bath dynamics, this unscaled form corresponds to rate one at each site. The observation parameter b instead measures Gaussian information with the precision rate defined below.

We use the following static conclusions established above, with exactly their stated probability qualifications: the sharp small-time covariance estimate (12.5), the estimate below the logarithmic threshold (12.8), and the positive-time assertions in Section 13. The latter include bounded covariance on each compact positive-time interval, a terminal observation gap, the stable-field implication (13.1), and the square-reweighting estimates (13.2) and (13.3).

In particular, the small-positive-time failure rate for stability can be any prescribed fixed multiple of $nb^{3/2}$, and the probability of an approximate zero with $q > u$ is at most $\exp(-cnu^3)$ for $C\sqrt{b} \leq u \leq u_*$, with a fixed small $u_* > 0$. Constants and residual tolerances may depend on a fixed compact positive-time interval. The proofs in Section 13 supply these statements at the critical parameter itself; no subcritical spectral-gap theorem is being applied at its endpoint.

14.1 The observation identities

The use of Gaussian observations here is a stochastic localization argument in the sense of Eldan [19]. The variance and entropy transfers are related to the localization methods of Eldan, Koehler, and Zeitouni [20] and the framework of Chen and Eldan [12]. We give the finite-state identities and the estimates needed here explicitly.

Choose the sufficiently large fixed terminal time T supplied by the static inputs. On the spectral event $\|J\| \leq 3$, fix $c_+ \geq 3$ and put $R = J + c_+I \succeq 0$. Continue the scalar observations after T for one unit of time with precision R . Set $E = T + 1$ and

$$\Lambda_b = I \quad (0 \leq b \leq T), \quad \Lambda_b = R \quad (T < b \leq E).$$

More explicitly, conditional on X , the field has independent increments with drift $\Lambda_b X db$ and covariance $\Lambda_b db$. The accumulated quadratic term in its likelihood removes the interaction by time E : at $b = T + \lambda$, the off-diagonal interaction is $(1 - \lambda)J^\circ$, and μ_E is a product law.

For a real function f with $\mu(f^2) = 1$, let $\nu = f^2\mu$. Use μ_b and ν_b for the conditional spin laws given the observation history, under the respective initial laws. For a fixed J , write $\mathbb{P}_J^\pi$ for the observation-history law with initial spin law π , and $\mathbb{E}^\pi$ for expectation under this conditional law. Define

$$\begin{aligned} w_b &= \mu_b(f^2), & m_b^{\text{post}} &= \mu_b(X), & \Sigma_b &= \text{Cov}_{\mu_b}(X), \\ a_b &= \nu_b(X) - \mu_b(X), & H_b &= \mathbb{E}^\nu \text{KL}(\nu_b \| \mu_b). \end{aligned}$$

Histories with $w_b = 0$ have zero probability under ν and can be omitted. The likelihood ratio of the two field-history laws is w_b , and $\nu_b = f^2\mu_b/w_b$. Consequently

$$H_b = \mathbb{E}^\mu \text{Ent}_{\mu_b}(f^2), \quad \text{KL}(\text{field law under } \nu \| \text{field law under } \mu) = H_0 - H_b.$$

The Gaussian filtering identities give, at every time away from the precision change and in integral form across it,

$$\begin{aligned} H'_b &= -\frac{1}{2} \mathbb{E}^\nu \|\Lambda_b^{1/2} a_b\|^2, \\ a_b &= \mathbb{E}^\nu \left[a_E + \int_b^E \Sigma_v \Lambda_v a_v dv \middle| \mathcal{F}_b \right]. \end{aligned} \tag{14.1}$$

Here $\mathcal{F}_b$ is the observation history, including J . To verify the first identity, the field drifts under the two laws differ by $\Lambda_b a_b$; the expected logarithm of their likelihood ratio therefore has derivative $\mathbb{E}^\nu \|\Lambda_b^{1/2} a_b\|^2/2$.

Subtract this from the preceding entropy decomposition. For the second identity, the ν -posterior mean is a martingale under ν , whereas the μ -posterior mean has drift $\Sigma_b \Lambda_b a_b$ under that law. Integrating their difference to E proves the assertion. These arguments also follow directly by Itô's formula for the finite posterior sums.

Two further identities will be used repeatedly. Since μ_E is product, binary Pinsker on each coordinate and entropy tensorization from below give

$$\mathbb{E}^\nu \|a_E\|^2 \leq 2H_E. \tag{14.2}$$

Also, conditioning additionally on the observations can only decrease an averaged conditional variance. Applied separately with X_{-i} fixed, this gives

$$\mathbb{E}^\mu \mathcal{D}_b(f) \leq \mathcal{D}_J(f), \quad \mathbb{E}^\nu \frac{\mathcal{D}_b(f)}{w_b} = \mathbb{E}^\mu \mathcal{D}_b(f) \leq \mathcal{D}_J(f). \tag{14.3}$$

14.2 A profile with a bounded test likelihood

Proposition 14.1 (Restricted support profile). *For every fixed $\omega > 0$ there are $c_\omega > 0$ and disorder events $\mathcal{G}_n(\omega)$ with probability tending to one such that, on $\mathcal{G}_n(\omega)$, simultaneously for every real f satisfying*

$$\mu(f^2) = 1, \quad 0 < p := \mu(f \neq 0) \leq \frac{1}{2}, \quad \|f\|_\infty^2 \leq 100p^{-2},$$

one has

$$\mathcal{D}_J(f) \geq c_\omega n^{-2/3-\omega} \log(1/p). \quad (14.4)$$

Proof. Choose the covariance accuracy $\epsilon > 0$ sufficiently small in terms of the requested ω , and obtain the corresponding $a, s_0, K_{\log}$ from [Theorem 12.3](#). This same accuracy is used in all three entropy ranges below. Put $h = \log(1/p)$. Entropy relative to the support and the upper bound on f^2 imply

$$h \leq H_0 = \mu(f^2 \log f^2) \leq 2h + \log 100. \quad (14.5)$$

On $\|J\| \leq 3$, every atom of μ is at least $2^{-n}e^{-3n}$, so $h \leq (3 + \log 2)n$. We treat three ranges of h .

Moderate entropy. For constants K' and c' fixed in the order below, first suppose

$$K' \log n \leq h \leq c'n, \quad r_h = (h/n)^{1/3}, \quad t_h = r_h^2.$$

The environment restrictions below are imposed on dyadic bins for h ; using an endpoint of a bin in place of h only changes fixed constants. Fix the low-likelihood exponent $C_2 > 0$ and a desired negligible error order n^{-M} . Choose C_{rate} with fixed slack beyond the likelihood exponent $4 + C_2$ and the subsequent Markov losses. The remaining constants will be chosen after the following static consequences are stated. For every required fixed rate C_{rate} , the preceding static estimates have the following consequences. All integrated probabilities in this list use planted disorder and reference observations with the spectral restriction imposed; the first also draws two conditionally independent replicas from μ_b .

1. For $0 \leq b \leq c_1 t_h$, where $c_1 \leq 1$, a reference posterior pair has $|\langle X^1, X^2 \rangle| \leq C_1 n r_h$, except with integrated probability at most $e^{-C_{\text{rate}} h}$.
Here C_1 need not increase as c_1 decreases.
2. On each fixed-ratio window $[c_1 t_h, L_1 t_h]$, with $0 < c_1 < L_1 < \infty$, the replica log-moment estimate (6.2) holds with time b and scale $n b^{3/2}$, with the required small off-diagonal tilts and any prescribed fixed relative error, outside integrated probability $e^{-C_{\text{rate}} h}$.
3. By choosing L_1 large enough, the sharp bound $\|\Sigma_b\| \leq (1 + \epsilon)/b$ on $[L_1 t_h, s_0]$ has the same failure rate. Beyond s_0 , use the bounded positive-time covariance, including the terminal extension. At E , the product field satisfies $\max_i |(h_E)_i| \leq C' \sqrt{h}$ with that rate, also when the exceptional probability is weighted by $1 + \max_i |(h_E)_i|$.

For completeness, the first assertion does not need a lower spherical width bound at very early times. Impose the upper width bound at scale t_h with an arbitrarily large fixed rate. The spherical length-deficit estimate then controls the pair integral relative to $Z_o(h_b)^2$, using (6.3), the field-norm bound, and the large-overlap comparison. Exact shell conditioning costs only a polynomial by the spectral and spherical estimates; translations improve the required small-ball bound.

The trace loss is estimated at shifts above the upper width bound. Orientations with $Z(h_b)/Z_o(h_b) < e^{-C_{\text{rate}} h}$ have at most that planted mass conditional on the eigen-coordinates, so division by the partition function is absorbed by an increase of the rate budget. The second and third assertions follow from the spherical width tails, (6.2), (12.5), and the positive-time inputs. For the last field maximum, Gaussian tails and a union bound apply; under planting, the RX term has Gaussian site marginals with bounded translates. Gaussian moment bounds give the assertion with the extra maximum-field factor.

Complete the fixed parameter choices before constructing the disorder event. Obtain C_1 at the chosen rate and then take c_1 small enough for (14.7); C_1 does not increase under this choice. Next enlarge L_1 . At the start of its sharp window the covariance exponent is

$$an(L_1 t_h)^{3/2} = aL_1^{3/2}h,$$

so L_1 can pay the required rate. Then choose the fixed replica number and its relative accuracy for the intermediate window. Finally decrease c' : a positive-time exception e^{-cn} has exponent at least $(c/c')h$, and this choice also keeps $L_1 t_h \leq s_0$ and all scales in their allowed ranges. Increase K' to absorb polynomials and the logarithmic threshold. We may require all discarded errors below to be $O(n^{-M})$ for the fixed sufficiently large M chosen above.

With these choices fixed, choose the disorder event on which the support profile will hold. Entropy levels have been placed in dyadic bins; in each bin integrate the conditional exceptions over its three time windows. This construction uses the bin and fixed tolerances, before any test function is chosen. For a field failure B_b , write $q_J(b) = \mathbb{P}_J^\mu(B_b)$ for its reference-field probability conditional on J . For a pair failure, $q_J(b)$ denotes the conditional field average of its $\mu_b^{\otimes 2}$ probability. The ordinary high-probability lower bound $Z(0) \geq n^{-C}e^{n/4}$, together with $\mathbb{E}Z(0) = e^{n/4}$, transfers planted exceptions on the spectral event to ordinary exceptions at polynomial cost. Indeed, with $L(J) = Z(0)/\mathbb{E}Z(0)$,

$$\mathbb{E}_{\text{ord}}[\mathbf{1}_{\{\mathcal{G}, L \geq n^{-C}\}} q_J(b)] \leq n^C \mathbb{E}_{\text{pl}}[\mathbf{1}_{\mathcal{G}} q_J(b)].$$

Integrate over each specified time window, apply Markov with unused rate slack, and take a union over the $O(\log n)$ bins.

This produces one typical disorder event, independent of f , on which all required conditional time-integral exceptions hold. There is enough unused rate to absorb the later likelihood factors. The chosen K' makes their products with any specified polynomial in n and $e^{C_0 h}$ negligible, for each of the finitely many constants C_0 used below. Endpoint tests are treated in the same way.

On the early window, the pair bound implies

$$\mathbb{E}^\nu \|a_b\|^2 \leq Cnr_h + \text{an error negligible in time integral.} \quad (14.6)$$

Indeed the square of a posterior mean is bounded by the posterior expectation of the absolute pair inner product. To apply this also to $\nu_b^{\otimes 2}$, discard $w_b < e^{-C_2 h}$: its probability under ν is at most $e^{-C_2 h}$, and its squared-drift contribution is at most $4ne^{-C_2 h}$. On the complement, the density of the ν -field and two ν_b replicas relative to the reference field and two μ_b replicas is exactly

$$w_b \frac{f^2(X^1)}{w_b} \frac{f^2(X^2)}{w_b} = \frac{f^2(X^1)f^2(X^2)}{w_b} \leq 10^4 e^{(4+C_2)h}.$$

The ordinary posterior mean evaluated under the ν -field law is controlled in the same way. Taking $c_1 > 0$ sufficiently small, (14.1), (14.5), and $nr_h t_h = h$ now give

$$H_{c_1 t_h} \geq h/2. \quad (14.7)$$

On the intermediate window, use l replicas in (6.2) with off-diagonal matrix entries δ/l , where $\delta > 0$ is sufficiently small and independent of l . The entropy variational inequality applied to $\nu_b^{\otimes l}$ gives a multiple of $b\|\nu_b(X) - m_o\|^2$ bounded by a constant times $\text{KL}(\nu_b\|\mu_b)$ and the log-moment cost divided by $\delta(l-1)$. Indeed the expectation of the tilt is exactly $b\delta(l-1)\|\nu_b(X) - m_o\|^2/2$, whereas the relative entropy is $l\text{KL}(\nu_b\|\mu_b)$.

The squared Hilbert–Schmidt norm of the tilt matrix is bounded by δ^2 , uniformly in l . Apply the same inequality to ordinary replicas, whose entropy cost is zero, and use $\|a_b\|^2 \leq 2\|\nu_b(X) - m_o\|^2 + 2\|\mu_b(X) - m_o\|^2$. After averaging over the ν -field law, this yields

$$b\mathbb{E}^\nu\|a_b\|^2 \leq C_\delta H_b + \zeta n b^{3/2} + \text{negligible integrated error.} \quad (14.8)$$

Here $\zeta > 0$ can be as small as desired by first taking a sufficiently large fixed l and then the relative log-moment error sufficiently small.

All exceptional field likelihoods are paid for by the same sup-norm bound. Gronwall on this fixed-ratio window, followed by a sufficiently small choice of ζ , gives

$$H_{L_1 t_h} \geq ch. \quad (14.9)$$

The positive constant may depend on the fixed ratio L_1/c_1 , which causes no loss of a power of n .

For later times let c_v be the good covariance bound: $c_v = (1 + \epsilon)/v$ up to s_0 , and a fixed constant afterwards. Set $M_v = \mathbb{E}^\nu\|\Lambda_v^{1/2}a_v\|^2$. Conditional Jensen in (14.1), then Cauchy–Schwarz including its endpoint term, gives

$$\begin{aligned} (\mathbb{E}^\nu\|a_b\|^2)^{1/2} &\leq \sqrt{2H_E} + \int_b^E c_v \|\Lambda_v\|^{1/2} \sqrt{M_v} dv + n^{-M} \\ &\leq \sqrt{2H_b} \left(1 + \int_b^E c_v^2 \|\Lambda_v\| dv\right)^{1/2} + n^{-M}. \end{aligned} \quad (14.10)$$

We used $2H_E + \int_b^E M_v dv = 2H_b$.

The deterministic bounds $\|\Sigma_v\| \leq n$, $\|a_v\| \leq 2\sqrt{n}$ control the discarded integrals at polynomial cost. Squaring introduces only another negligible error, since all remaining factors are polynomially bounded on this range. As $\Lambda_b = I$ below s_0 , the coefficient in the resulting differential inequality for H_b is at most

$$\frac{(1 + \epsilon)^2}{b} + C \quad (L_1 t_h \leq b \leq s_0),$$

and is bounded afterwards. Gronwall and (14.9) imply, with $\kappa = (1 + \epsilon)^2$,

$$H_E \geq ct_h^\kappa h - n^{-M+O(1)}. \quad (14.11)$$

For a Bernoulli law with smaller atom z , the elementary two-point inequality is

$$\text{Ent}(u^2) \leq C \log(1/z) \text{Var}(u).$$

One may first replace u by $|u|$. For comparable nonnegative values use $\text{Ent}(u^2) \leq \text{Var}(u^2)/\mathbb{E}u^2$; if the values are not comparable, bound the entropy by the larger value squared times binary entropy and compare the difference of the values. For a spin of field v , $\log(1/z) \leq C(1 + |v|)$. Tensorizing at the product endpoint and using (14.3) therefore gives

$$H_E \leq C\mathbb{E}^\mu \left[(1 + \max_i |(h_E)_i|) \mathcal{D}_E(f) \right] \leq C'\sqrt{h} \mathcal{D}_J(f) + n^{-M}. \quad (14.12)$$

To justify the correlated exceptional term, put $B_E = \{\max_i |(h_E)_i| > C'\sqrt{h}\}$. Pointwise, $\mathcal{D}_E(f) \leq n\mu_E(f^2) = nw_E \leq n\|f\|_\infty^2$. Hence its contribution is at most

$$n\|f\|_\infty^2 \mathbb{E}^\mu \left[(1 + \max_i |(h_E)_i|) \mathbf{1}_{B_E} \right]. \quad (14.13)$$

The weighted endpoint exception and the restricted supremum bound make this negligible. On B_E^c , pull out $C'\sqrt{h}$ and use (14.3).

Combining (14.11) and (14.12) yields

$$\mathcal{D}_J(f) \geq cn^{-2\kappa/3}h^{1/2+2\kappa/3}.$$

At $\kappa = 1$ the right side is $cn^{-2/3}h^{7/6}$. The initial choice of ϵ in terms of ω proves (14.4) in this range, since $h \geq 1$ and the extra exponent of n can be less than ω .

Small entropy. In this range the test likelihood is only polynomially large, so the bottom-scale estimate can absorb its exceptional contribution. We use variance rather than entropy; the accumulated covariance bound has the leading exponent needed for the support profile. For $h < K' \log n$, use $V_b = \mathbb{E}^\mu \text{Var}_{\mu_b}(f)$ up to T . Cauchy–Schwarz on the support gives $V_0 \geq 1 - p \geq 1/2$. Filtering and the covariance bound imply

$$-V'_b = \mathbb{E}^\mu \|\text{Cov}_{\mu_b}(f, X)\|^2 \leq c_b V_b + \text{negligible integrated error}.$$

Here the exceptional errors can be made arbitrarily small polynomially: the test bound is polynomial in n , (12.8) has any required inverse-polynomial failure rate, and (12.5) is used only above a sufficiently large logarithmic threshold.

Specifically take $b_* = (K'' \log n/n)^{2/3}$, with K'' enlarged as necessary. Below b_* use $c_b \leq Cn^{2/3}(\log \log(n + 1000))^C$ from (12.8); its integral is $o(\log n)$. Between b_* and s_0 use $(1 + \epsilon)/b$, and then the bounded positive-time input. Thus

$$\int_0^T c_b db \leq \frac{2}{3}(1 + \epsilon) \log n + o(\log n).$$

The same transfer and time-integrated Markov argument supplies one typical environment event for these assertions. Gronwall gives $V_T \geq n^{-2/3-O(\epsilon)-o(1)}$. The terminal gap and (14.3) give $V_T \leq C\mathcal{D}_J(f) + n^{-M}$. The initial choice of ϵ leaves enough power slack to absorb $h \leq K' \log n$ and obtain (14.4).

Entropy of order n . It remains to treat $h > c'n$. The maximum test likelihood no longer controls the fixed positive-time exceptions. Instead, we charge those exceptions to the relative entropy spent in changing the observation law. The resulting differential inequality forces that spent entropy to vanish at leading order. We show that, for every fixed $C_* > 0$, with probability tending to one no test function in this range can have $\mathcal{D}_J(f) \leq C_*n^{1/3}$. Taking $C_* = 1$ implies the desired estimate because $h \leq (3 + \log 2)n$.

The active class throughout this paragraph is still

$$\mu(f^2) = 1, \quad p = \mu(f \neq 0) \leq \frac{1}{2}, \quad \|f\|_\infty^2 \leq 100p^{-2}, \quad h = \log(1/p) > c'n.$$

We choose the diagnostics before selecting a violating function. Take a countable list of compact positive-time bands, integer rate multipliers A_* , the fixed dyadic outer-threshold grids from Section 13, and residual and mean tolerances tending to zero on each band. For each fixed finite collection, the integrated planted bounds, the size-bias lower bound, and Markov with unused rate slack give an ordinary disorder event of probability tending to one. Intersect it with the finite ordinary word diagnostics, the original spectral restriction used to define R , and the ordinary disorder event in Theorem 13.6. This intersection still has ordinary probability tending to one, and all its restrictions

are independent of f . On this event the corresponding conditional reference-field bounds hold outside time sets whose Lebesgue measure tends to zero.

If the desired conclusion failed with probability bounded away from zero, diagonal selection would therefore give sizes tending to infinity, environments satisfying successively more of these finite collections, and a violating restricted function on each. All diagnostics in this selection precede that function. Work along such a sequence and define

$$e_n(b) = \frac{H_0 - H_b}{n}, \quad 0 \leq b \leq T.$$

These functions vanish at zero, are nondecreasing, and are 2-Lipschitz by (14.1) and $\|a_b\| \leq 2\sqrt{n}$. Passing to a subsequence, assume $e_n \rightarrow e$ uniformly. We prove $e = 0$.

For the fixed environment J and $b \in [0, T]$, let $B \in \mathcal{F}_b$ be an event of the observation history through b with $\mathbb{P}_J^\mu(B) \leq e^{-nw}$, where $w > 0$. The relative entropy of the restrictions of the two field laws to $\mathcal{F}_b$, conditional on this same J , followed by binary relative entropy, gives

$$\mathbb{P}_J^\nu(B) \leq \frac{e_n(b) + n^{-1} \log 2}{w}. \quad (14.14)$$

Indeed this restricted field relative entropy is $ne_n(b)$, and its projection to the indicator of B is at least $\mathbb{P}_J^\nu(B)nw - \log 2$.

At small fixed positive times apply (13.1) with any fixed large rate multiplier A_* . Its bad event has $w = A_*b^{3/2}$ in (14.14), after using rate slack in the environment transfer. On its complement, (13.1) implies the whole-region stability at $A = V_y$ required by (13.2); that mean estimate uses no overlap lower bound. It makes $\mathbb{E}^\nu[\|a_b\|^2/n]$ arbitrarily small as $n \rightarrow \infty$ and then its tolerance tends to zero. To see the error scale explicitly, the additive energy error in that estimate has expectation at most

$$C_\zeta n^{-1/2}(1 + \mathcal{D}_J(f)) = O_{\zeta, C_*}(n^{-1/6})$$

by (14.3). Since $\|a_b\|/\sqrt{n} \leq 2$, this error is charged linearly when estimating the squared drift.

On the bad event use the approximate-zero estimate (13.3). Write Q_b for the supremum of $q(y)$ over the sufficiently-small-residual points occurring there. It yields, in squared units, an upper bound CQ_b plus the same vanishing error. For $C\sqrt{b} \leq u \leq u_*$, the outer estimate supplies $\mathbb{P}_J^\mu(Q_b > u) \leq e^{-cnu^3}$. On a compact positive-time band, and with finitely many thresholds, one may use a single residual tolerance smaller than all those provided: an approximate zero at this tolerance remains one at each larger tolerance.

Put $R_b = C'A_*^{1/3}\sqrt{b}$, taking C' large enough for the outer range and the initial interval short enough that $R_b \leq u_*$. If B_b is the stability failure, then $0 \leq Q_b \leq 1$ gives

$$\begin{aligned} \mathbb{E}^\nu[Q_b \mathbf{1}_{B_b}] &\leq R_b \mathbb{P}_J^\nu(B_b) + \sum_{\substack{j \geq 0 \\ 2^j R_b < u_*}} 2^{j+1} R_b \mathbb{P}_J^\nu(Q_b > 2^j R_b) \\ &\quad + \mathbb{P}_J^\nu(Q_b > u_*). \end{aligned}$$

On a fixed compact band the moving thresholds are bracketed by the finite fixed dyadic grid described in Section 13, losing at most a fixed factor in their cubic rates. Binary transfer makes the first term at most $CR_b e(b)/(A_*b^{3/2})$ in the limit, while the sum is at most

$$Ce(b) \sum_{j \geq 0} \frac{2^j R_b}{(2^j R_b)^3} \leq Ce(b)/R_b^2.$$

The final saturated tail uses the rate at u_* and contributes at most $Ce(b)/u_*^3 \leq C_{u_*}e(b)/R_b^2$. Thus no outer rate beyond the supported small range is used. Consequently, in integral form on every compact band in the initial interval,

$$e'(b) \leq CA_*^{-2/3} \frac{e(b)}{b}. \quad (14.15)$$

The constant C is independent of A_* . To see this, first fix A_* , then take $n \rightarrow \infty$, and then send the mean tolerance to zero. The stability margins and the refined mean constants, which may depend on A_* , disappear in this order. The remaining coefficient comes from the absolute leading constant 2 in (13.3), the absolute outer-tail constants, and the fixed u_* .

Choose A_* so large that $\theta = CA_*^{-2/3} < 1$. For $0 < a < b$ in this interval, Gronwall and $e(a) \leq 2a$ imply $e(b) \leq 2b^\theta a^{1-\theta}$; letting $a \downarrow 0$ proves $e(b) = 0$. On the remaining positive times, the fixed-rate version of (13.1), the good-field estimate (13.2), and (14.14) give $e' \leq Ce$ on each compact band. Therefore $e = 0$ through T .

The selection at the start of this paragraph licenses these inequalities in integral form on every selected compact band. The omitted time sets have vanishing measure, and the uniform Lipschitz bound on e_n makes their contribution vanish. Thus the limiting conclusion $e = 0$ applies to the selected sequence of violating functions.

As $e(T) = 0$, (14.5) gives $H_T \geq c'n - o(n)$. During the terminal extension the bounded covariance input holds with exponentially small exceptions even for an arbitrary input spin law. It therefore applies directly to the ν -field law. The proof of (14.10) on this fixed interval gives $H_E \geq cH_T - o(n) \geq c''n$. At the product endpoint it is enough to use $\max_i |(h_E)_i| \leq C_B\sqrt{n}$, with exception at most e^{-Bn} for any fixed B . This follows from $\|R\| = O(1)$, $\|X\| = \sqrt{n}$, and Gaussian tails.

Choose B to pay the restricted bound $f^2 \leq 100e^{2h}$. The deterministic estimate in (14.13), now with threshold $C_B\sqrt{n}$, and the corresponding weighted Gaussian tail control the correlated exceptional energy. The same product entropy estimate now gives

$$H_E \leq C\sqrt{n} \mathcal{D}_J(f) + o(1) \leq CC_*n^{5/6} + o(1) = o(n),$$

a contradiction. This completes all three ranges and the proposition. $\square$

14.3 Removing the bound on the test function

Theorem 14.2 (Uniform support profile). *For every fixed $\omega > 0$ there are $c_\omega > 0$ and disorder events of probability tending to one on which, simultaneously for every nonzero real function f with $p = \mu_J(f \neq 0) \leq 1/2$,*

$$\mathcal{D}_J(f) \geq c_\omega n^{-2/3-\omega} \log(1/p) \mu_J(f^2). \quad (14.16)$$

In particular the spectral gap for rate-one-per-site heat-bath dynamics is at least $c_\omega(\log 2)n^{-2/3-\omega}$ on the same event.

Proof. The restricted profile holds for every admissible test on one disorder event. We may therefore apply it to all dyadic pieces of an arbitrary function on that same event. Pieces too small to meet the likelihood bound will lose only a fixed fraction of the squared norm, and their combined heat-bath energy is bounded by the original energy.

It suffices to consider $f \geq 0$: replacing f by $|f|$ preserves its support and squared norm and cannot increase any conditional-variance form. For every integer j put

$$g_j = \min\{(f - 2^j)_+, 2^j\}, \quad A_j = \mu(g_j^2), \quad p_j = \mu(f > 2^j).$$

For a scalar $u \geq 0$ the same truncations satisfy

$$\frac{u^2}{4} \leq \sum_{j \in \mathbb{Z}} \min\{(u - 2^j)_+, 2^j\}^2 \leq u^2. \quad (14.17)$$

For $2^m \leq u \leq 2^{m+1}$ the sum is $4^m/3 + (u - 2^m)^2$, which verifies both bounds; $u = 0$ is immediate. Thus $S := \sum_j A_j$ is finite and lies between $\mu(f^2)/4$ and $\mu(f^2)$.

The truncation increments also obey, for $u \geq v \geq 0$,

$$d_j := \min\{(u - 2^j)_+, 2^j\} - \min\{(v - 2^j)_+, 2^j\} \geq 0, \quad \sum_j d_j = u - v.$$

The sum identity follows by integrating the indicators of the disjoint intervals $(2^j, 2^{j+1})$. Hence $\sum_j d_j^2 \leq (u - v)^2$. Using the representation of conditional variance by two independent conditional samples gives

$$\sum_j \mathcal{D}_J(g_j) \leq \mathcal{D}_J(f). \quad (14.18)$$

All sums are justified by nonnegativity and monotone convergence.

Discard indices with $A_j < 4^j p_j^2/100$, as well as $p_j = 0$. On $\{f > 2^j\}$ the preceding piece g_{j-1} is saturated, so $A_{j-1} \geq 4^{j-1} p_j$. Since $p_j \leq p \leq 1/2$, the discarded pieces satisfy

$$\sum_{j \text{ discarded}} A_j \leq \sum_j \frac{4^j p_j^2}{100} \leq \frac{1}{50} \sum_j A_{j-1} = \frac{S}{50}.$$

Every remaining normalized piece $g_j/\sqrt{A_j}$ has squared supremum at most $4^j/A_j \leq 100/p_j^2$. Proposition 14.1, on its single event valid for all test functions, therefore gives

$$\begin{aligned} \mathcal{D}_J(f) &\geq c_\omega n^{-2/3-\omega} \sum_{j \text{ remaining}} A_j \log(1/p_j) \\ &\geq \frac{49}{200} c_\omega n^{-2/3-\omega} \log(1/p) \mu(f^2). \end{aligned}$$

Renaming the constant proves (14.16).

Finally, for an arbitrary real u choose a μ -median m , so that $u_+ = (u - m)_+$ and $u_- = (m - u)_+$ both have support mass at most $1/2$. For each pair of values the squared increments of these two functions sum to at most the squared increment of u . Thus the support profile gives

$$\mathcal{D}_J(u) \geq \mathcal{D}_J(u_+) + \mathcal{D}_J(u_-) \geq c_\omega n^{-2/3-\omega} \log 2 \mu((u - m)^2) \geq c_\omega n^{-2/3-\omega} \log 2 \operatorname{Var}_\mu(u).$$

This is the asserted gap bound. □

The support profile and its Poincaré consequence complete the first functional route. Section 20 converts this endpoint to worst-start mixing in both time conventions.

15 Spectral caps and comparison with spherical spins

The cap route first compares partition functions and projected means with a Gaussian measure conditioned on a sphere. The comparison is uniform over macroscopic balls of edge fields. Its role is to control the values and slopes needed when differentiating a Gaussian observation convolution.

15.1 Observation and scale conventions

For a symmetric matrix A and field h , write

$$Z_A(h) = 2^{-n} \sum_{x \in \{-1,1\}^n} \exp\{\frac{1}{2}x^\top A x + h^\top x\}, \quad \phi_A(h) = \log Z_A(h). \quad (15.1)$$

We write $Z_p = Z_{J_p}$ once J_p has been defined, and use Z_A^{sp} for the analogous integral against uniform probability on the sphere of radius $\sqrt{n}$. Gradients and Hessians of ϕ_A are the Gibbs mean and covariance.

Here are the conventions for all cap sections. Every tolerance described as arbitrarily small is a fixed positive number chosen before $n \rightarrow \infty$. We write η_{err} for such a tolerance, which can be decreased at later finite stages of the parameter selection; η_{small} has the same meaning when a separate structural slack is needed. Neither symbol denotes a rate in n . Once the scale ratios, field ranges, and error tolerances have been fixed, the upper scale is chosen small enough, and only then is n taken large. Uniform $o(1)$ terms refer to this order of limits. High probability, unless otherwise specified, only means probability tending to one. Variables called fields are in energy units. A Gaussian observation with precision B is the field increment $Bx + N(0, B)$. Its effect on posterior log weights is to add the field and subtract B from the interaction matrix. One can use increasing precisions and independent noise increments (including Brownian observations). The logarithms of partition functions before an increment are obtained from those after it by log heat convolution. Additive constants in potentials don't affect gradients, heat-tilted transition probabilities or Hessians. We use Euclidean matrix and vector norms; empirical averages have brackets, and sometimes we use $|v|_n = |v|/\sqrt{n}$.

The edge coordinates are eigenvectors with gap $d_i = 2 - \lambda_i$, in order of increasing gap. We condition in spectral-coordinate arguments on the eigenvalues, leaving the frame Haar. The observation scheme is given by the spectral functions

$$P_p = 1_{\{d \leq p^2\}}, \quad k_p = np^3, \quad M_p = \sqrt{np}, \quad B_p = (p^2 - d)_+, \quad J_p = J - B_p. \quad (15.2)$$

Write $H_p = \text{ran } P_p$ and $d_p = \dim H_p = \text{Tr } P_p$; the spectral estimates below give $d_p \asymp k_p$. We use fields in H_p , and occasionally write P_p for this space. Field length units there are $p^2 M_p$; free-energy and projected-spin length units are k_p, M_p . The parameter p is small. Conditionally on J_p and the eigenvalues, refinements determining smaller caps use a fresh Haar rotation within P_p .

15.2 Spectral estimates at the bottom scale

The resolvent event of Lemma 5.1 covers the entire cap range. Indeed, if $np_{\min}^3 \asymp \sqrt{\log n}$, then

$$\frac{p_{\min}}{d_0} \asymp \frac{(\log n)^{1/6}}{\log \log(n + 1000)} \rightarrow \infty.$$

Thus the lemma applies at $d = \sqrt{u}$ for every $u \geq cp^2$, where $p \geq p_{\min}$ and $c > 0$ is fixed. In particular, on one ordinary disorder event of probability tending to one,

$$d_1 \geq -o(p^2), \quad \|J\| \leq 3, \quad \#\{i : d_i \leq u\} \leq Cnu^{3/2} \quad (p^2 \leq u \leq 4).$$

The count follows by applying the lemma on a dyadic grid in $\sqrt{u}$ and using monotonicity between grid points. The ordinary empirical spectral measure also converges to the semicircle law.

Write $m_J(u) = n^{-1} \text{Tr}(d+u)^{-1}$. The same lemma and $b_{\text{sc}}(2+u) = 1 - \sqrt{u} + O(u)$ give, uniformly for small $u \geq cp^2$,

$$m_J(u) = b_{\text{sc}}(2+u) + o\left(\frac{\sqrt{u}}{\sqrt{nu^{3/2}}}\right) = 1 - (1 + o(1) + O(\sqrt{u}))\sqrt{u}.$$

Here $nu^{3/2} \geq c^{3/2}np_{\min}^3 \rightarrow \infty$.

We will also use the precise edge counting measure. Put

$$\nu_p = \frac{1}{k_p} \sum_i \delta_{d_i/p^2}.$$

For fixed $v, w > 0$, its transform difference is exactly

$$\int \left(\frac{1}{x+v} - \frac{1}{x+w} \right) d\nu_p(x) = \frac{m_J(vp^2) - m_J(wp^2)}{p} = \sqrt{w} - \sqrt{v} + o(1) + O(p).$$

The limiting expression equals the same integral against $\pi^{-1}\sqrt{x}\mathbf{1}_{\{x>0\}}dx$. The count bound above controls the tail of the integrand, which is $O(x^{-2})$, by $O(R^{-1/2})$ beyond R . Since the negative support lies in $[-o(1), 0]$, the measures $(1+x)^{-2}\nu_p$ are tight. Uniqueness of their Stieltjes transforms therefore yields

$$\nu_p \longrightarrow \pi^{-1}\sqrt{x}\mathbf{1}_{\{x>0\}}dx \quad \text{on bounded intervals.}$$

All these statements are uniform in the cap range in the convention that $n \rightarrow \infty$ after the fixed upper cutoff on p is chosen, and then that cutoff may be decreased. For example, a failure of uniformity would give a sequence of cap scales on which the uniform resolvent estimate and the same tightness argument give the displayed limit, a contradiction.

The spectral event is imposed once for all scales. Later failures of size e^{-ak_p} on a geometric cap list are also summable:

$$\sum_{j \geq 0} e^{-a\gamma^{-3j}k_{\min}} = o(1), \quad k_{\min} \asymp \sqrt{\log n}.$$

15.3 Spherical saddles, norm densities, and tails

We compare several times with spins uniform on the Euclidean sphere of radius $\sqrt{n}$, with superscript sp for this change. Write $\mu_{p,h}$ and $\mu_{p,h}^{\text{sp}}$ for the cube and spherical Gibbs probabilities normalized by $Z_p(h)$ and $Z_p^{\text{sp}}(h)$. At the cap p with field $h \in H_p$ in any fixed bounded range of field units, write

$$R_{p,\alpha} = (\max(d, p^2) + \alpha)^{-1}, \quad a_{p,\alpha} = R_{p,\alpha}h, \quad n = \text{Tr } R_{p,\alpha} + \|a_{p,\alpha}\|^2. \quad (15.3)$$

This fixes the spectral parameter $\alpha = \alpha_p(h)$. At this solution we abbreviate $R_p = R_{p,\alpha}$ and $a_p = a_{p,\alpha}$. For any admissible α , let $f_{p,\alpha,h}^{\text{norm}}$ denote the density of $\|N(R_{p,\alpha}h, R_{p,\alpha})\|^2$. The spherical posterior is $N(a_p, R_p)$ conditioned on squared norm n . One has $-1 + c \leq \alpha/p^2 \leq C$ by the spectral facts. Conversely lengths for $\alpha = 0$ satisfy, with $b = 4/(3\pi)$,

$$n - \text{Tr } R_p = (b + o(1))np, \quad \text{Tr } R_p^2 = (2b + o(1))n/p. \quad (15.4)$$

Changing α by a small fraction of p^2 perturbs these by relative small amounts. For bounded arguments in scaled units the saddle and its inverse (squared field length as function of parameter, where nonnegative) are uniformly Lipschitz in scaled variables.

We record density-estimate details used below. Changing quadratic and linear tilts that retain a comparable spectral band, at fixed α , uses ordinary Gaussian mgfs times the ratio of squared-norm densities at n , by polar coordinates. At the cap saddle the density at n is comparable to $\sqrt{p/n}$. Indeed the norm-square variance is of order n/p ; in standard deviation units the characteristic function converges locally to $e^{-t^2/2}$ after centering (Taylor expansion, max Gaussian coordinate variance divided by norm-square standard deviation $O(1/\sqrt{k_p})$). Absolute values are bounded by $(1 + ct^2/k_p)^{-ck_p}$ from top components alone, also in the noncentral case. This proves the assertion by Fourier inversion. The sphere restriction is imposed by conditioning a Gaussian squared norm. To transfer an exponential-moment estimate through this conditioning, we need a lower density at the original saddle and an upper density after the additional tilt, both on the same scale. The characteristic function below supplies these two bounds. Its central limit expansion gives the density near the mean, while an entire spectral band supplies an integrable bound away from the origin.

In diagonal coordinates, for $Y_i = a_i + \sqrt{r_i}g_i$ the formula used is

$$\left| \mathbb{E} e^{it \sum_i Y_i^2} \right| = \prod_i (1 + 4t^2 r_i^2)^{-1/4} \exp \left\{ - \sum_i \frac{2t^2 r_i a_i^2}{1 + 4t^2 r_i^2} \right\}. \quad (15.5)$$

A band of order k_p coordinates with $r_i \asymp p^{-2}$ supplies the integrable majorant, and the remaining factors have absolute value at most one. The variance formula $\text{Var}(\sum_i Y_i^2) = 2 \text{Tr } R_p^2 + 4a_p^\top R_p a_p$ sets the scale n/p . This proves the centered lower density bound. It also gives the noncentral upper bound for a linear tilt, and for a quadratic tilt whenever the new covariance retains a band with the same size and comparable eigenvalues.

The same argument applies if an extra scalar multiple of $p^2 P_p$ is added to the interaction keeping comparable gaps at the saddle. Upper bounds suffice without the exact trace constraint or centering.

We will use the corresponding bound for a fixed number m of replicas. Let independent spectral rows $Y_i \in \mathbb{R}^m$ have laws $N(a_i, C_i)$ with $C_i \succ 0$, and put $\mathcal{Q}_j = \sum_i Y_{ij}^2$. For $T_t = \text{diag}(t_1, \dots, t_m)$, $H_i = C_i^{1/2} T_t C_i^{1/2}$, and $\beta_i = C_i^{-1/2} a_i$, the Gaussian quadratic formula gives

$$\left| \mathbb{E} e^{i Y_i^\top T_t Y_i} \right| = \det(I + 4H_i^2)^{-1/4} e^{-2\beta_i^\top H_i^2 (I + 4H_i^2)^{-1} \beta_i} \leq \det(I + 4H_i^2)^{-1/4}.$$

Suppose $L > 2$ rows satisfy $cs^{-1}I_m \preceq C_i \preceq Cs^{-1}I_m$. The singular values of H_i are at least cs^{-1} times those of T_t , even when T_t has both signs. Independence therefore yields

$$\left| \mathbb{E} e^{it \cdot \mathcal{Q}} \right| \leq \prod_{j=1}^m (1 + ct_j^2/s^2)^{-L/4}.$$

Each one-dimensional integral is at most $Cs/\sqrt{L}$. Fourier inversion then proves

$$\sup_q f_{\mathcal{Q}}(q) \leq C_m(s/\sqrt{L})^m \leq C_m(s^{1/4}/\sqrt{n})^m \quad \text{if } L \asymp ns^{3/2}. \quad (15.6)$$

Independent replicas at their original cap saddles have joint density at $n\mathbf{1}$ at least $c_m(p/n)^{m/2}$. Thus an additional Gaussian tilt that preserves the displayed band costs at most $C_m(\sqrt{s}/p)^{m/2}$ in the norm-density ratio.

The density in this last comparison must be the density under the normalized tilted law. Indeed, for $0 \leq F \leq 1$ with $\mathbb{E}F > 0$,

$$\mathbb{E}[F \mid \mathcal{Q} = n\mathbf{1}] = \mathbb{E}F \frac{f_{\mathcal{Q}}^F(n\mathbf{1})}{f_{\mathcal{Q}}(n\mathbf{1})}, \quad d\mathbb{P}_F = \frac{F d\mathbb{P}}{\mathbb{E}F},$$

where f_Q^F is the norm density under $\mathbb{P}_F$; at $n\mathbf{1}$ we use the representatives given by polar coarea. For a tail $F = \mathbf{1}_{\{\mathcal{T} \geq b\}}$, exponential Markov gives the useful polar-conditioning form

$$\Pr(\mathcal{T} \geq b \mid \mathcal{Q} = n\mathbf{1}) \leq e^{-\lambda b} \mathbb{E} e^{\lambda \mathcal{T}} \frac{f_Q^{(\lambda)}(n\mathbf{1})}{f_Q(n\mathbf{1})}, \quad d\mathbb{P}^{(\lambda)} = \frac{e^{\lambda \mathcal{T}} d\mathbb{P}}{\mathbb{E} e^{\lambda \mathcal{T}}}. \quad (15.7)$$

Here the conditional quantities are defined by polar disintegration. For a rowwise linear or quadratic $\mathcal{T}$, the tilted law remains Gaussian: a factor $e^{Y_i^\top A_i Y_i / 2 + v_i^\top Y_i}$ changes the covariance to $(C_i^{-1} - A_i)^{-1}$. A strict positive precision margin that keeps a full band comparable to $s^{-1}I_m$ permits (15.6) for $f_Q^{(\lambda)}$. For a general projected test, its spherical failure probability will instead be an explicit hypothesis in the test-transfer argument below.

We also record the exact linear moment bound at a cap saddle. Polar conditioning with the parameter frozen at $\alpha = \alpha_p(h)$ gives

$$\log \mathbb{E}_{\mu_{p,h}^{\text{sp}}} e^{v^\top X} = a_p^\top v + \frac{1}{2} v^\top R_p v + \log \frac{f_{p,\alpha,h+v}^{\text{norm}}(n)}{f_{p,\alpha,h}^{\text{norm}}(n)} \leq a_p^\top v + \frac{1}{2} v^\top R_p v + C_{\text{dens}}. \quad (15.8)$$

The denominator uses the centered lower bound and the numerator the noncentral upper bound with the same covariance. Thus C_{dens} is independent of n, p on the fixed bounded ranges used here.

Here are some tail estimates in this convention. For a fixed finite number of spherical replicas at a bounded cap tilt as above (also allowing the just-mentioned scalar modification), scaled projection norms $\|P_p x\|/M_p$ above large v cost $\exp(-ck_p v^6)$, up to a decrease in c ; pair overlaps above $r \geq Cp$, r small, cost $\exp(-c nr^3)$. The same norm tail applies for the completely uncapped zero-field model. For detail, change the Gaussian parameter upward by $s = \epsilon r^2$ for overlaps (take $r = v^2 p \leq 1$ for projection norms), ϵ small. The cost in log normalizer for this is $O(ns^{3/2})$ for $s \gg p^2$, by integrating the trace deficit; bounded cap fields add at most $O(k_p)$. At uncapped zero field one can start at parameter ρp^2 , fixed small $\rho > 0$: the cost in the lower partition bound via its norm density is at most Ck_p in addition to the log of the density scale $\sqrt{p/n}$ (fill the $O(np)$ deficit with one top Gaussian square, the other squares concentrating by Chebyshev).

In the changed Gaussian the mean contributes negligibly and the desired tails cost $c nr^3 \epsilon$, by exponential Markov with parameter $c's$ in the quadratic form (trace-square $O(n/\sqrt{s})$).

For this exponential tilt the replica precision on a band of order $ns^{3/2}$ rows remains comparable to sI , if the Markov parameter $c's$ is a sufficiently small multiple of s . Equations (15.7) and (15.6) therefore bound the joint norm-density ratio for m replicas by $C_{m,\epsilon}(r/p)^{m/2}$; for a pair it is $C_\epsilon r/p$. This introduces no separate factor in n when $r \asymp p$. For $r \geq p$ its logarithm is absorbed by the tail exponent because $nr^3 = k_p(r/p)^3$ and $k_p \rightarrow \infty$. More precisely, for fixed $a > 0$,

$$C_{m,\epsilon}(r/p)^{m/2} e^{-anr^3} \leq C_{m,\epsilon,a} e^{-anr^3/2}$$

uniformly once k_p is large. Taking ϵ small and then C large gives the assertions. For overlaps at cap saddles with zero field one also has below order p cost $\exp(O(1) - cnpr^2)$ by the same density/mgf argument at the saddle. For angular-moment comparisons at overlaps bounded away from zero we give details separately below.

15.4 Uniform comparison of values and means

Comparing the Ising and spherical models by rotation has precedents in Comets [14]; critical free-energy and overlap comparisons also appear in Du–Huang [16, 17]. The statement below includes the field and posterior-test uniformity required by the covariance argument.

Proposition 15.1 (Static cap comparison). *Fix a field radius $R < \infty$, an error tolerance $\eta > 0$, and a geometric ratio in $(0, 1)$. There is a sufficiently small upper scale $p_0 > 0$ such that, with probability tending to one, simultaneously over a geometric list of caps $p \leq p_0$ with $np^3 \geq c\sqrt{\log n}$ and all $h \in P_p \mathbb{R}^n$ satisfying $\|h\| \leq Rp^2 M_p$, one has*

$$|\log Z_p(h) - \log Z_p^{\text{sp}}(h)| \leq \eta k_p, \quad \|P_p m_p(h) - a_p(h)\| \leq \eta M_p, \quad m_p(h) = \nabla \log Z_p(h). \quad (15.9)$$

The projected norm tails proved above hold, with a decreased positive exponential constant, under these cube posteriors and under the uncapped zero-field cube posterior. The statement remains valid for any fixed finite collection of auxiliary geometric cap lists. The zero-field concentration and density estimates used in the proof also allow bounded scalar modifications of the cap interaction whose spherical saddle gaps remain comparable to p^2 .

Proof. It suffices to prove the value comparison on a slightly larger prescribed ball, with an error $\varepsilon_v k_p$ chosen below the requested value error. To see the mean implication, let $u \in H_p$ be a unit vector and choose $t > 0$ so that $h \pm tu$ stay in the larger ball. The two value bounds and (15.8) give

$$\log \mathbb{E}_{\mu_{p,h}} e^{\pm tu^\top (X - a_p)} \leq 2\varepsilon_v k_p + \frac{1}{2} t^2 u^\top R_p u + C_{\text{dens}}.$$

Jensen's inequality in the two signs yields

$$|\langle u, P_p m_p(h) - a_p \rangle| \leq \frac{t}{2} u^\top R_p u + \frac{2\varepsilon_v k_p + C_{\text{dens}}}{t}.$$

Set $t = \theta p^2 M_p$, and take u along the projected discrepancy. The saddle separation gives $p^2 \|R_p|_{H_p}\| \leq C$, while $p^2 M_p^2 = k_p$. Hence

$$\frac{\|P_p m_p(h) - a_p\|}{M_p} \leq C\theta + \frac{2\varepsilon_v + C_{\text{dens}}/k_p}{\theta}.$$

Choose θ from the desired mean tolerance, then $\varepsilon_v \ll \theta^2$, and finally k_p large. The larger ball provides the same fixed margin at boundary fields. We now prove the value comparison and the stated test transfers.

Replica Gram comparison. Haar first moments agree exactly for cube and sphere. For l -th moments, l fixed, the angular average once the replica Gram matrix is given is again the same. At off-diagonal normalized overlaps bounded by small ρ the cube Gram law is bounded by the spherical law, with factor

$$\exp(C_l + C_l n \rho^4), \quad (15.10)$$

using comparison cells of radius $O_l(1/n)$. Indeed the off-diagonal products of the signs in a row form a bounded lattice vector in fixed dimension $l(l-1)/2$, with identity covariance. Exponential tilting gives near the origin a local upper probability bound $C_l n^{-l(l-1)/4}$ times the Cramér exponential. The prefactor follows by Fourier inversion (off neighborhoods of the finite dual-lattice points in the torus one has strict decay, uniformly in small tilts). Its entropy rate agrees with $-\frac{1}{2} \log \det Q$, Q the Gram matrix with diagonal one, up to and including degree three: the only nonzero third moments are triangles, giving the cubic term minus the sum of triangle monomials. The spherical density proportional to $(\det Q)^{(n-l-1)/2}$, by successive projection of unit vectors, has the reciprocal-width prefactor. Cells of sidelength comparable to $1/n$ can be taken disjoint. Cholesky coordinates on the angular integral have bounded derivatives near the identity so change of the exponential weights on

the cells costs $O_l(1)$ in logarithm. Inequalities on projected samples used as restrictions there can correspondingly be thickened by $O_l(1/n)$ in unit-sphere norm.

For two replicas and $\rho \leq n^{-0.27}$, the binomial masses equal spherical masses on overlap cells of width $2/n$ to relative $o(1)$. At zero field the angular integrand is even increasing in absolute overlap (the sum/difference orthogonal coordinates have weights with coefficients $1+r, 1-r$); thus these cell comparisons can then use monotonicity instead of a constant-factor loss.

Zero field and overlaps bounded away from zero. Fix the eigenvalues and let $\mathbb{E}_O$ denote Haar averaging of the eigenframe. At uncapped zero field, and also for a small zero-field cap with the bounded scalar modifications above and comparable saddle separation, we have the following formulas. Here A denotes the uncapped interaction J or the indicated capped interaction.

$$\mathbb{E}_O Z_A(0) = Z_A^{\text{sp}}(0), \quad \frac{\mathbb{E}_O Z_A(0)^2}{(Z_A^{\text{sp}}(0))^2} = 1 + o(1).$$

For the second assertion, outside $n^{-0.27}$ up to a small constant, (15.10) costs less than the spherical tail gain on dyadic bands. For the uncapped sphere use the preceding overlap argument with lower-normalizer scale p taken minimal. At overlaps bounded away from zero the net exponential is strictly negative even with binomial weights. Here is one verification. In sum and difference orthogonal coordinates, upper exponents are bounded by the spherical free energies at $1+|r|$ and $1-|r|$; restriction to the orthogonal hyperplane does not change the bound by interlacing. In the limit at zero cutoff the spherical upper bound $F(t)$, exact at 1, has $F'(t) = t/2$ for $0 < t \leq 1$ and $F'(t) = 1 - 1/(2t)$ above 1. This follows by Gaussian change of density with precision $t(w - J)$, giving per-site bound $(tw - 1 - \log t - n^{-1} \log \det(w - J))/2$, optimized at $w = t + 1/t$ below one and at $w = 2 + 0^+$ above. The per-site density correction for the upper bound at strictly positive separation is $o(1)$, also in the hyperplane; equality at 1 follows by the deficit-filling lower bound. Hence $F(1+r) + F(1-r) - 2F(1) < r^2/2$, whereas the binomial cost is at least $r^2/2$. Continuity handles endpoints, and a small operator-norm change of the matrices preserves these strict bounds.

All errors giving convergence of the moment ratio are summable over our geometric list (Stirling errors negative powers of n ; tails for $|r| \geq n^{-0.27}$ exponentially small in a power of n). This proves zero-field concentration and transfers projection tails, by the first moment and Markov.

Positive moments and uniform upper bounds. The zero-field comparison supplies a denominator and removes pairs with macroscopic overlap. At a nonzero field, we first keep the replicas in the small-overlap region and compare their joint Gram law. The resulting high-moment estimates have two uses: they control the upper value of the partition function, and they transfer a posterior test whose spherical failure has a positive exponential rate. The latter use requires an additive estimate on discarded replicas after the denominator has been bounded below.

For bounded field lengths, high positive moments *restricted to small overlaps among all replicas* have normalized exponents as small as desired in units k_p , by (15.10) and the spherical tail at overlaps above Cp ; the relative error $O_l(np^4)$ here suffices. The restriction discards a negligible relative contribution with high probability in the quenched model, uniformly: pairs at overlap above fixed small cutoff have exponentially small probability at cap zero field by the preceding argument, and the field costs only $O(np^{5/2})$. One may take the upper scale small after fixing l .

Here is the normalized form of the test transfer used later. Fix the eigenvalues and a field u in the spectral copy of H_p ; in physical coordinates $J_p(O) = O J_p^{\text{diag}} O^\top$ and $h_O = Ou$. Let E_u be one of the projected-configuration events described by the Gram-cell comparison, with fixed geometric

slack. Write E_u^+ for its specified thickening: whenever a configuration satisfies E_u , all configurations in the same Gram cell satisfy E_u^+ . Define the tested, unnormalized sum

$$Z_{p,E,O}(u) = 2^{-n} \sum_x e^{x^\top J_p(O)x/2 + h_O^\top x} \mathbf{1}_{E_u}(O^\top x).$$

Its ratio to $Z_{p,O}(u) = Z_p(h_O)$ is the posterior tested probability. Assume that the spherical probability of E_u^+ is at most e^{-ck_p} . For l independent copies, let $\mathcal{Z}_{E,l}^{\text{small}}(O, u)$ be the sum defining $Z_{p,E,O}(u)^l$, restricted to pair overlaps below the fixed small cutoff. The Gram comparison and the spherical tail give

$$\frac{\mathbb{E}_O \mathcal{Z}_{E,l}^{\text{small}}(O, u)}{(Z_p^{\text{sp}}(h_O))^l} \leq \exp\{(-lc/2 + \eta_l)k_p + o_n(k_p)\}. \quad (15.11)$$

For fixed l , the exponent loss $\eta_l > 0$ can be made as small as prescribed by decreasing the upper scale; this includes the $O_l(np^4)$ Gram error. The remaining $o_n(k_p)$ is taken afterward. The spherical denominator depends only on the spectral field u . On overlap bands larger than Cp , Cauchy–Schwarz divides the spherical test cost by two and the remaining spherical overlap cost pays (15.10). This is the source of the factor $1/2$.

Let $b_l(O, u)$ be the posterior l -replica probability that some pair exceeds the fixed overlap cutoff. Directly from the decomposition of the replica sum,

$$\left(\frac{Z_{p,E,O}(u)}{Z_{p,O}(u)} \right)^l \leq \frac{\mathcal{Z}_{E,l}^{\text{small}}(O, u)}{Z_{p,O}(u)^l} + b_l(O, u).$$

On the quenched high-probability zero-field event, the pair estimate above, followed by the bounded-field change of density, makes b_l exponentially small in n , uniformly after the upper scale is chosen small for fixed l . Once the value lower bound proved below gives $Z_{p,O}(u) \geq e^{-\varepsilon_v k_p} Z_p^{\text{sp}}(h_O)$, Markov's inequality applied to (15.11) shows that

$$\Pr_O \left\{ \frac{Z_{p,E,O}(u)}{Z_{p,O}(u)} > 2e^{-c'k_p}, Z_{p,O}(u) \geq e^{-\varepsilon_v k_p} Z_p^{\text{sp}}(h_O), b_l(O, u) \leq e^{-2lc'k_p} \right\} \\ \leq \exp\{[-l(c/2 - c' - \varepsilon_v) + \eta_l]k_p + o_n(k_p)\}. \quad (15.12)$$

Choose $c' < c/2$ and $\varepsilon_v < c/2 - c'$. Increasing the fixed replica number l , and only then decreasing the upper scale, gives any prescribed fixed Haar failure rate. The separately bounded event for b_l is added after normalization, as the displayed decomposition requires. The untested positive moments give the upper value bounds.

The reusable test statement takes the spherical rate as a hypothesis. Linear and quadratic projection tests, including bounded spectral transformations, supply that rate through (15.7) when their exponential tilt retains the required precision margin. When a later concrete test also uses Gaussian observation noise, each replica receives an independent noise mark. Its application must check the same cellwise inclusion with those marks fixed; a small spherical probability alone does not extend the statement to an arbitrary bounded test.

Upper value bounds extend uniformly by convexity from a fine constant mesh on a larger ball. The zero-field bound (and evenness) then gives a Lipschitz constant in scaled units on inner balls. Posterior tests with slack and positive-rate tails can likewise use nets, by restricting the change of density to bounded projection lengths first (the removed large-length tails follow from the zero-field version and absorption). We only use nets of size $\exp(O(k_p))$.

Angular lower bounds and gradients. Upper moments alone do not provide a lower value at every field direction. We first obtain a set of directions on each length sphere where the value is large and the projected gradient is close to its prediction. The scalar modification of the cap turns a zero-field partition comparison into an average over such a sphere. A second-moment estimate then controls how small this set of directions can be.

First, on each fixed length sphere in field space, up to arbitrarily small log losses in scaled units its angular partition average has the predicted lower bound with high probability. To see this at nonzero length add a scalar interaction on P_p to J_p so that, at the parameter α predicted for the given length, the eigenvalue of R_p on P_p increases by $\|a_p\|^2/\text{Tr}P_p$. This restores the trace to n , with comparable separation. Decouple the addition by a centered Gaussian. Under the integrated spherical weight the field length thus generated concentrates exponentially at the specified length (the unconditioned generated field using the modified saddle is Gaussian, with mean zero and expected squared length precisely $\|h\|^2$; the norm constraint has at most polynomial cost in k_p , by density bounds, or by a small quadratic tilt with the bounds above). By first moment and the zero-field two-moment argument at the modified cap the same shell has nearly all the integrated cube weight. Narrowing the shell to any fixed tolerance and using the scaled Lipschitz bounds gives the claim.

Angular second moments of Z/Z^{sp} are at most $e^{\eta_{\text{err}} k_p}$ with high probability by the small-overlap comparison. Moreover the two-replica numerator cost is $e^{-c_\tau k_p}$ for

$$|(P_p x - a_p)^t (P_p x' - a_p)| > \tau M_p^2 \quad (15.13)$$

as follows. In the two independent canonical Gaussians, the exponential factor $e^{\pm \theta p^2 (P_p X - a_p)^\top (P_p X' - a_p)}$ changes each two-replica precision by an off-diagonal block of size θp^2 on H_p . For sufficiently small θ the cap band remains comparable to $p^2 I_2$, and the Gaussian log moment is at most $C\theta^2 k_p$. Equation (15.7) then gives exponent $-\theta \tau k_p + C\theta^2 k_p + O(1)$; choosing θ as a small multiple of τ gives the stated spherical rate. The Gram comparison transfers it to the angular numerator. Polynomial weights in the scaled projections are handled by the projected norm tails.

Thus there is a set of angular measure at least $e^{-\eta k_p}$, for arbitrarily small prescribed η , where the lower value holds up to error ηk_p and gradients differ from a_p in projection by any prescribed small fraction of M_p . Indeed discard the points with low value and those with large normalized mean discrepancy in the angular first moment (use the two-replica bound and Cauchy-Schwarz), then use the angular second moment. Losses in these arguments can be chosen in order after the tolerances. Length grids suffice. At these directions, the value and gradient estimates define supporting planes. Evaluating those planes at another field gives a lower bound there. The Gaussian moment-generating-function bound and the bounded saddle range show that the extra scaled loss is a small multiple of the distance plus a constant multiple of its square.

Refinement from a coarser cap. The remaining task is to reach every child field from the large-value sets just constructed. Freeze the child saddle parameter and use its canonical Gaussian observation as a reference bridge to a coarser cap. Under this reference law the parent length is predictable. Fresh Haar refinement makes the reference field close to the useful parent directions, and their supporting planes convert that distance estimate into a lower bound for the child value. The gain must pay for a net at the child scale; this is why the additional scale ratio is chosen small.

For this comparison it is useful to retain the exact normalization of the frozen Gaussian quadratic. For any admissible α , define

$$\begin{aligned} C_{n,\alpha} &= \log\left(2^{n/2} \Gamma(n/2) n^{1-n/2}\right) + \frac{n}{2}(2 + \alpha), \\ q_{u,\alpha}^{\text{fr}}(h) &= C_{n,\alpha} + \frac{1}{2} \log \det R_{u,\alpha} + \frac{1}{2} h^\top R_{u,\alpha} h. \end{aligned} \quad (15.14)$$

The determinant is on the ambient space. Coarea in squared radius gives, for every such α ,

$$\phi_u^{\text{sp}}(h) = q_{u,\alpha}^{\text{fr}}(h) + \log f_{u,\alpha,h}^{\text{norm}}(n). \quad (15.15)$$

This identity also follows by writing the Gaussian density on $\|x\|^2 = n$: its factor independent of x is $(2\pi)^{-n/2}(\det R_{u,\alpha})^{-1/2}e^{-n(2+\alpha)/2-h^\top R_{u,\alpha}h/2}$, and the squared-radius coarea factor is $\pi^{n/2}n^{n/2-1}/\Gamma(n/2)$.

To obtain a uniform lower bound at c , write $c = rp$ for a very small fixed ratio r , using an additional coarser cap list. Put $\Delta = B_p - B_c$, freeze $\alpha = \alpha_c(h_c)$, and abbreviate $R_u = R_{u,\alpha}$, $a_c = R_c h_c$. Since $R_c^{-1} = R_p^{-1} - \Delta$, Gaussian integration gives

$$q_{c,\alpha}^{\text{fr}}(h_c) = \log \int e^{q_{p,\alpha}^{\text{fr}}(h_c+z)} N(0, \Delta)(dz).$$

The corresponding probability law for the parent field is

$$\begin{aligned} \mathbf{G}_{c,p}^{h_c}(dy) &= e^{q_{p,\alpha}^{\text{fr}}(y) - q_{c,\alpha}^{\text{fr}}(h_c)} N(h_c, \Delta)(dy) \\ &= N(h_c + \Delta a_c, \Delta + \Delta R_c \Delta)(dy), \quad R_p Y \sim N(a_c, R_c - R_p). \end{aligned} \quad (15.16)$$

Equivalently, sample $X \sim N(a_c, R_c)$ without norm conditioning and set $Y = h_c + \Delta X + N(0, \Delta)$. Then $X \mid Y = y \sim N(R_p y, R_p)$. Conditioning this joint law on $\|X\|^2 = n$ gives the spherical transition and the exact density ratio

$$\frac{d\mathbf{T}_{c,p}^{\text{sp}}(y)}{d\mathbf{G}_{c,p}^{h_c}}(y) = \frac{f_{p,\alpha,y}^{\text{norm}}(n)}{f_{c,\alpha,h_c}^{\text{norm}}(n)}. \quad (15.17)$$

The numerator here is generally evaluated away from the parent's own saddle. Let $\hat{\alpha} = \alpha_p(y)$. From (15.15),

$$\log \frac{f_{p,\alpha,y}^{\text{norm}}(n)}{f_{p,\hat{\alpha},y}^{\text{norm}}(n)} = q_{p,\hat{\alpha}}^{\text{fr}}(y) - q_{p,\alpha}^{\text{fr}}(y).$$

The first α -derivative of $q_{p,\alpha}^{\text{fr}}(y)$ is $\frac{1}{2}(n - \text{Tr } R_{p,\alpha} - \|R_{p,\alpha}y\|^2)$, which vanishes at $\hat{\alpha}$. Its second derivative is $\frac{1}{4} \text{Var}_{p,\alpha,y}(\|X\|^2)$. Thus, with $d_\alpha = \alpha - \hat{\alpha}$,

$$\log \frac{f_{p,\alpha,y}^{\text{norm}}(n)}{f_{p,\hat{\alpha},y}^{\text{norm}}(n)} = -\frac{d_\alpha^2}{4} \int_0^1 (1-t) \text{Var}_{p,\hat{\alpha}+td_\alpha,y}(\|X\|^2) dt. \quad (15.18)$$

On bounded ranges with comparable gaps this variance is $O(n/p)$. Consequently a shell $|\hat{\alpha} - \alpha| \leq \tau p^2$ costs at most $C\tau^2 k_p$ in the logarithm. The centered parent and child densities have ratio of order $\sqrt{p/c}$, so their logarithmic cost is $O_r(1)$. This is a statement about the ratio: each individual density contains the factor $n^{-1/2}$.

The true heat convolution, rewritten using $\mathbf{G} = \mathbf{G}_{c,p}^{h_c}$, is

$$\phi_c(h_c) - q_{c,\alpha}^{\text{fr}}(h_c) = \log \mathbb{E}_{\mathbf{G}} e^{\phi_p(Y) - q_{p,\alpha}^{\text{fr}}(Y)}.$$

Jensen and (15.15) therefore give

$$\phi_c(h_c) - \phi_c^{\text{sp}}(h_c) \geq \mathbb{E}_{\mathbf{G}} [\phi_p(Y) - \phi_p^{\text{sp}}(Y)] + \mathbb{E}_{\mathbf{G}} \log \frac{f_{p,\alpha,Y}^{\text{norm}}(n)}{f_{c,\alpha,h_c}^{\text{norm}}(n)}. \quad (15.19)$$

We must bound both expectations from below, including the reference fields outside any fixed shell. We first prove that the normalization term satisfies

$$-C_{R,r} \leq \mathbb{E}_G \log \frac{f_{p,\alpha,Y}^{\text{norm}}(n)}{f_{c,\alpha,h_c}^{\text{norm}}(n)} \leq 0. \quad (15.20)$$

Here R is the prescribed child field radius. Since r is fixed and $k_c = r^3 k_p \rightarrow \infty$, this loss is $o(k_c)$ even at the bottom scale.

Averaged normalization loss. The child saddle bounds give $-1 + \kappa_R \leq \alpha/c^2 \leq C_R$, with constants depending on R before r is chosen. Taking r small makes $|\alpha|/p^2$ small. Put

$$v = (p^2 + \alpha)^{-1}, \quad D = n - \text{Tr } R_p.$$

Then $d_p \asymp k_p$, $v \asymp p^{-2}$, $D \asymp np$, and $\text{Tr } R_p^2 \asymp n/p$. For the deficit, changing the parameter from zero to α changes the trace by at most $C|\alpha|n/p = O_R(r^2 np)$, while (15.4) gives the positive deficit $(b + o(1))np$ at zero. The same estimates hold in a sufficiently small fixed p^2 -neighborhood of α .

We need a density lower bound valid for every $y \in H_p$. Split the squared norm of $N(R_p y, R_p)$ as $S_a + S_b$, where the independent active and bulk parts have laws

$$S_a = \|N(vy, vI_{H_p})\|^2, \quad S_b = \|N(0, R_p|_{H_p^\perp})\|^2.$$

Write $D_b = n - \mathbb{E} S_b = D + d_p v \asymp np$. Since $\text{Var } S_b \leq Cn/p$, Chebyshev gives

$$\Pr\{|S_b - \mathbb{E} S_b| \leq D_b/2\} \geq 1 - C/k_p \geq \frac{1}{2}.$$

On this event $t = n - S_b \in [D_b/2, 3D_b/2]$. Spherical integration in the active space, followed by Jensen with a uniform unit vector $\omega \in H_p$, gives

$$f_{S_a}(t) = e^{-v\|y\|^2/2} f_{v\chi_{d_p}^2}(t) \mathbb{E}_\omega e^{\sqrt{t}y^\top \omega} \geq e^{-v\|y\|^2/2} f_{v\chi_{d_p}^2}(t).$$

Here $t/(d_p v)$ lies in a fixed compact subset of $(0, \infty)$. Stirling's bound for the chi-square density therefore gives $f_{v\chi_{d_p}^2}(t) \geq C^{-1}(v\sqrt{d_p})^{-1}e^{-Ck_p}$. Convolving over the bulk event proves

$$f_{p,\alpha,y}^{\text{norm}}(n) \geq C^{-1} \sqrt{p/n} \exp \left\{ -Ck_p \left(1 + \frac{\|y\|^2}{p^4 M_p^2} \right) \right\} \quad (y \in H_p). \quad (15.21)$$

The child density is comparable to $\sqrt{c/n}$, and the noncentral upper bound from (15.5) bounds the parent density by $C\sqrt{p/n}$ for every y . Their common $n^{-1/2}$ factor thus cancels before taking logarithms. In particular the logarithmic ratio in (15.20) is integrable under the Gaussian bridge.

For its sharper average bound, set $T = R_c - R_p$ and $W = \|R_p Y\|^2$. The bridge and the child saddle equation give

$$\mathbb{E}_G W = \|a_c\|^2 + \text{Tr } T = D, \quad \text{Var}_G W = 2 \text{Tr } T^2 + 4a_c^\top T a_c \leq C_R n/c.$$

Indeed $\text{Tr } R_c^2 \leq C_R n/c$, $\|R_c\| \leq C_R c^{-2}$, and $\|a_c\|^2 \leq C_R n c$. Fix a sufficiently small $\delta > 0$ and put $\mathcal{C} = \{|W - D| \leq \delta np\}$. The Gaussian quadratic moment formula bounds $\log \mathbb{E} e^{t(W-D)}$ by $C_R t^2 n/c$ for $|t| \leq c_R c^2$. Exponential Markov in both signs consequently gives $G(\mathcal{C}^c) \leq C_{R,r,\delta} e^{-a_{R,r,\delta} k_p}$.

On $\mathcal{C}$, monotonicity of the saddle equation places $\hat{\alpha} = \alpha_p(Y)$ in the comparable-gap neighborhood above: the derivative of $\text{Tr } R_{p,s} + \|R_{p,s}Y\|^2 - n$ has negative magnitude at least $c_2 n/p$ there. Integrating that derivative to each endpoint first locates the root, and then gives

$$|\hat{\alpha} - \alpha| \leq C(p/n)|W - D|.$$

Along the intervening parameter segment, the norm-square variance is at most Cn/p . The centered density bounds and (15.18) imply on $\mathcal{C}$ that

$$\log \frac{f_{p,\alpha,Y}^{\text{norm}}(n)}{f_{c,\alpha,h_c}^{\text{norm}}(n)} \geq -C_{R,r} - C(n/p)|\hat{\alpha} - \alpha|^2,$$

$$\mathbb{E}_{\mathcal{G}}[\mathbf{1}_{\mathcal{C}}(n/p)|\hat{\alpha} - \alpha|^2] \leq C(p/n) \text{Var } W \leq C_R p/c.$$

For the complement, $\|Y\|^2/(p^4 M_p^2) \asymp W/(np)$ has bounded second moment. Cauchy–Schwarz and the preceding exponential tail show that the negative contribution allowed by (15.21) is at most $C_{R,r,\delta} k_p e^{-a_{R,r,\delta} k_p/2} = o(1)$. This proves the lower bound in (15.20). Integrability and (15.17) identify its expectation with $-D_{\text{KL}}(\mathcal{G} \parallel \mathcal{T}_{c,p}^{\text{sp}})$, which proves the upper bound.

Averaging the parent supporting planes. On a finite length grid near the typical parent length, let $\mathcal{U}$ be the closed set of fields z satisfying

$$\phi_p(z) - \phi_p^{\text{sp}}(z) \geq -\varepsilon_a k_p, \quad \|P_p m_p(z) - a_p(z)\| \leq \varepsilon_g M_p.$$

Choose $\varepsilon_a, \varepsilon_g$ using the preceding angular argument, so that these sets have angular measure at least $e^{-\eta k_p}$ on each grid sphere, with η as small as required. The set $\mathcal{U}$ uses only parent data and the prescribed radii, so it is fixed under the fresh refinement. These anchors lie in a fixed bounded parent field range. Convexity gives a lower supporting plane for ϕ_p at z . On the spherical side, (15.8), frozen at the own saddle of z , gives

$$\phi_p^{\text{sp}}(y) \leq \phi_p^{\text{sp}}(z) + a_p(z)^\top (y - z) + \frac{1}{2}(y - z)^\top R_{p,\alpha_p(z)}(y - z) + C_{\text{dens}}.$$

This holds for every $y \in H_p$: the denominator density is centered at z , and the numerator has the same covariance, so its noncentral upper bound is independent of y . Subtracting the two inequalities yields the global estimate

$$\phi_p(y) - \phi_p^{\text{sp}}(y) \geq -\varepsilon_a k_p - \varepsilon_g M_p \|y - z\| - C p^{-2} \|y - z\|^2 - C_{\text{dens}} \quad (y \in H_p, z \in \mathcal{U}). \quad (15.22)$$

Put $d(y) = \text{dist}(y, \mathcal{U})/(p^2 M_p)$ and $D_2 = \mathbb{E}_{\mathcal{G}} d(Y)^2$. Choosing a nearest anchor and averaging (15.22) gives

$$\mathbb{E}_{\mathcal{G}}[\phi_p(Y) - \phi_p^{\text{sp}}(Y)] \geq -k_p\{\varepsilon_a + \varepsilon_g \sqrt{D_2} + C D_2\} - C_{\text{dens}}. \quad (15.23)$$

In particular this average includes all Gaussian length tails.

To bound this distance, put $\ell_* = (p^2 + \alpha)\sqrt{D}$. The child saddle α depends only on the eigenvalues and $\|h_c\|$, since $h_c \in H_c$; thus ℓ_* is independent of the fresh orientation. The preceding variance estimate gives

$$\mathbb{E}_{\mathcal{G}} \left[\frac{(\|Y\| - \ell_*)^2}{p^4 M_p^2} \right] \leq \frac{C \text{Var } W}{D n p} = O_{R,r}(k_p^{-1}).$$

Let the grid mesh be at most $\xi p^2 M_p$. Under Haar averaging the direction of Y is uniform conditional on its length. Choose a grid radius within $\xi p^2 M_p$ of ℓ_* and apply spherical concentration to its anchor set of angular measure at least $e^{-\eta k_p}$. The radial bound and the triangle inequality give

$$\mathbb{E}_O D_2 \leq C_R(\eta + \xi^2) + O_{R,r}(k_p^{-1}),$$

where O denotes the fresh refinement. Hence the Haar average of $\sqrt{D_2}$ is as small as prescribed. Its Frobenius Lipschitz constant is at most $C\sqrt{r} + O_r(k_p^{-1/2})$. Indeed the reference field in parent units has mean norm $O(\sqrt{r})$ and covariance norm $O_r(1/k_p)$; the Gaussian L^2 norm of its change between two rotations bounds the change of its L^2 distance to the fixed set $\mathcal{U}$. Haar Lipschitz concentration therefore gives $D_2 \leq \epsilon r^3 + o(1)$ except with probability $\exp(-c_3 \epsilon r^2 k_p)$ (proper rotations suffice). The constants before choosing r depend only on the prescribed child field range. First choose ϵ from the requested child value error ε_v , then choose r sufficiently small that the failure rate pays for the fixed fine child net, whose size is $\exp(O(k_c)) = \exp(O(r^3 k_p))$. Next choose $\varepsilon_a, \varepsilon_g, \eta, \xi$ so that the Haar mean of $\sqrt{D_2}$ is smaller than $\frac{1}{2}\sqrt{\epsilon r^3}$ and the coefficient $\varepsilon_a + \varepsilon_g \sqrt{D_2} + C D_2$ is a sufficiently small multiple of $\varepsilon_v r^3$ on the retained event. Then choose the upper cap scale sufficiently small for the preceding angular estimates, and finally take n large with all parameters fixed. Equations (15.19), (15.20), and (15.23) give the child lower bound throughout the net. Extend by the scaled Lipschitz bound. All finitely many grids and extra lists, with positive-rate failures per k_p or the summable zero-field errors above, retain high probability. This proves (15.9). $\square$

16 Transport of the radial covariance deficit

We now propagate a covariance estimate from one cap to the next. The spherical saddle supplies a radial deficit, while a fresh Haar refinement reduces the transverse error. Section 18 constructs an initial potential and replaces it, after finitely many levels, by the true partition function. Work on successive levels $p, c = \gamma p$ with γ a very small fixed ratio. For $s > 0$, define

$$\mathcal{A}_p(s) = \{h \in H_p : |\alpha_p(h)| \leq s p^2\}; \quad \text{the induction uses } \mathcal{A}_p(\sigma) \quad (16.1)$$

with small fixed $\sigma > 0$. We will start with auxiliary smooth log potentials close in *gradient norm* by a *bounded amount* to the true ones there. This auxiliary phase lasts only a fixed number of levels, which can be arbitrarily large but is independent of n . Thereafter the potentials are the true ones. In either case write

$$K_p = \nabla_{H_p}^2(\phi_p|_{H_p}), \quad R_p^H = (P_p R_p P_p)|_{H_p} = (p^2 + \alpha_p(h))^{-1} I_{H_p}.$$

For the true potential, $K_p = \text{Cov}_{\mu_{p,h}}(P_p X)$ as an operator on H_p . Thus every cap curvature assertion below concerns H_p . In the auxiliary case we require smoothness only on the used annulus, and the starting auxiliary function can itself be a Gaussian convolution. Potentials at subsequent levels in the auxiliary phase are computed by heat convolution, using a semiconcave extension at each step, defined below. Additive calibrations of these potentials on (16.1) are harmless. The invariants are as follows; all uniformities on an annulus are pointwise everywhere. The saddle equation gives

$$\|h\| = (p^2 + \alpha) \sqrt{n - \text{Tr } R_{p,\alpha}} = (\sqrt{b} + O(\sigma) + o(1)) p^2 M_p \quad \text{when } |\alpha| \leq \sigma p^2.$$

The annulus is therefore separated from zero. In the frame radial and transverse to $e = h/\|h\|$, put $D = p^2(K_p - R_p^H) \in \text{Sym}(H_p)$. Then

$$\|D_{\perp\perp}\| \leq \delta, \quad \|D_{\perp e}\| \leq K_0 \sqrt{\delta}, \quad -K_0 I_{H_p} \preceq D, \quad D_{ee} \leq -d_0, \quad d_0 = 0.30. \quad (16.2)$$

Here K_0 is a sufficiently large absolute constant. We can take δ a sufficiently high fixed power of γ , and σ likewise. The negative radial sign also controls the full positive part. If $z = D_{\perp e}$ and $v = v_{\perp} + v_e e$, then

$$v^\top D v \leq \delta \|v_{\perp}\|^2 + 2|v_e| \|z\| \|v_{\perp}\| - d_0 v_e^2 \leq \left(\delta + \frac{\|z\|^2}{d_0} \right) \|v_{\perp}\|^2.$$

Hence $D \preceq C\delta I_{H_p}$ and $\|D\| \leq \max(K_0, C\delta)$. Throughout multiscale estimates constants polynomially bad in γ^{-1} from inverses with fixed spectral formulas are allowed (the powers do not depend on δ, σ); the small powers used for tolerances can be chosen with sufficiently large fixed exponents. Macroscopic value and gradient errors relative to the spherical predictions can be taken as tiny as subsequently required. This is by (15.9); in the auxiliary phase we use the gradient comparison just mentioned with the true model, which integrates to $o(k_p)$ value oscillation error on (16.1) since the scales in that phase are fixed constants. The starting potential, its auxiliary randomness if any, and the gradient error are intrinsic given the current cap data and independent of the unused Haar refinement.

The extension used in an auxiliary step is one function selected from the full parent annulus using only these parent data. It agrees with the parent potential on an inner annulus with fixed slack, has at most quadratic growth, and has a global upper Hessian bound $(1 + \gamma^4/100)I/p^2$ after the curvature tolerances are made sufficiently small. This is the semiconcavity bound that controls its Gaussian convolution. The supporting-quadratic construction below verifies these properties without using the child eigenframe. For a true parent the extension is used only to estimate the transition; the output remains the true child potential.

Proposition 16.1 (One cap step). *Fix a bounded child field range for the one-off evaluation conclusion below. There are a sufficiently large absolute K_0 and a sufficiently small ratio $\gamma > 0$ with the following property. Choose σ as a sufficiently high fixed power of γ , and then choose δ as a still higher fixed power, allowing these tolerances to depend on the prescribed field range. Let a parent potential at scale p be intrinsic to the capped data, satisfy the value and gradient comparisons above, and satisfy (16.2) at all fields in (16.1). Use the semiconcave extension just described when forming an auxiliary child potential. For a true parent, retain the true child potential. Condition on the parent data and used auxiliary randomness $\mathcal{F}_p$ described below, on which these hypotheses and the $\mathcal{F}_p$ -measurable parent-level conclusions of the static comparison hold, including the projected posterior tail and moment bounds for this capped parent. The separate $o(1)$ failure probability used to obtain these parent conclusions remains outside the conditional estimate below. Then, uniformly over such parent data, the probability over the unused Haar refinement O that (16.2) fails at $c = \gamma p$ anywhere on the child annulus is at most*

$$Ce^{-ak_p} + Ce^{-a'k_c}$$

for fixed positive a, a' . The same inequalities hold with strict slack on the complementary event. These failure probabilities are summable over the geometric cap list.

At intermediate ratios $c/p \in [\gamma, 1)$, if the frozen parameter stays inside the agreement domain with fixed slack, the transverse error and mixed block tend to zero with the parent errors, the total scaled Hessian is bounded, and the radial deficit is at least 0.26. The same deterministic evaluation bounds hold for $c/p \in [\gamma^2, \gamma]$ under this interior-parameter condition. For a true parent and the prescribed bounded child field range there is also a sufficiently small fixed evaluation-ratio interval on which the upper child curvature is at most C/c^2 , with C depending only on that field range. This last evaluation uses supports centered at the frozen parameter, constructed at the end of the proof. All statements are uniform down to $np^3 \asymp \sqrt{\log n}$.

The proof occupies the remainder of this section. Its probabilistic step is used only at ratio γ ; the intermediate-scale conclusions are consequences of the deterministic covariance bounds.

16.1 Semiconcave extensions and the observation bridge

We make the measurability in a refinement explicit. Conditional on the eigenvalues and the capped matrix J_p , let $\mathcal{F}_p$ include all auxiliary randomness already used to construct the parent potential.

The construction is required to be measurable with respect to $\mathcal{F}_p$; it does not reveal the orientation of the unequal eigenvalues within the flat eigenspace P_p . Choose a fixed orthonormal identification of this space with $\mathbb{R}^{d_p}$, where $d_p = \text{Tr } P_p$. The remaining refinement is an orthogonal matrix O which is Haar and independent of $\mathcal{F}_p$. All spectral projectors and increments below are obtained by conjugating fixed diagonal matrices by O . This independence is the reason for the auxiliary refinement used in Section 18.

After a step, the increment $\Delta = B_p - B_c$ is a function of J_p and J_c and is scalar on P_c . Thus the output auxiliary potential depends only on the new capped data and previously introduced independent seeds; it does not reveal the internal eigenframe of P_c . Conditional on these data the next finer rotation is Haar.

For $c = xp < p$, put

$$\Delta = B_p - B_c, \quad E = \text{ran } \Delta \subset H_p, \quad U = H_c \subset E.$$

Every inverse of Δ is taken on E , after omitting the zero increment coordinates. An inverse is taken before its block is compressed to U . In particular $[M^{-1}]_{UU}$ generally differs from $[M_{UU}]^{-1}$. The child increment is the scalar $\Delta_U = (1 - x^2)p^2 I_U$.

For a child field h_c , the exact convolution and its transition probability are

$$\phi_c(h_c) = \log \int \exp\{\phi_p(h_c + z)\} \mathbf{N}(0, \Delta)(dz), \quad (16.3)$$

$$\mathbf{T}_{c,p}^{h_c}(dh_p) = \frac{\exp\{\phi_p(h_p)\} \mathbf{N}(h_c, \Delta)(dh_p)}{\exp\{\phi_c(h_c)\}}. \quad (16.4)$$

For an auxiliary step, the function convolved in these equations is the selected extension $\bar{\phi}_p$ defined below; its logarithmic convolution defines the child potential. For the true partition function the equations follow by integrating the Gaussian linear tilt for each spin. Freeze the child saddle parameter α and put $R_u = R_{u,\alpha}$ for $u = c, p$. Replacing the convolved potential by $q_{p,\alpha}^{\text{fr}}$ from (15.14) gives the canonical probability $\mathbf{G}_{c,p}^{h_c}$. Under this law,

$$h_p = h_c + \Delta a_c + \zeta, \quad \zeta \sim \mathbf{N}(0, \Delta + \Delta R_c \Delta), \quad R_p h_p \sim \mathbf{N}(a_c, R_c - R_p). \quad (16.5)$$

These identities follow from $R_c^{-1} = R_p^{-1} - \Delta$, as in (15.16). The covariance $R_c - R_p$ is supported on E . The reference spin Gaussian is sampled without norm conditioning.

The parent Hessian estimate holds on an annulus, whereas the convolution integrates over the full increment space. We extend the potential to make the convolution globally controlled, while preserving it on the fields the bridge visits with high probability. The extension must be determined by the parent data alone: using the child refinement in its construction would consume the randomness needed for transverse compression.

Here is one parent-measurable extension with the needed buffers. The radius corresponding to a saddle parameter is

$$\ell_p(\alpha) = (p^2 + \alpha) \sqrt{n - \text{Tr } R_{p,\alpha}}, \quad \ell'_p(\alpha) = \sqrt{n - \text{Tr } R_{p,\alpha}} + \frac{(p^2 + \alpha) \text{Tr } R_{p,\alpha}^2}{2 \sqrt{n - \text{Tr } R_{p,\alpha}}} \asymp M_p.$$

Thus $\mathcal{A}_p(s)$ is the radial annulus between $\ell_p(-sp^2)$ and $\ell_p(sp^2)$. Put

$$S_0 = \mathcal{A}_p(\sigma/2), \quad S_1 = \mathcal{A}_p(3\sigma/4), \quad S_2 = \mathcal{A}_p(\sigma).$$

Half the distance from S_0 to $H_p \setminus S_1$, and from S_1 to $H_p \setminus S_2$, gives buffers $\rho_{01}, \rho_{12} \asymp \sigma p^2 M_p$. In particular $S_0 + B(0, \rho_{01}) \subset S_1$, and a segment of length at most ρ_{12} from a point of S_1 lies in S_2 .

Let $f_p = \phi_p|_{H_p}$ be the parent before extension and set

$$L_p = (1 + \zeta)p^{-2}I_{H_p}, \quad \zeta = \gamma^4/100.$$

We claim that every pair $z, w \in S_1$ satisfies

$$f_p(w) \leq f_p(z) + \nabla_{H_p} f_p(z)^\top (w - z) + \frac{1}{2}(w - z)^\top L_p (w - z). \quad (16.6)$$

For $\|w - z\| \leq \rho_{12}$, the segment stays in S_2 . There (16.2) and its positive-part bound give

$$K_p \preceq p^{-2}((1 - \sigma)^{-1} + C\delta)I_{H_p} \prec L_p$$

after choosing σ, δ below the excess ζ . Integrating the Hessian on the segment proves (16.6) in this case.

For $\|w - z\| \geq \rho_{12}$, put $d = w - z$ and freeze the own saddle at z . The value comparison, the gradient comparison, and (15.8) give

$$f_p(w) - f_p(z) - \nabla_{H_p} f_p(z)^\top d \leq \frac{\|d\|^2}{2(p^2 + \alpha_p(z))} + 2\varepsilon_v k_p + C_{\text{dens}} + \varepsilon_g M_p \|d\|.$$

In an auxiliary step, $\varepsilon_v, \varepsilon_g$ also include the calibrated auxiliary errors. With $\beta_{\text{buf}} = \rho_{12}/(p^2 M_p)$, the last three terms are at most

$$\frac{\|d\|^2}{p^2} \left(\frac{2\varepsilon_v + C_{\text{dens}}/k_p}{\beta_{\text{buf}}^2} + \frac{\varepsilon_g}{\beta_{\text{buf}}} \right).$$

Choose these errors after the buffers so that this quantity fits below the remaining excess of L_p . This proves (16.6) for every support pair.

Now define the single function

$$\bar{\phi}_p(y) = \inf_{z \in S_1} \left\{ f_p(z) + \nabla_{H_p} f_p(z)^\top (y - z) + \frac{1}{2}(y - z)^\top L_p (y - z) \right\}. \quad (16.7)$$

At $y \in S_1$, (16.6) bounds every term below by $f_p(y)$, and the term $z = y$ equals it. Thus $\bar{\phi}_p = f_p$ on S_1 , in particular on a neighborhood of S_0 . Subtracting $\frac{1}{2}y^\top L_p y$ leaves an infimum of affine functions, hence a concave function. The support set is bounded and its values and slopes are bounded, so this infimum is finite and has at most quadratic growth. Consequently $\nabla^2 \bar{\phi}_p \preceq L_p$ in the distributional sense. The construction uses only the parent annulus and parent function, so it is $\mathcal{F}_p$ -measurable and consumes no part of O .

On E the convolution has the strict margin

$$\|\Delta_E^{1/2} L_p \Delta_E^{1/2}\| \leq (1 + \zeta)(1 - x^2) \leq (1 + \zeta)(1 - \gamma^2) < 1.$$

The extended terminal density is therefore strongly log-concave in the whitened increment coordinates. In the true phase the extension is used only to estimate the transition; the output is the true child. If smoothing is needed, mollify the extension and pass to the limit at fixed n . Inside the buffered agreement region the mollifications converge to the original smooth potential with its derivatives.

We next record precisely how canonical estimates pass to a terminal transition. Let V_p be the potential actually convolved, either $\bar{\phi}_p$ in an auxiliary step or the true ϕ_p , and set

$$F(y) = V_p(y) - q_{p,\alpha}^{\text{fr}}(y).$$

For fixed parent data and child field, the exact likelihood ratio is

$$\frac{d\mathbb{T}}{d\mathbb{G}}(y) = \frac{e^{F(y)}}{\mathbb{E}_{\mathbb{G}} e^F}, \quad \mathbb{G} = \mathbb{G}_{c,p}^{h_c}. \quad (16.8)$$

Let $\hat{\alpha} = \alpha_p(y)$ and take a shell $\mathcal{S}_\tau = \{|\hat{\alpha} - \alpha| \leq \tau p^2\}$ lying inside the buffered agreement region. Under $\mathbb{G}$, its complement has probability at most $e^{-a_\tau k_p}$ for each fixed $\tau > 0$, by (16.5) and the saddle derivative estimate. On $\mathcal{S}_\tau$, the static comparison or its calibrated auxiliary version, together with (15.18), gives, for a separately chosen normalization shell $\mathcal{S}_{\tau_0}$ with $0 < \tau_0 < \tau$,

$$\sup_{\mathcal{S}_\tau} \frac{d\mathbb{T}}{d\mathbb{G}} \leq \mathbb{G}(\mathcal{S}_{\tau_0})^{-1} \exp\{[2\varepsilon_v + C(\tau^2 + \tau_0^2)]k_p + O_x(1)\}. \quad (16.9)$$

The denominator is bounded below by integrating e^F on the narrower shell. Choosing τ , then τ_0 , and then the comparison errors makes the exponent at most ηk_p for any prescribed fixed $\eta > 0$.

The shell complement requires its own estimate. Write $d(y) = |||y|| - \ell_p(\alpha)|/(p^2 M_p)$. The canonical bridge has

$$\mathbb{E}_{\mathbb{G}} \|Y\|^2 = \ell_p(\alpha)^2, \quad \|\text{Cov}_{\mathbb{G}}(Y)\| \leq C\gamma^{-2}p^2.$$

The first identity follows from $\mathbb{E}\|R_p Y\|^2 = \|a_c\|^2 + \text{Tr}(R_c - R_p) = n - \text{Tr } R_p$, since R_p is scalar on H_p . Gaussian concentration for the norm, together with $0 \leq \sqrt{\mathbb{E}\|Y\|^2} - \mathbb{E}\|Y\| \leq \sqrt{\|\text{Cov}(Y)\|}$, therefore gives

$$\Pr_{\mathbb{G}}\{d(Y) > v\} \leq 2e^{-c\gamma^2 k_p v^2}, \quad v \geq \frac{C}{\gamma\sqrt{k_p}}. \quad (16.10)$$

In particular this estimate applies at every fixed shell threshold once n is sufficiently large. For the extended potential, a support from the frozen typical sphere, with the calibrated value there, bounds $F(y)$ above by a fixed reference level plus

$$\eta_{\text{err}} k_p + \eta_{\text{small}} k_p d(y)^2.$$

Leaving $\mathcal{S}_\tau$ requires $d(Y) \geq c_0 \tau$ on the comparable-gap range. Choose $\eta_{\text{small}} < c\gamma^2/2$, then choose τ_0 and the comparison errors so that the normalization loss $2\varepsilon_v + C\tau_0^2$, together with η_{err} , is less than $(c\gamma^2 - \eta_{\text{small}})c_0^2 \tau^2/2$. The narrower shell has canonical probability tending to one for every fixed $\tau_0 > 0$. Integrating the tail against the upper support with this normalization budget gives, for every fixed integer j ,

$$\mathbb{E}_{\mathbb{T}}[(1 + d(Y))^j \mathbf{1}_{\{Y \notin \mathcal{S}_\tau\}}] \leq C_{j,\tau} e^{-a'_\tau k_p}.$$

For a true transition, on each bounded field range the static value comparison and the frozen numerator norm-density upper bound control F by its calibrated level plus $\varepsilon_v k_p + C$. The same normalization budget and (16.10) then control the bounded part of the shell complement. Beyond that range, the projected v^6 tail at parent zero field is multiplied by at most $e^{Ck_p(1+v^2)}$ from the added quadratic, so $e^{-ck_p v^6 + Ck_p(1+v^2)}$ bounds the large-length contribution and all fixed polynomial moments. Gaussian observation noise has the corresponding quadratic tail. These estimates make the error from replacing a true transition by its extension exponentially small in scaled covariance units. Indeed the two unnormalized terminal densities agree on S_1 . If each normalized law assigns at most ε to its complement, their normalization ratio lies between $1 - \varepsilon$ and $(1 - \varepsilon)^{-1}$; the complementary moment bounds then control the difference of their means and covariances.

The same statements hold for $x \in [\gamma, 1)$ whenever the frozen parameter lies inside parent agreement with fixed slack; one use is $|\alpha| \leq 2\sigma(\gamma p)^2$. As $x \uparrow 1$, the canonical increment covariance shrinks proportionally to $1 - x$, so the probability of leaving agreement is at most $Ce^{-ak_p/(1-x)}$. The

support and projection estimates give the same bound for the relevant true-transition errors. This exponential factor pays every fixed power of $1/(1-x)$. For derivative integrals on a true transition we use the projected trace bound

$$\mathrm{Tr}_{H_p} K_p(y) \leq \mathbb{E}_{\mu_{p,y}} \|P_p X\|^2.$$

The projected moment tails therefore leave only polynomial factors in k_p , rather than an ambient rank factor.

16.2 Upper and lower covariance transport

We use the Brascamp–Lieb variance inequality [10]; its relation to log-concavity and logarithmic Sobolev inequalities is discussed in [8]. The paired upper and lower covariance bounds also have the form used in [13, Lemma 7 and Appendix A]. The output Hessian is obtained from the covariance of the terminal field. We bound this covariance from both sides. Brascamp–Lieb keeps the inverse curvature correction and will preserve the useful negative radial term. The Fisher test is linear in the parent correction and supplies the lower bound needed for transverse and mixed blocks. We perform the lower test with cutoffs inside agreement, so uncontrolled curvature of the extension is not integrated.

Differentiating the logarithmic convolution in a direction of U gives the exact output Hessian for the potential V_p and its corresponding transition T :

$$K_c = \Delta_U^{-1} [\mathrm{Cov}_{\mathsf{T}}(Y)]_{UU} \Delta_U^{-1} - \Delta_U^{-1}. \quad (16.11)$$

The covariance is first computed on E . For Brascamp–Lieb, temporarily take a smooth mollification V_p^{ext} of the extension and denote its transition by $\mathsf{T}^{\mathrm{ext}}$. Put

$$K_p^{\mathrm{ext}}(Y) = \nabla^2 V_p^{\mathrm{ext}}(Y), \quad M_{\mathrm{ext}}(Y) = \Delta_E^{-1} - [K_p^{\mathrm{ext}}(Y)]_{EE}, \quad C_0 = (\Delta_E^{-1} - R_p|_E)^{-1} = \Delta + \Delta R_c \Delta.$$

The extension margin makes M_{ext} positive. Brascamp–Lieb on E gives $\mathrm{Cov}_{\mathsf{T}^{\mathrm{ext}}}(Y) \preceq \mathbb{E}_{\mathsf{T}^{\mathrm{ext}}} M_{\mathrm{ext}}(Y)^{-1}$. Combining this with (16.11) preserves the inverse before compression to U .

For the lower estimate under this smooth extension law, let ρ_{ext} be its density and $\mathfrak{s} = -\nabla_E \log \rho_{\mathrm{ext}}$ be the score. Integration by parts gives $\mathbb{E}[(Y - \mathbb{E}Y)\mathfrak{s}^\top] = I_E$ and $\mathbb{E}[\mathfrak{s}\mathfrak{s}^\top] = \mathbb{E}M_{\mathrm{ext}}$. The nonnegativity of the covariance of $Y - T\mathfrak{s}$ therefore gives, for every constant matrix T on E ,

$$\mathrm{Cov}_{\mathsf{T}^{\mathrm{ext}}}(Y) \succeq T + T^\top - T \mathbb{E}_{\mathsf{T}^{\mathrm{ext}}} [M_{\mathrm{ext}}(Y)] T^\top. \quad (16.12)$$

With $T = C_0$, the right-hand side is $C_0 + C_0 \mathbb{E}_{\mathsf{T}^{\mathrm{ext}}} [K_p^{\mathrm{ext}} - R_p]_{EE} C_0$, which is linear in the parent correction.

We use a localized version when only the agreement region has controlled classical curvature. This identity applies separately to either smooth law with density ρ , using its own score $\mathfrak{s}$, precision $M = -\nabla_E^2 \log \rho$, and expectations. Let χ be a smooth cutoff supported there and, for constant $t, w \in E$, put $b_\chi = \chi t^\top \mathfrak{s} - t^\top \nabla \chi$. Integration by parts yields

$$\mathbb{E}[(w^\top (Y - \mathbb{E}Y))b_\chi] = \mathbb{E}_\chi w^\top t, \quad \mathbb{E}b_\chi^2 = \mathbb{E}_\chi^2 t^\top M t + \mathbb{E}(t^\top \nabla \chi)^2.$$

The inequality $\mathbb{E}(w^\top (Y - \mathbb{E}Y) - b_\chi)^2 \geq 0$ becomes

$$\mathrm{Var}(w^\top Y) \geq 2\mathbb{E}_\chi w^\top t - \mathbb{E}_\chi^2 t^\top M t - \mathbb{E}(t^\top \nabla \chi)^2. \quad (16.13)$$

Choose $\chi = 1$ on a narrower shell and supported inside agreement. For an output test $v \in U$, take $w = \Delta_U^{-1}v$ and $t = C_0w = Nv$, where $N = (p^2 + \alpha)/(c^2 + \alpha)$. Only the scalar inverse Δ_U^{-1} appears in this test. The shell estimates above, including their polynomial moment bounds and the improved $e^{-ak_p/(1-x)}$ rate, make the cutoff errors exponentially small after any fixed powers of k_p and $(1-x)^{-1}$. Thus (16.13) applies to the true transition as well as a smoothed extension without integrating uncontrolled curvature outside agreement.

On the shell subtract the frozen cap resolvent, writing

$$D^\circ(y) = p^2(K_p(y) - (p^2 + \alpha)^{-1}I_{H_p}).$$

If $\hat{\alpha} = \alpha_p(y)$, then exactly

$$D^\circ(y) - D(y) = \frac{p^2(\alpha - \hat{\alpha})}{(p^2 + \hat{\alpha})(p^2 + \alpha)}I_{H_p}.$$

This is $O(\tau)$ on $\mathcal{S}_\tau$. In the following resolvents D° is restricted to E ; the restriction is suppressed in the notation. Put

$$S = p^{-2}C_0 = p^{-2}\Delta(I - R_p\Delta)^{-1}.$$

On the agreement shell, where the limiting extension Hessian is the parent Hessian, put $M = C_0^{-1} - p^{-2}D^\circ$. The exact identities there are

$$M^{-1} = C_0 + p^{-2}C_0D^\circ(I - SD^\circ)^{-1}C_0, \quad \Delta_U^{-1}(C_0)_{UU}\Delta_U^{-1} - \Delta_U^{-1} = R_c|_U.$$

Together with $C_0\Delta_U^{-1} = NI_U$, these identities give the extension bound. Pass to the mollification limit and, for a true output, use the preceding transition comparison. The cutoff and change-of-law errors are included in η_{err} . The resulting target bound is

$$c^2(K_c - R_c|_U) \preceq x^2N^2\mathbb{E}_\tau[D^\circ(I - SD^\circ)^{-1}]_{UU} + \eta_{\text{err}}I_U. \quad (16.14)$$

The constant Fisher test gives the corresponding lower bound $x^2N^2\mathbb{E}_\tau[D^\circ]_{UU}$, up to the same type of errors. Testing only $v_\perp$, now transverse to the child field, gives that transverse lower block with zero mixed block and radial block $-c^2(\Delta_U^{-1} + R_c|_U)$. The global upper curvature controls the rare-event remainder in (16.14); all expansions of D° use the good shell.

The matrix map in the upper bound preserves Loewner order on its positive domain. On E ,

$$D(I - SD)^{-1} = S^{-1/2}[(I - S^{1/2}DS^{1/2})^{-1} - I]S^{-1/2}. \quad (16.15)$$

It is increasing whenever $I - S^{1/2}DS^{1/2} \succ 0$. The extension margin supplies this condition for the actual correction. It will also hold for the upper comparison below: its positive part is $O(\delta) + \eta_{\text{err}}$, whereas $\|S\|$ is bounded by a fixed power of γ^{-1} . The stated tolerance order leaves a strict margin.

16.3 Regeneration of the radial deficit

Set $A = D_{\perp\perp}$, embedded with zero radial block, and $z = D_{\perp e}$. Completing the square in the mixed block, using a small fixed part of the radial deficit, gives

$$D^\circ \preceq A + Czz^\top - d_*ee^\top + \eta_{\text{err}}I, \quad (16.16)$$

where $d_* < d_0$ can be chosen as close to d_0 as desired and C depends on $d_0 - d_*$. The order identity (16.15) permits this substitution in (16.14). To see the effect of the negative rank, write $\mathcal{F}_S(D) = D(I - SD)^{-1}$ and $B = A + Czz^\top + \eta_{\text{err}}I$. Sherman–Morrison gives the exact formula

$$\mathcal{F}_S(B - d_*ee^\top) = \mathcal{F}_S(B) - \frac{v_Bv_B^\top}{d_*^{-1} + e^\top S(I - BS)^{-1}e}, \quad v_B = (I - BS)^{-1}e.$$

The matrices lie in the positive domain specified above. Since $\|B\| = O(\delta) + \eta_{\text{err}}$ and $\|S\| = O_\gamma(1)$, $\mathcal{F}_S(B) = A + Czz^\top + O_\gamma(\delta^2) + \eta_{\text{err}}I$, $v_B = e + O_\gamma(\delta) + \eta_{\text{err}}$, and the denominator differs from $d_*^{-1} + e^\top Se$ by $O_\gamma(\delta) + \eta_{\text{err}}$. After choosing the tolerances, the resulting upper bound retains the subtraction

$$-(1 - \eta_{\text{small}}) \frac{ee^\top}{1/d_* + e^\top Se} \quad (16.17)$$

up to $O_\gamma(\delta^2) + \eta_{\text{err}}$ in operator norm.

Put $m = R_p Y$ and $T = R_c - R_p$. On H_p , R_p is scalar, so $e = m/\|m\|$. Under the canonical bridge, m has mean a_c and covariance T . The shell likelihood and its complementary moment bounds transfer the corresponding linear and quadratic tail tests to $\mathbb{T}$, with arbitrarily small errors in scaled units. In particular on the shell $\|m\|^2 = (b + o(1) + O(\sigma) + O(\tau))np$, $\mathbb{E}_\mathbb{T} m$ is arbitrarily close to a_c , and $\mathbb{E}_\mathbb{T} m^\top T m$ is at most $a_c^\top T a_c + \text{Tr } T^2$ plus an arbitrarily small scaled error. The linear second-moment bounds are used separately for each tested direction in U , with numerical constants uniform in that direction.

At $\alpha = 0$, $S = p^2 T$. The denominator of the rank term after replacing e by $m/\|m\|$ is $\|m\|^2/d_* + p^2 m^\top T m$. The following calculation bounds its expectation and hence the radial subtraction.

Since $R_p = p^{-2}I$ on the support of T ,

$$\begin{aligned} \text{Tr } T &= (b + o(1))n(p - c), \\ \text{Tr } T^2 &= (2b + o(1))n(1/c - 1/p) - 2p^{-2}(b + o(1))n(p - c). \end{aligned} \quad (16.18)$$

The rank denominator before the factor x^{-2} is $\|m\|^2/d_* + p^2 m^\top T m$. Its expectation is bounded above by

$$\frac{bnp}{d_*} + p^2(a_c^\top T a_c + \text{Tr } T^2) + o(np^2/c). \quad (16.19)$$

For each test vector v , Cauchy–Schwarz in the form $\mathbb{E}[(v^\top m)^2/Q] \geq (v^\top \mathbb{E} m)^2/\mathbb{E} Q$ turns this into a rank-one matrix bound. Dividing the denominator by $bnp^2/c = x^{-2}\|a_c\|^2(1 + o(1))$ gives successively x/d_* , $1 - x^2$, and $2(1 - x)^2$. This proves that the radial deficit in output units is at least

$$\frac{1 - \eta_{\text{small}}}{x/d_* + (1 - x^2) + 2(1 - x)^2 + \eta_{\text{small}}}. \quad (16.20)$$

In the stated interior-parameter regimes, $x \geq \gamma^2$ and $|\alpha| \leq \sigma p^2$, so $|\alpha|/c^2 \leq \sigma \gamma^{-4}$. The high-power choice of σ therefore includes the resulting perturbation in η_{small} .

Cross-direction losses in making this a purely radial deficit can be arbitrarily small by the transverse mean and second-moment bounds above. Formula (16.20) exceeds 0.26 throughout, and exceeds 0.31 at $x = \gamma$. The small constant losses can be chosen to keep these strict inequalities. Remaining terms in the radial block are negligible for this estimate.

16.4 Compression under the fresh refinement

For the inductive transverse block at $x = \gamma$, first fix the complete orientation O and a child field. Expectations in this paragraph are under its target transition $\mathbb{T}_O$. Fix a smooth function $0 \leq \chi_0 \leq 1$, equal to one on $[-1/2, 1/2]$ and zero outside $(-1, 1)$, and use

$$\chi_h(y) = \chi_0\left(\frac{\alpha_p(y) - \alpha_c(h)}{\tau p^2}\right). \quad (16.21)$$

Here τ is a sufficiently small fixed power of γ , and the support lies inside agreement with fixed slack. At the current child field write $\chi = \chi_h$. Write $A_{\text{par}}, z_{\text{par}}$ for the parent quantities denoted by A, z above, and set $A = \chi A_{\text{par}}, z = \chi z_{\text{par}}$ in the compression estimates. The shell moment bounds make this truncation arbitrarily small. Put

$$g(y) = \nabla_{H_p} f_p(y) - (p^2 + \alpha)^{-1} y, \quad Q_s = \mathbf{1}_{\{\Delta_E \geq sp^2\}}.$$

On the support of χ , $\nabla_{H_p} g = D^\circ/p^2$ and $\|g\| \leq \varepsilon_{\text{mac}} M_p$, where the frozen-saddle error is included in ε_{mac} . With the score $\mathfrak{s} = -\nabla_E \log \mathbb{T}_O$, integration by parts gives the matrix identity

$$\mathbb{E}_{\mathbb{T}_O}[\chi D^\circ Q_s] = p^2 \mathbb{E}_{\mathbb{T}_O}[g(\chi Q_s \mathfrak{s} - Q_s \nabla \chi)^\top]. \quad (16.22)$$

The nuclear norm of this expectation is bounded by

$$p^2 (\mathbb{E}_{\mathbb{T}_O} \mathbf{1}_{\text{supp } \chi} \|g\|^2)^{1/2} (\mathbb{E}_{\mathbb{T}_O} \|\chi Q_s \mathfrak{s} - Q_s \nabla \chi\|^2)^{1/2}.$$

Summing the localized score identity used in (16.13) over an orthonormal basis of $Q_s E$ gives

$$\mathbb{E} \|\chi Q_s \mathfrak{s} - Q_s \nabla \chi\|^2 = \mathbb{E} \chi^2 \text{Tr}(Q_s M Q_s) + \mathbb{E} \|Q_s \nabla \chi\|^2 \leq C_s d_p/p^2 + C_s/(p^4 M_p^2) \leq C_s M_p^2.$$

Here $\Delta_E^{-1} \leq (sp^2)^{-1}$ on $Q_s E$, the invariant bounds $-K_p$ by C/p^2 on the cutoff support, and $\|\nabla \chi\| \leq C/(p^2 M_p)$ with fixed buffer constants. Thus (16.22) has nuclear norm at most $C_s \varepsilon_{\text{mac}} p^2 M_p^2 = C_s \varepsilon_{\text{mac}} k_p$.

The columns omitted by Q_s have rank $O(s)k_p + o(k_p)$, by the edge count in the strip $0 \leq \Delta < sp^2$, including the zero-increment boundary coordinates; their operator norm is bounded. Changing D° to D costs $C\tau d_p$ in nuclear norm on the shell. Finally, $D - A_{\text{par}}$ has rank at most two for each field and bounded nuclear norm. Its radial direction may change with the field, but

$$\|\mathbb{E}_{\mathbb{T}_O}[\chi(D - A_{\text{par}})]\|_* \leq \mathbb{E}_{\mathbb{T}_O} \|\chi(D - A_{\text{par}})\|_* \leq C.$$

This uses the nuclear triangle inequality, without assigning a fixed rank to the average. Choosing s first, then the macroscopic and shell errors, and finally k_p large proves

$$\frac{\|\bar{A}_O\|_*}{k_p} \leq \eta_{\text{err}}, \quad \bar{A}_O = \mathbb{E}_{\mathbb{T}_O} A. \quad (16.23)$$

The norm is outside the expectation. In particular the signed normalized trace is small.

The radial correction has the required sign, but a direct operator-norm estimate on the transverse block would be enlarged by the change of scale. The small signed average just proved allows a better estimate. We sample finitely many independent terminal fields, show that their average has small Frobenius norm, and compress it using the remaining Haar rotation. The sampled field directions are exceptional because each matrix depends on its own field; we estimate their span separately.

We need

$$\|\mathbb{E}_{\mathbb{T}_O} A\|_{U_\perp U_\perp} \leq o_{\gamma \rightarrow 0}(1) \gamma^2 \delta, \quad \mathbb{E}_{\mathbb{T}_O} \|P_U z\|^2 \leq CK_0^2 \delta H \gamma^3, \quad (16.24)$$

apart from tiny terms, where $H = C_1 \log(1/\gamma)$ with a sufficiently large fixed coefficient. These assertions hold simultaneously for child fields with high probability over the refinement. For the moment, keep O and the child field fixed and sample $Y_1, \dots, Y_l$ independently from $\mathbb{T}_O$. The deterministic function of these fields is $\mathcal{A}(Y_1, \dots, Y_l) = l^{-1} \sum_j A(Y_j)$. Independence gives exactly

$$\begin{aligned} \mathbb{E}_{\mathbb{T}_O^l} \|\mathcal{A}\|_F^2 &= l^{-1} \mathbb{E}_{\mathbb{T}_O} \|A\|_F^2 + \frac{l-1}{l} \|\bar{A}_O\|_F^2 \\ &\leq \delta^2 d_p/l + \delta \|\bar{A}_O\|_*. \end{aligned} \quad (16.25)$$

The last inequality uses $\|A\| \leq \delta$ and $\|\bar{A}_O\|_F^2 \leq \|\bar{A}_O\| \|\bar{A}_O\|_*$. By (16.23), the quotient of (16.25) by k_p is $O(\delta^2/l) + \eta_{\text{err}}$. We will combine this fixed- O target-law estimate with a Haar event first proved under canonical sampling.

Conditional Haar laws before tilting. We spell out the conditional laws because the actual transition is not itself Haar after its terminal fields have been sampled. Fix $\mathcal{F}_p$ and a child field in spectral coordinates, denoted $u \in U_0$. Its image is $h_c = Ou$, and $U = OU_0$. Sample l independent canonical terminal fields $\xi_1, \dots, \xi_l$ in spectral coordinates according to (16.5), independently of O . Their means are parallel to u . Denote this joint law by

$$\mathbb{P}_{\text{can}}(dO, d\xi_1 \cdots d\xi_l) = \text{Haar}(dO) \prod_{j=1}^l \mathbf{G}_{u, \text{diag}}(d\xi_j).$$

Its physical fields are $Y_j = O\xi_j$, whose conditional law for fixed O is $\mathbf{G}_O^l$. Conditional on these spectral fields, and on the images under O of u and any selected collection of the ξ_j , the remaining map on the orthogonal complements is Haar. Conditional on the image of ξ_j , the matrix $A_j = A(O\xi_j)$ is fixed: the parent potential, its radial splitting, and its derivatives are all measurable with respect to $\mathcal{F}_p$ and that image. Conditional on two images, $A_i A_j$ is fixed. The same statement applies to $z_j = z(O\xi_j)$, with $z_j \perp O\xi_j$.

Set $e_j = O\xi_j / \|\xi_j\|$, and let $e_0 = h_c / \|h_c\|$. The child annulus is separated from zero field, so e_0 is defined. All dimensions below are comparable to k_p or k_c by the spectral count. The elementary Haar estimate we use is

$$\|P_V O^\top T O P_V - \frac{\text{Tr } T}{d} P_V\| \leq C \left(\frac{\|T\|_F}{d} \sqrt{r+a} + \frac{\|T\|}{d} (r+a) \right), \quad (16.26)$$

with failure probability at most $2e^{-a}$, when O is Haar in dimension d , V is a fixed r -dimensional subspace, and T is symmetric and fixed. Replacing $r+a$ by a fixed larger multiple absorbs the size of a fixed-resolution unit net in V . To prove the estimate for a fixed unit vector, write its Haar image as $g/\|g\|$, diagonalize T , and use

$$\log \mathbb{E} e^{t \sum_i \lambda_i (g_i^2 - 1)} \leq C t^2 \sum_i \lambda_i^2, \quad |t| \max_i |\lambda_i| \leq \frac{1}{4}. \quad (16.27)$$

The same bound for $\|g\|^2$ handles its normalization. Polarization gives the bilinear and nonsymmetric versions used for products.

On the complement of the span of $e_0, e_1, \dots, e_l$, apply (16.26) with $T = \mathcal{A}$ and a a sufficiently large multiple of Hk_c . The deviation from its scalar trace is at most

$$C \left(\frac{\|\mathcal{A}\|_F}{\sqrt{k_p}} \sqrt{H\gamma^3} + \delta H\gamma^3 \right). \quad (16.28)$$

Conditioning only on the image of ξ_j , the vector z_j is fixed and perpendicular to e_j . The residual projection of U has rank at most $\dim U$, so the same Haar estimate gives $\|P_U z_j\|^2 \leq CK_0^2 \delta H\gamma^3$. Its failure rate can be $2Hk_c$ after increasing absolute constants.

The span of sampled directions. Write W for the span of the e_j after projection onto $e_0^\perp$. Under canonical sampling, the centered Gaussian vectors before direction normalization have squared norm comparable to one in field units, limiting orthogonal Gram matrix, U_0 -projection squared norm of order γ , and covariance operator norm at most $C/(\gamma^2 k_p)$. Gaussian quadratic tails and nets in the fixed l -dimensional coefficient space therefore give

$$\|P_{U \cap e_0^\perp} P_W\| \leq C \sqrt{H\gamma}, \quad \sum_{j=1}^l |\langle e_j, w \rangle|^2 \leq C \|w\|^2 \quad (w \in W). \quad (16.29)$$

The projected generating vectors have a synthesis map and an inverse on its range bounded by absolute constants. Any pair of the original directions has bounded Gram conditioning. These conclusions hold at the same $2Hk_c$ rate. Although l will be a large power of γ^{-1} , it is fixed before $n \rightarrow \infty$, so the fixed-dimensional nets cost no additional multiple of k_p in the limit.

Let $\bar{a} = \text{Tr } \mathcal{A}/d_p$. Then

$$\|P_W(\mathcal{A} - \bar{a}I)P_W\| \leq C\delta(\sqrt{H\gamma^3} + l^{-1/2}), \quad (16.30)$$

$$\|P_W\mathcal{A}^2P_W\| \leq C\|\mathcal{A}\|_F^2/k_p + C\delta^2(\sqrt{H\gamma^3} + l^{-1/2}). \quad (16.31)$$

To account for the dependence of each matrix on its own sampled direction, expand a test vector in the bounded generating frame of W . For the summand A_j , condition on e_0, e_j and apply (16.26) to the other generating directions in their orthogonal complement. The scalar term is $\text{Tr } A_j/d_p$ up to a fixed-dimensional trace correction, which is $o(1)$. Terms with a row or column in $\text{span}(e_0, e_j)$ have average bounded by

$$\frac{C\delta}{l} \sum_j \|P_{\text{span}(e_0, e_j)} w\| \leq C\delta l^{-1/2} \|w\|, \quad w \in W,$$

using $w \perp e_0$ and (16.29). This proves (16.30) by polarization and a fixed-dimensional net. For the second estimate expand $\mathcal{A}^2 = l^{-2} \sum_{i,j} A_i A_j$. Condition on e_0, e_i, e_j . The same Haar test gives scalar term $\text{Tr}(A_i A_j)/d_p$; the sum of these terms is $\text{Tr } \mathcal{A}^2/d_p$. The exceptional-row contribution is bounded by $C\delta^2 l^{-1/2}$, since

$$l^{-2} \sum_{i,j} \|P_{\text{span}(e_0, e_i, e_j)} w\|^2 \leq Cl^{-1} \|w\|^2.$$

The same bound includes the diagonal pairs. Summing the scalar terms and the exceptional-row errors proves (16.31). Before applying the quadratic Haar test to a product, symmetrize it or use polarization. The argument therefore does not require an individual $A_i A_j$ to be positive.

Conditional on all the sampled images, the residual rotation gives a factor $C\sqrt{H\gamma^3}$ on the nonexceptional side of the block from W to its complement. Together with (16.29), this is the product $C\sqrt{H\gamma}\sqrt{H\gamma^3}$, not merely the first factor. There is no child radial contribution, since the target tests lie in $U \cap e_0^\perp$. Combining the three blocks gives, in addition to the scalar trace,

$$\begin{aligned} C \left[\frac{\|\mathcal{A}\|_F}{\sqrt{k_p}} \sqrt{H\gamma^3} + \delta H\gamma^3 + \delta H\gamma(\sqrt{H\gamma^3} + l^{-1/2}) \right. \\ \left. + H\gamma^2 \left(\frac{\|\mathcal{A}\|_F}{\sqrt{k_p}} + \delta(H\gamma^3)^{1/4} + \delta l^{-1/4} \right) \right]. \end{aligned} \quad (16.32)$$

Choose $l = \lceil \gamma^{-16} \rceil$. Up to this point, (16.32) is a deterministic bound on the function $\mathcal{A}(Y_1, \dots, Y_l)$ outside a bad event under the joint canonical law $\mathbb{P}_{\text{can}}$. The fixed- O estimates (16.23) and (16.25) concern $\mathbb{T}_O^l$. We transfer the geometric event to that law before using those expectations.

Transfer to actual transitions and a field net. So far the Haar estimates concern canonical Gaussian samples. We next convert them into estimates for the actual transition while keeping track of the loss in exponential rate. On a narrow parent shell the density ratio costs an arbitrarily small fixed multiple of the parent dimension for each sampled field. We choose that loss only after fixing the sample count, so the remaining rate pays for the child field net. Shell complements already have uniform conditional bounds and are added separately.

For fixed O , let F_O denote $V_p - q_{p,\alpha}^{\text{fr}}$ in (16.8). Its normalizing denominator may depend on the remaining orientation even after the terminal images have been fixed. Thus the residual Haar statement above is used only under $\mathbb{P}_{\text{can}}$. On the shell, (16.9) bounds each likelihood ratio by $e^{\eta k_p}$. For the canonical bad event $\mathcal{E}$, the joint change of law gives

$$\mathbb{E}_O \mathbb{T}_O^l(\mathcal{E} \cap \mathcal{S}_\tau^l) \leq e^{\eta k_p} \mathbb{P}_{\text{can}}(\mathcal{E}).$$

Choose $\eta \ll H\gamma^3/l$. A canonical bound e^{-2Hk_c} then makes the right-hand side at most $e^{-3Hk_c/2}$. Markov's inequality in O gives the explicit conditional form

$$\mathbb{P}_O \left\{ \mathbb{T}_O^l(\mathcal{E} \cap \mathcal{S}_\tau^l) > e^{-Hk_c/2} \right\} \leq e^{-Hk_c}.$$

The complement of the narrow shell is handled separately. Its conditional probability and all bounded polynomial moment errors are exponentially small, uniformly at every child field, by the deterministic upper support bound and Gaussian distance tails already proved. This uniform estimate is not obtained by a union bound over orientations or field points, and its exponent need not be $2Hk_c$. The scalar signed-trace and sample independence calculations are also applied anew conditionally at each field.

For an orientation outside the displayed exceptional set, we may now take the $\mathbb{T}_O^l$ -expectation of (16.32). Equation (16.25) bounds the expected Frobenius term by $O(\delta/\sqrt{l}) + \eta_{\text{err}}$, and (16.23) bounds the expected scalar trace. The bad-event and shell-complement contributions are small because the matrices have bounded operator norm there. After division by $\gamma^2\delta$, the displayed remainder terms are bounded by constant multiples of $H\gamma$, $H^{3/2}\gamma^{1/2}$, $H^{5/4}\gamma^{3/4}$, and powers of γ from $l^{-1/4}$ and $l^{-1/2}$. They tend to zero as $\gamma \rightarrow 0$. This proves the first estimate in (16.24), with an arbitrarily small additive error. Applying the same change of law to the one-vector event gives the stated bound for $\mathbb{E}_{\mathbb{T}_O} \|P_U z\|^2$.

We extend this estimate to all fields by comparing geometric bad events, before taking their signed matrix expectations. Take a net of resolution ρ in child field units, with ρ a sufficiently high fixed power of γ . Its size is at most $\exp\{Ck_c \log(1/\gamma)\}$. Fix an orientation and neighboring fields with $\|h - h'\| \leq \rho c^2 M_c$, and keep the physical terminal samples fixed. Take ρ small enough that their joining segment still has its frozen parameter inside parent agreement. The untruncated parent matrices are then unchanged. The scaled saddle Lipschitz bound gives

$$\sup_y |\chi_h(y) - \chi_{h'}(y)| \leq C\gamma^2 \rho / \tau. \quad (16.33)$$

Thus the cutoff-weighted A changes by at most $C\delta\gamma^2\rho/\tau$ in operator norm, and its Frobenius norm divided by $\sqrt{k_p}$ changes by the same order. The corresponding bound for z has $K_0\sqrt{\delta}$ in place of δ . The child radial projection changes by $C\rho$, since the annulus is separated from zero. On the retained Gram event the generating frames and their inverses have bounded norms. Consequently the geometric inequalities leading to (16.32), with fixed slack in their constants, persist under these changes with an additional $C_\gamma\rho\delta$ error. The same holds for the one-vector test. The constant C_γ is at most a fixed power of γ^{-1} : the sample count, cutoff width, and frame bounds have already been fixed.

The transition densities require an exponential comparison. For a fixed orientation, on $\|Y\| \leq Cp^2 M_p$,

$$\left| \log \frac{d\mathbb{T}_h}{d\mathbb{T}_{h'}}(Y) \right| \leq C_\gamma \rho k_c. \quad (16.34)$$

Indeed the potential $V_p(Y)$ cancels in the ratio, and the Gaussian log ratio uses only Δ_U^{-1} . The normalizing factor obeys the same bound: differentiating the convolution gives $\nabla_U \phi_c = \Delta_U^{-1}(\mathbb{E}_{\mathbb{T}_h} Y_U -$

h), whose norm is $O_\gamma(M_p)$ by the parent shell and complementary moment bounds. Integrating this derivative along the short field segment proves the normalization assertion. This argument uses only the conditioned parent hypotheses, not a separate good event at the child.

Choose ρ so that the perturbation of the matrix tests is much smaller than $\gamma^2\delta$, and $lC_\gamma\rho < H/8$. Prove the net estimates with the stated strict slack. Outside their orientation exceptions, the conditional probability of a relaxed geometric failure at a neighboring field, with all samples in a narrower typical shell, is at most $e^{lC_\gamma\rho k_c}e^{-Hk_c/2} \leq e^{-3Hk_c/8}$. The cutoff and radial continuity above justify this event inclusion; the narrower shell stays in the shell used at the net field. The shell complements and large lengths have the separate uniform conditional moment bounds already proved. At the neighboring field apply (16.23) and (16.25) anew, and then take the expectation of the relaxed geometric inequality. This retains their signed cancellation and proves (16.24) everywhere on the annulus.

Choose the fixed exponents in τ, ρ and the sample count first, then C_1 in $H = C_1 \log(1/\gamma)$ to pay for the net, and then γ small. The separate uniform shell bounds are added without a field union bound. The resulting failure probabilities have the form $e^{-ak_p} + e^{-a'k_c}$ with fixed positive constants and are summable on the geometric list down to $k_p \asymp \sqrt{\log n}$.

16.5 Completion of one inductive step

We can now assemble the matrix estimates. Compression makes the transverse contributions in both transport bounds smaller than the allowed transverse error. Radial regeneration supplies strict slack in the upper radial block, and the Fisher test supplies a bounded lower radial block. Positive-matrix Cauchy–Schwarz between these two bounds then controls the actual mixed block. This closes the four parts of the invariant at the child scale. The intermediate-scale estimates use the same deterministic transport calculations; they do not require another random compression.

By (16.24), (16.14) has transverse surplus $o(1)\delta$, since the mixed-square term gains γ^3H . For the Fisher lower bound, let e_0 be the child radial direction. The canonical mean a_c is parallel to e_0 , and $v^\top(R_c - R_p)v \leq C_\gamma p^{-2}$ for each unit $v \in U \cap e_0^\perp$. The shell norm and transferred second moments from the radial calculation therefore give

$$\sup_{\substack{v \in U \cap e_0^\perp \\ \|v\|=1}} \mathbb{E}_\mathsf{T}(v^\top e)^2 \leq \eta_{\text{err}} + C_\gamma/k_p.$$

The bounded radial entry of D is negligible on these tests. Its mixed terms are negligible by Cauchy–Schwarz and $\|z\| \leq K_0\sqrt{\delta}$, after choosing the comparison error and then k_p . The transverse compression thus gives a Fisher deficit $o(1)\delta$ as well, including the fixed factor x^2N^2 . The transverse-only Fisher test gives a block lower matrix with bounded radial deficit, and the Brascamp–Lieb upper matrix has cross error $O_\gamma(\delta) + \eta_{\text{err}}$ after taking a purely radial gain as above. The two Loewner bounds therefore force an actual cross block of size $O(\sqrt{o(1)\delta}) + O_\gamma(\delta)$, by positive-matrix Cauchy–Schwarz. This proves (16.2) inductively, with slack. At intermediate scales in the stated interior-parameter regime the same calculations without (16.24) give transverse error and actual mixed block bounded by a quantity tending to zero (at fixed γ), bounded total Hessian, and radial deficit at least 0.26. For the lower mixed estimate near $x = 1$ use the full Fisher test; its predicted cross block is small by the same bridge projections. These intermediate bounds are deterministic and uniform under the parent hypotheses.

16.6 Deterministic evaluation at other cap ratios

Once a parent satisfies the invariant, evaluation at another scale uses its deterministic support and covariance bounds. No transverse compression is needed if the target is an upper curvature estimate with prescribed fixed error, rather than a renewed invariant with the same error.

Lemma 16.2 (Evaluation with the induction extension). *Under the parent hypotheses of Proposition 16.1, the deterministic intermediate-scale conclusions hold also for $x = c/p \in [\gamma^2, \gamma]$, whenever the frozen parameter is inside parent agreement with fixed slack. At fixed γ , the transverse and mixed errors tend to zero with the parent errors, the scaled Hessian is bounded, and the radial deficit is at least 0.26.*

Proof. The same extension has a positive convolution margin because

$$\|\Delta^{1/2} L_p \Delta^{1/2}\| \leq (1 + \gamma^4/100)(1 - \gamma^4) < 1.$$

All inverse bounds in the shell estimates and the upper and lower covariance transports remain bounded by fixed powers of γ^{-1} . Thus those deterministic estimates apply with enlarged constants. Formula (16.20) still gives the stated radial deficit, while the other blocks tend to zero with the input errors. The proof uses no additional orientation event and does not extend the probabilistic induction statement beyond ratio γ . $\square$

For a bounded child field ball, the saddle parameter can occupy a wider range in child units. We choose a coarser parent and center a new support annulus at that frozen parameter. This is a separate evaluation of the true potential; its extension is not propagated as the next inductive potential.

Lemma 16.3 (Supports centered at a frozen parameter). *Fix a child field radius R before choosing the curvature and comparison tolerances. Suppose a true parent at scale p satisfies the invariant and the static comparisons. There is a fixed $b_1 > 0$, depending on R, σ , such that the following holds for every $x \in [\gamma b_1, b_1]$, $c = xp$, and $h \in H_c$ with $\|h\| \leq Rc^2 M_c$. Put $\alpha = \alpha_c(h)$. With sufficiently small fixed tolerances, there is an upper-semiconcave extension agreeing with the true parent on a neighborhood of the frozen typical length sphere. Its convolution has a strict margin, and its terminal mean and covariance differ from those of the true transition by exponentially small errors in the corresponding scaled units. Consequently,*

$$K_c(h) \preceq C(R)c^{-2}I_{H_c}, \quad (16.35)$$

where $C(R)$ is independent of the chosen ratios and sharp curvature tolerances. The construction and estimates are uniform on the indicated fixed ratio interval, with n taken large after those choices.

Proof. The static saddle bounds give $\kappa_R, C_R > 0$, depending only on the field radius, such that

$$-1 + \kappa_R \leq \alpha/c^2 \leq C_R.$$

Choose b_1 so that $\max(1, C_R)b_1^2 < \sigma/2$. Then $|\alpha| < \sigma p^2/2$ throughout the ratio interval. Choose fixed numbers $e, w > 0$, also satisfying the tail and scalar-margin bounds below, with

$$C(w + \delta) \ll e \ll \kappa_R(\gamma b_1)^2, \quad w < \sigma/4,$$

and define nested shells

$$S_j^\alpha = \{y \in H_p : |\alpha_p(y) - \alpha| \leq t_j w p^2\}, \quad (t_0, t_1, t_2) = (1/2, 3/4, 1).$$

They lie inside the parent invariant annulus. The radius derivative computed above gives buffers comparable to wp^2M_p between them. Set

$$L_\alpha = ((p^2 + \alpha)^{-1} + ep^{-2})I_{H_p}.$$

On S_2^α , the invariant's positive-part bound gives

$$K_p(y) \preceq ((p^2 + \alpha)^{-1} + C(w + \delta)p^{-2})I_{H_p} \prec L_\alpha.$$

This proves the upper-support inequality for pairs in S_1^α whose joining segment remains in S_2^α .

For longer pairs, freeze the own saddle at the support point and use (15.8), the value comparison, and the gradient comparison as in (16.6). If the buffer divided by p^2M_p is $\beta \asymp w$, the excess over the frozen α quadratic is at most

$$\frac{\|y - z\|^2}{2p^2} \left[Cw + 2 \left(\frac{2\varepsilon_v + C_{\text{dens}}/k_p}{\beta^2} + \frac{\varepsilon_g}{\beta} \right) \right].$$

Take $\varepsilon_v + C_{\text{dens}}/k_p \ll ew^2$ and $\varepsilon_g \ll ew$. The bracket is smaller than e , so the upper-support inequality holds for every pair in S_1^α . Define

$$\bar{\phi}_{p,\alpha}(y) = \inf_{z \in S_1^\alpha} \left\{ \phi_p(z) + \nabla_{H_p} \phi_p(z)^\top (y - z) + \frac{1}{2}(y - z)^\top L_\alpha (y - z) \right\}.$$

The argument of (16.7) shows that this function agrees with ϕ_p on S_1^α , has at most quadratic growth, and satisfies $\nabla^2 \bar{\phi}_{p,\alpha} \preceq L_\alpha$ distributionally. Its dependence on the child field is through the scalar α , determined by the eigenvalues and $\|h\|$. No child angular refinement enters its construction. For this evaluation hold the selected extension fixed when differentiating its convolution at h . Different fields may select different extensions; no derivative of this selection, or of its parameter α , enters the Hessian formula.

We verify the complementary moments before comparing covariances. Let B_α denote the logarithm of the frozen norm density at a field on the sphere $\|y\| = \ell_p(\alpha)$. It is constant on that sphere. For $d(y) = \|\|y\| - \ell_p(\alpha)\|/(p^2M_p)$, the nearest support point on the sphere gives

$$\bar{\phi}_{p,\alpha}(y) - q_{p,\alpha}^{\text{fr}}(y) \leq B_\alpha + k_p(\varepsilon_v + \varepsilon_g d(y) + \frac{1}{2}ed(y)^2) \leq B_\alpha + k_p(\varepsilon_v + \varepsilon_g^2/e + ed(y)^2).$$

The canonical calculation leading to (16.10) now gives, with constants depending only on the bounded child saddle range,

$$\Pr_{\mathcal{G}}\{d(Y) > v\} \leq 2e^{-c_R x^2 k_p v^2}, \quad v \geq C_R/(x\sqrt{k_p}).$$

Require also $e < c_R(\gamma b_1)^2/4$. Normalize the extended transition on a shell $|\alpha_p(Y) - \alpha| \leq \tau_0 p^2$ with $\tau_0 \ll \gamma b_1 w$. Its canonical probability tends to one. The off-saddle identity (15.18) and centered density bounds give a logarithmic normalization loss at most $\varepsilon_v k_p + C\tau_0^2 k_p + O(1)$ relative to B_α . Choose the comparison errors and then n so that

$$2\varepsilon_v + \varepsilon_g^2/e + C\tau_0^2 + O(1)/k_p \ll (\gamma b_1)^2 w^2.$$

Leaving S_1^α requires $d(Y) \geq cw$. Integrating the Gaussian distance tail against the upper support therefore bounds every fixed polynomial moment of $1 + d(Y)$ on this complement by $C_j e^{-ak_p}$, for a fixed $a > 0$, uniformly in x, h .

The true transition needs its own complementary estimate. On each fixed bounded parent field range, static comparison and the frozen numerator density upper bound give

$$\phi_p(y) - q_{p,\alpha}^{\text{fr}}(y) \leq B_\alpha + \varepsilon_v k_p + C_{\text{dens}}.$$

The preceding normalization shell and Gaussian distance tail control the bounded part of the complement. For the remaining tail, sample the child spin by tilting the parent zero-field spin law with $\exp\{h^\top X + X^\top \Delta X/2\}$, then set $Y = h + \Delta X + N(0, \Delta)$. The tilt normalizer is at least one by zero-field symmetry and $\Delta \succeq 0$. With $v = \|P_p X\|/M_p$, its numerator is at most $e^{C_R k_p(1+v^2)}$. The parent projection tail $e^{-ck_p v^6}$ survives this factor, and the observation noise has a Gaussian tail in parent field units. Increasing the fixed outer radius therefore makes the outer contribution and all its fixed polynomial moments exponentially small. This argument uses the true parent projection tails, not the extension's Hessian bound.

The two unnormalized transition laws agree on S_1^α . Their normalization ratio and their complementary first and second moments thus show that their means and covariances differ exponentially in parent scaled units, as in the earlier transition comparison. Converting a covariance error in units $(p^2 M_p)^2$ to child Hessian units multiplies it by

$$c^2 \|\Delta_U^{-1}\|^2 (p^2 M_p)^2 = \frac{x^2 k_p}{(1-x^2)^2}.$$

The exact formula (16.11) therefore leaves an error in $c^2 K_c$ tending to zero uniformly on the fixed ratio interval.

Finally put $a = \alpha/c^2$ and $q = (1+x^2 a)^{-1} + e$. On U , the increment is $(1-x^2)p^2 I$, while the global support curvature is q/p^2 . Impose also $e \leq \kappa_R(\gamma b_1)^2/[4(1+\sigma/2)]$. Then the convolution has a strict margin. Brascamp–Lieb, first for smooth approximations and then by passage to the limit, and (16.11) give

$$c^2 K_c^{\text{ext}} \preceq \frac{1 + (1+x^2 a)e}{(1+a) - (1-x^2)(1+x^2 a)e/x^2} I_U \preceq \frac{8}{3\kappa_R} I_U.$$

The denominator is at least $3\kappa_R/4$, and the numerator is at most two. The exponentially small replacement error is absorbed by enlarging this constant once. This proves (16.35) with no dependence of its coarse constant on x or the sharp tolerances. All shell widths, comparison errors, and the ratio interval are fixed before $n \rightarrow \infty$; no estimate here requires taking $x \rightarrow 0$ while keeping these choices fixed. $\square$

17 Classical boundary fields

The cap construction starts at a fixed positive scale. At this boundary we need an approximation to the gradient in unnormalized Euclidean norm, a bounded covariance, and an estimate that remains useful after changing the spin law by a square density. We first specify the companion high-temperature input and its restricted endpoint extension. We then analyze small positive fields at the endpoint; it is this second analysis that supplies the curvature and value of the initial auxiliary potential. All scales and all error tolerances in this section are fixed before letting the dimension tend to infinity.

For a scale-one GOE matrix W , let $J_* = \beta W_{\text{off}}$ and $j = \beta^2$. The diagonal of W can be restored when convenient, because it is constant in the spin Hamiltonian and tends to zero in operator norm.

For $y, h \in \mathbb{R}^n$ set

$$m = \tanh y, \quad q = \frac{\|m\|^2}{n}, \quad r = 1 - q, \quad V = \text{diag}(1 - m_i^2), \quad F_h(y) = y + (jrI - J_*)m - h. \quad (17.1)$$

Write $V_0(z) = \text{sech}^2 z$, so $V = \text{diag } V_0(y_i)$. We write $\langle uv \rangle = n^{-1}u^\top v$ and $|u|_n = n^{-1/2}\|u\|$. A field passes the stability test with parameters $b, \eta, \epsilon_A, \rho > 0$ if the following implication holds for every $y \in \mathbb{R}^n$ and every diagonal A :

$$\left\{ \begin{array}{l} |F_h(y)|_n \leq 2\rho, \quad 0 \leq A \leq (1 + \eta)I, \\ n^{-1/2}\|A - V\|_F \leq \epsilon_A \end{array} \right\} \implies I + A^{1/2} \left(jrI - J_* - \frac{2j}{n}mm^\top \right) A^{1/2} \succeq bI. \quad (17.2)$$

The small enlargement of the residual region will allow later changes in the external field and the finite spin-flip neighborhoods needed in the residual identities.

17.1 The posterior input and its hypotheses

Proposition 17.1 (Classical posterior input). *Fix $0 < \beta < 1$, an operator-norm bound for J_* , and fixed positive parameters in (17.2), with $j(1 + \eta)^2 < 1$. On the fixed-length matrix-word events described below, every field satisfying (17.2) has a unique root y_* of F_h . If $m_* = \tanh y_*$ and*

$$\mu_h(x) \propto \exp\{x^\top J_* x / 2 + h^\top x\}, \quad \mathcal{D}_h(f) = \sum_i \mathbb{E}_{\mu_h} \text{Var}(f \mid x_{\neq i}),$$

then

$$\|\mathbb{E}_{\mu_h} x - m_*\| \leq C, \quad \|\text{Cov}_{\mu_h}(x)\| \leq C. \quad (17.3)$$

For every sufficiently small fixed $\epsilon > 0$, every real f with $N = \mathbb{E}_{\mu_h} f^2 > 0$, and $\nu = f^2 \mu_h / N$,

$$|\mathbb{E}_\nu x - m_*|_n \leq C\epsilon + \frac{C_\epsilon}{\sqrt{n}} \left(1 + \frac{\mathcal{D}_h(f)}{N} \right). \quad (17.4)$$

The event and constants are independent of h and f under the stated diagnostics and stability parameters. The required word length is finite and may depend on ϵ .

The same conclusions hold at $j = 1$ if, in addition, every approximate root in the residual region has $q \geq q_{\min} > 0$, and the inverse-word event is restricted to the diagonal class specified in Lemma 17.2. Constants may depend on $q_{\min}$. This endpoint assertion is a consequence of the proof below, rather than an application of a theorem with $\beta < 1$ at $\beta = 1$.

Here and below a word means an ordered product of bounded real diagonal matrices and copies of J_* , with at most one factor

$$K_A(z) = \left(I - zAJ_* + z^2 j \langle A \rangle A \right)^{-1}, \quad \langle A \rangle = n^{-1} \text{Tr } A, \quad 0 \leq z \leq 1.$$

Its prediction is obtained by the noncrossing contractions $J_* D J_* \mapsto j \langle D \rangle I$, after formally expanding the one inverse factor. For each fixed word length the diagnostic bounds are bounded operator norm, bounded off-diagonal entrywise ℓ^4 norm, and bounded Euclidean norm of the diagonal minus its prediction. Inverse words are only used when

$$I + z^2 j \langle A \rangle A - zA^{1/2} J_* A^{1/2} \succeq cI$$

for a fixed $c > 0$. For $j < 1$ and $j(1 + \eta)^2 < 1$, these are precisely the simultaneous diagnostics of [28, Lemma 3.1].

Lemma 17.2 (Restricted endpoint matrix diagnostics). *Fix $a_+ < \infty$ and $\kappa \in (0, 1)$. At $j = 1$, restrict inverse diagonals to*

$$\mathcal{A}_\kappa = \{A : A \text{ diagonal, } 0 \leq A \leq a_+ I, \|A\| \langle A \rangle \leq 1 - \kappa\}. \quad (17.5)$$

For each fixed word length and bounded diagonal coefficients, the preceding word diagnostics hold simultaneously over this class and all stable inverse parameters, with probability $1 - Ce^{-cn}$. The fixed-parameter bilinear equivalents of [28, Lemma 3.4] also hold, uniformly for deterministic $A \in \mathcal{A}_\kappa$ and finitely many deterministic bounded-norm test vectors. They are not asserted simultaneously over all test vectors. The regularized determinant bound of [28, Lemma 3.5] holds simultaneously for diagonal $0 \leq V \leq I$ with $\langle V \rangle \leq 1 - \kappa$, including its bounded-rank, bounded-norm perturbations.

Proof. We isolate the two places in the companion matrix proof [28, Section 3] where the temperature gap is used. The first is a lower bound on the symmetric inverse along its scalar parameter path; the second is a coefficient below one in the loop equation. The restricted diagonal condition supplies both margins at the endpoint. Once these margins are available, truncation and chaining give the simultaneous assertion by the same finite recursions.

For a deterministic A , the Gaussian process comparison in its Lemma 3.2 bounds the relevant supremum by $2d - d^2$, where now

$$0 \leq d \leq \sqrt{\langle A \rangle \|A\|} \leq \sqrt{1 - \kappa}.$$

It therefore gives the fixed margin $c_0 = (1 - \sqrt{1 - \kappa})^2/4$ along the whole path $0 \leq z \leq 1$, after concentration and a fixed mesh of that path. The constants are uniform over deterministic A in (17.5). Next, the loop equation for $u(z) = n^{-1} \mathbb{E} \text{Tr}(K_A(z)A) - \langle A \rangle$ has linear coefficient

$$z^2 n^{-1} \sum_i A_{ii}^2 \mathbb{E} K_A(z)_{ii}.$$

When u is small, the diagonal expectations are $1 + O(u) + O(n^{-1})$. Since $\langle A^2 \rangle \leq \|A\| \langle A \rangle \leq 1 - \kappa$, this coefficient stays below $1 - \kappa/2$. Continuity from $z = 0$ then gives $u(z) = O(n^{-1})$ throughout the path. All subsequent noncrossing loop recursions are finite and unchanged.

For clarity, the simultaneous assertion does not condition on a randomly chosen diagonal. Use the globally truncated inverse from [28, Section 3.2], with a smooth cutoff equal to x^{-1} on a neighborhood of $[\min(c, c_0)/2, C]$. Its word maps and their increments have Frobenius Lipschitz constants C and $C\|\theta - \theta'\|_\infty^{1/2}$, respectively. Cover the restricted parameter set by meshes with representatives in that set, or in $\mathcal{A}_{\kappa/2}$ when a buffer is needed. At resolution 2^{-l} there are at most $\exp(C(l+1)n)$ representatives. Gaussian concentration and chaining bound the centered word increments by $C\sqrt{l+1}2^{-l/2}$, with summable failure $Ce^{-c(l+1)n}$. The deterministic diagonal expectation error is $O(n^{-1})$ coordinatewise by the preceding loop calculation. This proves the word assertion even for diagonals selected from the observed matrix. The bilinear assertion follows by applying concentration to each fixed truncated bilinear form, exactly as in the cited lemma.

For the determinant take $A = V$. The fixed-diagonal margin and loop calculation give

$$\mathbb{E} \frac{1}{n} \log \det(L^\top L + \gamma I) \leq \langle V \rangle^2 + C_\kappa \gamma + O(n^{-1}), \quad L = I + (\langle V \rangle I - W)V.$$

The regularized normalized log determinant is $(n\gamma)^{-1/2}$ -Lipschitz in Frobenius norm, so its concentration has rate n^2 at fixed tolerance. A fixed mesh of the restricted diagonal class costs only $\exp(Cn)$; use its slight enlargement for continuity. The bounded-rank perturbation changes the answer by at most $\|E\| \sqrt{\text{rank } E/(n\gamma)}$. This proves the stated uniform determinant bound. Removing the GOE diagonal changes the truncated words by a bounded Frobenius amount and the stability matrices by $o(1)$ in operator norm, as in the companion proof. $\square$

Proof of Proposition 17.1. For fixed $j < 1$ the root assertion is [28, Lemma 5.7]; its proof is deterministic and allows any fixed $j > 0$. It gives a bounded inverse for F'_h and $\|y - y_*\| \leq C\|F_h(y)\|$ whenever $|F_h(y)|_n < 2\rho$; the later uses lie in a strictly smaller residual region. The last estimate in the proof of [28, Proposition 7.2], before centering its test function, is

$$|\mathbb{E}_{\mu_h}[G(x - m_*)^\top e]| \leq C \left(\mathbb{E}_{\mu_h} G^2 + \mathcal{D}_h(G) \right)^{1/2}, \quad \|e\| \leq 1. \quad (17.6)$$

Taking $G = 1$ gives the first assertion in (17.3). For covariance, apply the centered statement of that proposition to $G = e^\top x$, with e a unit eigenvector for the largest covariance eigenvalue λ . Since $\mathcal{D}_h(e^\top x) \leq \sum_i e_i^2 = 1$, it yields $\lambda \leq C\sqrt{\lambda + 1}$, hence $\lambda \leq C'$. The square-density estimate is exactly [28, Proposition 7.3], in the present rate-one-per-site normalization.

The endpoint word event is useful only if the diagonals actually produced by the posterior proof belong to its restricted class. We therefore inspect the implicit recipe, rather than substituting the endpoint into the companion statement. The root comparison places its primary vectors near the same root throughout a buffered residual region. This controls both the largest diagonal entry and the average diagonal entry, giving the strict product margin required by the endpoint inverse.

Its primary iterates are

$$m^0 = x, \quad h^1 = h + J_* x, \quad m^l = \tanh h^l, \quad b_l = 1 - |m^l|_n^2, \quad h^{l+1} = h + J_* m^l - j b_l m^{l-1}.$$

Write $R_l = m^{l-1} - m^l$ and $\Omega_{2\rho} = \{\max(\|R_k\|, \|R_{k-1}\|) \leq 2\rho\sqrt{n}\}$. The root comparison gives, independently of k ,

$$|h^k - y_*|_n + |m^{k-1} - m_*|_n \leq C\rho, \quad |1 - b_{k-1} - q_*| \leq C\rho.$$

The only inverse diagonal in the implicit recipe is

$$A_{ii} = \Phi(h_i^k; (y_*)_i, a), \quad a = j(1 - b_{k-1} - q_*), \quad \Phi(z; v, a) = \int_0^1 e^{a(1-u^2)/2} \frac{\cosh(u(z-v))}{\cosh z \cosh v} du.$$

Its secant-slope representation at $a = 0$ and its bounded derivatives give, on the entire buffer,

$$\|A\| \leq e^{C\rho}, \quad \langle A \rangle \leq 1 - q_* + C\rho, \quad n^{-1/2} \|A - V_{h^k}\|_F \leq C\rho.$$

Choose ρ after $q_{\min}$ and the stability margins so that $e^{C\rho}(1 - q_{\min} + C\rho) \leq 1 - q_{\min}/2$. Thus all these diagonals lie in the interior of $\mathcal{A}_{q_{\min}/4}$. At h^k , $\langle V_{h^k} \rangle = b_k$, so the same Frobenius bound gives

$$|\langle A \rangle - b_k| \leq n^{-1/2} \|A - V_{h^k}\|_F \leq C\rho.$$

Removing the negative rank from (17.2) and then changing the scalar coefficient from $j b_k$ to $j \langle A \rangle$ therefore gives

$$I + j \langle A \rangle A - A^{1/2} J_* A^{1/2} \succeq (b - j C \rho \|A\|) I.$$

Choose ρ to retain a fixed positive margin. This is the symmetric matrix for the actual recipe inverse $K_A(1)$; the path $0 \leq z \leq 1$ is used to establish its formal word prediction. Every later auxiliary recipe is ordinary; the closed differentiated formulas contain at most this one inverse.

These bounds must also hold at the finitely many neighboring spin configurations used by the residual identities. The recipe depth is fixed before the dimension grows. Consequently the total Euclidean change across its required flips is bounded independently of the dimension, while the residual buffer has radius proportional to its square root. Choosing the dimension last keeps every required coefficient inside the same restricted diagonal class.

The finite-flip buffers require no uniform bound over all depths. After fixing a recipe and its depth, let Q be the largest number of successive spin flips appearing in its residual induction, and let C_Q bound the one-flip Euclidean changes of all involved primary vectors. Both are finite by [28, Lemma 6.1]. Taking $n > (2QC_Q/\rho)^2$ places every required neighborhood of $\Omega_{3\rho/2}$ inside $\Omega_{2\rho}$. All inverse coefficients there retain the preceding trace margin. This is the buffer argument in the released version of [28, Section 6.3].

The order of choices is consequently: the root and diagonal stability margins; the restricted diagonal margin; a sufficiently small ρ ; the finite residual-selection depth; the resulting finite recipes and word length; and finally n . The residual-selection depth may increase when the accuracy in (17.4) decreases. None of these choices depends on f or on a directional seed. Ordinary words have no restricted inverse parameter at all. The proof of [28, Section 6 and Propositions 7.2–7.3] now applies term by term using Lemma 17.2, proving the endpoint version and its claimed uniformity.

For a compact coefficient interval strictly inside $(0, 1)$, choose the same strict diagonal margin throughout and include β as one more bounded scalar in the parameter meshes. The additional mesh cardinality does not change their exponential-in- n entropy, and all Lipschitz and loop constants have the same margin. The word and posterior conclusions are therefore simultaneous on that interval. In the restricted endpoint case the same argument applies to a sufficiently small coefficient neighborhood that preserves (17.5). $\square$

17.2 The positive-time endpoint stability test

Lemma 17.3. *For a critical GOE interaction the following assertions hold with probability tending to one over the disorder. For every fixed $q_0 > 0$ there is $\rho_0 > 0$ such that $|F_0(y)|_n \leq \rho_0$ implies $q(y) < q_0$. For every fixed $0 < t_b \leq T < \infty$, there are positive constants in (17.2) and $a > 0$ such that, conditionally on typical disorder, that test holds simultaneously along*

$$Y(t) = tX + B(t), \quad t_b \leq t \leq T, \quad X \sim \mu_0,$$

outside conditional probability e^{-an} . It also holds throughout a fixed normalized neighborhood of these fields. The conclusions persist for sufficiently small fixed changes of the interaction coefficient and sufficiently small normalized field displacements.

Proof. We adapt the conditional Gaussian, scalar-rate, and local-volume proof of [28, Section 5]. The adaptation discards a small- q region by an observation estimate, rather than using a temperature gap. Here are the details needed at the endpoint.

First the scalar entropy inequality in [28, Lemma 4.1] extends to $j = 1$ when $q > 0$. Its proof uses $k = jq/s \leq 1$ and $B = j(1 - q) < 1$, which remain valid. In its large- C case the quantity $C^2/(jk)$ is still strictly greater than one, since $C > 1$. The strictness and equality statements are therefore unchanged. The consistent Gaussian laws at $j = 1$ are $N(s, s)$ with

$$s = t + \mathbb{E}_{N(s,s)} \tanh Y.$$

Indeed both scalar one-Lipschitz comparisons in the proof of [28, Lemma 4.2] become strictly contracting on every nondegenerate compact interval: their Gaussian derivatives are strictly less than one. More explicitly, if $f(s) = \mathbb{E}_{N(s,s)} \tanh Y$, then $f(0) = 0$, $0 \leq f'(s) < 1$ for $s > 0$, and $f(s) \leq 1$. Thus $s - f(s)$ is strictly increasing from zero to infinity. The solution is unique, equals zero at $t = 0$, and is continuous by this monotonicity and compactness. The second scalar comparison forces equality of the consistent mean and variance because its Lipschitz inequality is strict between distinct nonnegative arguments. These facts replace the global contraction constant $j < 1$; no bound involving $(1 - j)^{-1}$ is used.

At $t = 0$, restrict the scalar-rate integral to $q \geq q_0/2$. Its variance is then bounded below, and no consistent equality law lies in this set. The compactness and Gaussian-tail argument of [28, Lemma 5.4] gives a strictly negative exponential rate, including small residual smoothing. The determinant diagnostic is only required on this retained set, where $\langle V \rangle = 1 - q \leq 1 - q_0/2$, so Lemma 17.2 supplies it. A small normalized ball around a witness with $q \geq q_0$ stays in the retained set. The local-volume lower bound of [28, Lemma 5.6] is deterministic and has no subcritical restriction. It turns the negative integral rate into exclusion of approximate zeros.

On $[t_b, T]$, very small q is excluded first. A sufficiently accurate approximate root satisfies $|y - Y(t)|_n \leq C\sqrt{q} + \rho$. Since $\tanh$ is one-Lipschitz, this would force $|\tanh Y(t)|_n$ to be small. For each fixed spin the coordinates of $tX + B(t)$ are independent Gaussians with means $\pm t$ and variance t , so their bounded squared hyperbolic tangents have an empirical average bounded below, with exponential probability uniformly on the compact time interval. A fixed time mesh and Brownian maximal bounds give the same conclusion along the path. We may therefore retain $q \geq q_{\min} > 0$ with slack.

On this retained set, the scalar equality curve and all scalar-rate arguments are compact and nondegenerate. We make the conditional matrix law explicit. During the Gaussian reduction use the temporary planted interaction $\tilde{J} = W + \mathbf{1}\mathbf{1}^\top/n$ with full GOE W . For fixed y , let

$$\begin{aligned} b &= 1 - q, & M &= \langle m \rangle, & d &= t + M, & s &= t + \sigma^2 + q, & S &= s + q, \\ a_y &= \langle (y - d\mathbf{1})m \rangle, & z_{\text{cond}} &= y - d\mathbf{1} + bm. \end{aligned}$$

The matrix factor in [28, Eq. (5.8)] is the Gaussian conditional law given

$$Wm + \sqrt{t + \sigma^2} g' = z_{\text{cond}}.$$

Here g' is independent standard Gaussian and $\sigma > 0$ is the residual smoothing. Put

$$u = \frac{m}{\sqrt{nq}}, \quad Z_1 = \frac{y - d\mathbf{1}}{\sqrt{ns}}, \quad P_u = I - uu^\top.$$

Writing $W = P_u W P_u + uw^\top + wu^\top + \alpha_c uu^\top$ with $w \perp u$, the elementary conditional Gaussian formulas give a representation using a fresh GOE W_0 :

$$\begin{aligned} P_u W P_u &= P_u W_0 P_u, \\ w &= \sqrt{q/s} Z_1 - (a_y/s)u + \sqrt{(t + \sigma^2)/s} P_u W_0 u, \\ \alpha_c &= \frac{2(a_y + bq)}{S} + \sqrt{(t + \sigma^2)/S} u^\top W_0 u. \end{aligned}$$

This is [28, Eq. (5.12)] at $j = 1$.

For the moment fix an admissible diagonal A , deterministic after y has been fixed. Set

$$\chi = \langle A \rangle, \quad S_A = I + \chi A - A^{1/2} W_0 A^{1/2}, \quad Y_A(v) = S_A^{-1/2} A^{1/2} v.$$

Lemma 17.2 supplies a positive lower bound for S_A . Its fixed-diagonal bilinear estimates apply to the three deterministic vectors $\mathbf{1}/\sqrt{n}, u, Z_1$, whose norms are bounded by the retained second-moment bound. Their transformed Gram entries are close to $v^\top A v'$ for the corresponding pair v, v' ; the extra vector $Y_A(W_0 u)$ has mixed entries close to $\chi v^\top A u$ and squared norm close to $\chi + \chi^2 u^\top A u$. These are Gram comparisons under the fresh W_0 law. At the limiting scalar equality law with $A = V$ and $\sigma = 0$, the whitened finite-rank test has determinant

$$(1 - b + 2g_1)(1 - b + g_1)^2 > 0, \quad g_1 = \mathbb{E}_{N(s,s)}[\tanh^2 Y \operatorname{sech}^2 Y].$$

The rank perturbation has at most one possibly nonpositive direction; the positive determinant excludes that direction, exactly as in [28, Lemma 5.5]. Compactness gives a uniform margin. For A close to V , choose η and its normalized Frobenius tolerance so that $\|A\|\langle A \rangle \leq 1 - \kappa$ on the retained range. To include data-dependent A , keep y fixed and round the entries exceeding a fixed sup-norm tolerance on their exceptional coordinate set. That set has fraction at most ϵ_A^2/ζ^2 , and the logarithm of the number of choices is at most

$$n\{H(\epsilon_A^2/\zeta^2) + (\epsilon_A^2/\zeta^2) \log(C/\zeta) + o(1)\}.$$

Choose ζ after fixing the vector-norm bounds, Gram tolerance, and fixed-diagonal exponent, then ϵ_A so that this counting rate is below that exponent. For this fixed y , the approximants are deterministic, so a union bound makes the conditional matrix comparison simultaneous in A . The omitted scalar mass has norm at most $(1 + \eta)|\chi - b| \leq (1 + \eta)n^{-1/2}\|A - V\|_F$ and is included in the reserved margin. The probability bound is uniform in the retained y -data, as required inside the scalar integral. Enlarging the retained ranges slightly before the local-volume step preserves all these margins.

The preceding calculations are under the usual size-biased planted law. The critical zero-field partition comparison in Proposition 15.1 gives $\log Z - \log \mathbb{E}Z = o(n)$ in probability. Thus negative exponential rates transfer to typical original disorder: restrict to $Z/\mathbb{E}Z \geq e^{-\delta n}$, choose δ below the rate, and use Markov's inequality. A finite time mesh and the Brownian maximal bound supply the whole path. Finally, F_h is one-Lipschitz in its field and its stability matrix does not use h ; coefficient and diagonal changes are uniformly continuous under the norm bounds. Reserving slack in all thresholds proves the neighborhood and perturbation assertions. $\square$

17.3 Noise diagnostics at a fixed small scale

We next work at $j = 1$ with the full GOE matrix W . A spectral Gaussian means $g = O\Sigma^{1/2}\xi$, where $W = OAO^\top$, ξ is standard Gaussian independent of O , and Σ is positive semidefinite and fixed after conditioning on the eigenvalues. We allow a fixed smooth compact family of such covariances with $\|\Sigma\| \leq Ct$. Write the field as

$$h = g + h_1, \quad |h_1|_n \leq Ct. \quad (17.7)$$

The shift may be selected after seeing the frame and noise in every frame estimate below. A lower bound for a shifted Gaussian projection has a different quantifier, specified separately in Lemma 17.4.

Let Π_d denote the projections onto dyadic bands for the gap $2 - W$, with rank at most $Cnd^{3/2}$. At a bottom label a , combine all smaller gaps into one band with that label. Separately, write $P_t^W = \mathbf{1}_{\{2-W \leq t\}}$ for the cumulative projection onto gaps at most t ; thus its cap-width parameter is $\sqrt{t}$. If a vector class obeys $|\Pi_d v|_n \leq b_d$, put

$$\mathcal{L}(v) = \sum_{d \geq a} b_d d^{3/4}. \quad (17.8)$$

Bounds for the individual band sizes can be chosen on arbitrarily fine fixed grids. A finite collection of known spectral-coordinate vectors, including the filtered noise, is included in the Gaussian comparison but is not charged this complexity. Its number is fixed independently of n ; smooth scalar spectral filters can be netted before n increases.

We record the two frame estimates used repeatedly. Up to logarithmic factors in the band count, for a vector of normalized size at most B ,

$$\sup_{|E| \leq fn} |\mathbf{1}_E v|_n \leq C\{B\sqrt{f \log(2/f)} + \mathcal{L}(v)\}. \quad (17.9)$$

Empirical averages of smooth clipped functions of finitely many vectors, normalized to have size at most one, differ from the corresponding centered Gaussian averages with the same Gram matrix by

$$C_\psi \sum_v \mathcal{L}(v). \quad (17.10)$$

Here the complexities are those of the normalized vectors and C_ψ is controlled by the first two derivatives of the clipped function. Both estimates permit an additional fixed slack and a failure allowance $C\sqrt{\mathfrak{h}/n}$, multiplied by the indicated size and derivative bounds, at failure rate $e^{-c\mathfrak{h}}$. We use $\mathfrak{h}$ of order $nt^{3/2}$.

For completeness, (17.9) follows by testing each band against a sparse coordinate unit vector. Nets of its coefficient sphere and of the chosen rows have logarithmic size $Cnd^{3/2} + Cnf \log(2/f)$, respectively. The Gaussian representation of a Haar unit vector bounds its scalar products at the square root of this entropy divided by n . Sum the band bounds and use Cauchy–Schwarz for the total-size term. To prove (17.10), first fix k spectral coefficient columns $C = (c^1, \dots, c^k)$, with k fixed and $|c^j|_n \leq 1$. Put

$$Q = C^\top C/n, \quad F_\psi(O, C) = n^{-1} \sum_i \psi((OC)_i).$$

For ω uniform on S^{n-1} , rotation invariance gives the exact one-row expectation

$$\mathbb{E}_O F_\psi(O, C) = \mathbb{E}_\omega \psi(\sqrt{n} Q^{1/2}(\omega_1, \dots, \omega_k)^\top) = \mathbb{E} \psi(G_Q) + o_n(1), \quad G_Q \sim N(0, Q).$$

The formula is interpreted on the support of a possibly singular Q . Thus the Gaussian uses the actual Gram matrix of the chosen columns. For the usual Frobenius metric on the orthogonal group,

$$\text{Lip}_O F_\psi \leq C_\psi \|C\|_{\text{op}}/\sqrt{n} = O(C_\psi), \quad \text{Lip}_O(F_\psi(\cdot, C) - F_\psi(\cdot, C')) \leq C_\psi \|C - C'\|_{\text{F}}/\sqrt{n}.$$

The first estimate follows by differentiating the row average and using $\|\nabla \psi(OC)\|_{\text{F}} \leq C_\psi \sqrt{n}$; the second uses the bounded second derivative. Haar concentration has exponent proportional to n at this Lipschitz scale. Chaining over the band balls therefore costs $C_\psi \sum b_d \sqrt{\text{rank } \Pi_d/n}$, and a failure budget $\mathfrak{h}$ costs $C_\psi \sqrt{\mathfrak{h}/n}$. This proves (17.10). The event is uniform over the coefficient balls, so vectors may be selected from those balls after the frame is observed.

We intersect the full-frame event with the following residual-rotation event. For this calculation fix the eigenvalues Λ , the low physical frame O_0 , and a high spectral multiplier D at a fixed scalar net point. Choose an isometry H onto the complement of the low frame and write the remaining columns as $O_{>} = HU$, where U is Haar in dimension m . Put $M = HUDU^\top H^\top$ and $P_\perp = HH^\top$. Its exact conditional mean is

$$\mathbb{E}[M \mid \Lambda, O_0, D] = \mu P_\perp, \quad \mu = \text{Tr } D/m.$$

For a finite net of vectors b, t fixed under this conditioning, let h_{net} be its logarithmic cardinality and $h_{\text{tail}} > 0$. The Haar quadratic-form bound and polarization give, with $\mathfrak{H} = C(h_{\text{net}} + h_{\text{tail}})$, outside conditional probability $2e^{-h_{\text{tail}}}$,

$$|\langle b, Mt \rangle - \mu \langle P_\perp b, P_\perp t \rangle| \leq C|b|_n|t|_n \left(\frac{\|D\|_{\text{F}}}{\sqrt{m}} \sqrt{\frac{\mathfrak{H}}{m}} + \|D\| \frac{\mathfrak{H}}{m} \right). \quad (17.11)$$

The probability is over the remaining U ; coefficient and sparse nets are included in h_{net} . This is the conditional estimate used for the high multipliers below. The low coordinate profiles remain fixed.

We specify the finite families included in the noise event. Scalar meshes cover the covariance parameters, r , the small values $s = q + \sqrt{t}$, and the derivative shift $0 \leq \ell \leq Cs$, with cutoffs $a = s^{7/5}$ and $u = s^{1/10}$. The uncharged spectral columns are the noise and the finitely many scalar resolvent filters used in the root and derivative bases, including $r^{-1}\mathbf{1}_{\{d>a\}}(r + r^{-1} - \Lambda)^{-1}\Sigma^{1/2}\xi$. The normalized root, derivative, base, correction, and band-test vectors range over their charged coefficient balls. Sparse tests cover the root and derivative tails, their union with a band-test tail, and the exceptional $A - V$ coordinates. The clipped recipe types are the root and derivative sources against one band test, the scalar functions $m^2, m^4, m^6, V'_0(y)w$, their low-input versions, and the differentiated functions in the single second-order Taylor expansion below. The number of these types and the number of columns in each average are fixed before n ; the possible directions within a type are paid for by the coefficient and sparse nets. For each type the event is uniform over every band ball whose radii lie on the chosen fixed grids; the later localization bounds select those radii.

Lemma 17.4 (Uniform shifts and conditional noise). *At every fixed sufficiently small $t > 0$, the spectral band, sparse-row, and clipped-average diagnostics just described can be imposed on an event depending only on (W, g) and the fixed parameter family. They hold there uniformly over all shifts in (17.7) and the coefficient balls, sparse supports, and recipe and filter types specified above. Their conditional failure probability tends to zero for typical W ; with fixed buffered tolerances it is at most $e^{-c_t n}$ after decreasing $c_t > 0$. The event is independent of any square-density test function.*

Suppose additionally that the covariance of g is at least ct on a spectral band contained in $\{2 - W \leq t\}$ of rank at least $cnt^{3/2}$. Then, for every fixed center $v \in \mathbb{R}^n$,

$$\mathbb{P}_g\{|P_t^W(g + v)|_n < c't^{5/4} \mid W\} \leq e^{-c''nt^{3/2}}. \quad (17.12)$$

The constants are uniform in the center. This is a probability bound for each center, not a simultaneous event over all centers. In particular, both types of estimate apply after averaging over any spin law that is fixed before the fresh Gaussian noise is drawn.

Proof. There are two different uniformity statements here. The frame estimates hold on one event for all coefficient vectors in the specified balls, so they admit shifts selected after the frame and noise are seen. The projection lower bound instead holds in probability for each fixed center, with a constant independent of that center. A spin sampled before the fresh noise can therefore be averaged into either statement. There is no simultaneous lower bound over every possible center.

The frame estimates are simultaneous coefficient-ball estimates. The shift contributes at most Ct to each unfiltered band norm; after a high-band inverse it contributes at most Ct/d . Thus every shift in the stated ball is already included in their deterministic domains. The Gaussian band norms and the norms of their finite smooth filtered families have exponential tails at the specified fixed scales. If $p(W)$ is the conditional noise failure probability and $\mathbb{E}_W p(W) \leq e^{-2c_t n}$, Markov gives $\Pr_W\{p(W) > e^{-c_t n}\} \leq e^{-c_t n}$. This gives the asserted disorder event and conditional bounds, without any reference to a test function. The same argument integrated over a fixed time interval gives the integrated conditional statement, when that is all that is needed.

For (17.12), restrict to the nondegenerate band of rank $r_t \geq cnt^{3/2}$. The Gaussian integral of a Euclidean ball is maximized when its center is the mean; alternatively, exponential Markov applied to $\exp(-\lambda\|g + v\|^2/t)$ gives a product bounded by $(1 + c\lambda)^{-r_t/2}$, since the noncentral exponential factors are at most one. Taking the ball radius squared to be a sufficiently small multiple of tr_t proves the claim. Finally, condition on each spin and then average this bound over its law. No union bound over spin configurations is involved, and no initial density bound is used. $\square$

17.4 The small-field TAP expansion

Proposition 17.5 (Boundary expansion). *Assume the preceding spectral and noise diagnostics, the zero-field approximate-root exclusion in Lemma 17.3, the field representation (17.7), and $|P_t^W h|_n \geq ct^{5/4}$. The test (17.2) holds with positive fixed slack; in particular F_h has a unique root $y_*(h)$. At all roots of F_h , and with arbitrarily small additional errors at sufficiently accurate approximate roots, set*

$$z = r + r^{-1}, \quad R = (z - W)^{-1}.$$

For sufficiently small fixed t and then sufficiently large n ,

$$\begin{aligned} q &\geq c\sqrt{t}, \quad |y - r^{-1}Rh|_n = o(\sqrt{q}), \\ K_{\text{TAP}} &= R - \frac{4q Ryy^\top R/n}{1 + 4q\langle yRy \rangle} + o(q^{-2}). \end{aligned} \quad (17.13)$$

Here $K_{\text{TAP}} = \partial_h \tanh y_(h)$, and the last error is in operator norm. The errors tend to zero as the boundary scales decrease, after which all dimensional errors and perturbation tolerances are chosen.*

If q is comparable to $\sqrt{t}$, define the TAP potential per site by

$$\mathcal{T}(h) = \langle hm \rangle + \frac{1}{2} \langle mWm \rangle - \langle I(m) \rangle + \frac{1}{4}r^2, \quad I(v) = \frac{1}{2}(1+v) \log(1+v) + \frac{1}{2}(1-v) \log(1-v).$$

With $\kappa_0 = 1/3$, the value expansion is

$$\mathcal{T}(h) - \frac{1}{2} \langle hRh \rangle = \frac{r^2}{4} + \frac{zq}{2} - \langle I(m) \rangle - \frac{\langle eRe \rangle}{2r^2} = \frac{1}{4} + \kappa_0 q^3 + o(q^3), \quad e = \tanh y - ry. \quad (17.14)$$

All assertions persist to any prescribed small macroscopic error under sufficiently small changes of the interaction coefficient, operator-norm perturbations, and normalized field displacements, with the TAP potential using the actual interaction and $jr^2/4$ at its actual root.

Proof. We first control the nonlinear source on the coordinates where a root is large, uniformly over every approximate root in the specified region. This gives initial localization bounds for the root and its derivative tests. A second gain is then necessary: an error of order q^2 would still be as large as the spectral separation. The later high-rotation calculation provides that gain before we invert on the low bands. Lemma 17.3 at zero, applied with the small field as an additional residual, first makes q as small as required. The field equation gives $|y|_n \leq C(\sqrt{q} + \sqrt{t})$. Put

$$s = q + \sqrt{t}, \quad a = s^{7/5}, \quad S = \{2 - W \leq a\}, \quad u = s^{1/10}, \quad P_0 = \mathbf{1}_{\{2-W \leq u\}}. \quad (17.15)$$

We only invert on $S^\perp$ until the lower bound for q proves separation. Dyadic bands there have gaps between a and a constant. In all estimates below $o_n(1)$ errors at fixed scales are chosen after the smallest scale and may be below any specified power of s .

The derivative estimates below test vectors w normalized by $|w|_n \leq 1$. For a diagonal A allowed by the stability test, put $v = Aw$ and $\dot{r} = -2\langle mv \rangle$. Equation (17.22) will specify the equation these tests satisfy; the present notation lets us account for their clipping errors before that calculation. The scalar variance profile satisfies $V'_0(z) = -2 \tanh z \operatorname{sech}^2 z$. The cutoffs below are applied to recipes built from this profile, including $V'_0(y)w$.

Clipping convention and its power budget. Write $T_s = s^{-\kappa}$, where $\kappa > 0$ is chosen after the finite clip-loss constant specified below. Use a smooth cutoff that is one on $[-1, 1]$, zero off $[-2, 2]$, and has fixed bounded derivatives. Its arguments are $y/(\sqrt{s}T_s)$, w/T_s , and a normalized linear test divided by T_s . The normalized root source $e/s^{3/2}$ and derivative source $[(q - m^2)w - \dot{r}y]/s$ have degree-three envelopes on this region; with a linear test they have degree-four envelopes. The normalized scalar recipes for m^2, m^4, m^6 , and $V'_0(y)w$ have envelopes of degrees 2, 4, 6, and 2, respectively. In these statements “degree” means that every derivative through the order used is bounded by a constant times that power of T_s . The frame comparison uses at most two derivatives, and its differentiated Taylor tests use at most three. Cutoff differentiation contributes a nonpositive power of T_s . Thus all the needed envelopes are polynomial in T_s .

The proof uses a fixed finite sequence of operations: band projection and dyadic summation, sparse inverses of norm at most two, two complexity absorptions, replacement of high inputs by P_0 inputs, and one second-order Taylor expansion. The scalar moment and value estimates use the same recipes and one further product or contraction. The polynomial derivative bounds above and the logarithmic sizes of the Gaussian and Haar nets therefore give each displayed error the form $Cs^\alpha T_s^{C_i}$, with a finite C_i independent of κ, s, n . The powers from a^{-1} and u^{-1} are displayed as powers of s ; they are not included in C_i .

There is a corresponding finite bound on the clipping exponents in the absorption coefficients. In an inequality $X \leq Cs^\alpha T_s^c + Cs^\theta T_s^d X$ with $\theta > 0$, the second term is absorbed once $\kappa d < \theta$ and the upper scale is small. Let C_{clip} dominate all the finitely many exponents produced by these operations, including those d . Choose

$$0 < \kappa < \frac{1}{1000(1 + C_{\text{clip}})}. \quad (17.16)$$

This choice precedes the upper boundary scale. Below, T_s^C denotes one of these fixed losses. The smallest raw gain in the displayed estimates is $1/20$, so (17.16) leaves a positive power in each absorption and in the final errors.

The natural-size term in every sparse restriction is kept as $CB\sqrt{f \log(2/f)}$, with no T_s^C multiplier. Only its complexity term incurs clipping losses. Thus threshold absorption uses $\sqrt{\log(2/f)}/T_s \rightarrow 0$ and is not circular. First separate fractions below an arbitrarily high power of s , whose contributions already lie below the target errors; otherwise $\log(2/f) = O(|\log s|)$. The finite remaining logarithms fit into a further fixed power of T_s . The failure allowance $\sqrt{\mathfrak{h}/n} = O(t^{3/4}) \leq O(s^{3/2})$ also lies below the errors for which it is used.

The first localization of the root. We first localize the root strongly enough to make the nonlinear source small on high spectral bands. This first estimate is intentionally weaker than the final low-band cancellation: it controls the tail set and makes a second approximation possible. The low-band root component and the filtered shift form the charged part of the base vector; the known Gaussian noise is included in the frame comparison without paying a vector-class complexity.

The exact high-band equation is

$$y = b_y + R'_{\text{hi}}e, \quad b_y = P_S y + r^{-1}R_{\text{hi}}h, \quad R' = r^{-2}R - r^{-1}I. \quad (17.17)$$

Both y and b_y have normalized size $O(\sqrt{s})$. The uncharged noise term has band norm $O(sd^{-1/4})$. The low-band vector and the filtered shift give

$$\mathcal{L}(b_y) \leq C\{\sqrt{s}a^{3/4} + s^2a^{-1/4}\} = C\{s^{31/20} + s^{33/20}\}, \quad (17.18)$$

up to logarithms. Let L_y be this complexity plus the complexities of the correction bands in (17.17). Initially $L_y = O(\sqrt{s})$ is sufficient.

Choose a smooth scalar cutoff $0 \leq \chi \leq 1$ equal to one on $[-1, 1]$ and zero outside $[-2, 2]$, and put $e_0(y) = \chi(y/(\sqrt{s}T_s))(\tanh y - ry)$ coordinatewise. Then $|e_0| \leq Cs^{3/2}T_s^3$ coordinatewise and $|e'_0| \leq CsT_s^2$. On the transition and tail coordinates, $\tanh y/y \leq r$ for sufficiently small s , so

$$e - e_0 = -D_E y, \quad 0 \leq D_E \leq rI, \quad E \subseteq \{|y| > \sqrt{s}T_s\}.$$

The sparse restriction estimate, with its first term absorbed against $\sqrt{s}T_s\sqrt{f}$, gives $\sqrt{f} \leq Cs^{-1/2}T_s^{-1}L_y$, apart from the already reserved tiny errors. Testing the clipped source on an arbitrary normalized unit vector in a band gives

$$|\Pi_d e_0|_n \leq CT_s^C \{s^{3/2}d^{3/4} + sL_y\}. \quad (17.19)$$

Here is the cancellation in this estimate. Gaussian integration by parts predicts its linear projection with coefficient $\mathbb{E}e'_0(Y)$, where $\text{Var } Y = |y|_n^2$. The identity $\langle q - m^2 \rangle = 0$ and the frame comparison bound the corresponding unclipped coefficient by $CT_s^C \sqrt{s}L_y$. For the cutoff error, the threshold estimate above and sparse restriction give

$$\sqrt{f} \leq Cs^{-1/2}T_s^{-1}L_y, \quad |\mathbf{1}_E y|_n \leq CT_s^C L_y.$$

Since the coarse input is $L_y = O(\sqrt{s})$,

$$|\mathbf{1}_E y|_n^2 + sf \leq CT_s^C L_y^2 \leq C\sqrt{s}T_s^C L_y.$$

This bounds the change in $\mathbb{E}e'_0(Y)$. Multiplication by $|\mathbb{E}[YU]| \leq C\sqrt{s}$ for the normalized Gaussian band test U gives the sL_y term in (17.19). The centered part has band-complexity cost $Cs^{3/2}T_s^C d^{3/4}$. Clipping a linear test first and then removing its clip costs its complexity times the size of the bounded source, again by (17.9).

Summing (17.19) with weight $d^{-1/2}$ yields

$$|R_{\text{hi}}^{1/2} e_0|_n + |e_0|_n \leq CT_s^C (s^{3/2} + sa^{-1/2}L_y).$$

The tail equation is coercive: on the high bands, $I + D_E^{1/2} R'_{\text{hi}} D_E^{1/2}$ dominates a fixed positive multiple of $I + D_E^{1/2} R_{\text{hi}} D_E^{1/2}$. Indeed, the initial bound $|y|_n \leq C\sqrt{s}$ already gives $f = |E|/n \leq CT_s^{-2}$. The uniform sparse restriction bound on S therefore implies

$$\|P_S \mathbf{1}_E\| \leq C\{a^{3/4} + \sqrt{f \log(2/f)}\} + o_n(1) = o(1),$$

before any improved tail estimate. Put $X_E = D_E^{1/2} R_{\text{hi}} D_E^{1/2}$ and $Q_E = D_E^{1/2} P_S D_E^{1/2}$. Since D_E is supported on E , $\|Q_E\| \leq r\|P_S \mathbf{1}_E\|^2 = o(1)$, and $R_{\text{hi}} \succeq cP_{\text{hi}}$ gives $X_E \succeq cD_E - o(1)I$. Thus, for a sufficiently small fixed $\theta > 0$,

$$\begin{aligned} I + D_E^{1/2} R'_{\text{hi}} D_E^{1/2} - \theta X_E \\ &= I - r^{-1}D_E + r^{-1}Q_E + (r^{-2} - \theta)X_E \\ &\succeq I + \{c(r^{-2} - \theta) - r^{-1}\}D_E - o(1)I \\ &\succeq c'I, \end{aligned}$$

where $c' > 0$ is fixed because $D_E \leq rI$ and $r \rightarrow 1$. This domination uses only the coarse tail fraction.

To apply it, put $u_E = D_E^{1/2}y$. Substituting $e = e_0 - D_E y$ into (17.17) gives the exact tail equation

$$(I + D_E^{1/2}R'_{\text{hi}}D_E^{1/2})u_E = D_E^{1/2}b_y + D_E^{1/2}R'_{\text{hi}}e_0.$$

Test against u_E and use the lower bound by a fixed multiple of $I + X_E$. Decomposing $R'_{\text{hi}} = r^{-2}R_{\text{hi}} - r^{-1}P_{\text{hi}}$ and applying Cauchy-Schwarz in that energy norm bounds the right side by the dual sizes $|D_E^{1/2}b_y|_n$, $|e_0|_n$, and $|R_{\text{hi}}^{1/2}e_0|_n$. Since $D_E \leq rI$, the resulting estimate is

$$|D_E y|_n + |R_{\text{hi}}^{1/2}D_E y|_n \leq C\{|D_E^{1/2}b_y|_n + |e_0|_n + |R_{\text{hi}}^{1/2}e_0|_n\}. \quad (17.20)$$

The first term is $o(1)L_y + C\mathcal{L}(b_y)$ by sparse restriction. The high correction complexity is bounded by the right side of (17.20); the coefficient $sa^{-1/2} = s^{3/10}$ absorbs after clipping losses. We conclude

$$\begin{aligned} L_y &\leq Cs^{3/2}T_s^C, & f &\leq Cs^2T_s^C, \\ |y - b_y|_n &\leq CT_s^C\{s^{3/2}a^{-1/4} + s^{5/2}a^{-1}\} \leq Cs^{11/10}T_s^C. \end{aligned} \quad (17.21)$$

For the last bound, the sparse part is tested bandwise using $\|\Pi_d \mathbf{1}_E\| \leq CT_s^C(d^{3/4} + s)$; the nonsparse part uses (17.19). This also shows $|e|_n \leq Cs^{3/2}T_s^C$.

Localization of derivative tests. Normalize $|w|_n \leq 1$, put $v = Aw$ and $\dot{r} = -2\langle mv \rangle$, and consider

$$w + (rI - W)v + \dot{r}m = h_v - \ell v, \quad |h_v|_n \leq Cs^2, \quad 0 \leq \ell \leq Cs. \quad (17.22)$$

This includes possible nonpositive stability eigenvalues, after a scalar shift, and the response to bounded sources. Set $R_\ell = (z + \ell - W)^{-1}$, $R'_\ell = r^{-2}R_\ell - r^{-1}I$, and $d_v = (A - rI)w - \dot{r}y$. Exact elimination gives

$$\begin{aligned} w &= b_w + R'_{\ell, \text{hi}}d_v, \\ b_w &= P_S w + r^{-1}R_{\ell, \text{hi}}h_v \\ &\quad + \dot{r}\{R_{\ell, \text{hi}}((r^{-2} - 1)y - e/r) - r^{-1}P_{\text{hi}}y\}. \end{aligned} \quad (17.23)$$

Using the noise band sizes, (17.21), and $|\dot{r}| \leq C\sqrt{s}$ gives

$$\mathcal{L}(b_w) \leq CT_s^C(a^{3/4} + s^{3/2} + s^2a^{-1/4}), \quad |P_{\text{hi}}b_w|_n \leq CT_s^C(s + s^2/a).$$

The first bound starts at $s^{21/20}$; this small gain will be retained. Let L_w include the complexity of the correction bands.

Split $d_v = d_0 + D'_E w + e_A$, where d_0 uses the bounded clipped coefficient $q - m^2$ and retains $-\dot{r}y$. Here D'_E is bounded and supported on the tail set; e_A is an arbitrarily small normalized error. To obtain this decomposition for all allowed A , fix a diagonal sup-norm tolerance δ_A , split $A - V$ outside that tolerance, and use

$$\frac{|\{i : |A_{ii} - V_{ii}| > \delta_A\}|}{n} \leq \epsilon_A^2/\delta_A^2.$$

At fixed bottom scale choose δ_A and then ϵ_A so that the first part is below every requested error and the added fraction is below every requested power of s . This choice is made before n , and the frame tests remain simultaneous over the added sparse set.

Uniformly on the high bands,

$$|\Pi_d d_0|_n \leq CT_s^C\{sd^{3/4} + sL_w + \sqrt{s}L_y\}. \quad (17.24)$$

For jointly centered Gaussian surrogates Y, G, U , Gaussian integration by parts gives the exact identity

$$\mathbb{E}[\text{sech}^2(Y)GU] = \mathbb{E}\text{sech}^2(Y)\mathbb{E}[GU] + \mathbb{E}[(\text{sech}^2)'(Y)G]\mathbb{E}[YU].$$

Centering by r removes the first term; $\dot{r}$ removes the second. The frame comparison for the latter scalar coefficient has natural size $\sqrt{s}$, and its unclipping cost is bounded by $C|\mathbf{1}_E y|_n |\mathbf{1}_E w|_n$. This proves (17.24); the diagonal error is handled by the preceding split. Removing clips on its linear factors uses sparse Cauchy–Schwarz, so it charges the sum of their complexities, not their full norms.

The compression of $M = R'_{\ell, \text{hi}}$ to the enlarged sparse set has operator norm $o(1)$. Its spectral mean is $O(\sqrt{a} + \sqrt{s}) + o_n(1)$, its Frobenius norm divided by $\sqrt{n}$ is at most $Ca^{-1/4}$, and its operator norm is at most C/a . Put $r_f = f \log(2/f) + \mathfrak{h}/n$. The Haar quadratic-form bound and the sparse nets give

$$\|\mathbf{1}_E M \mathbf{1}_E\| \leq \frac{|\text{Tr } M|}{n} + Ca^{-1/4} \sqrt{r_f} + Ca^{-1} r_f + o_n(1).$$

Since $f \leq Cs^2 T_s^C$, $\mathfrak{h}/n = O(t^{3/2}) = O(s^3)$, and $a = s^{7/5}$, the two deviation terms have raw powers at least $s^{13/20}$ and $s^{3/5}$. The clip choice makes them tend to zero. Restricting (17.23) to the sparse coordinates gives

$$(I - \mathbf{1}_E M \mathbf{1}_E D'_E) \mathbf{1}_E w = \mathbf{1}_E b_w + \mathbf{1}_E M(d_0 + e_A).$$

Because D'_E is bounded, the inverse on the left has norm at most two after the upper scale is decreased. The e_A term is below the reserved error. Bandwise testing of the other term gives

$$|\mathbf{1}_E w|_n \leq C \left[|\mathbf{1}_E b_w|_n + T_s^C \sum_{d \geq a} d^{-1} (d^{3/4} + \sqrt{f}) (s d^{3/4} + s L_w + \sqrt{s} L_y) \right]. \quad (17.25)$$

The factor multiplying $s L_w$ sums to $O(a^{-1/4} + s T_s^C a^{-1})$, before the displayed clipping loss. Multiplication by s thus gives a positive power; the sparse-source complexity costs only $C(1 + s a^{-1/4})$, apart from clips and logarithms. Together with the preceding bound for $\mathcal{L}(b_w)$ this absorbs the L_w terms and proves

$$L_w \leq C s T_s^C, \quad |P_{\text{hi}} w|_n \leq C T_s^C (s a^{-1/4} + s^2/a) \leq C s^{3/5} T_s^C. \quad (17.26)$$

A second gain from the independent high rotation. The first root and derivative estimates have not yet beaten the spectral separation s^2 . We now separate each vector into a part with smaller complexity and a structured high-frequency correction. The middle bands provide a power gain by summation. On the remaining high bands, replacing each source by a recipe of its low-frequency inputs leaves the high frame available for a signed estimate. Full operator-norm bounds would lose this gain, so the contractions are estimated before taking absolute operator norms.

We now prove a stronger decomposition, with some fixed $\zeta > 0$; one may take $\zeta = 1/100$ after (17.16):

$$\begin{aligned} y &= B_y + R'_{>u} t_y, & |t_y|_n &\leq C s^{3/2} T_s^C, & \mathcal{L}(B_y) &\leq C s^{3/2+\zeta}, \\ w &= B_w + R'_{\ell, >u} t_w, & |t_w|_n &\leq C s T_s^C, & \mathcal{L}(B_w) &\leq C s^{1+\zeta}. \end{aligned} \quad (17.27)$$

As before, the filtered noise is included but not charged in $\mathcal{L}(B_y)$. To specify the recipes, put $Y_0 = P_0 y$, $W_0 = P_0 w$ and let χ be the preceding scalar cutoff. We may take

$$\begin{aligned} t_y &= \chi \left(\frac{Y_0}{\sqrt{s} T_s} \right) (\tanh Y_0 - r Y_0), \\ t_w &= \chi \left(\frac{Y_0}{\sqrt{s} T_s} \right) \chi \left(\frac{W_0}{T_s} \right) \{(\text{sech}^2 Y_0 - r) W_0 - \dot{r} Y_0\}. \end{aligned} \quad (17.28)$$

All products here are coordinatewise. Small operator and sparse $A - V$ errors are assigned to B_w , using the tail equations below. Net the scalar parameters on their bounded ranges and let the low inputs vary over all vectors of the stated sizes in P_0 . Once these coefficients and the low columns have been fixed, the recipes are determined. The high columns are still free to rotate, which is the independence used in the next signed estimate.

First sum the source bounds on $a \leq d \leq u$. Both the weighted complexity and the norm restricted to a fraction $f \leq s^2 T_s^C$ gain, relative to the source size, the factor

$$T_s^C(u^{1/2} + sa^{-1/4} + s^2 a^{-1}) = T_s^C(s^{1/20} + s^{13/20} + s^{3/5}). \quad (17.29)$$

This follows by multiplying (17.19) and (17.24) by $d^{-1}(d^{3/4} + \sqrt{f})$ and summing dyadically. The same bounds with only the complexity weight prove the claimed gain in complexity. Above u we replace the bounded sources by (17.28). The necessary input estimates are

$$\frac{|P_{>u}y|_n}{\sqrt{s}} \leq CT_s^C(s^{19/40} + s^{3/5}), \quad |P_{>u}w|_n \leq CT_s^C s^{3/5}. \quad (17.30)$$

For the first, the filtered noise contributes $su^{-1/4}$ before division by $\sqrt{s}$, and (17.21) contributes $s^{11/10}$; the second follows from (17.26). The source derivatives have natural sizes s for the root source, s in the w variable for the derivative source, and $\sqrt{s}$ in its y variable after clipping w . Including the high inverse norm C/u , the replacement errors are at most

$$CT_s^C s^{15/8} \quad \text{for } y, \quad CT_s^C(s^{11/8} + s^{3/2}) \quad \text{for } w.$$

These are smaller than the gained sizes in (17.27). Clipping the linear w input at this stage uses its first sparse localization: the bounded coefficient has size sT_s^C , and the resulting error after division by u is $O(s^{19/10}T_s^C)$. Thus the refined joint tail estimate is not being assumed in constructing these recipes.

We justify the signed estimate for the structured high terms. Conditional on P_0 and its frame, the remaining rotation is Haar. For either high multiplier $M = R'_{>u}$ or $R'_{\ell, >u}$,

$$\left| \frac{1}{n} \text{Tr } M \right| \leq C(\sqrt{u} + \sqrt{s}) + o_n(1), \quad \frac{\|M\|_F}{\sqrt{n}} \leq Cu^{-1/4}, \quad \|M\| \leq C/u. \quad (17.31)$$

Net the low coefficient balls and the bounded scalar parameters before substituting the actual, possibly frame-dependent coefficients. At polynomial resolution in s , their logarithmic cardinality is $O(nu^{3/2}|\log s|)$. Here the fixed complement has dimension $m = n - O(nu^{3/2}) \asymp n$. Equation (17.11) bounds the deviation from the exact conditional mean $\mu\langle P_\perp b, P_\perp t \rangle$, with $\mu = \text{Tr } D/m$, by

$$C \left(u^{-1/4} \sqrt{u^{3/2}|\log s|} + u^{-1}u^{3/2}|\log s| \right) \leq Cu^{1/2}T_s^C$$

times the two vector sizes. The mean itself is bounded by $C(\sqrt{u} + \sqrt{s}) + o_n(1)$ from (17.31). The estimate is uniform over the nets and then over the full classes by their derivative bounds. Adding sparse coordinate unit tests costs $O(nf \log(2/f))$, which is smaller than this entropy. Applying this test to a sparse output gives a factor $s^{1/20}T_s^C$ relative to the source norm. Applying it to another low-input recipe gives the same factor relative to the product of the two norms. These are the two forms of the gain used below. The operator norm u^{-1} is not charged a second time to a signed contraction: it already appears in the second term of the quadratic-form estimate above.

The known high-noise column causes no conditioning problem here. It was included in the finite-column frame comparison, where its spectral coefficients are independent of the frame. For the

present signed tests, first replace each differentiated input by its P_0 version. Estimate the discarded part in full norm, including the high-noise contribution $s^{19/40}$ in (17.30). The power calculation below puts this error at higher order. The remaining recipe no longer uses the high-noise image. The retained recipes therefore depend only on the low frame. This distinction is essential: we do not condition on the image of the high noise and then call the remaining rotation unrestricted Haar.

It remains to improve the unclipped sparse sources. First separate the exceptional $A - V$ coordinates using the previously chosen diagonal tolerances; their contribution is below the errors in this paragraph. On the root tail E , the exact restriction is

$$(I + \mathbf{1}_E R'_{\text{hi}} \mathbf{1}_E D_E) \mathbf{1}_E y = \mathbf{1}_E b_y + \mathbf{1}_E R'_{\text{hi}} e_0.$$

The preceding sparse compression estimate includes $\ell = 0$, so the inverse on the left has norm at most two. This is the unweighted sparse inverse; the earlier energy estimate on $D_E y$ alone would not control $\mathbf{1}_E y$. The middle-band gain and the structured sparse-output estimate then give

$$|\mathbf{1}_E y|_n \leq C\sqrt{s}\sqrt{f \log(2/f)} + Cs^{3/2+1/20}T_s^C.$$

The threshold $\sqrt{s}T_s\sqrt{f} \leq |\mathbf{1}_E y|_n$ absorbs the first term, with its natural-size coefficient unchanged. Hence

$$\sqrt{f} \leq Cs^{1+1/20}T_s^C, \quad |\mathbf{1}_E y|_n \leq Cs^{3/2+1/20}T_s^C.$$

The derivative sparse equation uses this same root-coefficient tail, together with the separately controlled $A - V$ set. Its base restriction is now bounded by $C\sqrt{f \log(2/f)} + Cs^{1+1/20}T_s^C$. The inverse of norm at most two and the middle and structured output gains therefore give

$$|\mathbf{1}_E w|_n \leq C\sqrt{f \log(2/f)} + Cs^{1+1/20}T_s^C \leq Cs^{1+1/20}T_s^C.$$

For a sparse source of row norm B_E , its contribution to the weighted high-band complexity is at most $B_E T_s^C$ times

$$\sum_{d \geq a} d^{-1} d^{3/4} (d^{3/4} + \sqrt{f}) \leq C(1 + \sqrt{f} a^{-1/4}),$$

by the sparse band test and dyadic summation. Thus the improved row norms give the raw bounds

$$\mathcal{L}(B_y) \leq Cs^{3/2+1/20}T_s^C, \quad \mathcal{L}(B_w) \leq Cs^{1+1/20}T_s^C. \quad (17.32)$$

The middle-band and replacement errors have at least these powers. The clip choice weakens (17.32) to (17.27) with $\zeta = 1/100$.

Separate tail estimates are insufficient for the products used in a derivative test. We therefore clip the root, derivative vector, and low-band test on the union of their tail sets. The refined decomposition bounds all three vectors on this same set. Their natural-size terms can be absorbed against the clipping thresholds, leaving a positive power gain for every product discarded when the clips are removed.

If z_S is any normalized unit vector in S , then $\mathcal{L}(z_S) \leq Ca^{3/4} = Cs^{21/20}$. Let E be the union of the tails of $y/\sqrt{s}$, w , and z_S , each at threshold T_s , together with the sparse $A - V$ set. The first bounds put its fraction below $s^2 T_s^C$. The raw bounds (17.32), the structured sparse-output estimate, and sparse restriction give

$$\begin{aligned} |\mathbf{1}_E y|_n &\leq C\sqrt{s}\sqrt{f \log(2/f)} + Cs^{3/2+1/20}T_s^C, \\ |\mathbf{1}_E w|_n + |\mathbf{1}_E z_S|_n &\leq C\sqrt{f \log(2/f)} + Cs^{1+1/20}T_s^C. \end{aligned} \quad (17.33)$$

The threshold inequality, after removing the arbitrarily smaller $A - V$ fraction, is

$$T_s \sqrt{f} \leq |\mathbf{1}_E y|_n / \sqrt{s} + |\mathbf{1}_E w|_n + |\mathbf{1}_E z_S|_n.$$

The unmultiplied natural-size terms in (17.33) absorb. Therefore

$$\sqrt{f} \leq C s^{1+1/20} T_s^C, \quad |\mathbf{1}_E y|_n \leq C s^{3/2+1/20} T_s^C, \quad |\mathbf{1}_E w|_n + |\mathbf{1}_E z_S|_n \leq C s^{1+1/20} T_s^C. \quad (17.34)$$

For an explicit permitted diagonal tolerance, split $A - V$ at s^{20} and take its normalized Frobenius tolerance at most s^{40} . The exceptional fraction is then at most s^{40} ; its contribution is smaller than all the displayed errors. Across a fixed interval of boundary scales, use its smallest scale in these choices.

The low-band cancellations. All parts needed for the low-band estimate are now available. The base vectors have small enough complexity for comparison with Gaussian averages; the linear structured corrections use the signed high-rotation bound; and the quadratic Taylor remainder is smaller still. We list the powers separately before removing the joint clips. After these errors have been bounded, the Gaussian predictions cancel because the overlap and its derivative were defined from the same root.

$$|P_S e|_n = o(s^{5/2}), \quad |P_S d_v|_n = o(s^2). \quad (17.35)$$

We give the power ledger explicitly. Normalize the variables as $y/\sqrt{s}, w, z_S$. The base complexities in (17.27) are $s^{1+\zeta}$, $s^{1+\zeta}$, and $s^{21/20}$. Applying the frame comparison to the clipped averages therefore gains more than one power of s relative to the source sizes $s^{3/2}$ and s . The structured normalized perturbations have source norm $s T_s^C$ and full norm at most s/u times a clipping loss. Their second-order Taylor remainder consequently has relative size $s^{9/5} T_s^C$. For a linear term use the signed high contraction, giving relative size $su^{1/2} T_s^C = s^{21/20} T_s^C$. Replacing its differentiated recipe by a P_0 -input recipe has relative error at most $su^{-1}(s^{19/40} + s^{3/5}) T_s^C$. The resulting exponents, before clip losses, are

operation	relative power of s
base frame comparison	$1 + \min(\zeta, 1/20)$
structured linear contraction	$21/20$
second-order Taylor remainder	$9/5$
high-input replacement	$11/8$ and $3/2$.

(17.36)

The same calculation controls changes of Gram entries in the Gaussian predictions; the discarded high parts are included in the replacement error. The differentiated recipes are exactly derivatives, through order two, of ez_S and $d_v z_S$, and the scalar recipes $m^2, V'_0(y)w, m^4$. Their normalized polynomial envelopes have degree at most six. The sixth moment is needed only to leading order, for which the first localization and a crude perturbation bound suffice; no differentiated degree-eight recipe is required.

Removing the joint clips costs, by (17.34),

$$|\langle \mathbf{1}_E e z_S \rangle| \leq C s^{13/5} T_s^C, \quad |\langle \mathbf{1}_E d_v z_S \rangle| \leq C s^{21/10} T_s^C.$$

We used $|e| \leq C|y|$, boundedness of m, V, A , and $|\dot{r}| \leq C\sqrt{s}$. The sparse $A - V$ term obeys the second bound by the same joint row-energy product. Scalar unclipping uses $m^2 \leq y^2$ and $|V'_0(y)| \leq 2|y|$; its products obey the identical ledger. Gaussian unclipping has superpolynomially small tails because the normalized surrogate variances are bounded.

We have now bounded the comparison, Taylor, and unclipping errors. The Gaussian variables Y, G, U in the remaining calculation are centered and have the current empirical Gram matrix of y, w, z_S ; they are the full-frame surrogates, rather than the noncentral scalar equality laws used earlier. The scalar comparisons and the justified replacement of A by V give the precise target errors

$$|q - \mathbb{E} \tanh^2 Y| = o(s^2), \quad |\dot{r} - \mathbb{E}[V'_0(Y)G]| = o(s^{3/2}).$$

Indeed the natural scalar sizes are s and $\sqrt{s}$, and their comparison and unclipping errors gain more than one power of s . Gaussian integration by parts now gives

$$\begin{aligned} \mathbb{E}[(\tanh Y - rY)U] &= (q - \mathbb{E} \tanh^2 Y) \mathbb{E}[YU], \\ \mathbb{E}[(V_0(Y) - r)G - \dot{r}Y)U] &= (q - \mathbb{E} \tanh^2 Y) \mathbb{E}[GU] + (\mathbb{E}[V'_0(Y)G] - \dot{r}) \mathbb{E}[YU]. \end{aligned}$$

The Gram bounds are $|\mathbb{E}[GU]| \leq C$ and $|\mathbb{E}[YU]| \leq C\sqrt{s}$. The two displays therefore give $o(s^{5/2})$ and $o(s^2)$, respectively. Every entry of (17.36) is strictly greater than one, and each unclipping exponent is strictly above its target. By (17.16) the finite clipping losses leave positive margins. The supremum over unit S -tests proves (17.35). Quantitatively, after the fixed-scale dimensional errors are decreased, its two right-hand sides are bounded by $Cs^{5/2+1/200}$ and $Cs^{2+1/200}$, respectively; the smallest remaining margin is $\zeta - 1/1000 > 1/200$. This also proves the scalar comparison for m^4 with error $o(s^3)$ and for m^6 to leading order.

Separation, stability, and response. Suppose $q \leq \varepsilon\sqrt{t}$, where $\varepsilon > 0$ will be fixed below the field lower-bound constant. The exact field equation is

$$h = r(z - W)y + (r - W)e.$$

On the band $2 - W \leq t$, its first multiplier is at most $C(t + q^2)$ in norm. Also $s \asymp \sqrt{t}$ under the supposition, and $t \leq s^2 \leq s^{7/5} = a$ at small scale, so $P_t^W \preceq P_S$. The norm bound for y and (17.35) therefore give

$$|P_t^W h|_n \leq (C\sqrt{\varepsilon} + o_t(1) + o_n(1))t^{5/4}.$$

Choose ε so that $C\sqrt{\varepsilon}$ is below half the prescribed field lower-bound constant, then decrease the upper boundary scale, and finally the dimensional error. This contradicts the field lower bound, proving $q \geq c\sqrt{t}$. In particular s is now comparable to q , and the identity $z - 2 = q^2/(1 - q)$ gives the positive separation needed to use the full resolvent: $R > 0$ and $\|R\| \leq Cq^{-2}$. Inverting on all bands now gives $|y - r^{-1}Rh|_n = o(\sqrt{q})$ and $|m - Rh|_n = o(\sqrt{q})$. Replacing z by $2 + q^2$ incurs only a relative $o(1)$ error; consequently q approximately solves the scalar equation

$$q = |(2 + q^2 - W)^{-1}h|_n^2.$$

Its right-hand side is decreasing in $q > 0$; equivalently, $q - |(2 + q^2 - W)^{-1}h|_n^2$ has derivative at least one.

For derivative tests, use (17.23), (17.26), and (17.35) to obtain, uniformly under (17.22),

$$w = r^{-1}R_\ell h_v - 4qR_\ell y \langle yw \rangle + o(1) \quad \text{in normalized norm.} \quad (17.37)$$

To see the coefficient, substitute $v = rw + \dot{r}y + d_v$ into (17.22) and use $r - r^{-1} = -2q + O(q^2)$, $\dot{r} = -2\langle yw \rangle + o(\sqrt{q})$. The unenhanced term $\dot{r}y$ is $O(q)$, whereas the resolvent term has coefficient $2q\dot{r} = -4q\langle yw \rangle + o(q^{3/2})$. The low source errors are multiplied by at most Cq^{-2} ; their strict power gain therefore leaves an $o(1)$ error. The high-band errors are controlled by the preceding localization estimates. This justifies the remainder in the rank-response formula uniformly over the tests.

The rank-one operator $I + 4qR_\ell y y^\top / n$ has uniformly bounded inverse: its denominator is at least one, and its numerator correction has norm at most $4q\|R_\ell\|\|y\|_n^2 \leq C$. Thus (17.37) with zero source excludes a normalized unit solution. If the stability matrix had a nonpositive eigenvalue, increase $r - W$ by ℓI until its minimum eigenvalue reaches zero. The norm bound, $A \leq (1 + \eta)I$, and $q = O(s)$ ensure $0 \leq \ell \leq Cs$, choosing η sufficiently small at the fixed bottom scale. For invertible A , writing its eigenvector as $A^{1/2}w$ produces exactly the zero-source test above, a contradiction. Likewise a positive eigenvalue $\varepsilon \ll q^2$ at $\ell = 0$ produces the source $h_v = \varepsilon w$ and contradicts the same inverse estimate. Slightly smaller tolerances and continuity include noninvertible A . Hence the stability test has positive fixed slack. The deterministic root lemma now gives a unique root, and the implicit function theorem defines its response on this stable field family. For $A = V$ and arbitrary sources of size $O(q^2)$, (17.37) first bounds the response and then, by the rank-one inverse formula, gives (17.13); $|(I - V)w|_n = o(1)$ by the localization just proved.

The value of the potential. The root and response estimates determine the local curvature. To build a single extension at initialization we also need to compare values at separated fields. We therefore expand the stationary TAP potential through cubic order in the overlap. The scalar fourth and sixth moments come from the same Gaussian comparison, while the quadratic source correction uses the low-band cancellation and the signed high-rotation estimate already proved. Their coefficients must be combined before discarding the lower powers.

The stationary TAP expression has gradient m . Since $h = (z - W)m - e/r$, completing the square gives the first equality in (17.14) exactly. Joint tail control permits the entropy expansion

$$\langle I(m) \rangle = q/2 + \langle m^4 \rangle / 12 + \langle m^6 \rangle / 30 + o(q^3).$$

The scalar comparisons just established express these moments by their Gaussian Taylor expansions. If $\tau = |y|_n^2$ denotes the matching variance, then

$$\tau = q + 2q^2 + o(q^2), \quad \langle m^4 \rangle = 3q^2 - 8q^3 + o(q^3), \quad \langle m^6 \rangle = 15q^3 + o(q^3).$$

For the quadratic correction, the immediate goal is $\langle eR'e \rangle = o(q^3)$. The low spectral region contributes at most $Cq^{-2}|P_{Se}|_n^2 = o(q^3)$ by (17.35). On $a < d \leq u$, the clipped source e_0 contributes $O(q^3 u^{1/2} + q^5/a)T_s^C = o(q^3)$ by (17.19).

Keep the coordinate-sparse source $e_{\text{sp}} = e - e_0$ separate on all bands $d > a$. Its improved row norm is $|e_{\text{sp}}|_n \leq Cq^{31/20}T_s^C$. The bounded sparse compression of R_{hi} , together with $R' = r^{-2}R - r^{-1}I$, bounds its quadratic form by $O(q^{31/10}T_s^C) = o(q^3)$. The bandwise sparse estimate following (17.21) and the improved row bound make its cross terms on the middle bands $o(q^3)$. Here the spectral projection belongs to the operator: for example, $\langle P_{>u}e_{\text{sp}}, R'P_{>u}e_{\text{sp}} \rangle = \langle e_{\text{sp}}, R'_{>u}e_{\text{sp}} \rangle$. The vector tested by the sparse estimate is still e_{sp} .

Above u , write $e_0 = t_y + e_{\text{rep}}$, where $e_{\text{rep}} = e_0(y) - e_0(Y_0)$; the definition of t_y in (17.28) uses the same cutoff χ . For the structured term, $|t_y|_n \leq Cq^{3/2}T_s^C$, and the signed high test gives

$$|\langle t_y R'_{>u} t_y \rangle| \leq Cq^{1/20}T_s^C |t_y|_n^2 = o(q^3).$$

The high-input replacement estimates give $|R'_{>u}e_{\text{rep}}|_n \leq Cq^{15/8}T_s^C$ and $|e_{\text{rep}}|_n \leq Cq^{79/40}T_s^C$. Its cross term with t_y is therefore $O(q^{27/8}T_s^C)$, and its self term is of still higher order. For the cross term with e_{sp} , the structured sparse-output bound gives $|\mathbf{1}_E R'_{>u} t_y|_n \leq Cq^{31/20}T_s^C$, while the full replacement bound above handles $R'_{>u}e_{\text{rep}}$. Multiplication by $|e_{\text{sp}}|_n$ gives powers 31/10 and 137/40, respectively. All exceed three after the clipping losses. This proves $\langle eR'e \rangle = o(q^3)$. It follows from $R = r^2 R' + rI$ that

$$\langle eRe \rangle = r\langle e^2 \rangle + o(q^3) = \frac{2}{3}q^3 + o(q^3).$$

The last scalar coefficient follows by expanding $e = qY - Y^3/3 + O(Y^5)$ under the matched Gaussian law. Substitution cancels the quadratic terms and proves (17.14), with $\kappa_0 = 1/3$. This coefficient is used in the own-saddle support comparison at initialization.

Residuals at approximate roots and changes of coefficient or matrix add small normalized sources to the two exact equations used above. At a fixed bottom scale all inverses, clip derivatives, and tests have fixed bounds. Choose these perturbations after the displayed scales and error targets, and repeat the estimates with their reserved slack. The TAP expression then uses the actual J_* and $jr^2/4$. This proves the final perturbation assertion and completes the proof. $\square$

18 Initializing the cap induction and removing the auxiliary potential

The induction of Proposition 16.1 needs a negative radial correction at its first scale. We obtain this correction from the small-field TAP expansion, initially for an auxiliary potential. The auxiliary potential differs from the true cap potential by a bounded Euclidean gradient error. A finite number of cap steps contracts this error, after which a covariance comparison permits us to use the true potential at every remaining scale.

We retain the notation P_p, k_p, M_p, R_p, a_p of the cap construction and write

$$\mathcal{A}_p(s) = \{h \in P_p : |\alpha_p(h)| \leq sp^2\}, \quad \phi_p(h) = \log Z_p(h).$$

Here $\alpha_p(h)$ is determined by $n = \text{Tr } R_p + \|R_p h\|^2$. Whenever an auxiliary potential $\tilde{\phi}_p$ is used, its derivatives and its discrepancy from ϕ_p are restricted to P_p . An auxiliary object is called *intrinsic to the parent data* if it is measurable with respect to the capped matrix, the eigenvalues, and auxiliary randomness independent of the as yet unused finer eigenframe. This requirement is essential for the contraction argument below.

All parameters in the initialization and the finite comparison phase are fixed before n tends to infinity. Fix first a finite field radius L_0 for the eventual bounded-ball estimate. The construction tolerances may depend on L_0 ; the coarse constant supplied by Lemma 16.3 depends only on L_0 and is fixed before the sharp curvature tolerances. Choose a sufficiently large absolute C_* and the small excess η_* used below, and then the constant K_0 in (16.2). Choose the scale ratio γ sufficiently small, the annulus tolerance σ a sufficiently high power of γ , and δ a still higher power. Increasing the exponents is allowed, and the choice of δ is made last among these curvature tolerances. The finitely many error tolerances in the static comparison are chosen after these choices. Finally choose the starting scale $p_0 > 0$ sufficiently small. The coefficient $1 - \beta > 0$ introduced below is chosen still smaller, including the fixed-scale persistence requirement in Section 19.4; its scalar endpoint and buffers are fixed from p_0 before this choice. The coefficient is thereafter fixed. Constants in the companion fixed-temperature theorem may depend on this β and on p_0 .

18.1 A terminal TAP potential and its extension

Given the data at the starting cap $p = p_0$, restore the eigenvalues above the cap using an independent conditional Haar frame in H_p . Conditional on the capped matrix and eigenvalues, this auxiliary refinement and the original refinement are independent. Denote the resulting matrix by W and its cap subtraction by B_p^{aux} , so that $W - B_p^{\text{aux}} = J_p$. Its unconditional law is GOE. The sigma-field $\mathcal{F}_p^{\text{aux}}$ generated by the capped data, eigenvalues, and auxiliary Haar seed excludes the original finer frame. Set $t = C_* p^2$ and observe with precision

$$\Delta = tI + (1 - \beta)W - B_p^{\text{aux}}. \tag{18.1}$$

For C_* large and $1 - \beta$ sufficiently small this is positive definite. The terminal interaction is $\beta W - tI$, which gives the same spin law as βW_{off} . Scalar and Gaussian diagonal terms change only the partition-function normalization.

On the stable terminal region constructed below, let $\Psi(z)$ be the TAP potential at its unique root, with interaction $J_* = \beta W_{\text{off}}$ and $j = \beta^2$. Thus $\nabla \Psi(z) = m(z)$, the TAP magnetization. Propositions 17.1 and 17.5 give the following construction on an event of probability tending to one. We describe its domain carefully because it need not be convex.

For a starting field $h \in \mathcal{A}_p(\sigma)$, freeze its parameter $\alpha = \alpha_p(h)$. Write G_h for the canonical Gaussian terminal law

$$Z = h + \Delta a_p + \zeta, \quad \zeta \sim N(0, \Delta + \Delta R_p \Delta).$$

Its filtered field satisfies

$$R_f = (2 - W + t + \alpha + (1 - \beta)W)^{-1}, \quad a_f = R_f z, \quad a_f \sim N(a_p, R_p - R_f) \quad \text{under } G_h. \quad (18.2)$$

In particular,

$$\mathbb{E} \|a_f\|^2 = n - \text{Tr } R_f = (1 + o(1))n\sqrt{t + \alpha}.$$

Under G_h , the centered noise has covariance norm $O_{C_*}(t)$ and depends on the scalar α in a compact smooth family. Its shift $h + \Delta a_p$ has normalized norm $O_{C_*}(t^{5/4})$. The scalar and recipe nets in Proposition 17.5 cover this family and all input directions. The exact mean above, the spectral trace bounds, and a Gaussian quadratic deviation estimate give positive rates per k_p for the fixed filtered-length tolerances.

For comparison, let T_h be the true terminal law conditional on $\mathcal{F}_p^{\text{aux}}$ and h :

$$X \sim \mu_{p,h}, \quad Z = h + \Delta X + \sqrt{\Delta} \xi, \quad \xi \sim N(0, I) \text{ independently of } X.$$

The drift $h + \Delta X$ has normalized norm $O_{C_*}(t)$ for every spin. The noise diagnostics hold on one event for all permitted shifts. The low-band lower bound has a different quantifier: on a band with $2 - W \asymp t$ below t , condition first on X and apply the Gaussian small-ball bound to that fixed center, then average over X . This gives failure $e^{-cnt^{3/2}} = e^{-cC_*^{3/2}k_p}$ with no union over centers.

The exact filtered Gaussian law in (18.2) is not the law under T_h . For those tests the spherical reference is the canonical bridge conditioned on the prior spin norm. The tested norm-density calculation gives a positive rate for the named relaxed quadratic tests. The positive-moment transfer then attaches independent Gaussian noise to each replica and uses the specified Gram-cell enlargement with the noise marks fixed. Its field-uniform conclusion and parameter order are recorded after the terminal regions are defined. The same interface is needed for the actual finer frame at the starting cap.

The initial convolution uses one terminal potential for all input fields. We construct it on three nested regions $D_0 \subset D_1 \subset D_2$, determined by the parent data and the auxiliary frame. The region D_0 contains the predicted terminal fields, D_1 supplies the supporting quadratics, and D_2 is the larger region on which the TAP Hessian is controlled.

Fix $\eta_* > 0$ at the choice of C_* , sufficiently small for the spectral contraction and the Gaussian distance-tail comparison below. Choose a fixed geometric tolerance θ sufficiently small compared with η_* . The earlier finite choice of the annulus tolerance is made so that $\sigma \ll \theta$. To define membership from a terminal field, let $\Lambda = \{\text{obs}\} \sqcup [-\sigma p^2, \sigma p^2]$ label the noise covariances

$$\Sigma_{\text{obs}} = \Delta, \quad \Sigma_\alpha = \Delta + \Delta R_{p,\alpha} \Delta.$$

Let $\mathcal{G}_W$ be the set of labelled noise vectors (λ, g) passing the buffered diagnostics of Proposition 17.5 for every allowed shift. Use closed relaxed versions of these tests and a fixed Gaussian norm cutoff; their stricter versions retain the stated probability bounds. The diagnostic families are continuous on their compact coefficient sets, so $\mathcal{G}_W$ may be taken with a Borel graph and compact sections. The low-band lower bound is a test of the resulting field, not a condition imposed simultaneously on all shifts in $\mathcal{G}_W$.

Put $R_t = (2 - W + t)^{-1}$ and $P_t^W = \mathbf{1}_{\{2-W \leq t\}}$. Choose shift radii $L_0^{\text{shift}} < L_1^{\text{shift}} < L_2^{\text{shift}}$ inside the allowed range, with the first containing every drift $h + \Delta X$ above and with fixed margins between the radii. Choose lower thresholds $\ell_0 > \ell_1 > \ell_2 > 0$ below the small-ball threshold, and $(\theta_0, \theta_1, \theta_2) = (\theta/4, \theta/2, \theta)$. Define

$$D_j = \{z = g + v : (\lambda, g) \in \mathcal{G}_W, \quad |v|_n \leq L_j^{\text{shift}} t, \\ |P_t^W z|_n \geq \ell_j t^{5/4}, \quad ||R_t z|_n^2 / \sqrt{t} - 1| \leq \theta_j\}, \quad j = 0, 1, 2. \quad (18.3)$$

Membership means that at least one labelled noise and shift pair satisfies these conditions. All labels are included before h is chosen. These are nested parent-measurable closed sets: the defining graph has compact sections, and its projection under $(g, v) \mapsto g + v$ is compact. The geometric tolerances remain fixed throughout the argument.

The next lemma supplies the terminal tests at the fixed initialization scale. Its filtered-length estimate has a rate available before the replica order and upper scale are chosen. The noise diagnostics are then imposed at that chosen scale; their rate need only be positive there.

Lemma 18.1 (Terminal tests at the initialization scale). *Fix the preceding geometric tolerances and a filtered-length tolerance $\varepsilon_{\text{len}} > 0$. For a sufficiently small fixed $p = p_0$ and then sufficiently small fixed $1 - \beta > 0$, with probability tending to one in the capped data and auxiliary refinement, uniformly for $h \in \mathcal{A}_p(\sigma)$,*

$$\mathsf{T}_h(D_0^c) \leq e^{-ak_p}, \quad (18.4)$$

and

$$\mathsf{T}_h\left\{\left|\|R_f Z\|^2 - (n - \text{Tr } R_f)\right| > \varepsilon_{\text{len}} n \sqrt{t}\right\} \leq e^{-ak_p}. \quad (18.5)$$

Here $a > 0$ may depend on p_0 and the fixed parameters. For either exceptional event, the same conclusion holds, with constants $C_j, a_j > 0$, after inserting any fixed power of $1 + \|Z\|/\sqrt{n} + \|R_f Z\|/M_p$ in the expectation. A finite collection of stricter filtered-length tolerances may be imposed simultaneously. The same assertions hold for the actual refinement at this starting cap.

Proof. We first prove the filtered tests, keeping their rate uniform as the upper scale decreases. Under the canonical joint law, sample $X \sim N(a_p, R_p)$ and independent standard Gaussian ξ , and set $Z = h + \Delta X + \sqrt{\Delta} \xi$. With $A = R_f Z$ and $T = R_p - R_f$, Gaussian conditioning gives the joint representation

$$A \sim N(a_p, T), \quad X = A + \eta_f, \quad \eta_f \sim N(0, R_f) \text{ independently of } A. \quad (18.6)$$

In particular $\mathbb{E}\|A\|^2 = n - \text{Tr } R_f$. The spectral estimates, uniformly for $|\alpha| \leq \sigma p^2$ and $(1 - \beta)/t$ sufficiently small, give

$$\|T\| \leq Cp^{-2}, \quad \text{Tr } T^2 + a_p^\top T a_p \leq Cn/p.$$

The Gaussian quadratic moment formula therefore implies, for each fixed $\varepsilon > 0$,

$$\Pr\left\{\left|\|A\|^2 - \mathbb{E}\|A\|^2\right| > \varepsilon n \sqrt{t}\right\} \leq 2e^{-c_{\varepsilon, C_*} k_p}. \quad (18.7)$$

Indeed its centered logarithmic moment is at most $C\lambda^2 n/p$ for $|\lambda| \leq cp^2$; choose a sufficiently small fixed multiple of p^2 in each sign. These constants are independent of sufficiently small p .

We must retain this rate after conditioning $\|X\|^2 = n$. For a spectral quadratic tilt $e^{\lambda A^\top H A + v^\top A}$, the normalized tilted law of X is Gaussian with covariance

$$R_f + T^{1/2}(I - 2\lambda T^{1/2} H T^{1/2})^{-1} T^{1/2}.$$

Choose the tilt with its bracket between fixed positive multiples of I . This covariance is then comparable to $R_f + T = R_p$, regardless of the tilted mean. A band of order k_p eigenvalues remains comparable to p^{-2} . The characteristic-function estimate (15.5) bounds its squared-norm density by $Cp^2/\sqrt{k_p} = C\sqrt{p/n}$. The original saddle density is at least $c\sqrt{p/n}$. Thus the density ratio in (15.7) is bounded by a constant, and (18.7) holds with a decreased positive rate under the spherical terminal law. Independent marked replicas have the corresponding product bound; equivalently one may use (15.6).

This argument also covers the $R_t Z$ test defining D_0 . Write $R_t Z = (R_t R_f^{-1})A$. The spectral multiplier is bounded and differs from I by $O(|\alpha|/t + (1 - \beta)/t)$, so its quadratic tilt has the same precision margin. Its canonical mean square divided by $n\sqrt{t}$ is $1 + o(1) + O(\sigma/C_*) + O((1 - \beta)/t)$. The trace estimate and the resolvent identity give this comparison. Choose these errors below a fixed fraction of θ_0 and reserve fixed gaps between the tested and relaxed thresholds. The spherical failure rate for both filtered tests is consequently a fixed positive multiple of k_p before p_0 is chosen.

We now transfer these concrete tests to cube posteriors by (15.12). Give every replica its own independent standard Gaussian mark in spectral coordinates, and keep the marks fixed in the Gram-cell comparison. A cell of the size used there changes a spin representative by $O_l(n^{-1/2})$ in Euclidean norm. Hence it changes Z by $O_l(tn^{-1/2})$ and $R_f Z$ or $R_t Z$ by $O_{l,C_*}(n^{-1/2})$. The corresponding changes in the normalized filtered lengths tend to zero at the fixed scale. Norm inequalities in both directions show that a failure with tolerance ε implies failure with tolerance $\varepsilon/2$ throughout the cell, even for unbounded noise marks. This verifies the marked inclusion required after (15.12). Integrating the independent marks leaves the same cube-sphere Gram factor. The small-overlap moment bound thus has exponent $-lck_p/2 + \eta_l k_p + o_n(k_p)$; the discarded large-overlap term is added after normalization as in that equation.

For uniformity in h , take a mesh of fixed resolution ζ in $p^2 M_p$ units on a slightly enlarged annulus. Its logarithmic cardinality is at most $C_\zeta k_p$. Couple neighboring fields by using the same spin and mark. The scaled saddle is Lipschitz, and the resolvent identity gives

$$\|R_{f,h} Z_h - R_{f,h'} Z_{h'}\| \leq C_\zeta (M_p + \|R_{f,h'} Z_{h'}\|).$$

Also $|\text{Tr } R_{f,h} - \text{Tr } R_{f,h'}| \leq C_\zeta np$, so the centers of the squared-length tests respect the same fixed gaps. The R_t comparison is simpler, since its resolvent is fixed. Thus a sufficiently fine fixed mesh respects the threshold gaps. On $\|P_p X\| \leq VM_p$, changing the spin posterior between these fields costs at most $e^{C_\zeta V k_p}$, including the partition normalizer; use the static projected-mean bound on a prescribed ball containing the annulus and the joining segment. The complement has the uniform projected V^6 tail from Proposition 15.1. Choose V first to give the required tail rate and then ζ to fit both the threshold and likelihood budgets. In (15.12), choose a posterior exponent below $c/2$, the value error below the remaining margin, and then a fixed replica order l whose Haar exponent exceeds C_ζ . Only afterward decrease p_0 so that the Gram error $C_l np^4$ and all comparison errors fit this budget. The field net then proves both filtered tests uniformly with a positive exponent per k_p .

It remains to impose the other conditions of D_0 . The deterministic shift $h + \Delta X$ is within the chosen shift radius for every spin. Lemma 17.4 gives one conditional noise event for all these shifts, with failure at most $e^{-c_l n}$ for typical W . For the low-band lower bound, condition on each spin

and use that lemma's uniform-in-center small-ball estimate, then average. This bound is uniform in the spin law and in h , without a field net or a union over spins. At the chosen p_0 , the minimum of these positive rates and the filtered-test rate is a positive rate per k_{p_0} , proving (18.4).

At this fixed scale, the conditional Gaussian observation and $\|X\| = \sqrt{n}$ bound every fixed moment of $1 + \|Z\|/\sqrt{n} + \|R_f Z\|/M_p$, uniformly in h . Cauchy-Schwarz with a doubled moment gives the weighted exceptional bounds with a smaller exponent. The auxiliary refinement has marginal GOE law and is independent of the true finer frame conditional on the capped data. The event just proved is therefore measurable before that true frame is revealed. Repeating the same argument gives the actual-refinement event; it is imposed separately, without including it in $\mathcal{F}_p^{\text{aux}}$. A finite intersection proves the final assertion. $\square$

The true terminal field now lies in D_0 with the weighted error bounds needed for gradient transport. We next extend the TAP potential while preserving its values and derivatives on a neighborhood of these fields. A sufficiently small fixed $b_* > 0$ gives a ball of radius

$$b_* \sqrt{n} t^{5/4} \quad (18.8)$$

inside D_1 around every point of D_0 , and inside D_2 around every point of D_1 . Indeed, retain a witnessing (λ, g) and assign the displacement to v . A displacement of normalized length $b_* t^{5/4}$ changes the low-band norm by at most that amount and changes $|R_t z|_n^2 / \sqrt{t}$ by at most $Cb_* + Cb_*^2$. The fixed gaps between the shift, lower-bound, and filtered-length thresholds give both buffers.

We also reserve an outer shift and low-band margin beyond D_2 . Choose one residual radius ρ_{res} inside these margins, with $\rho_{\text{res}} \ll t^{5/4}$. If $e = F_{J_*, z}(y)$ and $|e|_n \leq \rho_{\text{res}}$, then $F_{J_*, z+e}(y) = 0$. The same noise witness remains admissible at $z+e$, and the realized low-band bound is preserved by the triangle inequality. The pre-stability part of Proposition 17.5 therefore applies to every such approximate root. Its zero-field exclusion first bounds q above, the low-band estimate bounds it below by $c\sqrt{t}$, and its scalar constraint, together with the filtered-length condition and monotonicity, places it in the common band used below. The derivative estimates then give the stability implication on this whole residual region, with common parameters. Only at this point do we apply the deterministic root lemma in Proposition 17.1: it gives a globally unique stable root for every $z \in D_2$. The root is measurable in the data and field, for example as the limit of measurably selected rational approximate roots; locally it is smooth by the inverse function theorem.

The local curvature and the comparison of distant support points require different arguments. To make the latter comparison without changing the geometric regions, use the fixed interval $I_t = [(1 - 2\theta)\sqrt{t}, (1 + 2\theta)\sqrt{t}]$ and define

$$R_q = (2 + q^2 - W)^{-1}, \quad F_q(z) = \frac{1}{2} z^\top R_q z + \frac{n}{3} q^3, \quad \partial_q F_q(z) = q(nq - \|R_q z\|^2). \quad (18.9)$$

The resolvents are positive on I_t for the fixed boundary scale and large n . The function $q - \|R_q z\|^2/n$ is strictly increasing. Its value at $\sqrt{t}$ has absolute value at most $\theta\sqrt{t}$ on D_2 , so it has a unique zero $\hat{q}(z)$ in the interior of I_t . Thus $\hat{q}(z)$ is the minimizer of $F_q(z)$ on this same interval for every $z \in D_2$.

The value calculation in Proposition 17.5 gives the exact coefficient $\kappa_0 = 1/3$. Together with its mean expansion it implies, uniformly on D_2 , for a field-independent constant c ,

$$\left| \Psi(z) - c - \min_{q \in I_t} F_q(z) \right| \leq \epsilon n t^{3/2}, \quad \left\| \nabla \Psi(z) - R_{\hat{q}(z)} z \right\| \leq \epsilon \sqrt{n} t^{1/4}. \quad (18.10)$$

Here ϵ is an approximation tolerance, separate from θ and the two buffers. It can be decreased after those geometric quantities are fixed. For clarity, the TAP resolvent initially has parameter $r + r^{-1} = 2 + q^2/(1 - q)$. Replacing it by $2 + q^2$ changes its quadratic value by $O(nq^4)$ and its mean

by $O(\sqrt{n}q^{3/2})$, uniformly on the coarse band $q \asymp \sqrt{t}$. The mean estimate and the definition of the TAP overlap give $q - \|R_q z\|^2/n = o(q)$. Strict monotonicity then gives $q - \hat{q}(z) = o(q)$. On this fixed band the first two scalar derivatives of F_q/n have orders q^2 and q , so this replacement changes the value by $o(nq^3)$ and the mean by $o(\sqrt{n}q^{1/2})$. These errors, and the perturbations from the interaction coefficient and diagonal completion, give (18.10): choose the diagnostic accuracies, then the boundary scale sufficiently small, then the coefficient perturbation sufficiently small, and finally n large. The noise diagnostic accuracy may improve, but assigning displacements to the allowed shift still preserves those diagnostics. These choices do not reduce the geometric tolerances or either fixed buffer.

By the resolvent identity and the fixed choice $\theta \ll \eta_*$, $R_q \preceq R_t + \eta_* I/(4t)$ on I_t . The local TAP Hessian, whose leading correction is negative semidefinite, therefore obeys the required upper bound on D_2 . Put

$$U = (2 - W + t)^{-1} + \frac{\eta_*}{t} I, \quad \|\Delta^{1/2} U \Delta^{1/2}\| \leq 1 - b_{**} \quad (18.11)$$

for a fixed $b_{**} > 0$. The last inequality follows in the common spectral basis: before the η_* and $1 - \beta$ perturbations, $1 - \Delta_i/(d_i + t) \geq c/C_*$ on both sides of the cap. The approximation tolerance is also chosen small enough that $\nabla^2 \Psi \preceq U$ on D_2 .

For $z, z' \in D_1$ at distance less than (18.8), their segment lies in D_2 ; integration of this Hessian bound gives the upper-support inequality. For more distant points put $d = z' - z$ and choose the minimizing parameter at z . Since $\min_{q \in I_t} F_q(z') \leq F_{\hat{q}(z)}(z')$, (18.10) gives

$$\Psi(z') - \Psi(z) - \nabla \Psi(z)^\top d \leq \frac{1}{2} d^\top R_{\hat{q}(z)} d + 2\epsilon n t^{3/2} + \epsilon \sqrt{n} t^{1/4} \|d\|.$$

At these distances the two error terms are at most $\epsilon(2b_*^{-2} + b_*^{-1})\|d\|^2/t$. Choose $\epsilon \ll \eta_* \min(b_*^2, b_*)$ after the fixed buffers. The curvature slack then bounds the right side by $d^\top U d/2$. This is a one-sided comparison using the saddle at the support point; it does not require an arbitrarily accurate absolute comparison with a single frozen quadratic on the whole geometric region.

Take the infimum of the upper supporting quadratics with curvature U over D_1 and call the result $\bar{\Psi}$. The preceding inequalities hold for every pair of support points. They show that the extension equals Ψ on D_1 , hence on a full fixed neighborhood of D_0 . The extension has global upper Hessian U in the distributional sense and at most quadratic growth; its support gradients have at most linear growth. One may mollify and pass to the limit at fixed n .

For later estimates we also need to transfer canonical tests to the heat convolution of this extension. At each input field, restrict the supports further to a stricter predicted subset H_h of D_0 where $q/\sqrt{t + \alpha}$ and the filtered norm at R_f have arbitrarily small fixed errors. This subset has canonical probability tending uniformly to one. On it, Ψ and its gradient agree with the frozen canonical quadratic up to an arbitrarily small error per k_p and per M_p , respectively. The distance to this subset, in $\sqrt{n} t^{5/4}$ units, has Gaussian upper tails with a positive rate per $nt^{3/2}$, uniformly as p decreases. Indeed, Gaussian enlargement uses only $G_h(H_h) \geq 1/2$ and the covariance bound $C_{C_*} t$; the large- n threshold for this mass bound may depend on the fixed p , while the distance-tail rate does not. The support comparison bounds the additional logarithmic weight by a small constant times k_p plus

$$C(\eta_* + \sigma) n t^{3/2} \text{dist}(z, \text{the subset})^2,$$

where distance is in those scaled units. Choose η_* below the distance-tail rate. Integrating the distance tails, or applying Hölder's inequality, transfers the required tests with a smaller positive exponent; the same subset gives the normalization lower bound. For tests excluding D_1 , the tolerances and exponents can be fixed independently of small p . Only then decrease p and $1 - \beta$ to control the approximation errors.

For the lower Fisher estimate, localize before differentiating the extension. Choose a cutoff equal to one on the stricter predicted subset in D_0 and supported strictly inside D_1 , where $\bar{\Psi} = \Psi$. This ensures that the calculation uses only the original smooth potential, even though D_1 need not be convex. The D_0 -to- D_1 buffer (18.8) gives polynomial derivative bounds for a distance cutoff. The larger domain D_2 is used for the local Hessian segments in the support comparison. Integration by parts with the cutoff and the score uses only the classical Hessian where the extension agrees locally. Negative distributional Hessian mass elsewhere is never integrated. The omitted terms have exponential probability and polynomial moments for the fixed parameters.

18.2 The initial radial deficit

Proposition 18.2. *For the preceding choices, with probability tending to one there is an intrinsic auxiliary cap potential*

$$\tilde{\phi}_p(h) = \log \mathbb{E}_g \exp \bar{\Psi}(h + \sqrt{\Delta} g), \quad h \in P_p,$$

whose Hessian satisfies (16.2) on the used annulus, with slack. Moreover, for a finite constant C independent of n ,

$$\sup_{h \in \mathcal{A}_p(\sigma)} \|\nabla \tilde{\phi}_p(h) - P_p \nabla \phi_p(h)\| \leq C. \quad (18.12)$$

The constant may depend on all the fixed initialization parameters.

Proof. We first verify curvature. Write $\tilde{T}_h$ for the terminal field law with density proportional to $e^{\bar{\Psi}(z)}$ relative to $N(h, \Delta)$. All expectations in the curvature calculation below use this law. An indicator $\mathbf{1}_{H_h}$ means that the integrand is integrated only over H_h , including when it is defined only there. The canonical identities in (18.2) enter through the transferred tests, rather than as identities under $\tilde{T}_h$.

On H_h , put $\nu = n/(4q)$; no value of q or ν outside H_h is used in these retained integrals. The ideal form of the TAP response there is

$$K_f^0 = R_f - \frac{R_f a_f a_f^\top R_f}{\nu + a_f^\top R_f a_f}.$$

Proposition 17.5 has the same two resolvent factors with the root vector y . Replacing its resolvent by R_f and $y = r^{-1}a_f + o(M_p)$ by a_f changes the response by an arbitrarily small amount in t^{-1} units; the scalar matching is detailed below. In the ideal calculation, $R_f^{-1} = R_p^{-1} + \Delta$ and rank-one inversion give the exact identity

$$\Delta^{-1}(\Delta^{-1} - K_f^0)^{-1}\Delta^{-1} - \Delta^{-1} = R_p - \frac{R_p a_f a_f^\top R_p}{\nu + a_f^\top R_p a_f}. \quad (18.13)$$

The left side is the Brascamp–Lieb covariance bound after the increment-covariance transformation. Thus the transported correction has two R_p factors, and its denominator contains R_p .

The transferred canonical tests give

$$\mathbb{E}[\mathbf{1}_{H_h} a_f] = a_p + o(M_p), \quad \|a_p\|^2 = (b + o(1))np,$$

and

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{H_h} a_f^\top R_p a_f] &\leq a_p^\top R_p a_p + \text{Tr}(R_p(R_p - R_f)) + o(n/p) \\ &\leq (3b + o(1))n/p. \end{aligned}$$

Here $b = 4/(3\pi)$, while $\mathbb{E}[\mathbf{1}_{H_h}\nu] = O(n/(\sqrt{C_*}p))$. On H_p , $R_p = (p^2 + \alpha)^{-1}I$. For $e_p = a_p/\|a_p\|$, Cauchy–Schwarz therefore bounds the retained expected radial subtraction in p^2 Hessian units below by

$$\frac{p^2}{(p^2 + \alpha)^2} \frac{(\mathbb{E}[\mathbf{1}_{H_h} e_p^\top a_f])^2}{\mathbb{E}[\mathbf{1}_{H_h} (\nu + a_f^\top R_p a_f)]} \geq \frac{1}{3 + o(1) + O(C_*^{-1/2})}. \quad (18.14)$$

The errors also tend to zero with σ . Taking C_* large leaves a strict margin above the radial constant 0.30.

For each fixed unit transverse direction $v \perp a_p$, the retained second moment $\mathbb{E}[\mathbf{1}_{H_h} (v^\top (a_f - a_p))^2]$ is $o(np)$, with bounds uniform in that direction. The Fisher lower bound and the Brascamp–Lieb upper bound therefore give arbitrarily small transverse error and a bounded radial lower error. The contribution from H_h in the localized Fisher calculation is linear in the terminal correction and retains its denominator $\nu + a_f^\top R_f a_f$; the upper bound uses the denominator in (18.13). The directional second-moment test controls both on transverse directions.

The reference resolvent must be matched at the actual frozen parameter. Put $s = t + \alpha$ and $q_0 = \sqrt{s}$. On H_h , the R_f filtered-length test is within $\epsilon_{\text{len}} q_0$ of its exact mean $1 - \text{Tr } R_f/n$, and the latter is within $\epsilon_{\text{tr}} q_0$ of q_0 . The TAP mean expansion gives $|q - |R_q z|_n^2| \leq \epsilon_{\text{TAP}} q$. Since

$$\frac{d}{dv} (v - |R_v z|_n^2) = 1 + \frac{4v}{n} z^\top R_v^3 z \geq 1, \quad R_f^{-1} = R_{q_0}^{-1} + (1 - \beta)W,$$

these tests imply

$$\frac{|q - q_0|}{q_0} \leq C \left(\epsilon_{\text{len}} + \epsilon_{\text{tr}} + \epsilon_{\text{TAP}} + \frac{1 - \beta}{t} \right).$$

Consequently the TAP parameter $q^2/(1 - q)$ differs from $t + \alpha$ by an arbitrarily small fraction of p^2 after the prescribed accuracy, scale, and coefficient choices. This uses the exact R_f test, so no fixed $O(\sigma)$ error remains. The exceptional set is already covered by the terminal transition bounds. The transverse error can therefore be made as small as required at fixed C_* and σ . Positive-matrix Cauchy–Schwarz applied to the two matrix bounds gives the mixed-block estimate in (16.2). The cutoffs described above justify the lower bound, and (18.11) controls the upper bound on the exceptional set.

The curvature invariant is now established for the auxiliary potential. We next compare its gradient with the true cap mean in unnormalized Euclidean norm. At the terminal field this is the uncentered consequence of the companion’s directional estimate. We transport that estimate backward through the fixed precision interval. This gives a bounded gradient error; it does not yet give a Hessian comparison.

At every terminal field in the agreement region the companion fixed- $\beta < 1$ posterior theorem gives

$$\|m_{\text{Gibbs}}(z) - m(z)\| \leq C. \quad (18.15)$$

Precisely, use [28, Section 7.2, proof of Proposition 7.2]: its estimate $|\mathbb{E}[G(X - t_*)^\top v]| \leq C\|G\|_*$, before centering G , with $G = 1$ and $\|v\| \leq 1$ gives (18.15). Its assumptions are the fixed-length word diagnostics and the stable-field implication through a fixed residual threshold. After fixing $\beta < 1$, restrict the permitted diagonal enlargement to a positive $\eta < \min(\eta_{\text{TAP}}, \beta^{-1} - 1)$, where η_{TAP} is the enlargement supplied by the boundary test. The stability implication remains valid on this smaller class, and $\beta^2(1 + \eta)^2 < 1$ supplies the companion’s strict diagonal margin. Impose its fixed-length word diagnostics with this choice. All constants are uniform in the terminal field and n under these diagnostics. No assertion about the constant as $\beta \uparrow 1$ is needed.

Transport (18.15) backward through the observation, using Lemma 18.3 below. The strict upper curvature bound (18.11) gives bounded amplification on this fixed precision interval. Outside

agreement, true observation probabilities and their polynomially weighted moments are exponentially small by Lemma 18.1. The extension gradients have at most linear growth, so their contribution there tends to zero. This proves (18.12). Finally the conditional auxiliary completion uses only the parent data and an independent Haar seed. Its high-probability properties may be imposed before revealing the true finer frame, which proves the asserted intrinsic measurability. $\square$

18.3 Transporting and contracting gradient errors

Lemma 18.3. *Consider a deterministic increasing matrix precision B_s , with derivative $G_s \succeq 0$, and a terminal potential extended with a strict semiconcavity margin relative to the total remaining precision. Let f_s be its backward log heat convolution and $K_s = \nabla^2 f_s$. Along the true observation process, write m_s for the posterior mean and $e_s = m_s - \nabla f_s$. For a starting direction v , let v_s solve*

$$dv_s = dB_s K_s v_s = G_s K_s v_s ds. \quad (18.16)$$

Then

$$e_{s_0}^\top v = \mathbb{E}[e_{s_1}^\top v_{s_1} \mid h_{s_0}],$$

whenever the indicated polynomial moments are finite. The same identity holds by approximation for the extensions used here.

Proof. The true posterior mean has zero drift. Apply Itô's formula to the approximate gradient along the same observation, and use the backward heat equation for f_s . Subtracting the two evolutions gives $de_s = -K_s G_s e_s ds + dM_s$. The finite-variation equation (18.16) cancels this drift in $e_s^\top v_s$. Localization and then the moment bounds give the identity. A global upper Hessian bound for the terminal potential gives a global upper bound for the convolved Hessian by Brascamp–Lieb. For a constant precision rate G this bounds amplification in the G^{-1} metric. On the cap path the Euclidean estimate applies instead, because the variation stays in the support of P_u , on which the rate is scalar. Mollification, the same bound, and dominated convergence justify the limiting extension. In (18.16), dB_s is a precision increment, not a Brownian increment. $\square$

Proposition 18.4. *Suppose the parent auxiliary potential is intrinsic, satisfies the curvature invariant with slack, and its true gradient discrepancy is bounded by D/p on its agreement annulus, where $D < \infty$ is fixed. Put $c = \gamma p$. Extend the parent potential as in Proposition 16.1 and take its log heat convolution to c . For the parameter choices above, with probability tending to one over the unused refinement,*

$$\sup_{h \in \mathcal{A}_c(\sigma)} \|P_c \nabla \phi_c(h) - \nabla \tilde{\phi}_c(h)\| \leq \frac{1}{c} \left(\frac{1}{2} D + o_n(1) \right). \quad (18.17)$$

This conclusion is uniform in the child field. For a fixed finite number of comparison steps, the additive errors tend to zero simultaneously.

Proof. Parameterize observation from c to p by $\log u$, so its rate is $2u^2 P_u$. A variation initially in P_c remains in P_u . Freeze the starting parameter α and write $a_u = R_u h_u$ along the canonical bridge. These vectors form a Gaussian martingale with independent increments and

$$\text{Cov}(a_u - a_s) = R_s - R_u \quad (c \leq s \leq u \leq p).$$

Norms and mixed products concentrate at their predictions, in particular $a_s^\top a_u = \|a_s\|^2 + o(M_p^2)$, with arbitrarily small fixed scaled errors. The inferred parameter is within an arbitrarily small fraction of u^2 of its frozen value. These assertions are uniform in u : use finite time grids and

Gaussian maximal bounds. In parent units the covariance of an increment has operator norm at most $O_\gamma((\text{Tr } P_p)^{-1})$ times its logarithmic time length and trace at most $O_\gamma(1)$ times that length. Transfer of path tests to the true law uses its terminal density. The exceptional-tail estimates are uniform conditionally on the starting field. Outside the retained paths the global semiconcavity estimate still bounds flow amplification.

If h_u^{lin} denotes the variation in (18.16), rescale it to $v_u = (c^2 + \alpha)h_u^{\text{lin}}/(u^2 + \alpha)$. The scalar reference curvature cancels. By the intermediate estimates of Proposition 16.1, the remaining equation, up to a small operator perturbation, is

$$\frac{dv_u}{d \log u} = -2d_u \frac{a_u a_u^\top}{\|a_u\|^2} v_u, \quad 0.26 \leq d_u \leq C. \quad (18.18)$$

The upper bound is absolute: use the variance-formula lower Hessian bound when $u/p \leq 0.9$, and the Fisher bound closer to the parent. The perturbation is bounded by $O_\gamma(\delta^{b'})$ for an absolute $b' > 0$, plus errors chosen arbitrarily small by the approximation tolerances. Variation of constants absorbs it after these choices. The starting parameter tolerance contributes no additional fixed scalar error, because the rescaling uses the actual frozen parameter. All intermediate support conditions have slack; near p use the parent agreement domain itself.

Let $V_u : P_c \rightarrow P_u$ be the pure flow (18.18), with the identity inclusion at c . It is a contraction. Its terminal radial component satisfies

$$\left\| \frac{a_p^\top V_p}{\|a_p\|} \right\| \leq C\sqrt{H} \gamma^{1.02} + \tau, \quad H = C_1 \log(1/\gamma), \quad (18.19)$$

where $\tau > 0$ can be chosen arbitrarily small before n is large. Removing the scalar reference growth leaves a flow that contracts only in the current radial direction. That direction changes along the bridge. To measure the cumulative contraction, test the flow against the terminal radial vector and use the bridge's mixed scalar products. The resulting scalar integral equation has an integrating factor bounded by a fixed power of the scale ratio. We use pathwise integration by parts, so no adaptedness claim is needed for that factor.

To see the exponent, put $G_s = a_s^\top V_s$. The mixed-product estimates in the integral equation give

$$G_s = a_s^\top V_c - \int_c^s 2d_u G_u \frac{du}{u} + \text{an arbitrarily small scaled error}.$$

Write $Q(p, u) = \exp(-\int_u^p 2d_s ds/s) \leq (u/p)^{0.52}$. The integrating-factor formula is

$$G_p = Q(p, c) a_c^\top V_c + \int_c^p Q(p, u) d(a_u^\top V_c) + \text{an arbitrarily small scaled error}.$$

The integral is interpreted by pathwise integration by parts; its integrand can depend on future values of d_s , so an adapted stochastic-integral interpretation is unnecessary. No differentiability of d_s is needed. On a logarithmic dyadic interval starting at u , increments and partial increments of $P_c a_s$, divided by $\|a_p\|$, are at most $C\sqrt{H} \gamma^{3/2} p/u$ with exponential probability. This follows from the covariance formula and $\text{Tr } P_c \asymp nc^3$. Integration by parts against the integrating factor gives a summable series with largest term $C\sqrt{H} \gamma^{3/2} \gamma^{-0.48}$. The initial term is at most $C\gamma^{1/2} \gamma^{0.52}$. This proves (18.19). Time and mixed-product errors have at most polynomial amplification in γ^{-1} .

We next prove the transverse gain, including its conditional law. Let $\mathcal{F}_p$ contain the capped matrix, the eigenvalues, and the entire intrinsic auxiliary construction. Write Q for the unused Haar map into P_p . For a fixed child-coordinate field, sample the canonical path in spectral coordinates

independently of Q . Condition on these path coordinates and on the terminal physical image $Q\xi_p = h_p$, where ξ_p is the terminal field in spectral coordinates. The terminal discrepancy, as a function of $(\mathcal{F}_p, h_p)$, is now fixed. Its transverse part $e_p^\perp$ is perpendicular to h_p , hence to a_p . The residual map from $\xi_p^\perp$ to $h_p^\perp$ is exactly Haar. We do not condition on the intermediate physical field images.

The elementary Haar projection bound, applied conditionally as just specified, therefore gives

$$\|P_c Q^\top e_p^\perp\| \leq C\sqrt{H}\gamma^{3/2}\|e_p^\perp\|, \quad |(e_p^\perp)^\top a_u|/\|a_u\| \leq C\sqrt{H}\gamma^{3/2}\|e_p^\perp\|$$

on a fixed polynomially fine time grid, with failure at most $\exp(-AHk_c)$ for a prescribed fixed A after increasing the constant. Here P_c in the first expression denotes the child coordinate subspace before applying Q . The bound uses $\text{Tr } P_c / \text{Tr } P_p \asymp \gamma^3$; adding finitely many grid directions does not affect it once n is large. Gaussian modulus tests fill the gaps with any prescribed fixed error. Since $\|V_u\| \leq 1$ and $d_u \leq C$, the integral equation for V_p yields

$$\|V_p^\top e_p^\perp\| \leq [C\sqrt{H}\gamma^{3/2}(1 + |\log \gamma|) + \tau]\|e_p^\perp\|. \quad (18.20)$$

This step also permits the coefficients d_u to depend on Q . The projection event controls the vectors appearing in the integral equation, while their coefficients use only the deterministic bounds already established. No additional independence of those coefficients is required.

The residual Haar statement is used only under canonical sampling. To pass to the actual heat-tilted path, its Radon–Nikodym derivative for each fixed Q is the terminal weight

$$\frac{\exp\{\phi_p(Q\xi_p) - q_p^{\text{fr}}(\xi_p)\}}{\mathbb{E}_{\text{can}} \exp\{\phi_p(Q\xi_p) - q_p^{\text{fr}}(\xi_p)\}},$$

where q_p^{fr} is the frozen quadratic. An additive calibration of the numerator is immaterial. On an arbitrarily narrow fixed terminal shell, the static comparison and the auxiliary gradient calibration bound this derivative by $\exp(\eta k_p)$ for any prescribed $\eta > 0$, and the same shell gives its normalization lower bound. This normalization may depend on Q . We transfer the canonical Haar failure estimate through the density ratio, rather than asserting that the tilted law is Haar. If a finite grid or replica transfer has r terminal factors, choose $r\eta \ll H\gamma^3$. More explicitly, if $\mathcal{E}_H$ is the high-rate Haar failure event on the retained terminal shell, the canonical bound and the r density factors give

$$\mathbb{E}_Q \mathbb{T}_{h,Q}^{(r)}(\mathcal{E}_H) \leq \exp\{-[AH - r\eta\gamma^{-3}]k_c\}.$$

Here $\mathbb{T}_{h,Q}^{(r)}$ is the joint actual path law for those r terminal factors at the fixed grid field. Choose $b_{\text{path}}, b_Q > 0$ with $b_{\text{path}} + b_Q < AH - r\eta\gamma^{-3}$ and $b_Q > C_{\text{net}} \log(1/\gamma)$, where the child-field net has logarithmic size at most $C_{\text{net}} \log(1/\gamma)k_c$. Markov's inequality then gives

$$\Pr_Q \left\{ \mathbb{T}_{h,Q}^{(r)}(\mathcal{E}_H) > e^{-b_{\text{path}}k_c} \right\} \leq e^{-b_Qk_c}.$$

The second exponent pays for the field net. The first remains as a conditional path bound and absorbs the polynomial discrepancy and flow moments at the fixed comparison scales. Here is the likelihood comparison needed to pass from the net to nearby fields. Let $\|h - h'\| \leq \rho c^2 M_c$ and keep the later physical grid fields fixed. Only the first Gaussian increment and the initial normalizer then change. Its inverse on P_c is bounded by a fixed multiple of c^{-2} once the first positive grid time is fixed. On bounded retained grid fields, the Gaussian exponent comparison therefore gives

$$\left| \log \frac{d\mathbb{T}_h^{(r)}}{d\mathbb{T}_{h'}^{(r)}} \right| \leq rC_\gamma \rho k_c.$$

The normalizer obeys the same bound: differentiate the exact full-interval convolution to obtain $\nabla_{P_c} \phi_c = \Delta_{P_c}^{-1}(\mathbb{E}Y_{P_c} - h)$, and use the uniform parent-field moment bound before integrating along the field segment. The constant C_γ may also depend on the fixed first grid time and buffers.

The intrinsic terminal discrepancy is the same function of the fixed terminal field in this comparison, so no derivative bound on it is required. Discarding an arbitrarily short initial grid interval costs an arbitrarily small operator error in the rank flow. The remaining projection and mixed-product tests, including their normalizations by nonzero annular lengths, have polynomial continuity in the fixed parameters. Choose a net of inverse-polynomial resolution ρ so that these changes fit the reserved test slack and $rC_\gamma\rho < b_{\text{path}}/2$. A relaxed bad event at a neighboring field is then bounded by its net-point probability times $e^{rC_\gamma\rho k_c}$, retaining exponent at least $b_{\text{path}}/2$. This comparison is made before taking conditional expectations. Its net still has logarithmic size $C_{\text{net}} \log(1/\gamma)k_c$, paid by the previously reserved orientation exponent. Modulus tests, mixed-product tests, and shell complements are controlled by the separate uniform conditional tail estimates. Their failure exponents need not equal the high Haar rate, and no union bound over the child net is applied to those tails.

Apply Lemma 18.3. The prefactor from undoing the rescaling is $O(\gamma^{-2})$. Multiplying the resulting bound by c , the radial and transverse terms in (18.19) and (18.20) are bounded by

$$C\sqrt{H} \left[\gamma^{0.02} + \gamma^{1/2}(1 + |\log \gamma|) \right] D.$$

Choose γ so this is below $1/4$, and choose the operator and fixed-tolerance errors so their contribution is below $D/4$. The exceptional sets contribute $o_n(1)$ in these units for fixed D and fixed scales. This proves (18.17). Agreement at each extension boundary is enough: true paths leave it only with the stated negligible probability. Integrating the bounded gradient discrepancy along annular paths and recalibrating an additive constant gives an $o(k_c)$ value oscillation error, since c is fixed during this finite phase. The next curvature step therefore retains all its macroscopic comparison hypotheses. $\square$

18.4 Replacing the auxiliary potential and evaluating all scales

After finitely many contractions the gradient discrepancy is small in the scale units needed for the next convolution. This is the point at which a Hessian comparison becomes available. The logarithmic density discrepancy is Lipschitz on the parent shell; strong log-concavity turns that Lipschitz bound into small thermal fluctuations. Changing density therefore perturbs the directional first and second moments of the terminal field by a small amount. The exact increment covariance formula then compares the two child Hessians.

Lemma 18.5. *Fix γ and the curvature tolerances. If the parent gradient discrepancy is at most D/p on the used annulus, then one further convolution makes the true and auxiliary output Hessians differ by $O_\gamma(D) + o_n(1)$ in child c^2 units. In particular, for D sufficiently small, the true output satisfies (16.2) whenever the auxiliary output satisfies it with the established slack.*

Proof. On the parent shell the difference of logarithmic potentials has Lipschitz constant at most CD/p . Annular paths have a bounded ratio of path length to chord length, so this is also a Euclidean Lipschitz bound up to an absolute factor. Extend the difference with the same bound, for example by the infimum formula for Lipschitz extensions. Under the strongly log-concave extended auxiliary bridge, its centered fluctuations are subgaussian with scale $C_\gamma D$. Indeed the bridge covariance scale is $O_\gamma(p^2)$ and its strong convexity bound applies to every Lipschitz function. Linear functions of the terminal field in thermal units have a bounded subgaussian scale as well.

If F is the centered logarithmic discrepancy and L such a normalized linear function, Cauchy–Schwarz and the subgaussian exponential moments give

$$\left| \frac{\mathbb{E}[e^F L^j]}{\mathbb{E}e^F} - \mathbb{E}L^j \right| \leq C_\gamma D, \quad j = 1, 2,$$

for D below a fixed constant. Center the linear observable in each tested direction before applying these inequalities. Their constants are uniform over unit directions, so the resulting variance estimates control the full covariance operator. The true bridge and the tilted extension agree conditionally on the shell. The true bridge has the preceding complementary moment bounds. For the discrepancy-tilted extension, Hölder’s inequality and the centered subgaussian bound on F transfer the auxiliary complementary moment bounds with a decreased positive exponent, including the required second moments. The increment covariance formula, with inverse increment $O_\gamma(p^{-2})$ on P_c , now proves the asserted Hessian comparison. $\square$

Starting with Proposition 18.2, write its finite error bound as D_0/p_0 . Apply Proposition 18.4 a fixed number of times until D is as small as Lemma 18.5 requires, and perform the final replacement step. The number of steps may be chosen after p_0 , β , and D_0 are fixed. There is no restriction on its size as a function of p_0 : it is finite and independent of n . The tolerances in the curvature induction and in the flow estimates depend only on the fixed ratios and curvature tolerances, not on this number of steps or on D_0 . Raw bounded gradient errors give vanishing macroscopic calibration errors at every one of these fixed scales. Thereafter all cap potentials in the induction are the true potentials. The resulting curvature conclusion depends only on the original disorder: if it fails for a given J , the joint construction fails for every auxiliary seed. Its disorder failure probability is therefore bounded by the joint failure probability.

Theorem 18.6 (Uniform curvature of the true cap potentials). *Fix a finite field radius L_0 . There is $C = C(L_0) < \infty$ such that for every $\epsilon > 0$ there are constants $p_{\text{top}} > 0$ and $s_0 > 0$ for which, with probability tending to one, the following assertions hold simultaneously for every*

$$p_{\min}(n) \leq p \leq p_{\text{top}}, \quad p_{\min}(n) = \left(\frac{\sqrt{\log n}}{n} \right)^{1/3}.$$

For all $h \in P_p$ with $|\alpha_p(h)| \leq s_0 p^2$,

$$P_p \nabla^2 \phi_p(h) P_p \preceq \frac{1 + \epsilon}{p^2} P_p. \quad (18.21)$$

For all $h \in P_p$ with $\|h\| \leq L_0 p^2 M_p$,

$$P_p \nabla^2 \phi_p(h) P_p \preceq \frac{C}{p^2} P_p. \quad (18.22)$$

The constants are independent of n, p, h . The field radius and all coarse constants may be fixed before the sharp curvature tolerances. The lower endpoint $p_{\min}(n)$ may be changed by a fixed factor.

Proof. Fix L_0 first and take the corresponding coarse constant from Lemma 16.3. Choose the construction tolerances after this radius and the requested sharp error ϵ ; the coarse constant is independent of ϵ . After the finite auxiliary phase, iterate Proposition 16.1 on a geometric list down to $np^3 \asymp \sqrt{\log n}$. Its positive-rate failure bounds are summable on this list. The static comparison

and spectral estimates hold on the same event. This proves the invariant for all retained true parent scales.

For (18.21), write the target scale as $p = xq$, where q is a true parent and $x \in [\gamma^2, \gamma]$. Choose s_0 small enough that the target frozen parameter lies strictly inside the parent agreement range. Lemma 16.2 applies on this entire ratio interval. Its transverse and mixed errors can be made arbitrarily small, while its radial correction is negative. Thus $p^2(K_p - R_p^H)$ has arbitrarily small positive part. Choose the fixed errors below the margin allowed by ϵ , including the replacement of $1/(p^2 + \alpha)$ by $1/p^2$. This proves (18.21) without another random compression.

The sharp estimate concerns a thin parameter annulus. A prescribed bounded ball may have a wider range of saddle parameters, so it requires a different evaluation ratio. Choose the parent sufficiently coarser that this entire range lies strictly inside the parent's agreement annulus. The child saddle remains separated from its pole, and an upper support with smaller excess than that separation gives the coarse ball bound. Each evaluation may use its own support extension, since no induction is performed with its output.

For the ball bound again write $p = xq$, now with $x \in [\gamma b_1, b_1]$, where b_1 is supplied by Lemma 16.3 with radius L_0 . The lemma constructs supports around the frozen typical length with excess below the child convolution margin, and proves the complementary moment estimates for both the extended and true transitions. Equation (16.35) therefore gives (18.22), with a coarse constant depending only on L_0 . The construction is uniform in the indicated fixed ratio interval; it uses no radial gain and need not agree with the extensions used for other targets.

Every target scale below a sufficiently small fixed p_{top} has a parent in each indicated ratio interval. The two evaluation lemmas are deterministic and uniform under the parent hypotheses. Thus the assertions include all intermediate scales, not just the geometric list. Fixed-factor enlargement of the lowest list interval covers the stated bottom endpoint. $\square$

19 Entropy transfer along the observation path

The curvature estimates of Theorem 18.6 control the information revealed by the edge observations. We now use that control to retain enough relative entropy until the posterior becomes a product measure. The product inequality then gives a lower bound on the original heat-bath Dirichlet form. All constants in this section may depend on fixed scale cutoffs and fixed tolerances, but not on the test function or on n .

For a probability measure η on the cube, write

$$\text{Ent}_\eta(u) = \eta[u \log u] - \eta[u] \log \eta[u], \quad \mathcal{D}_\eta(f) = \sum_{i=1}^n \eta[\text{Var}_\eta(f \mid X_{-i})].$$

Here $0 \log 0 = 0$, $\log_+ u = \max\{\log u, 0\}$, and $\mathcal{D}_0 = \mathcal{D}_\mu$. The interaction matrix is the GOE matrix J , with off-diagonal variance $1/n$; its diagonal does not change $\mu(x) \propto \exp(x^\top Jx/2)$.

19.1 A reduction to bounded tests

It is enough to prove the following estimate for every fixed $r > 0$:

$$\mathcal{D}_0(f) \geq n^{-2/3-r} NL, \quad N = \mu[f^2] = e^{-L}, \quad \nu = \frac{f^2 \mu}{N}, \quad (19.1)$$

whenever $0 \leq f \leq 1$, $0 < N$, $\mu(f > 0) \leq 1/2$, and

$$D(\nu \parallel \mu) \geq L/2, \quad L \leq Cn. \quad (19.2)$$

The probability event on which this holds must be independent of f . The observation argument is most effective for a bounded function whose squared-density tilt has entropy comparable to its logarithmic inverse mass. We first show that proving the energy estimate only for those tests is enough. Equal height layers turn it into a capacity estimate for sets; dyadic layers then control arbitrary half-supported functions. A median decomposition and an entropy shift recover every real-valued test. For this class, $\log 2 \leq L$ and $L/2 \leq D(\nu\|\mu) \leq L$, while $d\nu/d\mu = f^2/N \leq e^L$.

Lemma 19.1 (Layers and capacity). *Let η be a strictly positive probability measure on a finite state space, let $\mathcal{E}$ be a reversible Markov Dirichlet form, and suppose $\min_x \eta(x) \geq e^{-M}$, with $M \geq \log 2$. Suppose that, for some $A > 0$,*

$$\mathcal{E}(g) \geq A\eta[g^2] \log \frac{1}{\eta[g^2]}$$

for every $0 \leq g \leq 1$ supported on a set of measure at most $1/2$ and satisfying

$$D\left(\frac{g^2\eta}{\eta[g^2]}\middle\|\eta\right) \geq \frac{1}{2} \log \frac{1}{\eta[g^2]}, \quad 0 < \log \frac{1}{\eta[g^2]} \leq M.$$

Then, with an absolute constant C ,

$$\text{Ent}_\eta(f^2) \leq \frac{C(1 + \log_+ M)^2}{A} \mathcal{E}(f) \quad \text{for every real-valued } f. \quad (19.3)$$

Proof. Fix a nonempty set E of measure $a \leq 1/2$ and a test h equal to one on E and zero outside a set of measure at most $1/2$. Clipping h to $[0, 1]$ decreases its form. Set

$$K = \left\lceil \log_2 \left(1 + \frac{\log(1/a)}{\log 2}\right) \right\rceil, \quad g_i = K((h - i/K)_+ \wedge K^{-1}), \quad 0 \leq i < K.$$

Let $S_i = \eta(h > i/K)$ and $T_i = \eta(h \geq i/K)$. Some i satisfies $T_{i+1} \geq S_i^2$. Otherwise $T_1 < S_0^2 \leq 1/4$ and $T_{i+1} < T_i^2$, which would give $a \leq T_K < 2^{-2^K} < a$. For this layer put $N_i = \eta[g_i^2]$. Then $a \leq T_{i+1} \leq N_i \leq S_i \leq 1/2$ and

$$D\left(\frac{g_i^2\eta}{N_i}\middle\|\eta\right) \geq \log(1/S_i) \geq \frac{1}{2} \log(1/N_i).$$

The first inequality follows by conditioning on its support. Also $\log(1/N_i) \leq \log(1/a) \leq M$, and $\mathcal{E}(g_i) \leq K^2 \mathcal{E}(h)$. Since $s \log(1/s) \geq ca \log(1/a)$ for $a \leq s \leq 1/2$, we obtain

$$\mathcal{E}(h) \geq \frac{cA}{(1 + \log_+ M)^2} a \log(1/a). \quad (19.4)$$

This is the required capacity bound, where capacity is the infimum of $\mathcal{E}(h)$ over these admissible tests.

For $v \geq 0$ supported on a set of measure at most $1/2$, apply (19.4) to the layers $\phi_j = (v - 2^j)_+ \wedge 2^j$, $j \in \mathbb{Z}$, normalized by 2^j . Writing $a_j = \eta(v \geq 2^j)$ and omitting empty upper level sets gives

$$\mathcal{E}(v) \geq \sum_j \mathcal{E}(\phi_j) \geq \frac{cA}{(1 + \log_+ M)^2} \sum_j 4^j a_{j+1} \log(1/a_{j+1}).$$

For the first inequality, fix an edge and orient its endpoints so that the increment of v is nonnegative. The increments of all clipped layers are then nonnegative and sum to that increment. The sum of

their squares is at most its square; summing over edges gives the inequality. Denote the last sum by S . The pointwise dyadic sum $\sum_j 4^j \mathbf{1}_{\{v \geq 2^{j+1}\}}$ lies between $v^2/12$ and $v^2/3$, so $S \geq (\log 2)\eta[v^2]/12$. On $2^{j+1} \leq v < 2^{j+2}$, the inequalities $\eta[v^2] \geq 4^{j+1}a_{j+1}$ and $v^2 \leq 16 \cdot 4^j$ give

$$\log_+ \frac{v^2}{\eta[v^2]} \leq \log \frac{4}{a_{j+1}} \leq 3 \log \frac{1}{a_{j+1}}.$$

Consequently $\text{Ent}_\eta(v^2) \leq 48S$.

Choose a median b of f and apply these estimates to $v_+ = (f - b)_+$ and $v_- = (f - b)_-$. Their supports have measure at most $1/2$, their forms sum to at most $\mathcal{E}(f)$, and

$$\text{Ent}_\eta((f - b)^2) \leq \text{Ent}_\eta(v_+^2) + \text{Ent}_\eta(v_-^2).$$

Finally we use an entropy shift inequality of Rothaus type; compare [3, Lemma 4]. In the form needed here it is

$$\text{Ent}_\eta((u + b)^2) \leq C \text{Ent}_\eta(u^2) + C\eta[u^2] \quad (19.5)$$

for every u and every constant b . For completeness, normalize $\eta[u^2] = 1$. If $|b| \leq 2$, use $(u + b)^2 \leq 2u^2 + 8$ and the positive part of $s \log s$; its negative part costs at most an absolute constant. If $|b| > 2$, use

$$\text{Ent}_\eta((u + b)^2) \leq b^2 \eta \left[\Phi \left((1 + u/b)^2 \right) \right], \quad \Phi(s) = s \log s - s + 1.$$

On $|u| \leq |b|/2$ the integrand is at most Cu^2 by Taylor's theorem; on its complement it is at most $Cu^2(1 + \log_+ u^2)$. This proves (19.5), and hence (19.3). The case $\eta[u^2] = 0$ is immediate. $\square$

On $\|J\| \leq 3$, all Gibbs atoms are at least $\exp(-Cn)$. Lemma 19.1 therefore turns (19.1) into the desired log-Sobolev inequality, after absorbing $(\log n)^2$ by choosing r smaller than the final exponent allowance. It remains to prove the restricted estimate.

19.2 Information and energy under observations

This is the Gaussian observation form of stochastic localization [12, 19]. The conditional-variance contraction below is also the mechanism of [20, Lemma 9, arXiv version 2]. We give the identities directly for the present matrix precisions and keep the stopped-path estimates separate from any all-field assertion.

Fix a bounded test as in (19.2). For a deterministic increasing, piecewise smooth matrix precision B_s , let Q_μ and Q_ν be the observation-path laws obtained from the initial spin laws μ and ν , and let $\mathcal{F}_s$ be the observation filtration. The field has increments

$$dh_s = G_s X ds + G_s^{1/2} dW_s, \quad G_s = \dot{B}_s \succeq 0.$$

Let μ_s, ν_s be the respective spin posteriors, $N_s = \mu_s[f^2]$, and write

$$l_s = D(\nu_s \| \mu_s), \quad H(s) = \mathbb{E}_{Q_\nu} l_s, \quad m_{\mu,s} = \mu_s[X], \quad m_{\nu,s} = \nu_s[X].$$

The posterior interaction is $J - B_s$, and $\nu_s = f^2 \mu_s / N_s$. The posterior need not retain the original half-support or lower entropy-to-log-mass condition. The likelihood and averaged identities below replace those pointwise restrictions. These definitions also apply to precision jumps, by using the corresponding Gaussian channel.

Lemma 19.2 (Observation identities). *Along the continuous portions of this path,*

$$-\dot{H}(s) = \frac{1}{2} \mathbb{E}_{Q_\nu} \|m_{\nu,s} - m_{\mu,s}\|_{G_s}^2, \quad (19.6)$$

$$\mathbb{E}_{Q_\nu} \frac{\mathcal{D}_{\mu_s}(f)}{N_s} \leq \frac{\mathcal{D}_0(f)}{N}. \quad (19.7)$$

The continuous entropy-loss integral up to a stopping time, together with the information from preceding precision jumps, gives the relative entropy of the stopped observation laws. Moreover, for $v \geq 0$,

$$Q_\nu(l_s > L + v) \leq e^{-v}, \quad Q_\nu(E) \leq e^L Q_\mu(E) \quad (19.8)$$

for every observation event E , including stopped-path events.

Proof. The likelihood ratio of the observation laws through time s is

$$\left. \frac{dQ_\nu}{dQ_\mu} \right|_{\mathcal{F}_s} = \frac{N_s}{N} \leq e^L.$$

The conditional drifts are $G_s m_{\mu,s}$ and $G_s m_{\nu,s}$, with common innovation covariance G_s . The Gaussian likelihood formula and the chain rule for relative entropy give

$$D(\nu \parallel \mu) = D(Q_\nu|_{[0,s]} \parallel Q_\mu|_{[0,s]}) + H(s),$$

and then (19.6). The same calculation with the stopped likelihood proves the stopping assertion; zero-rate directions are omitted. The finite spin space and bounded finite precision remove any integrability difficulty.

To prove (19.7), first change the path expectation from the tilted law to the ordinary law:

$$\mathbb{E}_{Q_\nu} \frac{\mathcal{D}_{\mu_s}(f)}{N_s} = \frac{1}{N} \mathbb{E}_{Q_\mu} \mathcal{D}_{\mu_s}(f).$$

The remaining assertion is the conditional-variance decomposition, applied site by site under the joint ordinary law of the spin and observations. Conditioning additionally on the observations can only decrease the expected variance given X_{-i} . Finally $f^2 \leq 1$ implies $N_s/N \leq e^L$ and $l_s \leq -\log N_s$. Hence

$$Q_\nu(l_s > L + v) \leq \mathbb{E}_{Q_\mu} \left[\frac{N_s}{N} \mathbf{1}_{\{N_s/N < e^{-v}\}} \right] \leq e^{-v}.$$

□

We record the ordinary-path estimates used below. Under the original zero-field Gibbs law observed to precision B_p , the quantities $\|P_p X\|/M_p$ and $\|h_p\|/(p^2 M_p)$ have bounded-range tails with arbitrarily large fixed exponential rates per k_p , with the same control for polynomial moments. The parameter α inferred from the cap field satisfies

$$Q_\mu(|\alpha(h_p)| > \zeta p^2) \leq e^{-a_\zeta k_p} \quad (19.9)$$

for every fixed $\zeta > 0$, after reducing the upper cutoff on p . These are quenched estimates on events of probability tending to one. For integration over logarithmic scale intervals it suffices that the same bounds hold in the integrated form with a smaller positive exponent.

Here is the additional verification of (19.9). Put

$$R_\xi = (2 + \xi p^2 - J)^{-1}, \quad R_{p,\xi} = (2 + \xi p^2 - J_p)^{-1},$$

where $\xi > 0$ is a sufficiently small fixed constant. Under the unconditioned Gaussian reference, sample $X^G \sim N(0, R_\xi)$ and $h_p = B_p X^G + \eta$, with $\eta \sim N(0, B_p)$ independently. Then

$$a_{p,\xi} := R_{p,\xi} h_p \sim N(0, R_\xi - R_{p,\xi}), \quad \mathbb{E} \|a_{p,\xi}\|^2 = \text{Tr}(R_\xi - R_{p,\xi}).$$

This expected squared norm differs from $n - \text{Tr } R_{p,0}$ by $O(\sqrt{\xi} np) + o(np)$. The covariance has operator norm $O((\xi p^2)^{-1})$ and trace of its square $O(n/(\sqrt{\xi} p))$, so a fixed-tolerance deviation has exponent at most $-c_\xi \xi k_p$. The one-coordinate lower norm-density estimate for X^G costs $C\xi^{3/2} k_p + O(\log k_p)$ in the logarithm, besides the density scale $\sqrt{p/n}$. The quadratic tilt retains a spectral band whose upper norm-density bound has the same scale. Choose ξ small enough that the $O(\sqrt{\xi})$ target error fits the chosen saddle tolerance and $C\xi^{3/2} < c_\xi \xi/2$.

To connect this norm test to the saddle, use $h_p \in H_p$ and $R_{p,0} h_p = (1 + \xi) R_{p,\xi} h_p$. The monotone saddle equation and its scaled inverse bound then place every field with $|\alpha_p(h_p)| > \zeta p^2$ inside the relaxed quadratic-deviation event. The preceding conditioning calculation gives that event a positive spherical rate per k_p . The first-moment transfer and zero-field concentration in Proposition 15.1 prove the claim. The projection and field tails follow there in the same way, including polynomial moments. Markov's inequality and Fubini's theorem, summed over geometric scales, give the stated integrated versions. At a fixed finite number of constant-scale intervals, time nets and Gaussian maximal inequalities also give whole-path versions.

The two paths below use at most two fixed endpoint constants $T' > 3$, chosen from their path parameters before f . At each endpoint precision $T'I + J$, the posterior interaction is diagonal and its spin law is product. On the same kind of disorder event, imposed for this finite collection and simultaneously for $0 \leq L \leq Cn$,

$$Q_\mu \left(\|h_{\text{end}}\|_\infty > C\sqrt{\log n + L} \right) \leq e^{-2L} n^{-10}, \quad (19.10)$$

and the omitted tails admit polynomial moment bounds. To check this, condition on the eigenvalues in the Haar first moment and then on the spin and its spherical image. The component of JX parallel to X has bounded coefficient, and its perpendicular component is a uniformly rotated vector of norm at most $3\sqrt{n}$. Each coordinate thus has a Gaussian-type tail. Add the Gaussian observation noise, whose covariance has bounded norm, and take a coordinate union bound. Increase the rate before the first-moment transfer and use zero-field partition concentration. A scalar grid in L gives simultaneity, with stronger tails between grid points. The same calculation with tail integrals gives the polynomial moment assertion.

19.3 Retention at small and moderate entropy

Fix a small constant $\lambda > 0$, to be chosen after the curvature and ordinary-path tolerances, and first suppose that $L \leq \lambda n$. Choose p_1 so that, up to fixed endpoint factors,

$$k_1 = np_1^3 = \max\{k_{\min}, c_0 L\}, \quad k_{\min} \asymp \sqrt{\log n}, \quad (19.11)$$

where $c_0 > 0$ is a sufficiently small absolute constant. The coarse endpoint factors and c_0 are chosen before the sharp curvature tolerances.

The initial observation jump leaves the following amount of entropy:

$$H(p_1) \geq \begin{cases} cL, & k_1 = c_0 L, \\ e^{-C_0 k_1}, & k_1 = k_{\min}. \end{cases} \quad (19.12)$$

To prove the first line, let Q_0 be the pure-noise output of the precision- B_{p_1} channel. Convexity of relative entropy gives $D(Q_\nu \| Q_0) \leq \frac{1}{2} \nu[X^\top B_{p_1} X]$. By Jensen's inequality and $\mu[X] = 0$,

$$\log \frac{dQ_\mu}{dQ_0}(h) = \log \mu \left[e^{h^\top X - X^\top B_{p_1} X/2} \right] \geq -\frac{1}{2} \mu[X^\top B_{p_1} X].$$

Subtract the logarithmic likelihood bound from the pure-noise relative entropy estimate. The information revealed by the jump is therefore at most half the sum of these two quadratic moments. The projection tails, transferred also on an auxiliary dyadic grid, bound this sum by

$$Ck_1 \left(1 + (L/k_1)^{1/3} \right).$$

For example the v^6 tail and $d\nu/d\mu \leq e^L$ give $\nu[\|P_{p_1} X\|^2/M_{p_1}^2] \leq C(1 + (L/k_1)^{1/3})$. When $k_1 = c_0 L$, the displayed loss is at most $C(c_0 + c_0^{2/3})L$, which is smaller than $L/4$ after choosing c_0 ; the initial entropy is at least $L/2$.

This argument gives proportional entropy retention when the initial cap dimension is chosen proportional to L . At the smallest permitted cap that choice may be unavailable. We then use the support condition instead: the original law assigns a fixed positive mass to a bounded-projection set that the tilted law never visits. After a bounded field observation its posterior mass may be exponentially small in k_1 , but it remains positive and yields the weaker entropy bound required in the bottom branch.

For the second line, a set of μ -mass at least $1/4$ lies outside the support of f and has bounded scaled projection. On a sufficiently large fixed field ball its posterior mass is at least e^{-Ck_1} : the numerator is bounded below on these projections, and the denominator is bounded above by integrating the projection tails. The field ball has Q_ν -probability bounded below because $L \leq k_1/c_0$ and its ordinary complement can have arbitrarily large fixed rate per k_1 . On this ball ν_h assigns zero mass to the indicated complement of the support. Data processing gives $D(\nu_h \| \mu_h) \geq -\log(1 - e^{-Ck_1}) \geq e^{-C_0 k_1}$, after adjusting C_0 . These are all coarse constants.

Parameterize the cap observations by $s = \log p$. Their precision rate is $G = 2p^2 P_p$. Write $\delta m_h = m_{\nu, h} - m_{\mu, h}$. Suppose that, for every shift $u \in H_p$ in the neighborhood under consideration,

$$P_p \nabla^2 \phi_p(h + u) P_p \preceq \frac{C_{\text{ball}}}{p^2} P_p.$$

The required neighborhood contains all such shifts whose radius, in units $p^2 M_p$, is at most

$$\left(\frac{2l_h}{k_p C_{\text{ball}}} \right)^{1/2}. \quad (19.13)$$

Then the variational formula for relative entropy, tested against linear functions, gives

$$p^2 \|P_p \delta m_h\|^2 \leq 2C_{\text{ball}} l_h. \quad (19.14)$$

Indeed integrate the Hessian along a shift u to bound its centered log moment generating function by $C_{\text{ball}} \|u\|^2/(2p^2)$, and test in the direction of $P_p \delta m_h$. The trial shift of length $p\sqrt{2l_h/C_{\text{ball}}}$ is available precisely under (19.13); the resulting inequality also bounds the length of the unconstrained optimizer.

We next trade covariance control for a differential inequality on the remaining entropy. The variational argument requires an entire ball of linear field shifts, with radius depending on the conditional entropy. A coarse first interval makes the cap dimension large compared with that

entropy. On the later interval the required shifts fit inside the region of sharp curvature, apart from errors whose weighted drift contribution is exponentially small. Keeping those contributions inside the differential inequality is necessary for the bottom branch, where the retained entropy is itself small.

Here is the bound used for every discarded event. Put $Y_p(X) = \|P_p X\|/M_p$. Conditional Jensen gives

$$p^2 \|P_p \delta m_h\|^2 \leq 2k_p (\nu_h[Y_p^2] + \mu_h[Y_p^2]).$$

For any observation event E , including one depending on f , and any fixed $V > 0$, disintegration and the likelihood bound yield

$$\begin{aligned} \mathbb{E}_{Q_\nu} \left[p^2 \|P_p \delta m_h\|^2 \mathbf{1}_E \right] &\leq 4k_p V^2 Q_\nu(E) \\ &\quad + 4k_p e^L \mu \left[Y_p^2 \mathbf{1}_{\{Y_p > V\}} \right]. \end{aligned} \tag{19.15}$$

For example, the tilted posterior term averages to a moment under ν ; the ordinary posterior term uses the path density at most e^L . The projection tail pays the weighted term, so a probability estimate for E is used together with a moment estimate.

Let $k_2 = R_* k_1$ with R_* a sufficiently large fixed constant. On $[p_1, p_2]$, use the bounded-ball curvature bound of Theorem 18.6. Choose $A_\ell \geq A > C_0$, where C_0 is from (19.12). Discard fields outside a fixed ball and the event $l_h > L + A_\ell k_p$. Since $L/k_p \leq 1/c_0$, the latter event has probability at most $e^{-A_\ell k_p}$ by (19.8). Enlarge the eventual curvature constant to satisfy $C_{\text{ball}} \geq 1$. Before obtaining that constant, choose the ball to contain shifts of radius $\sqrt{2(c_0^{-1} + A_\ell)}$ in scaled units from every retained field; this contains the radius in (19.13). Choose its ordinary field-tail rate and the projection threshold in (19.15) larger than $A + 1/c_0$. After paying e^L , the discarded drift is at most $Ck_p e^{-A k_p}$. The field range and these coarse tail choices precede the sharp tolerances and R_* . Its curvature constant, even if large, causes only a fixed loss on this interval. Equations (19.6) and (19.14) give

$$\frac{dH}{d \log p} \geq -2C_{\text{ball}} H - Ck_p e^{-A k_p}.$$

The interval has fixed logarithmic length. Integrating this inequality therefore multiplies the initial lower bound in (19.12) by a fixed positive factor. The additive error is negligible relative to either branch of that lower bound.

On $[p_2, p_{\text{top}}]$, use instead the curvature $(1 + \epsilon)/p^2$ on the near-zero-parameter region. Choose an ordinary saddle tolerance ζ strictly inside that region, leaving a field buffer of radius r_{buf} in units $p^2 M_p$. Put $b_{\text{buf}} = (1 + \epsilon)r_{\text{buf}}^2/2$. The shift radius fits this buffer whenever $l_h \leq b_{\text{buf}} k_p$. At these scales $L/k_p \leq (R_* c_0)^{-1}$. Choose R_* so this ratio is below fixed fractions of both b_{buf} and the rate a_ζ in (19.9). Then

$$Q_\nu(l_h > b_{\text{buf}} k_p) \leq e^{-(b_{\text{buf}} - L/k_p)k_p}, \quad Q_\nu(|\alpha_p(h_p)| > \zeta p^2) \leq e^{-(a_\zeta - L/k_p)k_p}.$$

Equation (19.15), with another fixed projection threshold, leaves $Ck_p e^{-a k_p}$ for some fixed $a > 0$. Thus

$$\frac{dH}{d \log p} \geq -2(1 + \epsilon)H - Ck_p e^{-a k_p}.$$

Increase R_* again so that $aR_* > C_0$ with slack.

For either differential inequality, an exponent γ_H and error $Ck_p e^{-b k_p}$ give

$$H(p) \geq \left(\frac{p_a}{p} \right)^{\gamma_H} \left[H(p_a) - C \int_{p_a}^p \left(\frac{u}{p_a} \right)^{\gamma_H} k_u e^{-b k_u} d \log u \right].$$

The coarse error starts at k_1 with $A > C_0$; the sharp error starts at R_*k_1 with $aR_* > C_0$. Their weighted integrals are therefore negligible relative to either branch of (19.12). Solving yields

$$H(p_{\text{top}}) \geq c \left(\frac{p_1}{p_{\text{top}}} \right)^{2+2\epsilon} \begin{cases} L, & k_1 = c_0 L, \\ e^{-C_0 k_1}, & k_1 = k_{\min}. \end{cases} \quad (19.16)$$

Pointwise control of the discarded drift is not needed here. With the integrated bounds, the integrating factor charges each geometric interval only a fixed power of p/p_1 . Its error decreases exponentially in k_p , and $k_1 \rightarrow \infty$, so their weighted sum is still negligible. Finally choose λ small enough that $R_*c_0\lambda < p_{\text{top}}^3$, up to the fixed endpoint factors. Then $p_2 < p_{\text{top}}$ for all $L \leq \lambda n$.

19.4 The remaining fixed scales

We next transfer (19.16) to a product endpoint, allowing fixed multiplicative losses. The initialized scale is p_0 from Section 18, with $t = C_*p_0^2$. Starting from p_{top} , first traverse the finite remaining cap intervals to p_0 . Use the actual finer frame for the terminal bridge there. Its increment is

$$\Delta_{\text{base}} = tI + (1 - \beta)J - B_{p_0},$$

so the total precision is exactly $tI + (1 - \beta)J$. The posterior interaction is $\beta J - tI$, with the same spin law as βJ_{off} .

Now increase scalar precision along

$$B_u = uI + (1 - \beta)J, \quad t \leq u \leq T,$$

for a sufficiently large fixed T . The choice of the single coefficient $\beta < 1$ in Section 18 includes the persistence requirement for this interval. Namely, fix T and the normalized buffers for the ordinary critical test after p_0 is fixed and before choosing β . Adding an independent increment of precision $(1 - \beta)(4I + J)$ to scalar precision $(u - 4(1 - \beta))I$ gives B_u . The associated common-spin field displacement has normalized size $O((1 - \beta) + \sqrt{1 - \beta})$ outside an exponentially rare event. Time nets and Gaussian maximal bounds preserve the buffered tests along the interval. Lemma 17.3 then supplies a fixed whole-path rate e^{-an} and a fixed field neighborhood. The auxiliary removal is performed with this already fixed β ; it determines p_{top} , and λ is chosen afterward.

Next use

$$B_a = (T + 4a)I + (1 - \beta + a)J, \quad 0 \leq a \leq \beta.$$

Its rate is $4I + J \succ 0$, and it ends at $T'_\beta I + J$ with $T'_\beta = T + 4\beta > 3$. For T large, uniformly in the underlying spin, the field has large absolute coordinates except on a small site fraction: the main shift is TX , the other deterministic shifts have bounded normalized norm, and the noise has squared normalized norm $O(T)$ outside probability e^{-an} . At an approximate root the diagonal A in the stability test is therefore small outside a small site set. Gaussian nets give a small operator norm for every sufficiently sparse principal submatrix of a typical GOE matrix. Together with the full norm bound and the small normalized norm of $A^{1/2}m$, this verifies the stability test, again on a field neighborhood. While the coefficient of J belongs to a fixed compact subinterval of $(0, 1)$, the classical covariance bound and the entropy shift argument consequently give

$$\dot{H} \geq -CH - e^{-an}, \quad (19.17)$$

after reducing a and λ to absorb the path tilt e^L .

For sufficiently small interaction coefficients, the covariance is bounded globally. Indeed add a scalar diagonal to the interaction to obtain a matrix Q with $0 \prec Q \prec I/2$, and introduce a Gaussian

latent field with covariance Q . Its posterior log density has Hessian at most $-Q^{-1} + I \preceq -I$. Brascamp–Lieb bounds the covariance of the coordinatewise tanh by I , while the conditional spin variances contribute another I . Thus the spin covariance is at most $2I$. The variational argument now gives $\dot{H} \geq -CH$ for every field and every L , without any discarded set. Word and residual diagnostics along varying coefficients are consequently needed only on closed subintervals of $(0, 1)$.

The sharp cap estimate stops at p_{top} , below the initialization scales. There are only finitely many intervals left before the classical field, so bounded amplification and fixed losses suffice there. Starting from the terminal covariance bound, subtract the two mean-transport identities of Lemma 18.3. One term transports the difference of terminal means; the other compares expectations of the bounded terminal approximation error under two path laws. Pinsker’s inequality controls the latter by the square root of conditional entropy. This propagates the required quadratic mean estimate backward through the finite list of intervals.

At the terminal classical field, the covariance bound holds on a fixed normalized neighborhood outside an ordinary event of probability e^{-an} . The linear entropy test fits this neighborhood whenever $l_h \leq bn$ for a fixed $b > 0$. Choose $\lambda < \min(a, b)/2$ after these constants. The ordinary exception costs at most $e^{-(a-\lambda)n}$ under Q_ν , and (19.8) gives $Q_\nu(l_h > bn) \leq e^{-(b-\lambda)n}$. Since both spin means have norm at most $\sqrt{n}$, these exceptions have exponentially small quadratic cost. The neighborhood covariance estimate therefore gives, after reducing the positive exponent,

$$\mathbb{E}_{Q_\nu} \|m_\nu - m_\mu\|^2 \leq CH + e^{-an}.$$

Fix any current time s in one preceding interval and a current field h . Let $Q_{\nu,h}^{\text{cont}}$ and $Q_{\mu,h}^{\text{cont}}$ be the two laws of the future observations through the end of that interval, starting from the posteriors ν_h and μ_h . They live on the same path space, and data processing gives $D(Q_{\nu,h}^{\text{cont}} \| Q_{\mu,h}^{\text{cont}}) \leq l_h$. Let g be the gradient of one extended auxiliary potential evolved by heat convolution, and let U be its linear transport from time s to the interval end. The same path function U is evaluated under both continuation laws, and its amplification is bounded by a fixed constant. Subtracting the two transport identities gives, in every direction v of the current rate subspace,

$$\begin{aligned} \langle v, m_\nu - m_\mu \rangle &= \mathbb{E}_{Q_{\nu,h}^{\text{cont}}} \langle Uv, m_{\nu,\text{end}} - m_{\mu,\text{end}} \rangle \\ &\quad + (\mathbb{E}_{Q_{\nu,h}^{\text{cont}}} - \mathbb{E}_{Q_{\mu,h}^{\text{cont}}}) \langle Uv, m_{\mu,\text{end}} - g_{\text{end}} \rangle. \end{aligned}$$

On agreement events the second observable is bounded by a constant times $\|v\|$, by the raw gradient comparison, including at the classical terminal field. Truncate it there and use the displayed continuation entropy bound and Pinsker’s inequality to bound its expectation difference by $C\sqrt{l_h}\|v\|$. To bound the complements after averaging, let $W \geq 0$ be a terminal cost, such as the squared raw error outside agreement. The amplification bound lets us dominate the transported exceptional terms by such costs. Disintegration and the observation likelihood give

$$\begin{aligned} \mathbb{E}_{Q_\nu} \mathbb{E}_{Q_{\nu,h}^{\text{cont}}} W &= \mathbb{E}_{Q_\nu} W \leq e^L \mathbb{E}_{Q_\mu} W, \\ \mathbb{E}_{Q_\nu} \mathbb{E}_{Q_{\mu,h}^{\text{cont}}} W &= \mathbb{E}_{Q_\mu} \left[\frac{N_s}{N} \mathbb{E}_{Q_{\mu,h}^{\text{cont}}} W \right] \leq e^L \mathbb{E}_{Q_\mu} W. \end{aligned}$$

The ordinary terminal agreement estimates and their polynomially weighted versions therefore give exponentially small averaged costs after choosing λ below the finitely many fixed rates. This uses full-path and current-prefix likelihood bounds, separately from conditional Pinsker. Apply Cauchy–Schwarz first conditionally on the current field and then over the tilted observation path. The terminal entropy satisfies $H_{\text{end}} \leq H(s)$. If P_{rate} projects onto the current rate subspace, these bounds give

$$\mathbb{E}_{Q_\nu} \|P_{\text{rate}}(m_{\nu,s} - m_{\mu,s})\|^2 \leq CH(s) + e^{-an}.$$

The transported directions end in the appropriate terminal rate subspace on cap intervals; for a full-rank bridge $P_{\text{rate}} = I$. Since s was arbitrary, this estimate supplies the differential inequality throughout the interval.

For the bridge use the actual-refinement TAP extension; for cap intervals use the auxiliary or true-cap extension supplied by the induction. At each cap endpoint, (19.9) gives agreement outside an exponentially rare event, and the extension has polynomial gradient moments. For the bridge, apply Lemma 18.1 on the agreement annulus at p_0 ; its ordinary complement is paid by the same saddle test and polynomial moments. The amplification bound holds from every intermediate time, while these terminal estimates enter through the averaged comparison above. No agreement condition at the current field is needed in the subtraction, since its common approximate gradient cancels. Iterating through this finite list, even when p_{top} is interior to one interval, proves (19.17). Its constants may depend on the fixed initialization length. Solving it proves (19.16), up to constants, at the product endpoint.

19.5 The product estimate

At precision $T'I + J$ the posterior has independent coordinates with fields h_i . A two-point measure at field h_i has inverse log-Sobolev constant at most $C(1 + |h_i|)$ in conditional-variance normalization. One direct proof centers the test, uses (19.5), and bounds the entropy of the centered square by its second moment times the logarithm of the inverse smaller atom; that logarithm is at most $C(1 + |h_i|)$. Tensorization and the posterior entropy identity first give

$$l_h \leq C(1 + \|h\|_\infty) \frac{\mathcal{D}_{\mu_h}(f)}{N_h}.$$

Averaging this bound with (19.7) gives

$$H_{\text{end}} \leq C\sqrt{\log n + L} \frac{\mathcal{D}_0(f)}{N} + O(n^{-3}). \quad (19.18)$$

To justify the truncation implicit here, apply tensorization on the event in (19.10). On its complement use

$$l_h \leq \log \frac{1}{\min_x \mu_h(x)} \leq Cn(1 + \|h\|_\infty).$$

The moment version of (19.10), followed by the tilt bound e^L , makes its contribution $O(n^{-3})$, uniformly for $L \leq Cn$.

Combining this with (19.16) proves (19.1) for $L \leq \lambda n$. Here is the exponent bookkeeping. In the first case, $p_1^3 \asymp L/n$ and $L \gtrsim \sqrt{\log n}$, so

$$\frac{H_{\text{end}}}{\sqrt{\log n + L}} \geq cn^{-(2+2\epsilon)/3} L^{\frac{L^{(2+2\epsilon)/3}}{\sqrt{\log n + L}}}.$$

The last quotient is bounded below by a negative fixed power of $\log n$ when $L \leq \log n$, and by a constant when $L \geq \log n$. In the second case $k_1 \asymp \sqrt{\log n}$, $p_1^3 = k_1/n$, and $L \leq C\sqrt{\log n}$; hence

$$\frac{H_{\text{end}}}{\sqrt{\log n + L}} \geq n^{-(2+2\epsilon)/3 - o(1)}.$$

Both lower bounds dominate $n^{-2/3-r}L$ by choosing ϵ small relative to r . They also dominate the additive $O(n^{-3})$ error.

19.6 Large entropy and stopping

The remaining entropy levels are proportional to the dimension. At these levels an ordinary rare-event bound cannot simply be multiplied by the initial density bound. We instead argue under the proposed small energy ratio. The posterior mean estimate then makes the revealed information small. Relative entropy of the stopped observation laws transfers the ordinary stopping probability to the tilted law, and bounded spin means control the loss after stopping. This retains order- n entropy and conflicts with the product estimate.

Now suppose $\lambda n \leq L \leq Cn$. We prove the stronger estimate $\mathcal{D}_0(f)/N > n^{-2/3}L$ by contradiction. Thus assume

$$\frac{\mathcal{D}_0(f)}{N} \leq n^{-2/3}L \leq Cn^{1/3}. \quad (19.19)$$

The objective is to retain a fixed positive multiple of λn until the product endpoint of this path. Formula (19.18) would then bound that retained entropy by $O(n^{5/6}) + O(n^{-3})$, a contradiction.

We specify the uniformity required at the start. The small-positive-time statement in Proposition 17.5, with the noise interface of Lemma 17.4, uses diagnostics of the centered Gaussian noise and admits every additional shift of normalized norm at most Ct . Its separate low-band condition is a Gaussian small-ball bound uniform in the center, not a simultaneous event for all centers. More precisely, on a disorder event independent of f , the noise-only diagnostics fail with conditional probability $o(1)$, integrated over each fixed interval $[t_*, t_b]$, and

$$\sup_{z \in \mathbb{R}^n} \mathbb{P}_g \left(\|P_t^J(g_t + z)\| < c\sqrt{n}t^{5/4} \right) \leq e^{-c'nt^{3/2}}. \quad (19.20)$$

Here $P_t^J = \mathbf{1}_{\{2I - J \leq tI\}} = P_{\sqrt{t}}$ is the cumulative projection used in Lemma 17.4, with $W = J$. The covariance on a band of rank comparable to $nt^{3/2}$ is comparable to tI ; projection onto that band and the Gaussian ball bound prove (19.20). Translation cannot increase the mass of a centered Gaussian Euclidean ball. For fixed J the spin is independent of the observation noise. Since its drift shift has normalized norm $O(t)$ for every $X \in \{-1, 1\}^n$, the deterministic shift allowance and the uniform-in-center probability give the same $o(1)$ integrated failure bound under every initial spin law, including $f^2\mu/N$. One first uses Fubini and Markov for the noise-only diagnostic event to select the disorder event; no union over the 2^n spin centers and no factor e^L occurs in this step. The exclusion-at-zero and matrix-word events are also selected independently of f .

First fix the large scalar endpoint T and the interaction coefficient below which covariance is globally bounded. Next choose $t_b > 0$ sufficiently small for Proposition 17.5, and obtain the buffered ordinary test and its failure rate $a > 0$ on $[t_b, T]$ and the subsequent weakening interval. These margins are uniform for all sufficiently small changes of the interaction coefficient from one. Fix a target integrated drift budget η_1 sufficiently small relative to $a\lambda$. Choose $0 < t_* \ll t_b$ so that Ct_* fits that budget, and finally choose $1 - \beta' > 0$ sufficiently small relative to these fixed scales and t_* .

After fixing β' , reduce the permitted diagonal enlargement in the stability test to a positive $\eta_{\text{stab}} < (\beta')^{-1} - 1$. The stability implication persists on this smaller diagonal class, and $(\beta')^2(1 + \eta_{\text{stab}})^2 < 1$ supplies the strict margin required by Proposition 17.1.

Once the resulting stability margins are known, choose the residual tolerance and finite diagnostic depth in the square-reweighted mean estimate of Proposition 17.1 to realize the target budget. The target η_1 is distinct from that proposition's accuracy parameter: the leading constant multiplying its accuracy may depend on the fixed stability margins. Its remaining constants may be arbitrarily large but are all fixed before $n \rightarrow \infty$. The factor $n^{-1/2}$ in the posterior estimate will therefore still make the energy-dependent error vanish under (19.19).

Start directly with precision $t_*I + (1 - \beta')J$, increase scalar precision to T , and then weaken the interaction at rate $4I + J$ for duration β' . The endpoint is $T'_{\beta'}I + J$ with $T'_{\beta'} = T + 4\beta'$, the second

fixed endpoint included in (19.10). Positivity of the initial precision follows from choosing $1 - \beta'$ sufficiently small after t_* . For comparing this path with the ordinary critical scalar observation, one may add an independent increment of precision $(1 - \beta')(4I + J) \succ 0$ and increase scalar noise time by $O(1 - \beta')$. This produces a small normalized field displacement outside an exponentially rare event, so the buffered ordinary tests remain valid with rate a after a fixed decrease.

Let s_g be the onset of the globally bounded-covariance portion. On the interval $[t_*, t_b]$, the uniform noise and small-ball argument just given verifies the stability hypotheses with integrated failure tending to zero under Q_ν . On their validity set the tilted mean estimate from Proposition 17.1, together with (19.7), yields an arbitrarily small integrated bound on $\mathbb{E}_{Q_\nu} |m_\nu - m_\mu|_n^2$. Indeed the relevant error is bounded by

$$C\eta + \frac{C_\eta}{\sqrt{n}} \left(1 + \frac{\mathcal{D}_{\mu_h}(f)}{N_h} \right),$$

and its average is $C\eta + O_\eta(n^{-1/6}) + O_\eta(n^{-1/2})$ by (19.19). This estimate initially controls a first moment. Since $|m_\nu - m_\mu|_n \leq 2$, its square is at most twice its first power, giving the needed quadratic bound. The integrated failure probability contributes only $o(1)$ by the same uniform bound.

For times at least t_b , use the fixed buffered ordinary stability test, selected independently of f before observing the path. Let $\tau \leq s_g$ be its first violation, taking $\tau = t_b$ if it already fails there and $\tau = s_g$ if no violation occurs. Up to τ the same integrated tilted-mean bound applies deterministically on the diagnostic event. Write Q_ν^τ, Q_μ^τ for the laws of the observations stopped at τ . The initial jump leaks at most Ct_*n , by the pure-noise comparison above without the projection improvement. The stopped version of (19.6) then gives

$$D(Q_\nu^\tau \| Q_\mu^\tau) \leq (Ct_* + \eta_1 + o(1))n, \quad (19.21)$$

where $\eta_1 > 0$ can be made arbitrarily small by the preceding fixed accuracy choices. This includes the possibility of immediate stopping at t_b .

Let E be the event that a violation occurs before the globally controlled portion. Its ordinary probability is at most e^{-an} . Data processing for the indicator of E gives

$$D(Q_\nu^\tau \| Q_\mu^\tau) \geq Q_\nu(E) \log \frac{1}{Q_\mu(E)} - \log 2.$$

Consequently

$$Q_\nu(E) \leq \frac{Ct_* + \eta_1 + o(1)}{a}. \quad (19.22)$$

On a path that stops, use the spin norm bound for the remaining portion of the observation. Every mean has norm at most $\sqrt{n}$, and the precision rate has bounded norm over a bounded duration. Its entropy-loss integral is consequently at most Cn . Multiplying by the stopping probability bounds the averaged cost of these paths. By the constant choices, (19.21) and (19.22) make the total loss before s_g smaller than $\lambda n/4$. Initially $H(0) \geq L/2 \geq \lambda n/2$, so $H(s_g) \geq c\lambda n$. The subsequent global covariance estimate propagates a fixed fraction to the product endpoint. This is the claimed contradiction with (19.18).

All disorder, word, and stability events in this argument have fixed thresholds selected independently of f . The only function-dependent objects are the tilted law and the entropy level L . The proof uses the exponential ordinary failure rate through stopped-path relative entropy, and therefore remains valid when L is of order n .

19.7 The log-Sobolev bound

We have proved (19.1) in both entropy ranges. Lemma 19.1 now gives the promised functional inequality.

Theorem 19.3 (Critical log-Sobolev upper bound). *For every fixed $\rho > 0$, with probability tending to one over the Gaussian couplings, the critical zero-field Gibbs measure satisfies*

$$\text{Ent}_\mu(f^2) \leq n^{2/3+\rho} \mathcal{D}_0(f) \quad \text{for every } f : \{-1, 1\}^n \rightarrow \mathbb{R}. \quad (19.23)$$

The form is normalized for the generator $\mathcal{L} = \sum_{i=1}^n (Q_i - I)$, where Q_i is conditional expectation given X_{-i} .

Proof. Apply the preceding argument with an exponent allowance strictly smaller than ρ . On $\|J\| \leq 3$ the value M in Lemma 19.1 is Cn . Its fixed constant and $(1 + \log M)^2$ factor are absorbed in the remaining power of n . $\square$

Section 20 derives the mixing and relaxation bounds from this inequality and combines them with the companion's lower bounds. The cap comparison also gives the following direct linear test. It explains the same worst-start lower exponent with weaker quantifiers.

Remark 19.4 (An independent linear lower test). The static comparison also gives a direct, weaker-in-quantifiers route to the matching worst-start lower exponent. Fix $0 < \epsilon < 1/3$ and take $p = n^{-1/3+\epsilon}$. In the uncapped zero-field spherical model, $\|P_p X\| \geq cM_p$ with probability tending to one. To see this, compare with the Gaussian of precision $2 + \xi p^2 - J$. A spectral band with gaps comparable to p^2 already has this lower bound outside e^{-ck_p} . The lower norm-density estimate and the bounded density of the remaining coordinates transfer it through norm conditioning, when $\xi > 0$ is sufficiently small. Haar first moments and the zero-field second-moment comparison in Proposition 15.1 transfer the conclusion to the cube with high Gibbs probability for typical disorder. Spin symmetry and rank $P_p \leq Cnp^3$ imply that some Euclidean unit vector a has

$$\text{Var}_\mu(a^\top X) \geq cp^{-2}, \quad \mathcal{D}_0(a^\top X) = \sum_i a_i^2 \mu[\text{Var}(X_i | X_{-i})] \leq 1.$$

Thus the rate-one spectral gap is at most Cp^2 . Testing a gap eigenfunction from a state where its absolute value is maximal gives $t_{\text{mix}}(1/4) \geq cp^{-2}$; the discrete gap is smaller by n . Letting the fixed ϵ decrease proves the same lower exponents. The realized-start statement and the bounds for every arbitrarily slowly diverging M_n remain the separate conclusions of Theorem 1.2.

20 From the functional inequalities to mixing

We now turn the two functional endpoints into worst-start mixing bounds. Both deductions use the unscaled heat-bath form and keep the factor n between the two clocks explicit. The support-profile proof is the finite-state truncation mechanism of [22, Theorem 1.1 and Lemma 2.1]. The classical logarithmic Sobolev proof uses the hypercontractive smoothing of [15, Theorems 3.5 and 3.7], followed by spectral-gap decay.

20.1 The support-profile route

Lemma 20.1. *Let μ be a full-support law on the cube, with heat-bath form $\mathcal{D}$, generator $\mathcal{L}$, and one-attempt kernel P . Suppose that for some $\alpha > 0$ and every nonnegative f with support mass $0 < p = \mu(f > 0) \leq 1/2$,*

$$\mathcal{D}(f) \geq \alpha \mu(f^2) \log(1/p). \quad (20.1)$$

Write $t_{\text{mix}}^{\text{ct}}(\varepsilon)$ and $t_{\text{mix}}^{\text{dt}}(\varepsilon)$ for the worst-start total-variation mixing times of $e^{t\mathcal{L}}$ and P^k , respectively. Put

$$\mu_{\min} = \min_x \mu(x), \quad M = \max\{8, \mu_{\min}^{-1}\}, \quad k_0 = \left\lceil \frac{2n}{\alpha} \log \frac{\log(M/4)}{\log 2} \right\rceil.$$

Then, for every $0 < \varepsilon < 1/2$,

$$t_{\text{mix}}^{\text{ct}}(\varepsilon) \leq \frac{1}{\alpha} \log \frac{\log(M/4)}{\log 2} + \frac{1}{2\alpha \log 2} \log \frac{7}{4\varepsilon^2}, \quad (20.2)$$

$$t_{\text{mix}}^{\text{dt}}(\varepsilon) \leq k_0 + \left\lceil \frac{n}{\alpha \log 2} \log \frac{7}{4\varepsilon^2} \right\rceil. \quad (20.3)$$

In particular, if $\log(1/\mu_{\min}) \leq Cn$, then for every fixed $\varepsilon > 0$ the worst-start mixing time is $O_{C,\varepsilon}(\alpha^{-1} \log(n+1))$ in per-site time and $O_{C,\varepsilon}(n\alpha^{-1} \log(n+1))$ attempts.

Proof. First obtain a Poincare inequality, which will handle the final part of mixing. Subtract a median from a real function and take the positive and negative parts f_+, f_- . Both supports have mass at most $1/2$. For each edge of the cube, splitting into these two parts can only decrease the sum of squared increments. The two-point representation of each conditional variance therefore gives $\mathcal{D}(f_+) + \mathcal{D}(f_-) \leq \mathcal{D}(f)$. Applying (20.1) to the two parts yields

$$\text{Var}_\mu(f) \leq \mu[(f - \text{med}_\mu f)^2] \leq \frac{\mathcal{D}(f)}{\alpha \log 2}.$$

Thus the gap is at least $\alpha \log 2$.

If $g \geq 0$ is a probability density relative to μ and $V = \mu(g^2)$, let $u = (g - V/4)_+$. Markov's inequality and the elementary inequality $(x - a)_+^2 \geq x^2 - 2ax$ for $x, a \geq 0$ give

$$\mu(u \neq 0) \leq 4/V, \quad \mu(u^2) \geq V/2.$$

The truncation is a contraction, so $\mathcal{D}(g) \geq \mathcal{D}(u)$. For $V \geq 8$, (20.1) consequently implies

$$\mathcal{D}(g) \geq \frac{\alpha}{2} V \log(V/4) \geq \frac{\alpha V}{6} \log V. \quad (20.4)$$

For the continuous-time semigroup with generator $\mathcal{L} = \sum_{i=1}^n (Q_i - I)$, where Q_i is conditional expectation given X_{-i} , the density from a point evolves by $g_t = e^{t\mathcal{L}} g_0$. Reversibility gives $V_t' = -2\mathcal{D}(g_t)$. While $V_t \geq 8$, setting $z_t = \log(V_t/4)$ in (20.4) yields $z_t' \leq -\alpha z_t$. Put

$$t_0 = \alpha^{-1} \log \frac{\log(M/4)}{\log 2}.$$

Since $V_0 = 1/\mu(x) \leq M$, by time t_0 we have $V_t \leq 8$ for every starting point.

Afterwards the gap bound gives $V_{t_0+t} - 1 \leq 7e^{-2\alpha(\log 2)t}$. The inequality $\|g\mu - \mu\|_{\text{TV}} \leq \frac{1}{2}\sqrt{V-1}$ therefore proves (20.2).

For attempted updates we obtain the squared-norm decrease directly from positivity of the average of conditional-expectation projections. The transition operator is $P = n^{-1} \sum_i Q_i = I + \mathcal{L}/n$. Each Q_i is an orthogonal projection in $L^2(\mu)$, hence $0 \preceq P \preceq I$ and $P^2 \preceq P$. If $g_{k+1} = Pg_k$, then

$$V_k - V_{k+1} = \langle g_k, (I - P^2)g_k \rangle_\mu \geq \langle g_k, (I - P)g_k \rangle_\mu = \frac{1}{n} \mathcal{D}(g_k).$$

While $V_k \geq 8$, this and (20.4) give $V_{k+1} \leq V_k[1 - \alpha \log(V_k/4)/(2n)]$.

If the next value is still at least 8, taking logarithms gives

$$z_{k+1} \leq (1 - \alpha/(2n))z_k \leq e^{-\alpha/(2n)}z_k, \quad z_k = \log(V_k/4).$$

The factor is positive: the gap bound and $-\mathcal{L} \preceq nI$ imply $\alpha \log 2 \leq n$. In addition, whenever this recurrence is used its bracket must be positive since it bounds the positive value V_{k+1} . Thus after at most k_0 steps the squared density norm is at most 8.

On centered functions the same positive-operator inequality and the gap imply $V_{k+1} - 1 \leq (1 - \alpha \log 2/n)(V_k - 1)$. The total-variation inequality above now gives (20.3). This proves the transfer for deterministic numbers of attempts.

If $\log(1/\mu_{\min}) \leq Cn$, then $\log(\log(M/4)/\log 2) = O_C(\log(n+1))$. The two explicit bounds give the stated big- O consequences for fixed $0 < \varepsilon < 1/2$. For $\varepsilon \geq 1/2$, the bound at accuracy $1/4$ suffices by monotonicity of mixing time in the accuracy threshold. $\square$

The support bound in Theorem 14.2 holds simultaneously for all tests on its good-disorder event, so it applies throughout the density evolution in the lemma. Fixed constants and logarithmic losses are absorbed by choosing a slightly smaller exponent tolerance first. For the big- O consequence used here, the remaining hypothesis is a lower bound on point masses. On $\|J\| \leq 3$, the Hamiltonian oscillation is at most $3n$, and

$$\mu_{J,\min} \geq \exp[-(3 + \log 2)n]. \quad (20.5)$$

This proves the asserted worst-start upper exponents through the support-profile route.

20.2 The logarithmic Sobolev route

The classical inequality gives a second, independent smoothing argument. With our convention $\text{Ent}_\mu(f^2) \leq C_{\text{LS}}\mathcal{D}(f)$, finite-chain hypercontractivity takes the form

$$\left\| e^{t\mathcal{L}} f \right\|_{q(t)} \leq \|f\|_p, \quad q(t) - 1 = (p - 1)e^{4t/C_{\text{LS}}}, \quad p > 1. \quad (20.6)$$

Here the factor four is the reversible-chain normalization in [15, Theorem 3.5(ii)]: its energy-to-entropy constant is $1/C_{\text{LS}}$ for our unscaled form. For the density of a point mass, choose $p = 1 + 1/\max(1, \log(1/\mu_{\min}))$. Its L^p norm is at most e . The time at which $q(t) = 2$ is $O(C_{\text{LS}} \log(2 + \log(1/\mu_{\min})))$. Expansion of the logarithmic Sobolev inequality at $f = 1 + ug$, with $\mu g = 0$, gives

$$2 \text{Var}_\mu(g) \leq C_{\text{LS}}\mathcal{D}(g). \quad (20.7)$$

The remaining L^2 decay to any fixed TV accuracy therefore costs $O_\varepsilon(C_{\text{LS}})$.

To transfer this bound to attempted updates, compare the two semigroups in L^2 . If $\theta \in [0, 1]$ is an eigenvalue of P , then $\theta^{2k} \leq e^{-2k(1-\theta)}$. Expanding a centered initial density in an orthonormal eigenbasis gives

$$\left\| P^k(g - 1) \right\|_2 \leq \left\| e^{(k/n)\mathcal{L}}(g - 1) \right\|_2. \quad (20.8)$$

Thus the same argument, with a factor n in time, proves the discrete upper bound. Using Theorem 19.3 and (20.5) establishes both upper exponents by the second route as well.

20.3 Sharp exponents and preservation of the lower quantifiers

The linear tests already give lower bounds for the full functional constants: by definition $R_J \geq R_{\text{lin},J}$. Applying [Theorem 1.2](#) and (20.7) gives

$$C_{\text{LS},J} \geq 2R_J \geq 2R_{\text{lin},J} = n^{2/3+o(1)}$$

as a lower-exponent statement in disorder probability. The median consequence of [Theorem 14.2](#) gives the upper bound for R_J , while [Theorem 19.3](#) gives the upper bound for $C_{\text{LS},J}$ (and, by (20.7), another upper bound for R_J). Intersecting these finitely many fixed-tolerance disorder events gives the functional assertions of [Theorem 1.1](#).

For a lower mixing bound at every fixed threshold below $1/2$, choose a nonconstant eigenfunction g of $-\mathcal{L}_J$ with eigenvalue $1/R_J$. Center it and normalize $\|g\|_\infty = 1$. At a point where $|g| = 1$, its expectation decays exactly as e^{-t/R_J} , whereas its stationary expectation is zero. Testing TV against this bounded function gives

$$\max_x \left\| e^{t\mathcal{L}_J}(x, \cdot) - \mu_J \right\|_{\text{TV}} \geq \frac{1}{2} e^{-t/R_J}.$$

The corresponding eigenvalue of P_J is $1 - 1/(nR_J)$, so the discrete lower bound is $\frac{1}{2}(1 - 1/(nR_J))^k$. For every fixed $\varepsilon < 1/2$, these inequalities give mixing lower bounds of order R_J and nR_J , respectively. Together with the upper bounds, they prove [Theorem 1.1](#).

The realized-start assertions in [Theorem 1.2](#) apply to every deterministic little-oh time sequence. Its covariance and linear-Rayleigh assertions apply to every diverging deterministic sequence M_n . These quantifiers are retained from the companion; no stronger upper quantifier or cutoff statement is inferred from the new exponent estimates.

20.4 Uniform relative-entropy decay

The spectral-gap and classical logarithmic Sobolev bounds together determine the critical time scale for uniform relative-entropy decay. We state the normalization explicitly, since constants called modified logarithmic Sobolev constants use several conventions. For the per-site generator, define

$$\mathcal{E}_J(u, v) := -\mu_J[u\mathcal{L}_J v] = \sum_i \mu_J[\text{Cov}_{\mu_J}(u, v \mid X_{-i})], \quad \mathcal{E}_J(f, f) = \mathcal{D}_J(f),$$

and let

$$C_{\text{EP},J} := \sup_{\substack{g>0, \mu_J g=1 \\ \text{Ent}_{\mu_J}(g)>0}} \frac{\text{Ent}_{\mu_J}(g)}{\mathcal{E}_J(g, \log g)}.$$

This is the inverse constant for the logarithmic entropy-production inequality. The standard comparison and semigroup interpretation appear in [9, Propositions 3.5–3.6 and Theorem 2.4]; we give the short derivation in the present normalization.

Corollary 20.2 (Entropy-production exponent). *For every realization of the disorder,*

$$\frac{R_J}{2} \leq C_{\text{EP},J} \leq \frac{C_{\text{LS},J}}{4}.$$

Consequently, for the critical zero-field Gaussian model, $C_{\text{EP},J} = n^{2/3+o(1)}$ in disorder probability, with the fixed exponent-slack meaning of [Theorem 1.1](#). For every initial probability law ν on the cube and every $t \geq 0$,

$$\text{KL}(\nu e^{t\mathcal{L}_J} \parallel \mu_J) \leq e^{-t/C_{\text{EP},J}} \text{KL}(\nu \parallel \mu_J) \leq e^{-4t/C_{\text{LS},J}} \text{KL}(\nu \parallel \mu_J).$$

Proof. For positive a, b , $(a - b)(\log a - \log b) \geq 4(\sqrt{a} - \sqrt{b})^2$. Apply this inequality to the two-point conditional representation of each term of $\mathcal{E}_J$. The classical inequality then gives

$$\mathcal{E}_J(g, \log g) \geq 4\mathcal{D}_J(\sqrt{g}) \geq \frac{4}{C_{\text{LS},J}} \text{Ent}_{\mu_J}(g),$$

which proves the upper comparison. For a centered real function f , $g = 1 + uf$ is a positive density for sufficiently small $|u|$. On the finite cube,

$$\text{Ent}_{\mu_J}(1 + uf) = \frac{1}{2}u^2 \text{Var}_{\mu_J}(f) + O(u^3), \quad \mathcal{E}_J(1 + uf, \log(1 + uf)) = u^2\mathcal{D}_J(f) + O(u^3).$$

Letting $u \rightarrow 0$ and then taking the supremum over f proves $C_{\text{EP},J} \geq R_J/2$. For a requested exponent slack δ , apply [Theorem 1.1](#) with slack $\delta/2$ and then absorb the fixed factors for sufficiently large n .

If g_t is the density of $\nu e^{t\mathcal{L}_J}$ relative to μ_J , reversibility gives $g_t = e^{t\mathcal{L}_J}g_0$ and

$$\frac{d}{dt} \text{Ent}_{\mu_J}(g_t) = -\mathcal{E}_J(g_t, \log g_t).$$

The chain is irreducible, so $g_t > 0$ for $t > 0$ even when g_0 vanishes at some states. The defining inequality and Gronwall give the displayed decay; continuity includes $t = 0$. Conversely, differentiating any uniform exponential entropy bound at zero for positive initial densities recovers its entropy-production inequality. Thus $1/C_{\text{EP},J}$ is the optimal rate c in bounds of the displayed form $\text{KL}(\nu e^{t\mathcal{L}_J} \parallel \mu_J) \leq e^{-ct} \text{KL}(\nu \parallel \mu_J)$ that hold for every initial law and every time. $\square$

If $\text{KL}(\nu \parallel \mu_J) \leq H$ for a fixed H , Pinsker's inequality therefore gives total-variation accuracy $\varepsilon > 0$ in $O_{H,\varepsilon}(C_{\text{LS},J})$ per-site time. A point mass has initial relative entropy $\log(1/\mu_J(x)) = O(n)$ under (20.5), recovering the logarithmic factor in the worst-start upper bound. The exponent for $C_{\text{EP},J}$ concerns this uniform rate; it does not assert that every initial law relaxes at the slowest rate.

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